

Rosa Scherson · Daniel P Faith *Editors*

Phylogenetic Diversity

Applications and Challenges in
Biodiversity Science

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Rosa A. Scherson • Daniel P. Faith
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Science

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Introduction

“Biodiversity” means the variety of life. Research on the biology of biodiversity is about understanding the scope and importance of this living variation. Given that society clearly faces a biodiversity crisis, it is ironic that this meaning of the term “biodiversity” perhaps has become less clear. As a popular buzzword, it sometimes has been linked to all of nature, to a specific species of interest, or to multiple aspects of ecology as a way to link to ecosystem services. But the *biology* of biodiversity is fundamentally a story about the *variety* of life, and its values to society fundamentally are the values of maintaining such *variety*.

Phylogenetic diversity (referred to as “PD”; Faith 1992) over the past 25 years has become an important focus of research on living variation and its value. Studies on PD have resulted in a particularly rich area of research activity globally and, remarkably, have spanned nearly all major taxonomic groups, including microbes. PD research has covered global to local scales. In any given week, one new study may report a global map of PD losses due to climate change, and another may report on the consequences of the PD loss in the microbial community of the human gut. Faith’s much-cited original paper on phylogenetic diversity (Faith 1992) began a transformation in biodiversity science. Prior to 1992, a handful of papers had used the term “phylogenetic diversity”, largely descriptively. Following Faith’s (1992) paper (and other early papers on PD), there are now thousands of papers in *Web of Science* with the keyword “phylogenetic diversity.”

One aspect of this increased popularity of the term “phylogenetic diversity” is that it has taken on a range of meanings. For example, among ecologists, “phylogenetic diversity” now typically refers to a phylogenetic analogue of any of the vast number of so-called ecological “diversity” indices developed at the species level over the past century. In this book, we will focus on phylogenetic diversity as it relates to assessments of biodiversity interpreted as the variety of life, following the original use of the term in Faith (1992). This accords also with the perspective (reviewed in Faith 2017) that any measure of “biodiversity” indicates a variety because it (at least

implicitly) “counts up” some units of interest (for PD, typically these units are regarded as “features”). This may seem restrictive, but in fact PD, as a measure of biodiversity that counts up units, means that we also have a whole family of other associated calculations that we can carry out on those same units (e.g., calculating “complementarity”: the gain in biodiversity when an additional species is conserved). As illustrated by the importance of complementarity in decision-making, this family of associated calculations is important to biodiversity conservation policy and priority setting. We will see many of these PD-related calculations in the chapters of this book.

Because PD links phylogeny to a measure of variation (e.g., “feature diversity”), it also creates a clear link to the early rationale that we should conserve biodiversity because it represents a storehouse of possible future benefits (“option value”; for discussion, see Faith 2017). For these reasons, PD now is regarded as “a resonant symbol of the current biodiversity crisis” (Davies and Buckley 2011) and is used in important global conservation research and assessments. For example, the high-profile EDGE of Existence program is a global conservation initiative focusing on those threatened species that represent a significant amount of unique PD (Isaac et al. 2007). This PD-based research, in the last 5 years, has expanded to carry out research and conservation work in over 20 countries. Another high-profile global initiative, building in part on PD research, is OneZoom (www.onezoom.org), which displays large phylogenetic trees and provides extensive PD information, supporting not only research but also education about the biology of biodiversity. Meanwhile, microbial ecologists have continued to pioneer PD applications, highlighting the importance of biodiversity conservation in this context. Widely available software tools such as UniFrac (Lozupone and Knight 2005) and Qiime (Caporaso et al. 2010) have established PD as part of the standard assessments in the analyses of microbial biodiversity and human diseases.

A goal of this book is to bring together the philosophical, technical, and practical perspectives across these different scales of application, from global to local. Our book addresses the challenges and future opportunities in PD research and applications. With this in mind, we have divided the book into three sections: a section on general issues regarding measures of PD, a second section on methodology and tools, and a third section on case studies that have used PD in different areas of the world, over a variety of taxonomic groups.

Historical perspectives on these developments are important in helping us understand why phylogeny is relevant to conservation and ecology. The first section addresses these aspects. The first chapter traces the roots of phylogenetic considerations and biodiversity value back to IUCN 1980. The third chapter shows how not only scientists but also the public can be made aware of the importance of considering evolution for conservation and can better understand the concept of the tree of life. The second chapter in this section turns to perhaps one of the success stories in engaging the public, the EDGE program, which successfully uses PD as a criterion for setting conservation priorities and acting on them. Together, these chapters document the reasons why phylogeny is important in conservation, and also document the increasing awareness of this by scientists and the general public.

The second section describes important steps in the development of techniques, addressing challenges and applications and even policy relevance. The first paper follows on from the important early work, on EDGE and its precursors, focused on averting expected loss of PD. It shows that we can create useful indicators with limited data, providing useful reporting on the option value of PD. The second chapter follows the theme of practical issues by looking at the range of resolution and metrics. This is important in recognizing that the choice of indices depends upon the purpose. The third chapter reinforces the importance of the availability of a range of indices relevant to PD. This chapter discusses applications ranging from regional to global scales. Together, these chapters show that the methods are still evolving, and that the methods increasingly are well tuned-in to the application context and to the need for robustness to limited data.

Our third and final section turns to case studies, which not only address specific needs but also are feeding back in providing general lessons for our tools and methods. The first chapter illustrates this; a case study explores how well patterns can be revealed even with limited data, suggesting that relevant studies can and should be done even when comprehensive data are not yet fully available. The next two chapters show how the increase in data and the growth of knowledge have been integrated successfully in the developing story of the PD of two of the world's hotspots. In considering case studies, it is apparent that successful integration of PD benefits from our understanding of evolutionary processes. Our final chapter shows that this is also true for ecological processes. This chapter explores how, at a more localised scale, PD is relevant to ecological restoration.

Together, the papers in these sections illustrate well the increasing interaction between previously separate arenas of application of PD methods, producing a vibrant discipline of phylogenetic diversity conservation, with shared ideas, values, and tools.

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Contents

1	Phylogenetic Diversity and Conservation Evaluation: Perspectives on Multiple Values, Indices, and Scales of Application	1
	Daniel P. Faith	
2	The Use of EDGE (Evolutionary Distinct Globally Endangered) and EDGE-Like Metrics to Evaluate Taxa for Conservation	27
	Nick J.B. Isaac and William D. Pearse	
3	Biodiversity, the Tree of Life, and Science Communication	41
	James Rosindell and Yan Wong	
4	Indicators for the Expected Loss of Phylogenetic Diversity	73
	Daniel P. Faith, Simon Veron, Sandrine Pavoine, and Roseli Pellens	
5	Phylogenetic Resolution and Metrics of Biodiversity and Signal in Conservation	93
	Nathan G. Swenson and Samantha J. Worthy	
6	Phylogeny-Based Measurements at Global and Regional Scales	111
	Shawn W. Laffan	
7	Phylogeny-Based Measures of Biodiversity When Data Is Scarce: Examples with the Vascular Flora of Chile and California	131
	Rosa A. Scherson, Taryn Fuentes-Castillo, Rafael Urbina-Casanova, and Patricio Pliscoff	
8	Priority Areas for Phylogenetic Diversity: Maximising Gains in the Mediterranean Basin	145
	Simon Veron, Daniel P. Faith, Roseli Pellens, and Sandrine Pavoine	

9	Evolutionary Diversity Patterns in the Cape Flora of South Africa	167
	Félix Forest, Jonathan F. Colville, and Richard M. Cowling	
10	Effects of Phylogenetic Diversity and Phylogenetic Identity in a Restoration Ecology Experiment	189
	Andrew L. Hipp, Mary-Claire Glasenhardt, Marlin L. Bowles, Mira Garner, Bryant C. Scharenbroch, Evelyn W. Williams, Rebecca S. Barak, Amy Byrne, Adrienne R. Ernst, Emily Grigg, Meghan G. Midgley, Hayley Wagreich, and Daniel J. Larkin	
	Index	211

Chapter 1

Phylogenetic Diversity and Conservation Evaluation: Perspectives on Multiple Values, Indices, and Scales of Application



Daniel P. Faith

Abstract “Phylogenetic diversity” and its abbreviation “PD” have now become popular terms describing a fundamental aspect of biodiversity based on phylogeny. After more than 25 years of work on PD (following the 1992 paper in *Biological Conservation*), methods and applications have explored a wide range of taxonomic groups and geographic scales. PD provides a way to address biodiversity at the level of features or characteristics of species, based on its well-corroborated model linking phylogeny and feature variation. The quantification of feature diversity justifies PD as a measure of option value – the value of living variation in keeping options open for society. This justification for PD in biodiversity conservation gives attention to often-neglected arguments for the value to society of biotic diversity. These largely global option values are complemented by the “insurance” value of PD at the local ecosystem scale. Microbial applications of PD, particularly in human health studies, have successfully implemented a range of PD calculations, including PD dissimilarities among samples. Reduced microbial PD in the human body may indicate reduced resilience, and it is now associated with many human diseases. “Macrobial” ecology has been less successful in integrating PD into a consistent coherent approach. Here, the traditional recognition of many “diversity” indices has been extended to phylogeny. PD as a “biodiversity” measure is confounded with the multitude of phylogeny-based diversity indices describing various ecological factors. Greater integration among the different areas of PD application could better establish PD as a core biodiversity measure, with a shared toolbox providing a range of PD-related calculations.

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Background

My title echoes the title of the paper, “Conservation evaluation and phylogenetic diversity” (Faith 1992) which proposed “phylogenetic diversity” (“PD”) as a natural phylogeny-based measure of biodiversity. The definition of PD provided in that paper was deceptively simple: the PD of a subset of species (or other taxa) from a phylogenetic tree is the total length of the branches spanning that subset on the tree. I say “deceptively” for two reasons. First, that simple definition corresponded to a not-so-simple framework to quantify the relative “feature diversity” of subsets of taxa, equating features with the units of biodiversity that we would like to count up. Second, feature counting meant that the simple definition of PD did not just define one simple “index.” Instead, it established a basis for a whole family of associated calculations (the PD “calculus”) that could be used for practical decision-making in biodiversity conservation. As I will discuss in this perspectives paper, these not-so-simple aspects of PD continue to be debated in the many studies that now apply, critique, and extend this popular approach.

As my title also indicates, the many published PD studies now range widely across taxonomic groups and scales of application. Prior to 1992, a handful of papers had used the term “phylogenetic diversity,” largely descriptively. Following the Faith (1992) paper, there have been about 5000 papers in *Web of Science* with the key term “phylogenetic diversity.” At the time of this writing, the 1992 paper is the most-cited paper ever in *Biological Conservation* from the more than 7000 papers in that journal spanning more than 40 years. While there are some partial reviews of this work, for example, by Porter and Hajibabaei (2018) in the genomics context, in a recent book (Pellens and Grandcolas 2016), and in Faith (2015, 2016, 2017) (see also Chap. 7), there appears to be no single review covering the full range of PD studies.

Some exciting areas of PD application have departed from organismal phylogeny. As examples, PD is now applied to conservation of “phylogenetic language diversity” (Gavin et al. 2013), and PD is used as an analogous biodiversity measure for hierarchies other than phylogenies (environmental hierarchies, e.g., Faith 2013, and functional trait hierarchies, “FD”, Petchey and Gaston 2002). As I will discuss below, more communication is needed among all these different areas of PD application. This need includes better integration of the PD work in microbial and macrobial ecology.

As preparation for discussion in this chapter of the ongoing applications of PD, it is useful first to look back at PD’s “prehistory” (the history of a term before it was invented; see Faith 2017). This reveals how PD naturally emerged from some important precursors, including early discussions about biodiversity’s value, about “taxonomic distinctiveness,” and about how phylogeny informs us about characters or features.

It is commonly thought that the development of methods for using taxonomy and phylogeny in biodiversity conservation was an “explosive radiation,” given that several notable papers appeared at about the same time (e.g., May 1990;

Vane-Wright et al. 1991; Faith 1992; Weitzman 1992; Crozier 1992). The reality is that there was significant foundational thinking on this topic over the decade prior to the appearance of those papers. Foremost among this work was the “World Conservation Strategy” report by IUCN (1980). This single report foreshadowed later thinking on biodiversity option values, on ecological integrity, and on taxonomic distinctiveness (all relevant to this chapter) and also on the integration of distinctiveness and extinction threat (relevant to Chap. 4). These early IUCN influences complemented other foundations for PD, including early influential work in systematics that explored the links from phylogeny to features (discussed in Faith 1992).

As part of this Background section, I will discuss these precursors below, under the headings “Taxonomic Distinctiveness,” “The Link from Phylogeny to Feature Diversity,” “Option Values,” and “Insurance Value/Ecological Integrity.” With these in mind, the main body of this chapter then will explore some current challenges in applications of PD. Here, my discussion of microbial and “macrobial” ecology applications will complement the discussion of PD at the regional/global conservation scale, found in other chapters in this book (Chaps. 4 and 8).

Taxonomic Distinctiveness

The development of PD involved sorting out what we mean by “distinctiveness” as compared to “diversity.” For IUCN (1980), the reference to taxonomically “distinct” species implied differences among species, presumably in characters. IUCN referred to “the greater the gap between the nearest related family (or genus) and therefore the more distinct” and noted:

The size of potential genetic loss is related to the taxonomic hierarchy because, ideally at least, different positions in this hierarchy reflect greater or lesser degrees of genetic difference, and hence difference in such variables as morphology...

Following IUCN, this basic argument about taxonomically distinct species appears to have inspired early work on related priority setting. Faith (1994a), for example, noted that several state agencies in Australia had been working on taxonomic distinctiveness and priorities during the late 1980s. Later important work focused specifically on quantitative measures of taxonomic distinctiveness (Weitzman 1992; Vane-Wright et al. 1991). The Faith (1992) reference to distinctiveness was more oblique, but it very much motivated the basics of an emerging PD calculus, as described below. PD was from the outset much more than a single index of biodiversity; it also provided a set of associated calculations that quantified distinctiveness and complementarity (gains and losses in biodiversity). Faith (1992) established an important perspective on distinctiveness and diversity by linking these to the implicit counting up of features of species. Distinctiveness of a species then was not a static quantity, but would depend on the context:

Species that are taxonomically distinct will be expected to make a large contribution to some overall measure of diversity of any subset of the total set of species. This is apparent if species are replaced as the basic units (or attributes) of biological diversity by features of

species. Taxonomically distinct species then contribute more to the diversity of a given subset because they contribute different “features”.

Thus, the Faith (1992) discussion of distinctiveness and features echoed the IUCN reference to “genetic difference” and morphology.

Faith (1992) referred to the contribution of the terminal phylogenetic branches to the total PD as the “length of a terminal branch (corresponding to the number of uniquely derived features for the terminal taxon)” (Faith et al. (2004) later described this as the “PD endemism” or the “unique PD contribution” of a species). More generally, this complementarity contribution is calculated as the gain in representation of feature diversity if another species or another area is added to the set (Faith (1992) simply called this “G” for gain). G is the increment in the PD estimate of the number of new features resulting from the addition of a new taxon to a subset.

Thus, we not only might calculate the total PD, say, of the mammal species represented in a set of protected areas but also use PD calculations to add and subtract and compare sets. This discussion reveals how “biodiversity” – here, an estimated total number of features – is related to companion calculations such as distinctiveness or complementarity that reflect *additional* features relative to some existing set. This family of calculations is important; to select a *set* of species that collectively captures maximal biodiversity, we cannot just select individual species that were designated at the start as individually “distinctive.” We must use dynamic distinctiveness or complementarity (a process nicely illustrated in examples in Faith 1992 and later papers).

This consideration is fundamental to conservation policy. I note that the World Conservation Conference (WCC 2012) proposed distinctiveness as an important conservation criterion, resolving to halt the loss of evolutionarily distinct lineages. However, any efficient program for preserving PD must utilize the dynamic forms of distinctiveness, within a program perhaps resolving to halt the loss of our evolutionary heritage.

The Link from Phylogeny to Feature Diversity

PD is one example of a general biodiversity framework (Faith 1994b, 2017) in which a process-based model is used to make inferences about the relative number of different “units” contained in any given set of objects. For example, the “ED” (“Environmental Diversity”) strategy infers the relative number of ecological traits among different sets of species, based on a process-based model related to adaptations to environmental gradients and “niche space” (see e.g., Faith 1994b, 2015, 2017). PD was based on the well-established idea that phylogenetic pattern is informative because shared features (the “units” of biodiversity) are accounted for by shared ancestry (Faith 1992, 1994a,b). This means that the basic PD calculation of gains and losses of feature diversity is based on summed phylogenetic branch lengths.

Phylogeny is informative about features generally. Not only do we not know, in general, the future value of different features (discussed in the next section), but also, we cannot even list the features for most species. Phylogenetic pattern provides one way to estimate and quantify this variation at the feature level. A species complements others in representing additional evolutionary history (Faith 1994a,b), as depicted in the branches of the estimated phylogeny.

As noted above, the degree of complementarity reflects the relative number of additional features contributed by that species. For example, given some subset of species that are well-protected, and two species in that taxonomic group that are endangered, the priority for conservation investment may depend on the relative gains in feature diversity (the complementarity values) expected for each species. We do not know in practice what all the actual features are, but we can make predictions about these relative gains and losses.

How well can phylogeny inform us about feature diversity? Faith (1992) did not propose an untested model or assumption for this link. In fact, the PD assumption that shared ancestry explains shared features had reasonable justification based on already accomplished tests. In this context, an established form of testing was noted by Faith (1992): the PTP (permutation probability test of Faith and Cranston (1991)) evaluates whether the shared-ancestry model explains the given feature/character data better than expected by chance (e.g., compared to randomized characters). Faith (1992) cautioned against relying on trees (or “cladograms”) that do not pass PTP and so do not provide corroboration (Faith and Cranston 1992) of this shared-ancestry/shared-features model:

Cladograms based on a small number of characters, or on characters that exhibit large amounts of homoplasy (convergences and reversals in the derivation of features), are probably less reliable, as indicated by recently developed statistical approaches for evaluating cladograms. (Faith and Cranston 1991, and references therein)

The PTP test of Faith and Cranston (1991) evaluates how well a set of observed characters or features all can be explained by shared ancestry on a single phylogeny. PTP counts up the total number of “steps” on the tree – if a shared feature is accounted for by shared ancestry, only one step is needed. The total count (“parsimony” value) is compared to parsimony value scores under a null model where the features are randomly assigned to species. A significant result supports the PD model. PTP therefore is not just any arbitrary test for “phylogenetic signal”; it specifically corresponds to evaluating the PD shared-ancestry model.

The nature of these tests is important. Because PD does not assume that every feature is accounted for by model (Faith 1992), it is not strictly falsifiable. However, the PTP test is well-established as a method for corroboration assessment (see Faith and Cranston 1991, 1992). Corroboration of a hypothesis is found when the apparent supporting evidence for the hypothesis could not have been found easily by chance alone. For a set of characters/features, the count of the number of shared features explained by shared ancestry is evidence for the PD model, and this provides corroboration when that count could not have been found easily by chance alone.

Since 1990, there have been many PTP tests using actual character data sets, over many different taxonomic groups. How well have these supported the PD model? While Faith (1990) documented a dramatic case where the characters failed the PTP test, most character sets have passed the test. For example, Slowinski and Crother (1998) examined PTP tests on 40 published and unpublished character data sets, and all but one character set passed. Wilkinson et al. (2002) surveyed existing tests and reached the same conclusion, noting:

PTP is a relatively conservative test statistic. Over a range of numbers of taxa, characters, character states, and relative proportions of character states the Type 1 error rate is mostly <5%.

Thus, PTP tests often “reject the null hypothesis” and this rarely is an incorrect rejection. We can conclude that the shared-ancestry model linking PD to feature diversity is well corroborated based on many published PTP tests that corroborate the PD model.

From the outset, Faith (1992) also emphasized that PD should not be expected to magically make inferences about every favorite character or feature. While we have the intuitive, fundamental, link between shared ancestry and shared features, the history of phylogenetics shows that homoplasy (“extra steps” on the phylogeny, as for convergent evolution) also is common. Faith (1992) cautioned that “the diversity patterns for some features (namely those incongruent with the cladogram in implying extra steps) may be poorly predicted by phylogenetic diversity.” Faith (1992, 1994a,b, 2015) suggested that alternative patterns and models (including the “ED” approach; see Faith 1994b) are more effective for inferences about such features (or ecological traits).

The early development of PD as a measure of biodiversity at the level of features recognized several important properties that are relevant to issues often raised in PD research and applications. First, while PD is the measure of biodiversity at the level of features, that inference of features allows many associated companion calculations. Indeed, any index conventionally defined at the species level (endemism, dissimilarity, expected loss, etc.) has a PD counterpart (Faith 1994b; see also Faith 2013, 2017). Second, PD drew upon a link between phylogeny and features that was already well tested and corroborated. Third, shared convergently derived ecological traits were highlighted as not well predicted by PD, and an alternative approach based on a shared-habitat/shared-traits model was proposed (Faith 1989, 1992, 1994b; see also Faith 2015, 2017).

Option Value

PD provides a useful calculus at the level of features for conservation decision-making, but why should we worry about conserving all that feature diversity? Faith (1992, 1994a,b) proposed that PD as a measure of feature diversity provided a fundamental current value of biodiversity – called “option value.” This reflects the

idea, as Faith (1992) described it, that “society may be able to benefit (e.g. economically) from features of these species in response to future needs.” IUCN (1980) not only provided foundations for PD’s recasting of distinctiveness but also provided a rationale for “option value.” However, as described below, IUCN (1980) also reflected insights on such values from the previous decades.

The term “biodiversity” refers to the variety of life, and the earliest discussions of its values (in contrast to discussions of the value of individual species) logically reflect the value to society of such living variation. In fact, such discussions precede the actual invention of the term “biodiversity” (around 1985). Early discussions of the value of variety referred to “biotic diversity,” for example. Faith (2017) sketched what he called the “prehistory” of the term “biodiversity” – basically the history of that term before it was invented.

A message emerging strongly in the 1970s was that the crisis of disappearing species could be thought of as a loss of biotic diversity and that this loss matters because it means a loss of options for future generations (e.g., Ehrenfeld 1970; Iltis 1972; Haskins 1974; Roush 1977; see also reviews by Mazur and Lee 1993; Farnham 2007). For example, back in 1972, the botanist, H. H. Iltis, argued to “preserve sufficient diversity of species and of ecosystems” because “we will never reach a point where we shall know which organisms are going to be of value to man and which are not” (Iltis 1972). Myers (1976) argued that loss of what he called “the spectrum of species” “will affect generations into the indefinite future, whose options to utilize species in ways yet undetermined should be kept open.” A clear message from these early prehistory papers was that biotic diversity or living variation provides what we can call “option value” or “maintenance of options.”

Later, IUCN (1980), in making it clear that conservation “is for people,” argued that “preservation of genetic diversity” (their stand-in for the not-yet-defined “biodiversity”) “is both a matter of insurance and investment to keep open future options.” The investment aspect is “prudent,” they argued, because “we cannot predict what species will become useful to us.” This nicely echoes the arguments from the previous decade about the option value of biotic diversity. In the next section, I will return to this quote to consider the companion “insurance” aspect.

Option value of biodiversity typically refers to a global benefit for people. Several chapters in this book look at regional/global PD conservation (see Chaps. 4 and 8). Much of this regional/global PD work links to its option value rationale (see also Faith and Richards 2012; Jetz et al. 2014; Thuiller et al. 2011, 2015; Laity et al. 2015; Mouillot et al. 2016; González-Orozco et al. 2016; Pollock et al. 2017). This rationale also extends below the species level. Faith (1992) also proposed within-species geographic priority setting using PD on phylogenetic patterns for within-species genetic variation. Carvalho et al. (2017) developed this further as “the intraspecific phylogeography of multiple species into spatial prioritization.”

The PD study of Mouillot et al. (2016) nicely summarized the recognition of PD as indicating global biodiversity option value:

“Global marine protected areas do not secure the evolutionary history of tropical corals and fishes”: “... phylogenetic diversity, represented by the Tree of Life, is becoming an increasingly important component of conservation science since it represents the breadth of

evolutionary history and supports biodiversity benefits and uses, often unanticipated, for future generations.”

Larsen et al. (2012) highlighted the challenge in biodiversity conservation to find a “robust proxy” for global option values, and they concluded that “maximizing the retention of phylogenetic diversity (PD) should also maximize option value.”

Forest et al. (2007) carried out a further evaluation of PD as a measure of feature diversity and option value. Using a phylogenetic tree, and a handbook for useful genera found in the Cape hotspot of South Africa, they demonstrated that if we did not know about these medicinal, food, and other uses, then preserving sets of species with high PD would be a good way to preserve these unknown benefits.

Support for PD as a measure of option value is found also in philosophy of science work (e.g., Maclaurin and Sterelny 2008; Faith 2017) and among economists (e.g., Nehring and Puppe 2004). Option value of biodiversity has been promoted well by conservation groups (Gascon et al. 2015) in presenting many examples of surprising benefits from biodiversity. Gascon et al. also noted the “measurement” problem and pointed to “phylogenetic diversity” as a likely good measure of option value.

Chapter 4 of this book describes the important assessment, by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES), of PD as a measure of maintenance of options. The Asia-Pacific IPBES assessment (Davies et al. [in press](#)) complemented the assessment of PD status with examples of recent unexpected benefits in the region (e.g., noting how the venom of an Australian funnel web spider was found to be the unlikely source for a drug to ward off brain damage caused by strokes; Chassagnon et al. 2017).

PD measures option value, but this does not mean that its conservation utilizes measures only quantifying total PD. The calculus of PD is a calculus of option value. The Laity et al. (2015) study illustrated how the most useful measures for planning and decision-making typically will employ calculations based on PD (endemism, complementarity, etc.; see also Mouillot et al. 2016; Pollock et al. 2017; Veron et al. 2016) rather than total PD. Studies continue to move beyond the local scoring approaches that simply score total PD (e.g., of a grid cell). Scherson et al. (2017) applied some important recent extensions of the PD calculus for spatial analyses, in their study “Spatial Phylogenetics of the Vascular Flora of Chile.” PD of individual grid cells, which may not on its own be informative for regional conservation, was complemented by maps showing places with high PD endemism. The calculus has been important in highlighting how PD and species richness calculations for decision-making, such as complementarity, are disconnected (see Forest et al. 2007).

This fundamental point about the PD calculus is still not universally appreciated. For example, Lindegren et al. (2018) used within-area PD for marine conservation priority setting at the global scale, ignoring the marine PD global planning of Mouillot et al. (2016), based on complementarity. Lindegren et al. claimed that lack of correspondence among areas between PD and richness of other aspects of biodiversity or of ecosystem services implies conflicting objectives requiring trade-offs in designing a protected areas system. In fact, when we consider what matters in

designing a protected areas system – complementarity – it is apparent that an efficient system could require no trade-offs, even when richness values lack correspondence.

This section has documented the very early roots of biodiversity option value (even before the term “biodiversity” was invented) and how this is increasingly appreciated as the motivation for PD conservation, particularly at the regional and global scale. The next section examines a closely related biodiversity value that can be associated with PD, providing a possible stronger link to the ecological theme of this chapter.

Insurance Value/Ecological Integrity

In contrast to the previous section, this section will not highlight an early connection to PD. Faith (1992, 1994a,b) did not consider the within-ecosystem community level and did not really consider insurance value (beyond the basic idea that PD is associated with evolutionary potential). Biodiversity insurance value has a range of interpretations but generally relates to resilience in the face of unanticipated environmental changes; viewed anthropocentrically, this maintains ecosystem functions and services. I referred above to the IUCN (1980) arguments that “preservation of genetic diversity is both a matter of insurance and investment.” I noted their argument that the investment aspect, related to option value, is “prudent” because “we cannot predict what species will become useful to us.” I now consider the other half of their statement, that insurance aspect is prudent as well, because we cannot predict which species will maintain ecosystem services or other benefits in the face of change.

IUCN (1980) nicely combined both of these aspects of biodiversity value, in describing arguments for the conservation of “the range of genetic material found in the world’s organisms”:

we may learn that many species that seem dispensable are capable of providing important products, such as pharmaceuticals, or are vital parts of life-support systems on which we depend.

This value dualism as described by IUCN seems to have been a precursor for later summaries of these dual values. For example, Bartkowski (2017) descriptively linked option value/investment to “demand” and insurance value to “supply”:

the two perspectives – insurance and options – are inherently interlinked; however, they depend on different types of uncertainty (supply vs. demand), which makes the differentiation sensible.

Support for these ideas also can be found in the early history of discussions of the “insurance” value of biodiversity, preceding IUCN (1980). This history overlaps with that of option value. Both Roush (1977) and Ehrenfeld (1970), in discussing the value of biotic diversity, distinguish between a within-ecosystem functioning/stability argument and a very different option value argument. Norse and McManus

(1980) similarly discussed both global biodiversity and within-ecosystem diversity for functioning.

Later, Ehrlich and Wilson (1991) echoed these ideas in listing three reasons why we care about biodiversity. One was “moral responsibility,” and the second was the future unanticipated benefits (option value). The third was the support for the “recognized ecosystem services provided by natural ecosystems.” In the latter case, they made an indirect link to biodiversity in arguing that “diverse species are the key working parts” of such ecosystems. This suggests that this value dualism can be summarized as the value of biodiversity – all the components or units – for unanticipated benefits (option value) or for unanticipated systems support (insurance value).

All this seems straightforward, but insurance value of “biodiversity” cannot be discussed without considering the definition of “biodiversity” and how it relates to ecological “diversity.” It is revealing that the Ehrlich and Wilson (1991) insurance argument that “diverse species are the key working parts” echoes the IUCN (1980) insurance phrasing pointing to “vital parts of life-support systems.” Both phrases are vague; there are lots of ways that “vital parts of life-support systems” and “diverse species” as “key working parts” might provide insurance that functions and services continue in the face of unanticipated change. Generally, these turn out to be more about ecology than about “biodiversity” in the sense of counting up units.

As an example, Bullock et al. (2017), in discussing insurance value, argued that “resilience is contained in the dynamic and innumerable interactions.” Similarly, Pascual et al.’s (2015) statement that the insurance value of soil biodiversity could be linked to “particular soil organisms” also clearly departed from biodiversity as variety. Both of those examples of ecological aspects we might want to measure could fall under the broad banner of “ecological diversity” measures.

One might anticipate that these ecological aspects of insurance (abundance of key species, interactions among species, etc.) logically then would not be characterized as insurance derived from biodiversity as variety. However, this highlights what is perhaps the biggest conundrum in current biodiversity science. On the one hand, “biodiversity” can be interpreted as counting up some nominated units (as advocated by Faith 2017). On the other hand, “biodiversity” might be defined or interpreted as including all those ecological factors, so providing a self-fulfilling prophecy that “biodiversity” is critical to producing and insuring ecosystem services (for discussion, see Faith 2017).

If “biodiversity” is accepted strictly as variety and a counting-up, this has an intuitive link to a natural companion of insurance value – the precautionary principle (basically, the idea that it is prudent to keep all the units, in the absence of knowledge about what the importance of different individual units will be in the face of change). Here, there is a nice parallel with option value and unanticipated benefits. With good knowledge, we may guess the kinds of species or other units likely to deliver unanticipated benefits, but in the absence of such insider knowledge, it is prudent to maintain the biodiversity (all the units), as a kind of investment (to use IUCN wording). Just as we have a *precautionary* principle that addresses insurance, we have a *pre-beneficiary* principle that addresses option value/investment. Biodiversity

is valuable because it helps us to be ready for unanticipated “cautions” (changes amounting to disbenefits) and helps us to be ready for unanticipated benefits.

Biodiversity, in the sense of counting up units, therefore has a clear role as insurance, but if we accept that simple perspective, what then happens to all those ecological factors that also may well play a role in insurance of functions and services? One possible reconciliation is that we recognize that these ecological aspects typically are not “biodiversity” but are related to “ecological integrity.” Here again, IUCN (1980) has provided early guidance that is reflected in its current definition of “ecological integrity”:

maintaining the diversity and quality of ecosystems and enhancing their capacity to adapt to change and provide for the needs of future generations. (https://www.iucn.org/downloads/en_iucn_glossary_definitions.pdf)

Ecological integrity can include lots of ways in which “vital parts” and “key working parts” contribute – including single keystone species, key interactions among species, and many other aspects of ecology. If we know the system well, measures of integrity and insurance can reflect this ecological knowledge, but in the absence of such knowledge, it is precautionary to maintain biodiversity as insurance.

Thus, one way to view the current biodiversity conundrum is that we have a choice. Either (1) (as above) biodiversity is seen as counting up units, with “ecological integrity” complementing that by capturing the rich ecological story of resilience and insurance value, or (2) alternatively, the ecological factors (often expressed as “diversity” measures) are all taken to be part of “biodiversity.”

This conundrum is evident already at the species level, but it has now imposed itself on the phylogenetic level – where every classic species-level ecological diversity measure has a phylogenetic counterpart. Are these measures all measures of “biodiversity” and, indeed, “phylogenetic diversity”? Or are they measures that sometimes might be “biodiversity” but mostly are related to “ecological integrity”?

In the sections below, I will examine the concepts and applications of PD in two sections about ecosystems and community ecology: microbial and “macrobial.” Microbial workers have established a coherent framework that uses PD as a primary biodiversity measure while also taking advantage of the PD calculus. Macrobial workers, in contrast, have struggled to find a coherent conceptual framework, with most any phylogenetic index called “phylogenetic diversity” and with no awareness of the recognized utility of the PD model and the PD calculus.

Microbial Community PD

Applications of PD in microbial ecology owe much to the pioneering work of Lozupone and Knight (2005) and Caporaso et al. (2010), who have provided microbial workers with guidelines and tools for applying PD and its calculus (see also Faith et al. 2009).

A review by Lozupone and Knight (2008) refers not only to overall PD but also to “G” (see above) as part of the PD calculus: “PD tells us about the overall diversity of the lineages within a given community, G is a measurement of which communities contain the most previously unseen diversity (i.e. are the most phylogenetically unique).” Thus, early on, microbial workers appreciated the extension of distinctiveness to dynamic, context-dependent, complementarity values.

“G” and complementarity also linked strongly to Lozupone and Knight’s “unique fraction” concept of dissimilarity:

The unique fraction metric, or UniFrac, measures the phylogenetic distance between sets of taxa in a phylogenetic tree as the fraction of the branch length of the tree that leads to descendants from either one environment or the other, but not both . . . This measure thus captures the total amount of evolution that is unique to each state. . .

Lozupone and Knight (2008) noted that UniFrac “falls within a broader family of divergence-based β diversity measures related to PD that differ in the details of how the unique and shared branches are combined.”

Both PD and UniFrac PD dissimilarities have been used in thousands of microbial studies and contribute to the growing evidence that resilience for microbial communities is linked to a range of aspects of biotic diversity (Shade et al. 2012; Lozupone et al. 2012). In this chapter, I will focus on the human health aspects of microbial ecology and PD. It is now well-known that many so-called “noncommunicable diseases” are associated within reduced microbial diversity in the human microbiome (for review, see Convention on Biological Diversity and World Health Organization 2015; see also Jason Lloyd-Price et al. 2016).

I will focus here also on some examples drawn from studies on the human digestive system. Many of these studies reveal the sensitivity of PD relative to other indices in distinguishing disease factors. These example studies discuss possible interpretations of observed reduced PD as relating to resilience and health. For example, Lindheim et al. (2017) in the study “Gut Microbial Diversity in Women with Polycystic Ovary Syndrome” found affected individuals to have reduced microbial PD of the gut microbiome. This was interpreted as an indicator of reduced health and stability of the microbial community. Lindheim et al. found no significant pattern for an alternative diversity measure (Pielou evenness index), suggesting that PD and phylogeny best captured differences in the microbial communities.

Human Inflammatory Bowel Disease and the Loss of Resilience

Bassett et al. (2015) explored intestinal microbial communities and inflammation, as part of a study of human inflammatory bowel disease (IBD). Bassett et al. noted that IBD “is a chronic intestinal disease where the resident microbiota contributes to disease development, yet the specific mechanisms remain unclear.” They examined changes in the caecal microbiota, including changes in PD, associated with colonic inflammation. They concluded that reduced PD implies a greater risk of

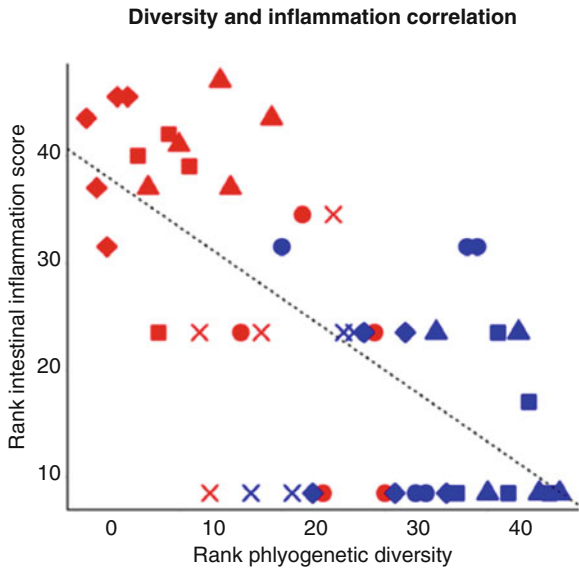


Fig. 1.1 X axis is PD amounts and Y axis is inflammation rating. Blue points indicate less susceptible individuals and red points indicate more susceptible individuals. Shapes of the points indicate treatment groups. Overall, the plot shows that increased inflammation was associated with a decrease in caecal microbial PD. For further information, see Bassett et al. (2015). Figure reproduced from Bassett et al. (2015)

inflammation because the microbial community of the intestine is less able to recover from disruptions to normal “homeostasis” (Fig. 1.1). Thus, PD was associated with a kind of insurance value.

There is support for this interpretation from other PD studies on IBD. Lozupone et al. (2012) similarly concluded “High functional response diversity in human gut-adapted bacteria is likely because phylogenetically disparate microbes often perform similar metabolic functions,” and that loss of PD therefore is a loss of resilience. Zhang et al. (2017) noted “it remains unclear whether the observed changes in phylogenetic composition are causative for the development of IBD or simply a consequence of an altered intestinal environment during the progression of IBD” and concluded that “a precautionary approach is advocated by many.” Thus, microbial PD studies illustrate a case where our limited current knowledge justifies a precautionary approach, maintaining high PD for insurance and resilience (PD as “bowel-logical” diversity).

The biodiversity (PD) conservation actions also may involve restoration of PD. In a study on gut microbial and inflammatory bowel disease (IBD), Khanna et al. (2016) found fecal microbiota transplantation (FMT) in patients restored microbial phylogenetic diversity (PD) to levels more typical of a healthy person.

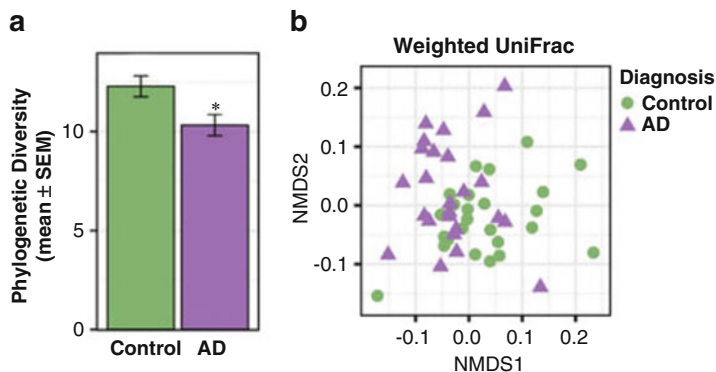


Fig. 1.2 “AD” indicates participants with diagnosis of dementia due to Alzheimers disease. **(a)** PD is decreased in the microbiome of AD participants. $*p < 0.05$. **(b)** Non-metric multidimensional scaling (NMDS) plot of weighted UniFrac analysis of composition. The NMDS analysis reveals differences in composition in AD versus Controls. Figure reproduced from Vogt et al. (2017)

Autism

Similar PD restoration strategies have been explored for autism. Kang et al. (2013) documented lower microbial PD in the intestines of autistic children. Kang et al. (2017) explored restoration or “rebalancing” of human gut microbiota through fecal microbiota transplant (FMT), to treat this autism spectrum disorder (ASD). Their study concurred that, based on PD, gut bacteria were significantly less diverse in children with ASD at baseline. Importantly, at the end of treatment, in contrast, bacterial diversity PD significantly increased in children with ASD.

Alzheimer’s Disease (AD)

A study by Vogt et al. (2017) compared the diversity and composition of the gut microbiome (bacterial 16S rRNA gene sequencing on DNA isolated from fecal samples) in participants with and without a diagnosis of dementia due to Alzheimer’s disease (AD). PD was found to be significantly reduced in the microbiome of the AD participants (Fig. 1.2a). In their study, much of the discussion of possible causes of the links and possible diagnostic strategies focused on compositional changes as summarized using UniFrac (Fig. 1.2b). Vogt et al. concluded that much further work is needed to determine any cause-effect relationship between gut microbiota and AD. The study nevertheless illustrates the complementary information derived from looking at total PD and at PD dissimilarities through UniFrac.

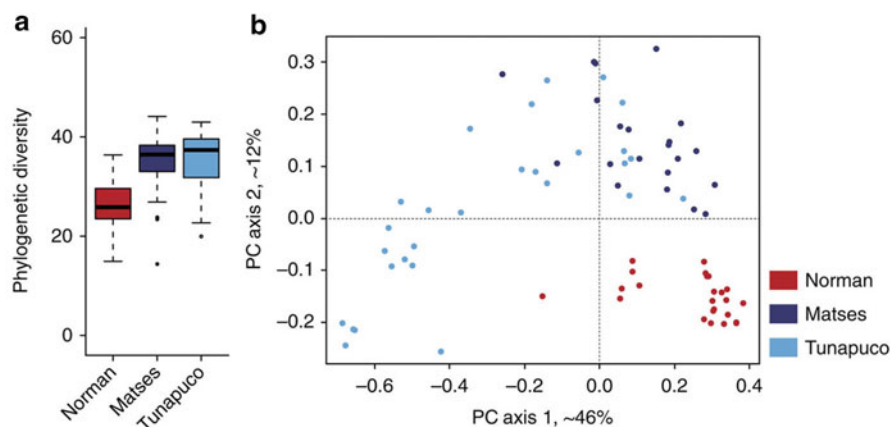


Fig. 1.3 (a) Microbial PD comparison of the gut microbiota of traditional hunter-gatherers, traditional agriculturalists, and urban-industrialized peoples. The urban population (Norman) has significantly lower microbial PD compared with the two rural populations. Whiskers in the boxplot represent the range of minimum and maximum PD values within a population, excluding outliers. (b) Principal coordinates analysis of UniFrac distances which were weighted by abundance values. The rural and the urban populations (Norman) show clear separation, indicating compositional differences in phylogenetic branches as indicated by UniFrac PD dissimilarities. (a) Figure reproduced from Obregon-Tito et al. (2015)

Subsistence Strategies in Traditional Societies Distinguish Gut Microbiomes

Obregon-Tito et al. (2015) applied high-throughput 16S ribosomal RNA (rRNA) gene amplicon sequencing and shotgun metagenomic sequencing to compare the PD of the gut microbiota of traditional hunter-gatherers, traditional agriculturalists, and urban-industrialized peoples. The two rural communities were the Matses, a remote hunter-gatherer population from the Peruvian Amazon, and the Tunapuco, a traditional agricultural community from the Andean highlands. Obregon-Tito et al. hypothesized that dietary differences would correspond to differences in the gut microbiome. They noted that:

Studies of peoples maintaining traditional subsistence practices are critical for understanding ... how the human microbiome responds to urbanism and Westernization, especially regarding diseases of civilization, such as obesity and chronic inflammatory disorders.

They found that the urban population (Norman, OK, USA) had significantly lower microbial PD compared with the two rural populations (Fig. 1.3a) and also had PD compositional differences (Fig. 1.3b).

Obregon-Tito et al. (2015) found stronger differences for PD, compared to estimated “species” richness, again supporting PD and phylogeny as useful for summarizing microbial community differences. However, this contrast is not a universal pattern. For example, Montassier et al. (2015) found significant reduced

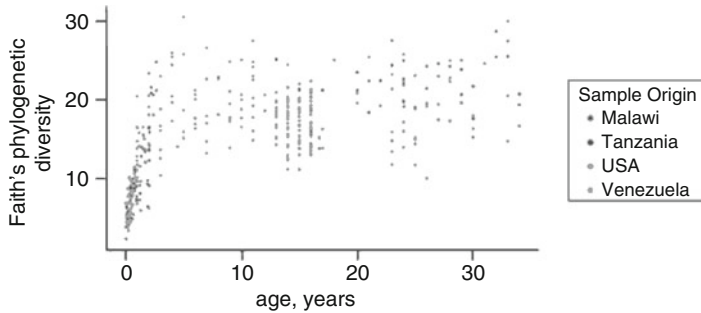


Fig. 1.4 Scatterplot of faecal microbiota of individuals plotted by phylogenetic diversity against age of the Hadza hunter-gatherers from Tanzania ($n = 16$, green), agrarians from Malawi ($n = 81$, red) and Venezuela ($n = 78$, purple) and Americans ($n = 213$, blue). Figure reproduced from Sonnenburg et al. (2016)

intestinal microbial PD following chemotherapy, and this loss of diversity was also clear for estimated number of species.

More on Diet and Microbial PD: Regional Extinctions

The review by Zhang et al. (2017) summarized the normal changes in intestinal PD with age:

The phylogenetic diversity of the intestinal microbiota increases with growth and development and ultimately leads to a complex and relatively stable community of microorganisms at the age of roughly 2–3 years.

Zhang et al. referred to diet as a well-established factor in microbial PD loss. Sonnenburg et al. (2016) examined this issue by comparing the microbiota of hunter-gatherers in Tanzania, agrarians from Malawi and Venezuela, and “Westerners” from the United States. They found that Western populations have lower PD (see Fig. 1.4) from birth through childbearing years. Sonnenberg et al. argued that such loss can amount to an extinction crisis and concluded that:

Our data support a model in which consuming a modern diet low in fiber contributes to the loss of taxa over generations and may be responsible for the lower diversity microbiota observed in the industrialized world compared to present-day hunter-gatherers and rural agrarians. The data we present also hint that further deterioration of the Western microbiota is possible.

Sonnenberg et al. also concluded that not only would a low-fiber diet result in a progressive loss of PD but also that reintroduction of a high-fiber diet would not be enough to recover this biodiversity. They concluded that restoring the microbiota to its original state “requires the administration of missing taxa” and predicted that such “microbiota reprogramming” will be important as “more diseases are linked to the Western microbiota and the microbiota is targeted therapeutically.” As in the case of

the other studies discussed here, a precautionary approach calls for maintaining microbial PD and its resilience/insurance value.

Discussion

A growing number of human diseases are associated with reduced microbial PD. However, it is revealing also to identify the rarer cases where a disease appears associated with *higher* microbial PD. Lappan et al. (2018) studied recurrent ear infection, a common childhood disease. They found that a healthy microbial community in the ear appears protective (“protective commensal bacteria”). Thus, PD again provides insurance and resilience. However, the PD of affected subjects was significantly *higher* because the presence of pathogens meant increased overall PD. Thus, in their study, PD on its own was not informative, and their most informative PD calculations (UniFrac) were those assessing compositional change.

The use of the PD calculus in microbial ecology means that neither overall PD nor PD dissimilarities and compositional change are assumed to have primary analytical importance. With this in mind, future work can further improve the toolbox. One opportunity relates to making better use of robust analyses of PD dissimilarities. Figure 1.3b illustrates a common “horseshoe” distortion that may obscure interpretations. This can be corrected with robust ordination methods (for discussion, see, e.g., Faith et al. 1987, 2009; Faith 2013). The toolbox for microbial PD then may consider also some of the additional analyses enabled by these robust methods (see, e.g., Faith 2015).

Further, other aspects of PD calculations may find use for the first time, including the homogeneity, distinctiveness, and endemism parts of the PD calculus (see Faith 2013). Overall, microbial ecology is well-placed in embracing the PD calculus. This combines appreciation of overall PD of a community, for insurance/resilience, with various compositional PD calculations, for addressing the range of questions related to understanding change. In the next section, I look at the relatively less successful integration of PD and its calculus into macrobial community ecology.

Macrobial Community-Level PD

Pioneering work by Webb (2000; see also Webb et al. 2002) laid foundations for community phylogenetics, or “ecophylogenetics” (for reviews, see Pearse et al. 2014; Cadotte and Davies 2016; Cadotte et al. 2017). However, foundational work also could be regarded as much older, given that phylogenetic community ecology inherited all the decades of development of ecological “diversity” indices. As illustrated in the studies by Chao et al. (2016), some indices created at the species level could be extended to phylogenies. Much of the work in this area following Webb’s pioneering work has created more and more phylogenetic “diversity” indices. Following Pavoine and Bonsall (2011), these metrics are often classed into richness,

divergence, and regularity categories. In this spirit, PD has been investigated as one of many possible phylogenetic indices, as an analogue to species richness.

In this section, I will try to focus particularly on those macrobial studies about the importance and value of “biodiversity.” Here, there have been useful studies using PD, with an interesting transformation from the conventional PD focus on “features” to a focus on ecological “traits.” For example, there now is support from multiple studies for the idea that phylogeny and PD can provide a surrogate for general functional trait diversity (Cadotte et al. 2009, 2010; Flynn et al. 2011; Davies and Buckley 2012; Mason and Pavoine 2013; Meynard et al. 2011). Cadotte and Davies (2010) concluded that PD can be expected to capture feature diversity, “including unmeasured, but ecologically important traits.” The Cadotte et al. (2017) review concluded that:

PD has been shown to be a powerful predictor of quantitative measures of ecosystem functioning (Cadotte et al. 2008; Srivastava et al. 2012; Cadotte 2013; Pu et al. 2014)

These findings may support the role for PD as a proxy, but not necessarily the role for PD as providing insurance. Faith (2017) described three values associated with PD and feature diversity – option value, insurance value, and proxy value. The latter covers cases, as in the findings above, where high PD means high functions, high productivity, etc. Here, the current feature diversity represented by PD acts as an effective proxy for a wide variety of current functions/benefits. PD conservation then is not so much a focus on insurance as on the conservation of possibly unknown current functions and benefits.

While these studies noted above represent important findings, there does not appear to be a coherent framework emerging that describes how PD fits into the broader use of phylogenetic measures in macrobial ecology. A coherent conceptual framework could help particularly in linking phylogeny to “biodiversity” and its value in community ecology. I see this limitation of the current research as due to several persisting confusions and misrepresentations. Noting the history of PD development described in the Background section, it is particularly unfortunate that macrobial ecology has ignored this progress. This sad neglect includes ignoring PD fundamentals (see Background), including ignoring:

- The definition as not tied to within-community, but including the global scale
- The shared-ancestry model as the PD link to feature diversity
- The extensive existing support (“tests”) for that link to feature diversity
- The fact that PD is not a single index, but is a calculus based on features
- The stated reasons why functional traits often will not be captured by PD
- The alternative method for explaining functional trait diversity

Here, I will present examples of these misrepresentations. I will begin with the basic issue of definitions and use of terms. The false statement is frequently made that Faith (1992) *defined* PD as within community. For example, the Mouquett et al. (2012) review of the emerging field of “ecophylogenetics” defines “Phylogenetic diversity: the amount of evolutionary history represented in the species of a particular community (Faith 1992).”

Similarly, Winter et al. (2013) attributed to Faith (1992) this misdefinition: “Phylogenetic diversity (PD). Calculated as the sum of branch lengths between root and tips *for a community*.” The unfortunate influence of that confusion is seen, for example, in Gudde and Venditti (2016) where they repeat the false definition, “Phylogenetic diversity represents the total of branch lengths that connect all species found *in an area* (Faith 1992).” (similarly, see Ahrendsen et al. 2016; Davies et al. 2016).

Definition problems extend to the use of “phylogenetic diversity” and even “PD” for other phylo-diversity measures, without explanation. For example, Mouquett et al. (2012) says: “Commonly used measures of phylogenetic diversity are the total branch length of a phylogenetic tree that contains all species present in a community, or the sum of pairwise distances between species weighted by their relative abundances.” Winter et al. (2013) interpreted “phylogenetic diversity” as derived from any between-species biodiversity distance, based on phylogeny.

Confusions have been made worse in macrobial community ecology by the popular use of “PD” for lots of other phylogenetic indices. For example, in Pio et al. (2014), “PD” is used to refer to any diversity measure linked in any way to phylogeny. They refer to a variety of published studies on the performance of “PD”, but it is not clear if this is Faith’s (1992) PD or some other measures. Pio et al. go on to apply the actual PD method in their analyses, but without reference to that as the Faith (1992) method.

It is also unfortunate that the justification for PD in conservation is falsely portrayed as depending on its recovery of functional traits. Winter et al. (2013), in questioning the conservation relevance, refer to the supposed issue that “the generality of the assumption that phylogenetic diversity can indeed be used as a proxy for functional diversity is unclear.”

Mazel et al. (2017) similarly also illustrate the misconception that PD’s justification for biodiversity conservation depends on functional trait recovery. Mazel et al. (2017) also incorrectly claim that the most prominent argument for PD conservation “is that conserving phylogenetic diversity (PD) will also conserve diversity in traits and features (functional diversity [FD]).” This not only ignores the warnings about functional traits, it ignores the actual motivation for PD conservation – option values. Mazel et al. claim to evaluate the rationale for PD, but nowhere refer to option value. Winter et al. (2013) claimed to consider the proposed justifications for conservation of phylogenetic diversity, but also did not even list option value. Davies et al. (2016) similarly ignore option value in arguing that the rationale for PD is countered by a finding that some branches of the same length differ in their functional trait importance.

The Mazel et al. (2017) study also illustrates the common misconception that PD has not been tested and that tests must focus on functional traits. Some workers have suggested that PD has no clear justification and/or that the link to feature diversity is either untested or has failed limited tests (see, e.g., the working group report at <https://www.idiv.de/?id=427>).

Some evaluations of PD have misrepresented the PD model (of shared ancestry). For example, Kelly et al. (2014) portray the model as distance-based, rather than

shared ancestry-based. Misrepresentation of the PD assumptions also has meant that the PD calculus based on features has been ignored. Tucker et al. (2016) had the goal of a unified framework, but ignored the existing PD calculus, which would have unified many measures and helped to clarify the status of a core biodiversity measure as compared to associated calculations for decision-making. In ignoring the calculus, Tucker et al. confused PD with its associated calculations. They assumed that “richness metrics sum up the quantity of phylogenetic differences present in an assemblage” and so lumped together PD and its endemism and distinctiveness calculations as equivalent “richness” dimension measures. Unfortunately, they went on to say: “high redundancy amongst metrics within a dimension can make selection among them somewhat arbitrary” (see also the similar error in Vellend et al. 2010).

The relationship between PD and functional trait diversity has raised many confusions and misrepresentations. Faith (1992; see also Faith 1989) made it clear (along with an example case study (Faith 1989); see also Faith 2015) that conserving phylogenetic diversity sometimes can be a poor strategy for conserving functional diversity. Now, more than 25 years later, ecologists are claiming to have now discovered this (e.g., Mazel et al. 2017), “Conserving phylogenetic diversity can be a poor strategy for conserving functional diversity”), without making any reference to the original insights. These early insights that have been ignored included an effective functional trait diversity approach (Faith 1989, 2015).

Mazel et al. (2017) evaluated PD links to functional traits by adopting a popular convex hull measure of trait diversity. Supposedly PD has justification for conservation if it recovers this trait diversity. However, this convex hull measure of trait diversity is troubling as a conservation target. Suppose, for example, all traits fit the phylogenetic tree perfectly, and one lineage evolved lots of traits for life in one habitat type, while other lineages evolved lots of traits for other habitat types. These traits are reflected in the branch lengths. PD applied to this tree would capture lots of feature/trait diversity, with high contributions from each lineage. However, the trait space based on all those traits, and the convex hull volume, would not reflect that high diversity of traits. Podani (2009) documented the key problem associated with convex hull volume: the presence of lots of correlated traits, as in this example, will not be reflected in a higher volume. It is apparent that convex hull volume is not something that would deserve to be conserved if one is interested in trait diversity, so why should PD try to capture it? We are left to wonder why Mazel et al. ignored the Faith (1992) warning that PD sometimes will not represent functional trait diversity and that another model and method (“ED” discussed earlier) provides a way to quantify and conserve functional trait diversity. Indeed, a review of functional trait diversity measures (Pla et al. 2012) notes that this is perhaps the oldest such measure. We can conclude that progress will be made in PD macrobial ecology applications when the lessons of history, across the different area of study, are better appreciated.

Conclusions

The Faith (1992) proposal of PD built on a rich history of discussions about taxonomy, phylogeny, features, distinctiveness, and also about the core values of biodiversity including option value and insurance value. Some of this appears in IUCN (1980), but in addition earlier papers, some largely ignored, have explored these issues. Faith (1992) in considering primarily regional and global and whole-clade applications emphasizes PD links to option value. In contrast, ecological community-level studies will be more focused on insurance value and resilience. These ideas have perhaps had the most compelling development in microbial ecology. Here, I described just one portion of that discipline, relating to human health. The study of various diseases suggests that insurance value and resilience are linked to microbial PD, but these studies also showed that the PD calculus (e.g., dissimilarities) is an equally important tool.

Insurance value and resilience no doubt will be equally important in macrobial ecology, and PD can be expected to be informative. However, progress in this area has been slow. Part of the problem is that ecology, even at the species level, remains uncertain about its definition of “biodiversity.” A helpful approach may be to separate biodiversity and other important aspects such as ecological integrity. In addition, there is an urgent need clearly to develop meaningful ways to look at functional trait diversity. This may include both phylogenetic and other models as equal status explanations of trait diversity patterns. All these issues make it clear that progress will depend on not only a shared toolbox across areas of study but also shared conceptual frameworks.

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Chapter 2

The Use of EDGE (Evolutionary Distinct Globally Endangered) and EDGE-Like Metrics to Evaluate Taxa for Conservation



Nick J. B. Isaac and William D. Pearse

Abstract The idea of setting conservation priorities using phylogenetic information has been in the literature for decades. Since 2007 it has been implemented by the EDGE of Existence program at the Zoological Society of London. There are now “EDGE lists” of mammals, birds, amphibians, sharks, and corals: all these lists prioritize species that are both highly threatened and evolutionarily distinct (i.e., they have few close relatives). In this chapter, we review the metrics used to calculate priority scores for species on EDGE lists. As the EDGE of Existence program enters its second decade, this is an appropriate moment to reflect on how the process of selecting species could be improved or streamlined. A suite of metrics have been proposed as alternatives or improvements to the original EDGE metric, each differing in a number of important ways, such as how they handle uncertainty in their underlying data. Perhaps the most profound differences among these metrics reflect differences in how threat status and phylogenetic information are scaled to calculate the prioritization metric and in whether (and how) they incorporate the principle of complementarity. We discuss these choices of metric in the context of the properties of each metric and what they have to say about balance between how much we value the existence of each species (its phylogenetic position) and the risk that it would be lost due to extinction.

Introduction

While species are the natural units for biodiversity and conservation, the very notion of species has long been problematic to evolutionary biology (Hey et al. 2003; Isaac and Purvis 2004). One problem is that species differ markedly in age (Purvis and Hector 2000), so the amount of evolution, or phylogenetic diversity, that each

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species represents is different. Thus, incorporating evolutionary principles into conservation biology requires metrics to assess the contribution of individual species to phylogenetic diversity.

Typically, species' contribution to phylogenetic diversity is expressed in terms of evolutionary isolation within a phylogeny. Vane-Wright and colleagues (Vane-Wright et al. 1991) proposed measuring the reciprocal of the number of nodes between the root and tips of a phylogeny, without using information from branch lengths.

The number of metrics for evolutionary isolation has continued to grow with time. Redding et al. (2014) reviewed eleven metrics and compared their statistical properties along two axes: uniqueness (the amount of evolutionary history that is uniquely represented in the focal species) and originality (the average distance from the focal species to all others). Redding et al. (2014) pointed out that both uniqueness and originality are properties of metrics that might be desirable, from a conservation or evolutionary perspective.

Conservation biology is often characterized as a crisis discipline in which scientists construct a triage system for nature, so prioritization has become a pervasive theme in the conservation literature. Species-based conservation has generally focused on prioritizing those species at risk of extinction, e.g., those in the IUCN critically endangered category over endangered and vulnerable species. This approach implicitly assumes that the extinction of any species would be a tragedy of equal magnitude, i.e., the amount of biodiversity represented by each species is the same. Adopting an evolutionary perspective challenges this practice, because evaluating a species' contribution to phylogenetic diversity implies that the extinction of certain species would represent a greater loss than others. From this, it follows that we may allow evolutionary isolation to trump extinction risk when setting priorities for conservation.

This notion of trading-off risk and value is controversial in conservation but is a standard approach in economics known as cost-benefit analysis. Therefore, it is perhaps unsurprising that it was an economist, not a biologist, who first proposed a formula to combine species' value and risk. Weitzman (1992, 1998) sought to define the set of species that would maximize diversity at some point in the future. When placed into a phylogenetic context (Faith and Walker 1996; Witting and Loeschcke 1995), Weitzman's approach provides a means to derive the expected gain in diversity that would be achieved through the successful conservation of a species or the expected loss that would result from its extinction.

Over the past decade, notions of species' value and risk in conservation biology have become embodied by the acronym EDGE, which stands for evolutionary distinct and globally endangered. EDGE has become a global vehicle for promoting the idea of preserving evolutionary history as a conservation narrative. In this chapter, we review the development of EDGE, and metrics related to it, and how they are used to set conservation priorities. We then explore in detail the issues around how best to incorporate uncertainty into EDGE lists, how to scale the value and risk components with respect to each other, and how complementarity among species should be assessed.

EDGE Metrics and the EDGE of Existence Program

EDGE consists of a set of algorithms and metrics derived from Weitzman's ideas, as well as a conservation program (the "EDGE of Existence") run by the Zoological Society of London (ZSL). Here we describe the metrics and how they've been used for prioritization of conservation action.

Calculating EDGE Scores

The algorithms and metrics underpinning EDGE were laid out in a paper by Isaac et al. (2007) and illustrated by application to the entire class of mammals. A species EDGE score is a composite of two components: ED (evolutionary distinctiveness) and GE (global endangerment, representing risk).

Isaac et al. (2007) defined a metric for calculating each species' contribution to phylogenetic diversity (evolutionary isolation, or value). They called this metric ED, for evolutionary distinctiveness, but it is also known in the literature as "fair proportion" (Redding et al. 2014) because each species' score is made up of shares of the branch lengths from which it is descended, with each branch length divided by the number of descendant species. Moreover, for large phylogenies, ED is equivalent and identical to a theoretically derived metric known as the Shapley value (Hartmann 2013).

The global endangerment part of EDGE is simply an ordinal representation of the IUCN Red List categories, with scores increasing from 0 (least concern) up to 4 (critically endangered). The two components are then combined to create an "EDGE score" for each species:

$$\text{EDGE} = \ln(1 + \text{ED}) + \text{GE}^* \ln(2)$$

There are several details in this equation that merit further explanation. First, the ED and GE components are added together because both are measured on, or converted to, the log scale. So EDGE scores can be thought of as the logarithm of the product of two numbers measuring value and risk. Second, the ED component has a one added to it (before being logged) for the simple expedient of avoiding negative EDGE scores (for species with $\text{ED} < 1$). Third, the GE component is multiplied by the natural log of 2 in order that EDGE scores can be interpreted as the (logarithm of) the expected loss of diversity, in millions of years, under a particular interpretation of how the IUCN Red List categories map onto extinction risk. Specifically, it assumes that each increase in IUCN category represents a doubling of extinction risk. In this way, EDGE scores are consistent with the logic of Weitzman (1992, 1998) while also maintaining a transparent mapping against the IUCN categories.

Around 6 months before Isaac et al. (2007) was published, a paper appeared by Redding and Mooers (2006) describing essentially the same idea, with applications to birds. While the overall aims and concepts of the two papers were broadly similar, there were some important differences. The key difference was in the algorithm used for both the ED and GE metrics. The ED part was calculated by sharing the branch lengths equally among descendent branches (known as “equal splits” – see Redding et al. 2014 for more details), such that branch lengths are divided from root to tip of the phylogeny, rather than from tip to root in Isaac et al. Redding and Mooers (2006) used published estimates of extinction risk from population viability analyses for GE, rather than the arbitrary doubling approach in Isaac et al. – see Mooers et al. (2008) for a detailed exploration of this issue. Another distinction is that Redding and Mooers emphasized the conceptual advance over the resultant conservation priorities, which were treated largely hypothetical. In part this reflected the bird phylogeny available at the time, which was based mostly on taxonomy and lacked credible estimates of divergence dates. By contrast, Isaac et al. (2007) explicitly derived priorities from their scores, which they justified using a series of modifications and sensitivity analyses to account for various uncertainties in the data (notably missing species and unresolved nodes, or polytomies). These uncertainties are explored further in later sections of the chapter.

The EDGE of Existence Program

The EDGE of Existence program aims to conserve phylogenetic diversity (Faith 1992) by prioritizing the conservation of species that are both evolutionarily distinct (i.e., valuable, from a phylogenetic perspective) and globally endangered (i.e., at high risk of extinction). The EDGE scores defined above are used in this prioritization exercise, although other factors are factored in.

The EDGE program originated in 2007, when the first list of priority mammals was published alongside the EDGE algorithm (Isaac et al. 2007). The mammal list was subsequently updated by Collen et al. (2011). EDGE amphibians first appeared in 2008 but underwent several iterations before a peer-reviewed list was published (Isaac et al. 2012). EDGE corals followed in 2012, EDGE birds in 2014 (Jetz et al. 2014), and most recently an EDGE-like list for sharks (Stein et al. 2018). EDGE lists for reptiles and plants are currently in development.

The Edge of Existence program has adopted a multi-tiered approach to select species for conservation. The first criterion is that species should be both evolutionarily distinct and globally endangered (i.e., “EDGE species”). Historically, EDGE species have been defined as those that are both above the median ED for the clade in question and listed on the IUCN Red List as vulnerable, endangered, or critically endangered. Note that species listed as extinct in the wild are not considered within scope, so the focus is explicitly on in situ conservation efforts. Given this definition, and the proportion of species listed on the Red List, there are many hundreds of

EDGE species in each of the three vertebrate classes for which a list has been prepared.

To focus attention onto a manageable subset, the second tier is simply the set of species (typically 100) with the highest EDGE scores in each group. Each species in the “top 100” has a page on the EDGE of existence website, with information about their ecology, phylogeny, biogeography, threats, and current conservation action. This last piece of information informs the selection of ten “focal species” for which targeted action would make the biggest difference to its conservation status, i.e., those currently receiving little or no action at present. The focal species list is the primary means by which the EDGE of Existence program raises funds and awareness about the status of species that are both evolutionarily isolated and most in need of conservation intervention.

In theory, the algorithm devised by Isaac et al. (2007) allows non-EDGE species to appear within the top 100. This has not happened yet: among amphibians and mammals, the most evolutionary distinct species that is classified as near threatened is roughly 150th, when ranked in terms of EDGE score. However, if the armadillo (*Dasypus novaeboracensis*) or platypus (*Ornithorhynchus anatinus*) were reclassified from least concern to near threatened, their EDGE scores would be comfortably within the top 50 of all mammals. At present, both species have EDGE scores that are higher than approximately 250 species classified as EDGE mammals. Resolving this inconsistency is clearly a desirable goal for the future transparency of the program.

The principle mechanism that EDGE of Existence delivers on-the-ground conservation is via an annual program of “EDGE fellowships”. Typically, EDGE fellows are early career researchers in developing countries: they receive funding to undertake a 2-year field project on a top-ranked EDGE species (typically a species in the top 100), as well as 6 weeks of training in conservation management, ecological surveys, and leadership. Over 10 years, the fellowship has trained 79 fellows working on 74 EDGE species in 40 countries. In this way, the program delivers tangible capacity building in addition to improvements in conservation status.

A Decade of Debate on EDGE-Like Metrics

In the 10 years since the first EDGE list was published, a number of criticisms with the approach have been raised. We believe that the debate would be advanced by distinguishing clearly between *properties* of species scoring and *performance* of scoring approaches. Numerous EDGE-like approaches (e.g., HEDGE, Steel, Mimoto, and Mooers 2007; ADEPD, Nunes et al. 2015; EDAM, Pearse et al. 2015, iHEDGE, Jensen et al. 2016) have been suggested, each *performing* perfectly with respect to what they were intended to do, but as each was designed to have different *properties* the resulting lists differ. There is no “best” flavor of ice cream, simply flavors that tend to be preferable, and in much the same way there is no “right” conservation approach, rather approaches that prioritize properties that more people find preferable. Each of them shares the property of somehow combining

species' evolutionary distinctiveness (ED) with their (global) endangerment (GE); thus, in the discussion below, we refer to the output of these as EDGE metrics and lists.

Uncertainty

Conservation is, by its very nature, an urgent task, and so there is a trade-off between the risk of delaying action (“fiddling while Rome burns”) and acting too early on the wrong species (“saving the wrong city”). It is commonly thought that accounting for, and measuring, uncertainty helps manage this trade-off, but doing so practically is challenging (Forest et al. 2015).

EDGE lists are affected by uncertainty in both the phylogeny (including branch lengths) and extinction risk (including the lack of precision imposed by using IUCN categories). It would be worrisome if uncertainty in one component was substantially greater than the other, but to our knowledge, no one has quantitatively studied this.

Isaac et al. (2012) randomly perturbed both phylogenetic topology and extinction risk categories, which they used as a general check on the accuracy of their results. Such randomizations are useful in that they allow for tests of “worst-case” scenarios and can provide reassurance that prioritizations are robust to uncertainty, but they do require the investigator to decide reasonable parameters for those randomizations. An alternative is to calculate EDGE lists across the range of uncertainty for ED and GE in each species. This is now a standard, if computationally intensive, approach to dealing with phylogenetic uncertainty but is challenging to implement for extinction risk status.

Phylogenetic Uncertainty

Modern phylogenetic reconstruction algorithms do not produce a single tree, but rather a set of candidate trees with differing topologies and branch lengths. These candidates are referred to either as a posterior distribution of trees or as bootstrapped trees, depending on how they were constructed. In this situation, it is relatively straightforward to calculate ED across all possible trees (Collen et al. 2011; Jetz et al. 2014). Such *posterior predictive methods* are common in comparative biology and are intuitively appealing as they incorporate uncertainty in phylogenetic inference directly. However, posterior predictive methods cannot account for uncertainty outside the scope of the original phylogenetic search (e.g., few loci for the search, missing fossil dates, etc.) and so may give the illusion of false precision.

A variety of work-around solutions have been developed for dealing with situations where phylogenetic information is less well-developed. Unresolved nodes (polytomies) are widespread in “consensus” phylogenies and where taxonomic

hierarchy has been used to approximate the phylogenetic structure (e.g., the amphibian tree of Isaac et al. 2012). Naïve treatment of polytomies leads to biased estimates of ED scores, so the nodes must either be resolved through phylogenetic imputation (e.g., Kuhn et al. 2011) or a correction factor applied (e.g., distribute the “missing” branch length among presumed close relatives; Isaac et al. 2007). Either way, a model of evolution is required to impute the branch lengths and thus estimate the likely range of possible ED scores for species descended from polytomies. The choice of evolutionary model underpinning this imputation requires careful consideration, because the largest part of a species’ ED score derives from the terminal branch length and the available models differ greatly in the distribution of branch lengths from root to tips (Mooers et al. 2012). At present, we lack a clear understanding of how to choose the right model for any given situation, so this should be a priority for future research.

A “special case” of such phylogenetic imputation is the addition of species that are missing from the phylogeny; such species are typically added into the phylogeny using taxonomic information, and then the structure of that clade resolved through phylogenetic imputation (see above). Such imputation should be free of bias, although this would evidently not be the case if the evolutionary model (see above) or taxonomy were estimated with error. Ultimately, however, such imputed scores should be used with care and certainly with an understanding of the uncertainty associated with them.

Further, as Pearse et al. (2015) showed, *rogue taxa* (species whose correct placement is unknown but is one of a few discrete possibilities) can have fundamentally different placements, and it is unclear how these species should be prioritized. Such cases, and all cases where phylogenetic is uncertain, can be handled elegantly with methods based around phylogenetic networks. Phylogenetic networks were developed for situations where hybridization and horizontal gene transfer are widespread (Huson and Bryant 2006), such that a hierarchical branching structure no longer describes the relationships among taxa.

Uncertainty in Extinction Risk

Uncertainty in extinction risk presents a number of challenges that are poorly understood, compared with the large literature on phylogenetic uncertainty. Within the Weitzman framework of expected loss (or gain) in diversity (Weitzman 1992, 1998), the parameter requiring estimation is the probability of extinction, $pExt$. These probabilities can be estimated using population viability analysis, but these are data-intensive and are rarely applied across entire species’ ranges.

The IUCN Red List categories of extinction risk provide a convenient and shortcut to estimating $pExt$ that is both widely accepted and for which large quantities of data are available: as of 2016, over 76,000 species had been assessed and categorized. However, there are substantive issues around the use of Red List categories; the most widely debated is the values of $pExt$ chosen for each category.

The values chosen have a substantial impact on the resulting list of priority species (Mooers et al. 2008). This choice arises because of the categorical nature of the IUCN Red List: it is implausible that all species within any category share the same value for their probability of extinction. A further source of uncertainty arises from the fact that $pExt$ increases nonlinearly with the time window over which it is measured, and it is difficult to translate probabilities that are measured over different periods (Mooers et al. 2008). Furthermore, the uncertainty in any one estimate of $pExt$ probably increases with the time window, at least within the decadal scales used in conservation management (over geological timescales, the extinction of all species is more or less guaranteed).

It would be possible to incorporate uncertainty in $pExt$ into EDGE scores using a distribution of $pExt$ for each category of risk, if a sufficiently large dataset of population viability analyses (PVA) results were assembled. In theory this could be extended to produce distributions that were specific to the criterion under which the species was listed or that propagate uncertainty in the raw data up to uncertainty in the final Red List category.

Particular problems are associated with data deficient (DD) and not evaluated (NE) species. EDGE lists have generally ignored these species, which are clearly undesirable. Simply treating all DD or not evaluated (NE) species as if they were threatened may be defensible in some situations. A better approach would be to produce a parallel list of “potential EDGE species” based on the EDGE scores that would be achieved if they were threatened. This list might even be used to prioritize the collection of data for a Red List assessment to be made. A more sophisticated approach would be to predict extinction risk based on species’ traits (Bland et al. 2015). For example, Stein et al. (2018) used body size and depth characteristics as a proxy for extinction risk in their EDGE-like prioritization of sharks. Such approaches are powerful, but obviously data-intensive.

Dealing with Uncertainty in Practice

Although the sources of uncertainty in the GE component of EDGE scores are more numerous and intractable than for the ED component, it is arguable that they matter less. The IUCN Red List is inherently precautionary in nature: it is designed such that it is hard to justify a high-threat category for species about which there is substantial uncertainty in the underlying data. We can therefore be fairly confident that all threatened species deserve conservation attention, even if their precise position on the EDGE list is uncertain. By contrast, relatively minor phylogenetic uncertainty has the potential, if unchecked, to vastly inflate species’ ED score to the point that undeserving species could be listed as priorities.

In theory it would be possible to generate measures of uncertainty on each species’ EDGE score by first estimating and then propagating the uncertainty in the ED and GE components. The practical choice about what to do with this information is essentially a value judgment. One option would be to down-weight

species with highly uncertain scores, in order to ensure that limited resources are spent on species that actually deserve attention. Conversely, one might argue that uncertain species should be up-weighted, in order to prevent the loss of *any* species that *might* be a priority. This line of reasoning would be appropriate if the goal were to ensure that “safe” species with high ED scores don’t become threatened in the future (“LEDGE” species: Faith 2015a, b). Ultimately, the choice comes down to one’s attitude to risk and the budget available for action. However, we note that uncertainty itself is likely to display a high degree of phylogenetic autocorrelation, thus further complicating the choices available.

Complementarity

Perhaps the most fundamental of issues with EDGE-like approaches is *complementarity*. Complementarity has a number of definitions, but all of them share in common the idea that some species, when added to a *set* of species already being conserved, add more diversity – are more complementary – than others. The original EDGE approach, which was designed to rank individual species, can never truly have the property of complementarity because it does not select sets of species – it merely gives all species within a group value.

Faith’s original PD framework (Faith 1992) embodies the complementarity principle: given a number of species to be preserved, it is possible to select the set of species that would preserve the greatest branch lengths (i.e., be the most complementary to each other). Faith later extended this framework to propagate the probability of extinction of each species throughout a phylogeny (Faith 2008), effectively estimating the extinction risk for each branch in the tree. This method, which is mathematically equivalent to that of HEDGE (“Heightened” EDGE: Steel et al. 2007), naturally gives a higher score to species whose close relatives are likely to go extinct. As such it is only partially consistent with the complementarity principle: species’ rankings are, in part, a consequence of other species’ rankings, but sets of species are not chosen. In such cases the deeper phylogenetic structure tends to contribute very little to the final score, because most major clades tend to have at least one “safe” representative (i.e., low extinction risk) such that ancient branches are rarely under direct threat.

It is possible to create complementary sets of species from HEDGE-type approaches through some form of iterative updating procedure. In I-HEDGE (Iterative-HEDGE: Jensen et al. 2016), a HEDGE list is calculated, the top-ranked species marked as conserved (and so safe), the HEDGE list recalculated, and then the procedure repeated, possibly with jackknifing or bootstrapping to accommodate uncertainty (see above), in order to generate a set of priority species for conservation. Similarly, ADEPD produces sets of species based on the investment that would be required to move or maintain a species’ threat category over time (Nunes et al. 2015).

Note that ADEPD and I-HEDGE contain markedly different concepts about the outcome of conservation actions on the set of top-ranked priority species. In

I-HEDGE, the algorithm assumes that the *pExt* of species in the conserved set has been reduced to zero; in ADEPD it is assumed that the required investment will be guaranteed to maintain *pExt* at its current level. In practice, conservation practitioners may be uncomfortable with either of these extremes, which perhaps makes complementarity the most difficult property of EDGE-like approaches to successfully apply, despite it being something that, in our experience, most conservation biologists seem to intuitively agree with.

Scaling

Recall that EDGE scores can be thought of the logarithm of the product of value and risk. The ED component has natural units (millions of years) and is approximately normally distributed for most clades (after log transformation). Recall also that measuring *pExt* (the GE component) is fraught with uncertainty and that the choice exerts a large impact on the resulting EDGE scores (Mooers et al. 2008). While objective measures of GE remain elusive, the choice may be motivated by a desire to ensure that both ED and GE contribute meaningful amounts of information to the final set of EDGE scores. In technical parlance, this is referred to as the *scaling* of ED and GE.

The EDGE of Existence program scales GE by treating each category as having double the extinction risk of the previous category. This choice produces a list of priority species with a mixture of categories, rather than being dominated by critically endangered species (Mooers et al. 2008). In the original mammal list (Isaac et al. 2007), the top 100 contained eight species in the vulnerable category, and the top 10 contained three species in the endangered category. Down-weighting the critically endangered category in this way was a deliberate, subjective, and value-based decision based on what was seen as desirable scaling of ED and GE. While the original EDGE list (Isaac et al. 2007) was intended to have roughly equal scaling, it is not clear that this is a goal that everyone necessarily shares.

There are several factors that make it difficult to achieve a consistent scaling across all taxonomic groups, particularly if the mapping of GE scores to threat categories remains fixed. First, GE scores are bounded, but ED scores are not: older phylogenies will, all things being equal, tend to have species with greater ED values than species in younger phylogenies. Second, scaling is a function of the variance in both ED and GE components: the variance in ED is determined by phylogenetic tree shape, with unbalanced trees having higher variance; variance in GE is determined by the distribution of species among threat categories, particularly the proportion that are critically endangered.

All these factors mean that the relative contribution of changes in ED to GE is not equal (i.e., a doubling of ED will not have the same effect on EDGE as a doubling of GE) and the strength of this effect will vary across EDGE lists (Pearse et al. 2015). In the plant example Pearse et al. used, the importance of ED vs. GE was more than double (i.e., to match a doubling in ED, a species would require a quadrupling of GE

in order to achieve the same EDGE score). They suggested that a z-score transformation could be used to achieve equal scaling, with weighted (i.e., multiplicative) variants to explicitly weight the ED and GE components in order to fit the desired conservation goal. For example, a weighting of 2:3 would scale the EDGE scores such that that 60% of the information in the priority list derived from the GE component (Pearse et al. 2015).

Future Directions

Over the past decade, the size and coverage of phylogenetic has increased dramatically, as a result of advances in phylogenetic software, computing power, and DNA sequencing. This means we have more information on the evolutionary distinctiveness of species than we do of their threat status. Thus the utility of EDGE and EDGE-like approaches has, arguably, never been greater: in a world where we need to prioritize on the basis of *something*, phylogeny is perhaps one of the few things we have to choose. Conservation biology, as a whole, needs more data, and phylogeny-based conservation needs only a conceptual synthesis of what *properties* we wish the metrics we use to have. Conceptual issues are, arguably, easier to solve than global data limitations.

EDGE-like conservation prioritization strategies represent a unique opportunity for evolutionary biologists to contribute to active conservation. The EDGE of Existence program has delivered tangible benefits for the conservation of many highly threatened species. EDGE-like metrics also have potential to contribute to conservation in indirect ways, for example, by offsetting the disruptive influence of taxonomic change on conservation priorities. Splitting one threatened species into two both increases the need for conservation and spreads the resources more thinly (Isaac et al. 2004; Agapow et al. 2004), but weighting each by ED reduces the priority attached to each new species.

A major challenge remains to apply the results of EDGE-type approaches in mainstream conservation. EDGE listing forms part of a conservation narrative of the preservation of species that both represent unique evolutionary history and/or unique ecosystem services that humans need. By working with conservation practitioners, we can develop tools and provide the data they need to generate new narratives for promoting the conservation of phylogenetic diversity.

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Chapter 3

Biodiversity, the Tree of Life, and Science Communication



James Rosindell and Yan Wong

Abstract The usual approach to organising human knowledge in any field is to invent an intuitive, often hierarchical system of classification. Perhaps uniquely to natural history, humans did not so much invent such a system as discover one: the tree of life. This now iconic object has utility, not only in biodiversity research but also in conservation and science communication. For example, engagement in conservation requires highlighting the scale of biodiversity and what we stand to lose; this can be communicated very effectively by showing people the complete tree of life. Here we discuss the challenges involved in making the complete tree of life accessible to a broad audience. The first challenge relates to data curation, which is marred by the lack of a universally recognised database of species names. Different species can have the same name, different names can refer to the same species, and spelling errors are rife. The second challenge is tree visualisation. While many visualisation methods already exist, few are able to represent the complete tree of life. We argue that scaling up any method to the extent necessary for showing the whole tree of life is bound to expose inescapable trade-offs in the prominence given to each aspect of the underlying data. For example, should topology or species richness be prioritised when displaying clades that contain an insignificant proportion of species but which occupy a prominent position in the tree topology? We conclude with an optimistic outlook for using the tree of life in science communication.

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Why We Should Care About the Tree of Life?

Why Use Phylogenies?

As a rule, human understanding involves some form of classification. What colour is this object? Will it hurt me? Is it something I can eat? Such categorisations are at the root of our interaction with the world around us. Most examples, such as the Dewey Decimal System for libraries, are hierarchical. For species of conservation concern, various such classifications might be imagined, for example, based on species traits, population distributions, perceived threats, or the ecosystem functions and services that species provide. However, perhaps uniquely in the sciences, the study of biodiversity is in the privileged position that nature already provides an entirely objective hierarchical classification. All the species we study are the result of an evolutionary process that has left behind a historic scaffolding waiting to be discovered, a set of almost perfectly hierarchical relationships, the ‘tree of life’. Echoes of this natural classification, or ‘phylogeny’, can be seen in the systems produced by Linnaeus and his successors, but wholesale incorporation of phylogenetic thinking into biological taxonomy only started in the latter half of the twentieth century. As DNA sequencing has become cheaper and robust methods developed to infer evolutionary history from these sequences, phylogenies have started to assume a pivotal role in rigorous scientific studies of biodiversity. Despite their ubiquity and utility however, we contend that phylogenetic trees are not typically used by members of the public to frame their thinking about biodiversity and conservation.

Apart from the fact that phylogenies can be thought of as an actual historical record and are a key component of scientific studies, there are several supplementary reasons why we believe a phylogenetic approach should underpin attempts to discuss biodiversity in popular science. Perhaps most importantly, the tree of life is universal. Each new organism has a relatively comprehensible place in the phylogenetic system, even organisms which are unknown to a general audience, or indeed unknown to science (Shalchian-Tabrizi et al. 2008; Brown et al. 2015). This applies not only to animals, plants, fungi, and other eukaryotes, but even (with some caveats) to organisms such as archaea and eubacteria (Hug et al. 2016). In other words, the tree of life is a fundamental object onto which we can pin all natural history. Compared to the Linnaean system, phylogenetic trees also provide a more detailed classification. At the broad scale, unranked phylogenetic groups are supplementing (and even replacing) higher-level Linnaean groups, such as in the widely used APG classification of plants (Chase et al. 2016). At the finest scale, the very definition of a species is often established by methods, such as cross-coalescence (Nater et al. 2017), that are explicitly based on evolutionary trees (albeit of DNA sequences rather than whole species). Finally, phylogenies also provide an objective way to predict the traits we expect to see. If a mammal is a marsupial, it probably has a pouch. If it is a monotreme, it probably lays eggs. Critically, these predictions also apply to traits that we may not even have considered, including in cases where external appearances may be deceptive. For example, we might expect development in tunicates to possess similarities to that in vertebrates and the

biochemistry of coelacanths to be closer to that of tetrapods than of (ray-finned) fishes. In this sense, a phylogenetic tree can be thought of as predicting the degree of nonindependence between species. Statistically, we can think of the diversity of life being composed of ecological specialisations layered over a default expectation given by phylogeny.

An additional benefit of a phylogenetic approach over classifications based on, say, ecological or endangerment criteria, is that it is not constrained to extant species. Long-extinct species such as dinosaurs provide examples where phylogenetic thought has entered the popular domain: any number of schoolchildren will be able to explain to a passing adult that birds actually are dinosaurs. The mass move to molecular phylogenies has also started to include ancient DNA from recently extinct species – often victims of neolithic conservation management. Not only can ancient DNA phylogenies change our understanding of the history of extinct species (Mitchell et al. 2014), it may even allow us to reconstruct previously unknown features of their biology.

Despite these strong advantages, for members of the public to engage with phylogenies and the scientific rationale behind them, we need to represent them in a manner that is intuitive and easy to explore. What is more, for effective outreach, we also need to map additional data – common names and images certainly, but also ecological information – onto existing phylogenetic trees.

Conveying the Tree of Life

An important part of communicating the concept of biodiversity is conveying the grand scope of all life on earth, the sheer number of species, and the mind-boggling timescale of evolution. For most of this chapter, we will concentrate on visualising this diagrammatically, but there are other interesting possibilities. One novel example is to actually walk a phylogenetic tree of life. The Ancestor's Trail is a community hiking event which imagines paths through the tree of life as being mapped to real footpaths through a landscape (see Fig. 3.1). Each step taken corresponds to 10,000 years; in their first stride, participants can imagine the entire history of human civilisation as having flashed past them. The scale later changes so that each step corresponds to 100,000 and finally one million years. A day of walking in this manner results in a much better appreciation for the vast amount of time that the process of evolution has required to shape today's biodiversity. A more tactile demonstration of biodiversity that we have used focuses on numbers of species rather than time. We give a feel for the size of different parts of the tree of life by representing each species by a grain of wheat. Mammals correspond to a small handful of grain, while other well-known groups can be introduced in hessian sacks of increasing size, concluding with insects which are spread over two sacks weighing over 40 kilograms.

These examples notwithstanding, the predominant method of conveying knowledge about the tree of life is visualisation. Imagery is an essential science

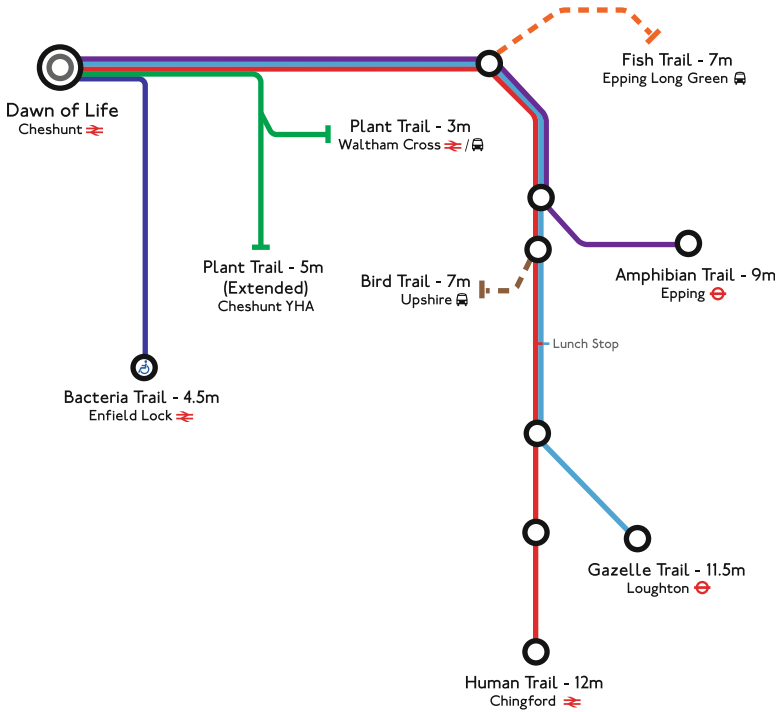


Fig. 3.1 A London underground style map of the Ancestor's Trail created by Damian Belson, showing multiple starting points for trekkers who ultimately coalesce at the origin of life later in the day

communication tool to maximise public engagement in many subjects. We argue that this is particularly true for phylogeny, due to its complexity, and biodiversity conservation, due to the inherent visual appeal of many species and environments. The importance of visualisation also applies to experts: scientific research is, for the time being at least, a pursuit directed by humans (despite computer-generated manuscripts superficially fooling some audiences (Stribling et al. 2005)), so human intuition guides which analyses to perform and enables ‘sanity checks’ that the data and findings are valid. Such intuition depends on seeing the data ourselves, which often requires advanced visualisations to accompany our analysis tools. As the amount of data increases, such visualisations become both more important and harder to develop.

Why Use the Complete Tree?

Although science communication does not require a complete tree of life (i.e. down to the species level), now that we have a draft phylogeny of all cellular organisms,

the Open Tree of Life (Hinchliff et al. 2015), we should surely aspire to visualise it in its entirety. There are many further reasons why an explorable depiction of the complete tree is desirable.

Few previous phylogenetic visualisations have aimed to show every species of life on earth. Most choose only a subset of organisms, which creates the problem of choosing which to include or omit. A pruned phylogenetic tree must either be biased (e.g. a tree of just mammals) or leave out iconic species that most people would expect to see on a tree of life. A complete tree avoids this pitfall and, because of its scale, encourages people to choose where to explore for themselves. For science communication, we have observed that such freedom of discovery results in a more engaging user experience and enables people to search for any life they can imagine without being disappointed (unless they are looking for unicorns¹). It also implicitly introduces a convincing body of evidence for the evolutionary process. This is a compelling alternative to trotting out a few stock examples like peppered moths or banded snails, of particular relevance to the problem of acceptance of evolution in various countries. We will return to the role of the tree of life in evolutionary education later.

A final and compelling reason to consider the complete tree of life is that, like the Ancestor's Trail walk and sacks of wheat mentioned earlier, it conveys the tremendous scale of biodiversity, and in an elegant yet rigorous manner. As Mark Westneat put it: 'just like Google Earth changed the way people look at geography, a sophisticated tree of life browser could really change the way we look at the life around us' (Zimmer 2009). Indeed, the initial motivation for OneZoom, our visualisation project which we detail later and from which many of our examples will be drawn (Rosindell and Harmon 2012), was to provide a way for anyone to easily explore the complete tree of life, just as they can explore Google Earth.

Conservation and the Tree of Life

As well as the need to preserve ecosystem functions and services, central to many conservation narratives is the beauty and wonder inspired by the sheer number and diversity of biological species. The reasoning parallels arguments for preserving important examples of original art or architecture: seeing and appreciating them enriches people's lives. In contrast to art however, biological diversity can seem like a permanent feature of the world around us, with some ability to heal and replicate itself. Unfortunately, this makes it all too easy to take biodiversity for granted, despite the fact that it is being lost at a worrying rate (Tittensor et al. 2014). Engaging people in conservation requires us to highlight what we stand to lose. If it is possible to do this over the entire tree of life, it should provide a more comprehensive picture

¹Actually, even then they would find many points of interest including unicorn fishes, unicorn beetles, unicorn plants, and 'Unicorn' – a genus of goblin spiders.

than just focussing on a selected list of species. We argue that an effective and underutilised way to raise awareness is to provide an easily accessible complete tree of life.

One approach to conservation, pioneered by the EDGE of Existence programme at the Zoological Society of London (Isaac et al. 2007), attaches value not only to the species at the tips of the tree of life but also to their evolutionary history – in other words the unique branches that they share (see Chap. 2). This results in a higher priority being given to species that are evolutionarily distinct and consequently represent a greater amount of biological (phylogenetic) diversity (Chap. 1 in this volume describes the history of thinking about phylogenetic diversity and conservation). An analogy can again be made to artwork; imagine a curator running through a gallery in the event of a fire, deciding which paintings to rescue in a limited time. They would probably try to save a wide selection of works, representing different artistic periods, movements, or bodies of work, taking care to rescue particularly rare and distinctive examples. In biological conservation, the equivalent is conserving distinct branches of the tree of life, each of which may contain a suite of unique evolutionary features. To incorporate anything like this approach into conservation biology requires appropriate phylogenies so that we can identify dangerously depauperate branches which may need conservation effort.

A final and more philosophical motivation for curating a complete tree of life, especially one enriched with comprehensive information, is that it may count as a form of ‘digital conservation’ in its own right. Of course we would rather preserve biodiversity in living form as part of a healthy ecosystem. However, if biodiversity loss continues as feared, then it may be of some small consolation to preserve a testimony of what once was. This could also serve as an important warning to future generations and inspire them to conserve what is left. In the gallery analogy, it is like generating a high-resolution scan of important paintings: not only can they be appreciated second-hand by those not able to visit the museum, but in the event of a disaster, at least we retain a compelling record.

Data Availability and Curation

Names and Other Identifiers

The first challenge we encounter when creating a tree of life is to acquire the necessary data. Even in an age of extensive online databases, this is extremely difficult. It is not just phylogenetic information: simply compiling a list of known species is surprisingly problematic. The root of the problem is that unlike registering cars, people, or domain names, there has been no historical requirement to notify a central authority of new species names and descriptions. True, there are official naming rules which must be followed within different fields, and in some, such as zoology, official registers have been recently introduced (ZooBank website). But although applying the rules is compulsory, registration, with a few exceptions, is not

(Hawksworth et al. 2009). Even today, descriptions of new species can be published anywhere in the world, without central notification, which means that comprehensive species lists can only be compiled by scouring the Internet for potential sources of names. Lack of registration also means, for example, that the multiple formal descriptions of the same species can exist (the earliest should take precedence, but this will only happen if the synonymy is detected). The same shortcoming applies to name changes. For example, it is common to reassign whole swathes of species to different or novel genera, creating new names for the original species. The result is widespread duplication because there is no centralised system to register the old names as redundant synonyms (Patterson et al. 2010).

The need to collect species names from multiple, often inconsistent sources leads to another problem: most trace back to hand- or type-written originals. Transcribing these into databases frequently results in the creation of new, erroneous species names. These range from simple misspellings to the creation of nonsense entries such as the genus *Not* (incorporating species such as *Not* used, *Not* listed, and *Not* known sp. (Mesibov 2018)). In summary, even before we consider whether a species is biologically valid, there are reams of spurious naming errors to deal with. The advent of computational Taxonomic Name Resolution Services (TNRS) (Boyle et al. 2013) is starting to help, but errors continue to persist. We are still in a situation where not only are there many species we do not know about but we also do not know how many we *do* know about.

Any project dealing with species from across the tree of life will hit these issues, and most end up using a system of unique species identifiers instead of taxonomic names. For example, genetic sequences submitted to GenBank must be mapped to the NCBI taxonomy, where species are given a unique, nonreusable number, and a team of curators attempts to rationalise synonyms, misspellings, and taxonomic revisions (Federhen 2011), but even with its restricted remit and coverage (only a few hundred thousand species), this is not a trivial task. Similar unique IDs are implemented by more comprehensive global biodiversity projects such as GBIF (over 1.6 million species in their Catalogue of Life), the Encyclopedia of Life (EoL, over 2.2 million species), the WikiProject Taxonomy on Wikidata (~2 million species), and (most relevant to this chapter) the Open Tree of Life (over 2.1 million species). Although these have differing aims, all attempt to unify taxonomic names from multiple sources (e.g. the NCBI taxonomy) into their own numbering system. Unfortunately, by merging multiple sources for comprehensive coverage, all global biodiversity projects accidentally promulgate, and sometimes compound, errors in the original data sets. A major ongoing task is thus to implement curation methods to help clean these data sets, for example, sophisticated parsers for scientific names (GlobalNames website). Outreach can play a major role here, since many taxonomic errors are obvious to humans but not to computers (e.g. at OneZoom we regularly get users providing taxonomic corrections for data sourced from elsewhere, and we endeavour to pass these back to the original source). In the long term, we suspect that Wikidata probably stands the best chance of capturing the ‘many eyeballs’ needed for the mixture of automated and manual curation required for this volume of data

(also note that the WikiProject is uncommonly fastidious in its approach, e.g. see discussions at the WikiProject Taxonomy web page).

Given that comprehensive biodiversity inventories require automation, we must accept that any complete tree of life visualisation will contain spurious species and other naming errors. Moreover, successful public outreach is likely to require coordinating information such as phylogenetic placement, common names, images, and so on, from multiple sources, each likely to contain their own errors. This coordination requires the use of some sort of identifiers, be they taxonomic names or numbers, to match up organisms from different databases. We suggest that the key here is to avoid compounding problems by implementing yet another bespoke TNRS. For this reason, our OneZoom project reuses identifiers from a single project (the Open Tree) and creates mappings to entries in other projects (most importantly EoL, Wikidata, and IUCN) by comparing IDs from the source data sets, rather than using species names. A robust mapping to sources such as Wikidata not only allows rich information to be added to the tree but also enables more ambitious projects, such as obtaining the frequency of Wikipedia visits to various taxa, which allows us to calculate an index of popularity for all species and taxa over the entire tree of life (available at OneZoom.org).

A separate issue is the problem of accessing and sharing biodiversity information. It is generally accepted that taxonomic names should be made publicly available and reusable (Patterson et al. 2014), but this may not apply to the published phylogenies that show relationships between taxa (Stoltzfus et al. 2012). Alongside taxonomic name resolution, availability and reusability of phylogenetic tree data is now one of the main factors which hinder us from constructing a tree of life that correctly reflects the full extent of scientific knowledge.

Taxonomies and Phylogenies

Given a comprehensive list of species, it is reasonably straightforward to present a taxonomic classification of all life, since nearly all sources provide a rough taxonomic placement for their names. However, in Section 1 we gave strong arguments for presenting an accessible phylogeny of all life, rather than a simple taxonomy. Clearly this is a much more difficult task. Indeed, by a rough estimate, over 90% of species are not in any published phylogeny (Rees 2018), so all attempts to present a phylogenetic arrangement of life must be a combination of taxonomy and phylogenies (for instance, treating all members of a single genus as a single unresolved clade). An impressive trail was blazed by the Tree of Life Web Project (ToLweb), first made public over 20 years ago (Maddison and Maddison 1996). However, it relies upon manual construction and curation of trees by authorities in the relevant area, and although some expert assessment of phylogenetic data is highly desirable, it is hard to see how projects which take this approach alone can scale to trees with millions of tips. It is for this reason that we believe the Open Tree of Life (Hinchliff et al. 2015) currently has the greatest potential to provide the key data required for a

complete tree of life browser. Although some expert curation and feedback can help select trusted phylogenies and resolve problems such as taxonomic name errors, in general anyone can add new studies to the Open Tree, and these are automatically combined into a summary, ‘synthetic’ tree of life. While there is no space to describe the project in detail here, what stands out is the automated combination of phylogeny and taxonomy, the relatively low level of expertise needed to add new data to the tree, and the focus on providing referenced sources for topology and conflict. Offsetting these benefits is the complex process required to create the synthetic tree and the subsequent rather infrequent update cycle (months, not hours). Additionally and in common with projects such as TreeBASE (one of the principal sources for phylogenies in the Open Tree), there does not seem to have been a wholesale move by taxonomists to provide data and curation expertise to the Open Tree project – which is what is needed to turn it into a central repository of phylogenetic information. Along with the non-permanent funding for this project, these are the critical issues which we think must be addressed if we wish to create an accurate public portrait of the current scientific understanding of biodiversity and the tree of life.

Dates

It is perfectly possible to produce a phylogenetic classification without any mention of dates or timescales on the tree. However, one of the main benefits of using detailed trees rather than simple taxonomies for outreach is that they are directly interpretable in terms of evolutionary history. This becomes much more intuitive if there are estimated dates at branch points. Nevertheless any complete tree is bound to contain many undated nodes (e.g. from taxonomy), so any presentation must be able to seamlessly combine both dated and undated nodes – and ideally communicate measures of the uncertainty in dates too.

Obtaining trustworthy dates is complicated by the fact that reconciling dates from multiple phylogenetic studies is much less straightforward than reconciling the phylogenies themselves. Not only may studies disagree about the existence of key nodes, but there is also the problem that time estimates are rarely independent, often being based on the same fossil calibrations or estimates of rates of molecular evolution, and it is difficult to infer nonindependence between studies from digitised summaries. There have been some efforts to provide expert summaries of key dates on the tree, notably the TimeTree project (Kumar et al. 2017), but its requirement for extensive manual curation, an absolute focus on molecular time estimates, and restrictions on harvesting the information for other projects combine to make this a problematic source of dating information for a species-level complete tree of life.

One promising effort to provide reusable dates for a complete tree of life is the DateLife project (O’Meara 2017), which is at a relatively early stage of development (but see (Stoltzfus et al. 2013)). Node dates in our OneZoom tree thus currently come from a manually curated backbone combined with specific single-dated phylogenies,

such as for birds (see the OneZoom website for details of data sources). This is clearly not scalable, and we see automated large-scale estimation of evolutionary dates from multiple studies as a major area for future research.

Phylogenetic Tree Visualisation

The Broad Range of Possible Choices

A huge range of methods have been proposed for visualising phylogenies and more generally for visualising hierarchies. For example, at the time of writing, the website treevis.net (Schulz 2011) indexes over 300 generalised visualisation approaches, some more intuitively ‘tree-like’ than others. In addition, Page (2012) provides a comprehensive review of many approaches specific to phylogenies. The fundamental reason for this variety is that nodes can be thrown down in any position (in two or more dimensions) and then combined with any imaginable method depicting associations between nodes to result in a valid representation of the tree. This provides a huge subjective freedom to design visualisations that emphasise different features of the tree for different purposes. For instance, a maze without bridges and cross-links is a tree; hence, we can visualise all or part of the tree of life as a maze. This creates a novel type of educational evolutionary puzzle: the reader is invited to solve the example laid out in Fig. 3.2.

There are broader issues with the idea of visualising the tree of life. Firstly it may not be a single tree at all, due to hybridisation, incomplete lineage sorting, and deeper cross-links created by endosymbiosis or other forms of horizontal gene transfer. Furthermore, there is uncertainty around many parts of the tree, including fundamentals such as where to place the root (something which may never be known) (Gouy et al. 2015). These are weighty problems, and we show some example visualisations in Fig. 3.3 from DensiTree (Bouckaert 2010) and ToLweb (Maddison and Maddison 1996) that attempt to overcome the issues. In this chapter, however we restrict ourselves to visualisation of phylogenies that can be represented as objects satisfying the mathematical definition of the tree: a set of nodes, joined by a set of links, such that there is exactly one possible route between any pair of nodes.

Existing Methods

The methods most commonly used by researchers in phylogenetics are simple two-dimensional representations where a node corresponds to a point and an edge or branch to a continuous line. Often we wish to convey branch lengths, where the connections between nodes have an associated evolutionary distance, either time or some measure of genetic distance. For unrooted (technically ‘undirected’) trees, the connecting edges are usually straight lines whose length is a direct measure of

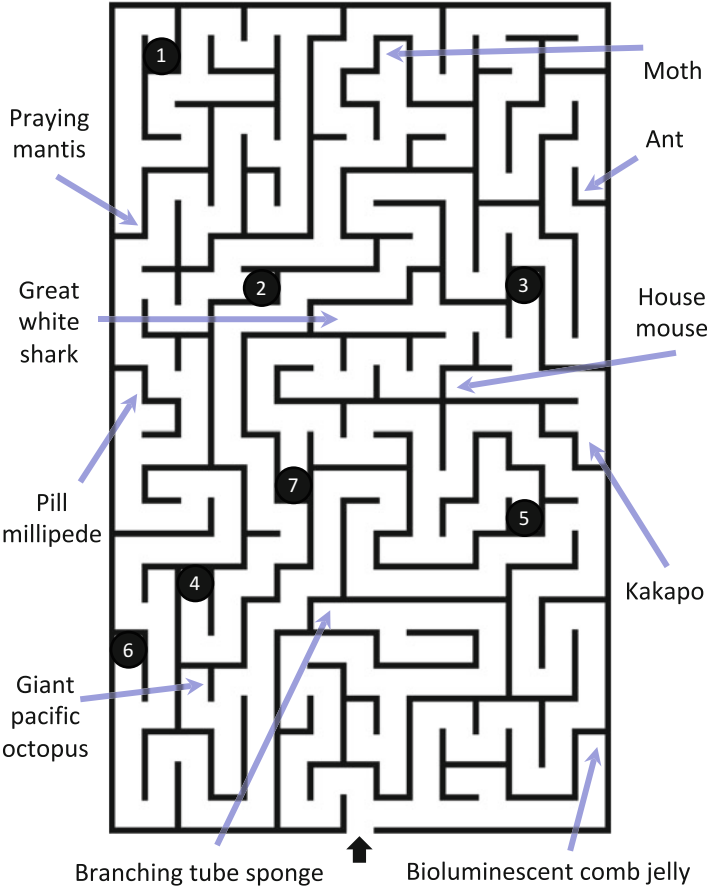


Fig. 3.2 An example ‘maze of life’ puzzle. Can you find numbers to match with the following species names? Human, house mouse louse, olympic torrent salamander, tardigrade, blue-black spider wasp, chocolate chip starfish, Pacific dampwood termite, and cactus sea squirt. Solutions are found at the end of the chapter

evolutionary distance. For rooted (directional) trees, the projection of lines in one dimension is usually made proportional to branch length, so that one axis corresponds to time or genetic distance, while the other dimension is used to avoid crossed lines (Felsenstein 2004). Three-dimensional approaches can add excitement or additional data, but although some magnificent 3D trees have been produced (Sanderson 2014), in general, we argue that a compelling reason is needed to justify adding a third dimension, given the extra difficulty in displaying and exploring such trees, including the almost inevitable problem that some parts of the tree will overlap others. One example that may justify the use of a third dimension is visualising geophylogenies, where the leaves of a relatively small phylogeny sit as points on a two-dimensional map of geographical regions and the third dimension shows connections from those sites to common ancestors (Shapiro et al. 2006). Another

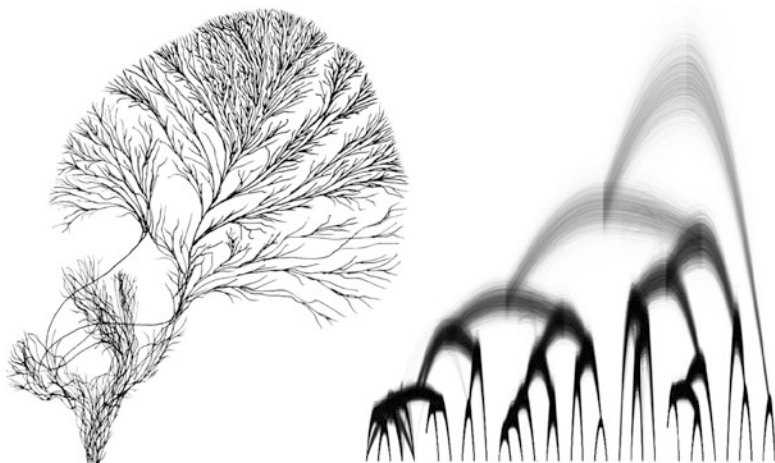


Fig. 3.3 Trees that are not trees. Left panel: a figure from the ToLweb project (Maddison and Maddison 1996) showing endosymbiosis. Right panel: a figure from the DensiTree software (Bouckaert 2010) showing tree uncertainty

possible direction is to use the third dimension of height as a way to embellish an essentially two-dimensional visualisation, much like a 3D topographic map. As long as this is rendered from a high viewpoint, then the problem of overlap will be much less severe, and exploration should still be possible with only panning and zooming.

Choosing from the plethora of alternative tree visualisations will be subjective and depends on intended use; there is no ‘one-size-fits-all’ solution. Here we describe some of the considerations in this choice. In the context of phylogenetics, extra information beyond tree topology is important, for example: evolutionary distance, taxonomic identity on leaves and nodes of the tree, and the total diversity present in different clades. There are also general principles of visualisation to consider such as appropriate use of space, visual appeal, avoidance of clutter, and ease of picking out important information. These factors can directly pull against one another in trade-offs that we discuss later. Another problem is that scientists immersed in phylogenies on a day-to-day basis may be biased towards the standard scientific visualisations they are used to, even if such visualisations are inherently worse for general communication purposes. Hence the opinions of the wider public and those in full-time education, especially children, are valuable in choosing which visualisation methods are best for the purposes of outreach.

Dealing with Very Large Trees

One body of work has focused on the problem of scale and exploration of large trees; the recent release of the 2.2 million tip Open Tree of Life (Hinchliff et al. 2015) has rendered this particularly relevant. Four broad classes of approach were identified by

Page (2012), though we must keep an open mind to the possibility of new innovations.

First, the *large media* approach simply allows visualisations to spread over multiple monitors or poster-sized prints, but this quickly runs into limitations. Take a complete tree of life with roughly two million tips and suppose (generously!) that we have some way to visualise the tree that uses only 1 square centimetre for each tip including all necessary branches. Such a tree would cover 200 m², just over the size of a tennis court. At least triple this space would be needed just to print the labels for each species in a legible font.

Second, the *simplification* approach redesigns the visualisation so that only a smaller tree needs to be visualised at any one time, for example, by allowing the user to expand and collapse different parts of the tree or by showing the tree in many smaller sections each of which is essentially an independent visualisation. One instance is the native browser for the Open Tree of Life project, where pages are generated to represent small sections of the tree, connected by hyperlinks (Hinchliff et al. 2015). This is the easiest way to enable exploration of a large tree. However, from the visualisation perspective, it feels like a cheat: it never does visualise a large tree, it only redefines the problem so that we do not need to.

Third, the *focus and context* approach distorts the view to expand one area of interest while shrinking (rather than hiding or collapsing) the context outside it (Lamping and Rao 1996). This is used by hyperbolic tree viewers, which can be intuitive and visually impressive. However, we have yet to see this applied to a large navigable tree of life. Scaling to this size is likely to present technical challenges, as well as requiring a rethink of the method by which the user chooses a focus. For example, if focus is chosen based on the position of a mouse pointer on a conventionally sized screen, a tiny move of the mouse would result in a dramatic and disorientating jump in the visualisation.

Last, the *panning and zooming* approach has a user interface equivalent to that of interactive geographical maps. The user zooms in for details and out for context; very small items become effectively hidden. In our view this approach is most promising for the largest trees: it incorporates an intuitive user interface to which people are already accustomed, enables essentially limitless space, and provides a single seamless visualisation.

It is noteworthy that none of the above approaches are suitable for distribution on paper, or even for publication in the electronic offerings of traditional scientific journals. This is one limitation to the publication process that will need to be addressed in the future with movement towards richer and more interactive digital offerings.

Use of Fractals

A fractal object is one which appears similar at different scales and is therefore well-suited to being explored using a panning and zooming user interface. Certain fractals

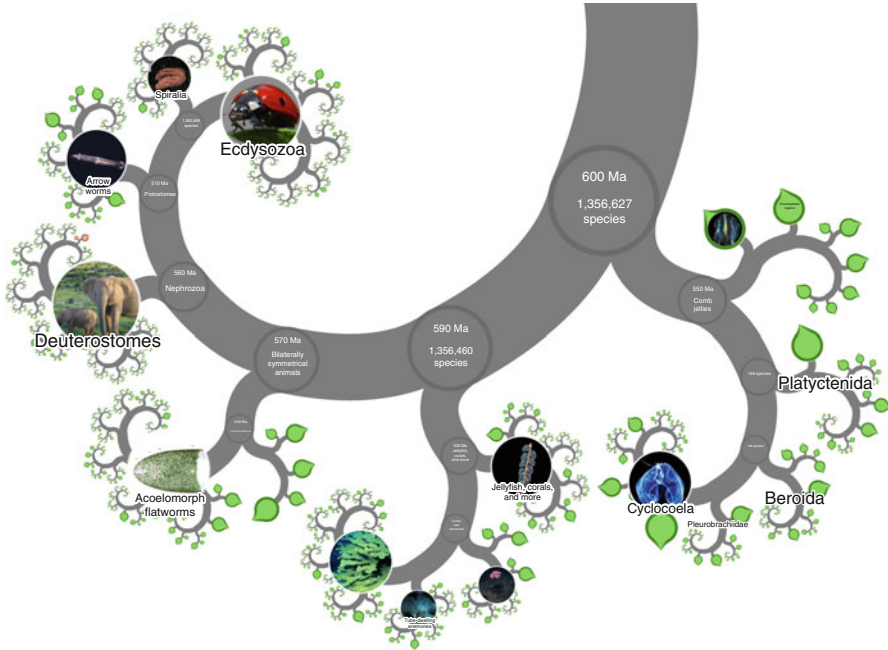


Fig. 3.4 A screenshot from [OneZoom.org](https://onezoom.org) illustrating a fractal viewer for the tree of life with a section of the animal phylogeny (over 1.3 million species)

can be used to represent connected nodes and edges, including hierarchical tree structures (Koike and Yoshihara 1993). Once embellished with useful data, we term these ‘Interactive Fractal Inspired Graphs’. To the best of our knowledge, all attempts to visualise the complete tree of life, or even a large part of it, have used this approach (Rosindell and Harmon 2012; Block et al. 2012; de Vienne 2016) (see Fig. 3.4). The fractal-like layout of leaves and nodes means that the tree is comfortable to interpret at all scales of zoom and also results in highly memorable visualisations (Borkin et al. 2013). One usually wishes such a fractal tree to be nonoverlapping in space, which can be guaranteed using a nested method of fractal construction. Other more general methods of fractal construction can achieve identical results but require careful parameterisation to do so.

Note that we use the term ‘fractal’ somewhat loosely, because in a tree of life visualisation, the fractal structure does not carry on indefinitely. After enough zooming, terminal branches of the tree are revealed (corresponding to individual taxa), which end the self-similar behaviour of the visualisation. This is similar to fractals observed in nature, such as the surface of an animal lung, where fractal behaviour ends at the scale of the alveoli. Visual comparison to a true mathematical fractal may also fail because, except in special cases such as a perfectly balanced tree with $2n$ tips, the termination of the fractal structure is not uniform across the tree but happens where the empirical data dictates.

There are several technical challenges in visualising a fractal tree of life with millions of nodes on a zoomable computer canvas. Firstly, the coordinates of graphical elements (and the user's view of the tree) cannot be described with reference to a common anchor point, because the amount of precision required for a precise layout after deep zooming is beyond the machine error of normal variables. OneZoom (Rosindell and Harmon 2012) and DeepTree (Block et al. 2012) independently arrived at a similar solution: the tree may have a fixed root, but the graphics of the visualisation do not. Consequently each graphic element as well as the user view into the tree must be described in reference to a close by 'anchor' elements of similar size. A second issue is that the amount of data in the tree is typically too large to be stored in local memory during tree exploration. Consequently, a dynamic loading system must be developed to swap the relevant parts of the tree in and out of cache as the user explores. In OneZoom, this is managed in two parts, the logical topology of the tree is sufficiently small to be stored in local memory, while the graphical structure of the tree itself expands and contracts dynamically, requesting further information on each leaf and node of the tree from a remote server. A third software platform, Lifemap (de Vienne 2016), is based on core technology from OpenStreetMap. This approach renders the tree as a number of tiles at differing resolutions that are loaded in and out of local cache and provided by a server as needed, based on the underlying fractal generation methods.

Is the Tree of Life Itself Fractal?

This is a question that predates fractal-based visualisations of the tree of life (Green 1991), and is often put to us in connection with OneZoom – with the tacit assumption that the answer will be 'yes'. However, the analogy is not obvious. In mathematics, a fractal is an object which has a noninteger 'fractal dimension'. Imagine the example, similar to the surface of a lung, of a geometrical object embedded in three-dimensional space in such a convoluted manner that it is somehow more than just surface (two dimensions) but less than a completely solid object (three dimensions). Its fractal dimension will be somewhere between two and three. A raw tree topology, however, is embedded in space only with a visualisation: the resulting object may have a fractal dimension, but that is a property of the visualisation rather than of the tree. It is therefore unclear whether the tree itself is fractal. One approach to answering the question is to consider another key property of fractals: self-similar patterns tend to recur as we change scale. Tree objects have a somewhat analogous property because an edge branches into two or more smaller edges which can then branch again, and so on, resulting in similarity at different scales. If the tree of life is fractal-like in this manner, we would expect to be able to see tree shape parameters that show similar patterns or properties at different scales and positions in the tree. For example, if we think about balance, which measures the difference in richnesses of branches descended from the same node, this might be static across the tree as a

whole, implying some kind of self-similarity, or it could vary, for example, as a function of date or taxonomic level (Purvis and Agapow 2002). Another possible avenue for future testing self-similarity in a large tree would be to slice the tree into small chunks of a fixed size and look for constant features of those smaller trees, such as fixed associations between topological properties and measures of (say) age or diversity. One complicating factor to consider is whether we would expect self-similarity in the tree of all species, both extinct and extant, or in the pruned tree of only living species. The difference between these two trees will certainly be time dependent, since a greater proportion of old lineages will be extinct, and we would not expect to see self-similarity in both trees. Self-similarity seems intuitively more likely to be manifested in the tree that includes extinct species.

Understanding Inherent Limitations

Building visualisations for small trees is relatively easy and can thus give the impression that there is no need for compromise. But the human mind only has a limited amount of attention, and for large trees, emphasising one aspect of data often requires reducing emphasis on another. We suggest that to move forwards with representations of very large trees we must accept the inevitable need to compromise. Various visual trade-offs exist, and choosing a balance depends on subjective priorities.

‘Projections’ of the Tree of Life onto a Zoomable Canvas

Analogies between large phylogeny visualisation and geographic maps have often been pointed out (Page 2012) and are particularly relevant here. Geographic maps require projection from the surface of a sphere onto a plane, and there are many ways of doing so, each with their relative merits (Snyder 1987). For example, it is possible to preserve the areas of regions or the distances between points, but not both. In an analogy we consider different ‘projections’ of the tree of life onto a zoomable canvas by using different possible fractal formulas. Here, the problem is tougher because in geographical maps a point at least has a known location on the surface of a sphere. In contrast, in a generalised tree, there are no predefined coordinates to use, and so they must be decided upon using other visualisation principles. DeepTree (Block et al. 2012) is reminiscent of a classic cladogram, adjusted for use in a zooming environment. OneZoom (Rosindell and Harmon 2012) uses several projection models and aims to provide a framework for constructing more (see Fig. 3.5). The OneZoom ‘natural’ and ‘polytomy’ projections as well as the view in Lifemap (de Vienne 2016) all show more substantial variability in their fractal structures at different scales, compared to simpler fractal forms. This is because the angle and size of deeper parts directly encode information about the diversity of groups. We find that

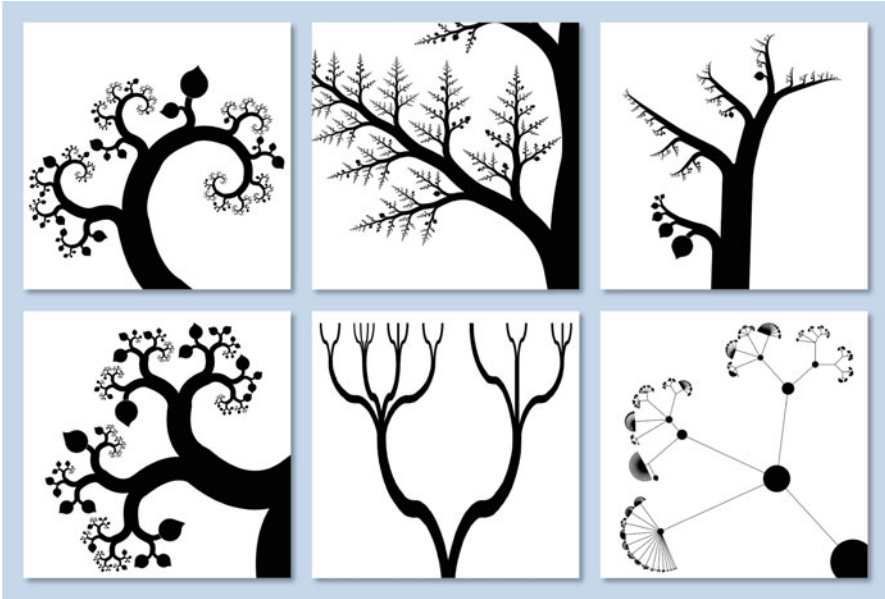


Fig. 3.5 A range of different fractal projections, top left to bottom right: spiral, fern, natural, balanced, cladogram-like, and polytomy. The cladogram-like projection was mocked up following DeepTree. All other views are silhouetted screenshots from OneZoom; the polytomy view was partly inspired by Lifemap and follows a related fractal form

these differences in shape make patterns at different scales prominent when exploring the tree. For example, in the ‘natural’ view, it is clear that the tree consists of lots of imbalanced nodes, where present day is unevenly split between descendent branches, followed by an occasional balanced node, where diversity is split fairly evenly.

Many tree of life visualisations, especially those based on a taxonomy rather than a phylogeny, have to deal with polytomies, where a node can have more than two descendants. Such views have fewer nodes to travel through before reaching a destination, making the tree seem less daunting. However, if there are very large polytomies in the data, this causes a different problem as they are also difficult to navigate due to lack of structure. Think of how you organise files on your computer: for ease of exploration, it is suboptimal either to have two items or hundreds of items in a single folder. That there are polytomies at all is a feature of the underlying data and not of the visualisation, so we are lucky if such data is dominated by relatively small polytomies which are more intuitive for human reading. In the case of balanced bifurcating tree data, a polytomy view (de Vienne 2016) would look very like the ‘balanced’ view, and in the case of bifurcating trees that are consistently slightly unbalanced, it would be reminiscent of the ‘spiral’ view (Fig. 3.5).

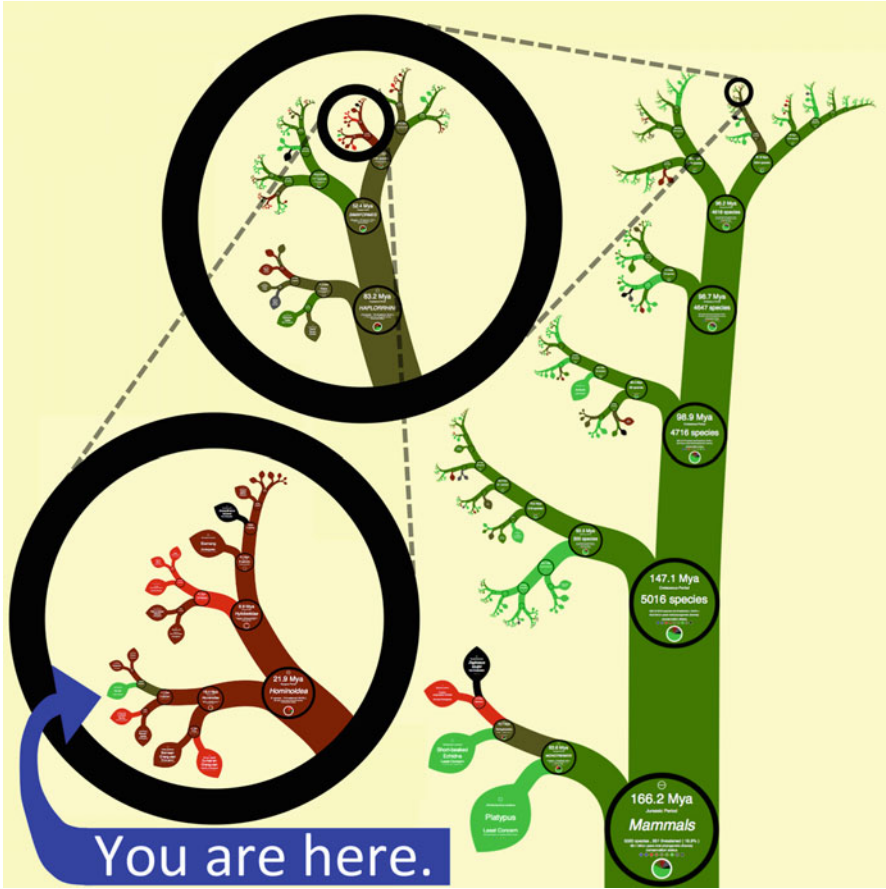


Fig. 3.6 Compound screenshot first published in 2012, built using the OneZoom ‘natural’ projection model

A separate aspect of projections is that the children of nodes can be ordered according to any logic. For example, in many of the OneZoom projections, we chose to put the more diverse group on the right of any split. As a result, humans are not always in a predictable or ‘special’ position on the tree, helping us maintain a pleasing sense of perspective. This can be highlighted with a ‘You are here’ label (Fig. 3.6), inspired by Douglas Adams’ *The Hitchhiker’s Guide to the Galaxy* series (Adams 1995) in which a ‘Total Perspective Vortex’ shows victims their insignificance on the scale of the entire universe as a form of psychological torture. We feel that this message is worth remembering as the human species tries to get a sense of perspective in the context of the rest of life on earth, much of which is under threat by our actions.

Emphasising Topology Versus Species Richness

One of the most important balancing acts in choosing a projection of the tree of life involves how to convey the number of species in a group, the ‘species richness’. In classic cladograms, an equal width is allocated to each species (tip) on the tree. The richness of any group is then clear from the width of space it occupies. When the tree is small enough for every tip to be visible, this works well, but for larger phylogenies, it can present a problem. In particular, if the visibility of a clade is directly proportional to its richness, clades with very few species become essentially invisible. This is not a major problem for deeply nested clades, which simply become subsumed into larger, visible groups. But in unbalanced trees, it can actually alter the perceived topology of the tree. An extreme example is seen in the phylogeny of animals, which includes a deep branch leading to a single described species, *Trichoplax adhaerens* (Srivastava et al. 2008). There is no way to give all species equal prominence while still displaying this branch – in fact, in most trees designed for public outreach, this branch is omitted, and the incompleteness of the tree topology is simply accepted. The only way to maintain a complete topology in this case is to make this branch visible, while fitting other groups, such as insects, into a smaller space than their richness would suggest. It is worth noting that a somewhat analogous situation occurs in geographical mapping, where tiny islands may need to be shown at a larger scale (Crahen 2016). In the case of large phylogeny visualisation, a common solution to reducing the space requirements for a whole branch is to collapse it into a single (often visually different) tip which can be expanded as required. Clearly, this means that the richness of groups on the tree is no longer intuitively obvious. It is an illustration of the unavoidable trade-off between emphasis of topology and richness.

One workaround to the ‘invisible *Trichoplax* problem’ is to add labels to the tree as a separate layer over the top of the underlying graphics. The tree graphics themselves could therefore scale as we wish, for example, proportionately to species richness, while text would label the presence of groups that are invisible on the graphical layer. However, this approach is unlikely to scale to large, unbalanced trees and simply shifts the problem of trade-off and bias onto our choices of which labels to show and where to show them.

Another common approach is to abandon the idea of completeness altogether and make all the tips of the tree visible by replacing larger groups with a few ‘representative’ species. Although all tips on the tree can then occupy equal space, such visualisations also give a misleading view of richness because some tips inevitably end up representing more diverse groups than others. In essence the pruning decisions (rather than the visualisation algorithm) implicitly define a position on the richness-topology continuum. In fact, even the act of selecting representative tips often incorporates a topology-richness trade-off. For instance, imagine selecting a set of eight ‘representative’ animals. If we choose proportionately to species richness, all but one should be arthropods, and most of those should be insects. For members of the public, indeed for most biologists, this is unlikely to be considered a

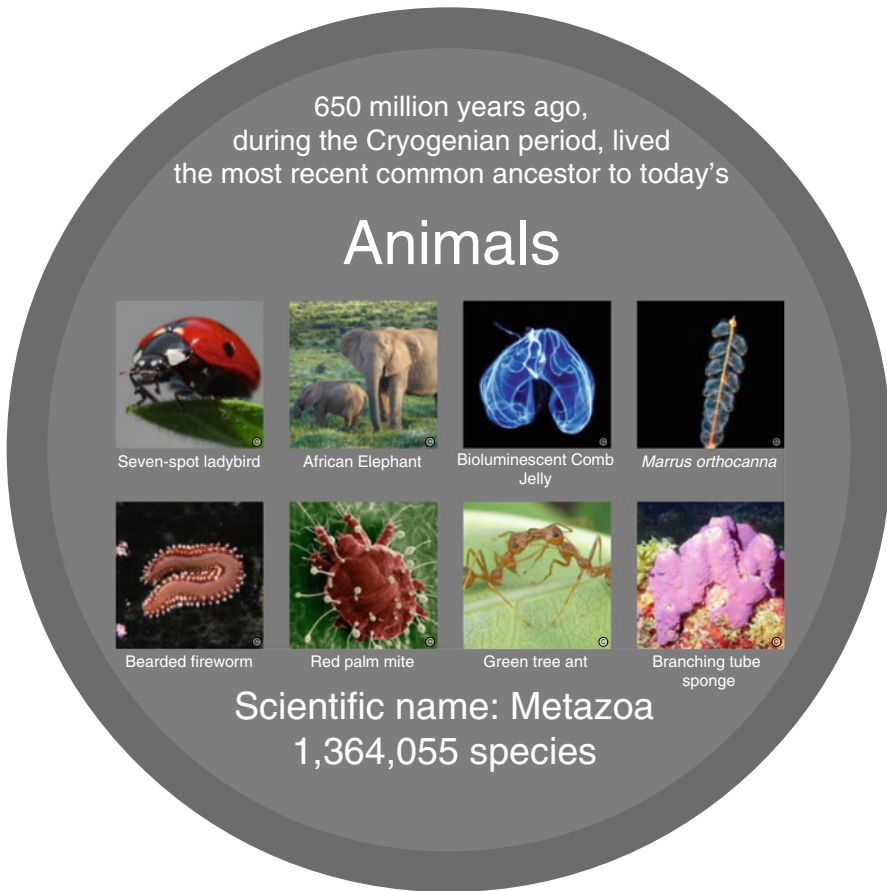


Fig. 3.7 An example interior node for all animals from [OneZoom.org](https://onezoom.org). The eight images are selected by an automated algorithm. This balances the goals of showing high-quality images, high phylogenetic diversity (there is at least one sponge), and reasonable representation from species-rich taxa (there are three arthropods and two insects)

reasonable representation of animals as a whole. A more acceptable approach is to use the topology of the tree to help reflect the phylogenetic diversity of the group. The example in Fig. 3.7 shows the result of such an approach implemented in OneZoom: a comb jelly, cnidarian, sponge, vertebrate, and worm have been automatically selected as representative animals, at the cost of fewer arthropods. Because this approach incorporates phylogeny rather than requiring selection of representative species by hand, it is automatable and able to produce reasonable representative selections of species for all nodes on the complete tree of life.

Let us consider two potential fractal projections onto a zoomable canvas representing opposite ends of the richness-topology continuum. To emphasise richness, a ‘treemap’ (Balzer et al. 2005, Vliegen et al. 2006) has area directly proportionate to richness, but at the cost of an opaque topology. In contrast,

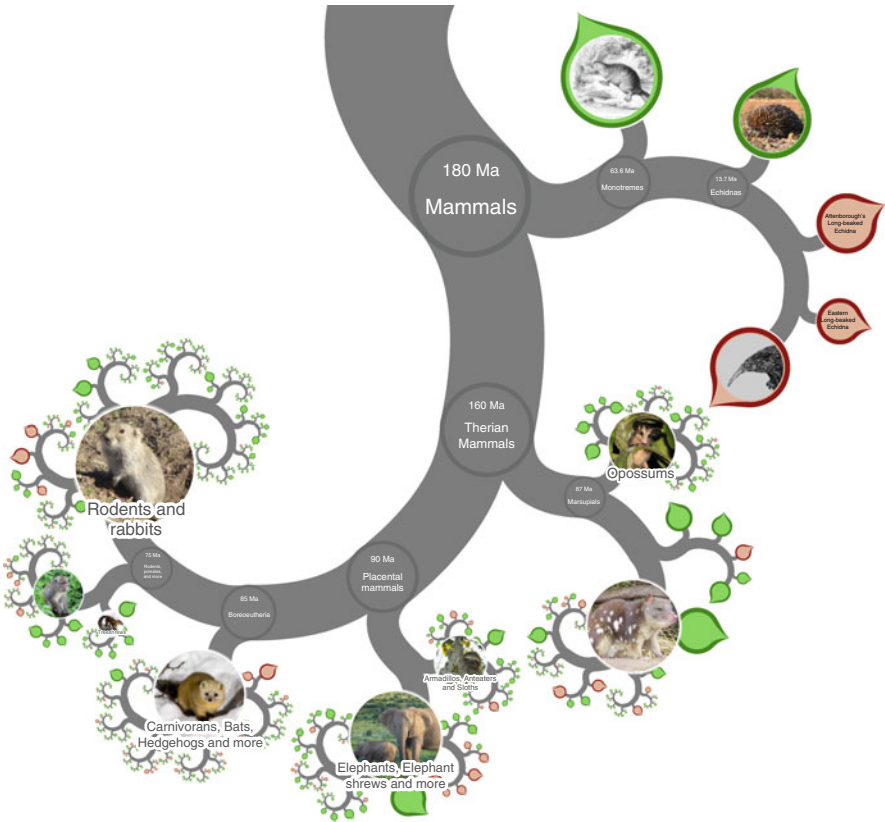


Fig. 3.8 A screenshot from the ‘spiral’ projection, emphasising topology over richness resulting in Evolutionarily Distinct (ED) species occupying more space. Globally Endangered (GE) species are coloured in red. Overall, this emphasises the three high-ranking EDGE species (long-beaked echidnas) at the top right that are Evolutionarily Distinct and Globally Endangered (EDGE)

something like the ‘spiral’ view on OneZoom (see Fig. 3.8) allocates space to descendant branches at a fixed ratio. Richness plays a very minor role here: although the richer group always branches to the right, we cannot tell how much richer it is. Topology-biased views naturally highlight evolutionarily distinct species. When combined with endangerment status, this can produce visualisations which emphasise leaves that are Evolutionarily Distinct and Globally Endangered (EDGE species, as discussed earlier). Intermediate solutions sit closer to the centre of the trade-off, for example, by scaling species richness by square root (de Vienne 2016) or in log scale (Fig. 3.5 bottom right panel) rather than linearly. Another possible direction is to draw a treemap but reserve a large amount of the total area to draw a tree of connections between the different treemap sections. For example, it would be interesting to plot a zoomable treemap where half of the area is allocated proportionately to species richness and the other half proportional to total branch length (i.e. phylogenetic diversity) and used to display topology explicitly.

Emphasising Topology Versus Time

At present, there is no dated version of the complete tree of life, though excellent progress has been made (Kumar et al. 2017; O'Meara 2017). Consequently, the need to perfect zooming fractal visualisations of such trees is less pressing. We believe such visualisations to be possible with alternative projections. Indeed, visualisations with polytomies are really just a special case of dated tree visualisation where certain branches are of length zero. One subjective choice in a dated tree is whether to emphasise the length of a very long branch or the identity of what lies at the end. Whatever the emphasis, it may be necessary to sacrifice perfect scaling of branch lengths with time, just as appears to be the case with richness and space. In fact, even standard phylogeny visualisations do this: a neat arrangement of branches is achieved by allowing branch lengths to vary arbitrarily along one of the two axes of the visualisation, provided that their length along the other axis scales with estimated time or evolutionary distance. If we are using both axes for topological arrangement, one useful enhancement may be to add what we name 'tempographic lines'; analogous to topographic lines, these join points of equal date on the tree. The resulting visualisation shows date in a similar manner to height on a geographic map (Dawkins and Wong 2016) (Fig. 3.9); indeed another possibility is to actually plot a fractal-inspired graph onto a 3D surface, with the third dimension (height) reflecting geological time.

Some visualisations reasonably aim to incorporate labels on interior nodes as significant features of the tree. However, doing so highlights a large disparity in the number of nodes passed through on a journey from the root of the tree to different tips. This disparity is real, in the sense that the tree is not balanced, but showing this risks perpetuating the misunderstanding that there is an evolutionary ladder with a 'main branch' of evolution (no doubt with humans at the top) and minor (read less important) side groups. The more attention we draw to interior nodes, the greater emphasis we place on showing the true imbalance of the tree and the more distorted our perception of time. Paths to leaves of the tree will not appear equal because they pass through different numbers of prominent nodes along the way. Returning to the analogy with geographical mapping, consider the map of routes on a public transport system, such as the London underground. Such maps often ignore true distance and just show the stops, connections, and (occasionally) journey times as text. As a result of catering for the user, who cares primarily about the network topology and number of stops or changes, the apparent distance of some routes is artificially altered. In tree of life visualisations, the equivalent subjective trade-off relates to the importance we attach to interior nodes of the tree. At the scale of the complete tree of life, some kind of interior node labels are essential to navigate the diagram, and it will be challenging to achieve this without leading to some perceived distortion from imbalance.

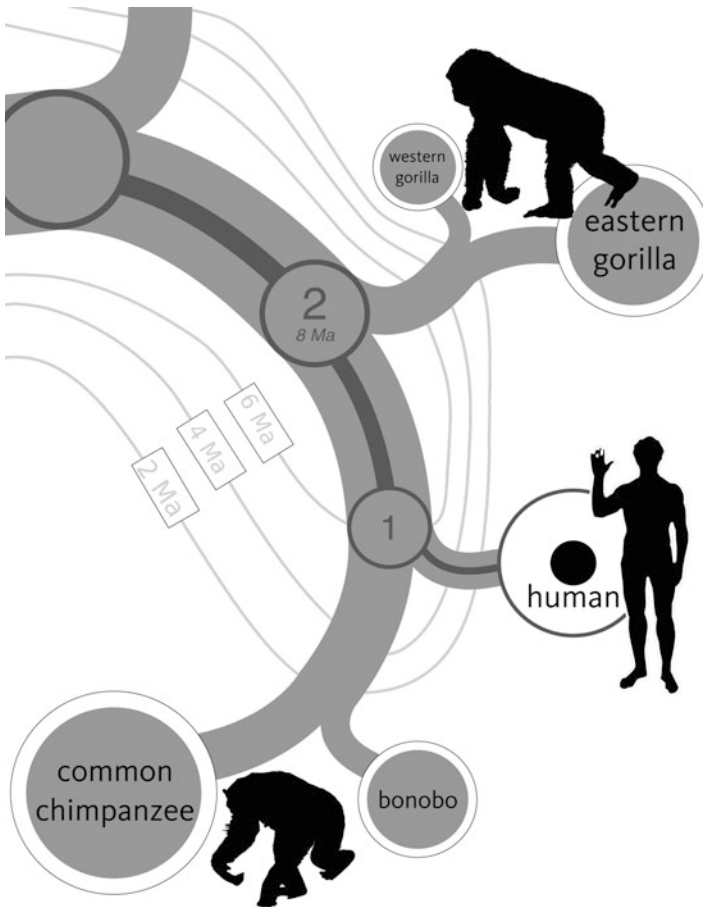


Fig. 3.9 Imagery modified from *The Ancestor's Tale* (Dawkins and Wong 2016) showing 'tempographic' lines analogous to topographic lines in geographic mapping, but here indicating passage of time

Aesthetic Appeal

Aesthetic appeal, like science communication in general, is easily forgotten if we concentrate only on obvious scientific utility to researchers, a tiny proportion of the potential audience. Researchers may get excited by a simple phylogeny revealing some interesting finding, but it is unlikely to capture the interest of a member of the public. For outreach, it is therefore important to have an immediate visual impact that conveys a scientific point while staying true to the underlying science. In analogy to geographic maps, think of the map of a theme park or public attraction that you visited. At the pay gate, you are not handed a sterile, scale representation of what you are about to experience. Instead, you see a basically accurate arrangement

of features, of varying scale, highlighting points of interest: perhaps the big loop of an important ride is drawn with a distorted scale so that it playfully jumps out of the page at you. There is a reason for these visual choices – they get people more easily excited – and we should perhaps learn from these graphical design principles when constructing images of the tree of life for public consumption: this could be a case where thoughtful 3D visualisation techniques could be put to use. We hope to see more outreach work around the tree of life with these ideas in mind, but some of the principles have already been used. For example, we intended the aesthetic of the original OneZoom design to invoke a feeling of a botanical tree – in analogy to the ‘tree’ of life. Indeed, patterns generated by growing plants have inspired visualisation of large hierarchies more generally (Kleiberg et al. 2001). Fractal visualisations including polytomies as a star arrangement (de Vienne 2016) also have aesthetic appeal of a kind that does not invoke a botanical tree. Furthermore, we wanted to give a prominence to appealing and varied images of the species themselves. These decisions were independent of the fractal nature of the visualisation (we also mocked up numerous other options). They may also have a downside, for example, adding visual embellishments might be perceived as less rigorous or distracting. However we judge them useful for our target audience, and of course, it is easy to turn off such embellishments in software to cater for different user tastes. It is our experience that the overlap between art and science is important in engagement; many applications of OneZoom have focused on this theme. For example, interactive kiosks at ‘Beautiful Science’ (British Library), hardback cover art for popular science (Dawkins and Wong 2016), and a projected light installation at the Harn Museum in Florida (see Fig. 3.10).

A Vision for the Future of Tree of Life Exploration

At the evolution conference in the summer of 2012 when OneZoom was unveiled, we laid out a vision for the future of big tree visualisation based on fractal zooming visualisation techniques and described the role this could have in both research and science communication. After 6 years we have a working visualisation of the complete tree of life, richly embellished with images, common names, and other information and explorable using a zooming user interface and fractal-based structure, based on publicly usable code at github.com/OneZoom. One can now fly through the tree and see information on extinction risks and variety of life on earth. The OneZoom.org website has attracted over a million unique visitors, and there are now a number of other websites based on the same ideas, each with their own merits, for example, DeepTree (Block et al. 2012) and Lifemap (de Vienne 2016). What do we hope will happen in 5 more years? Which major challenges remain to be resolved? We can imagine many incremental improvements, for example, richer data in the tree, but in this closing discussion, we outline more ambitious ideas for the future of tree of life exploration and science outreach.

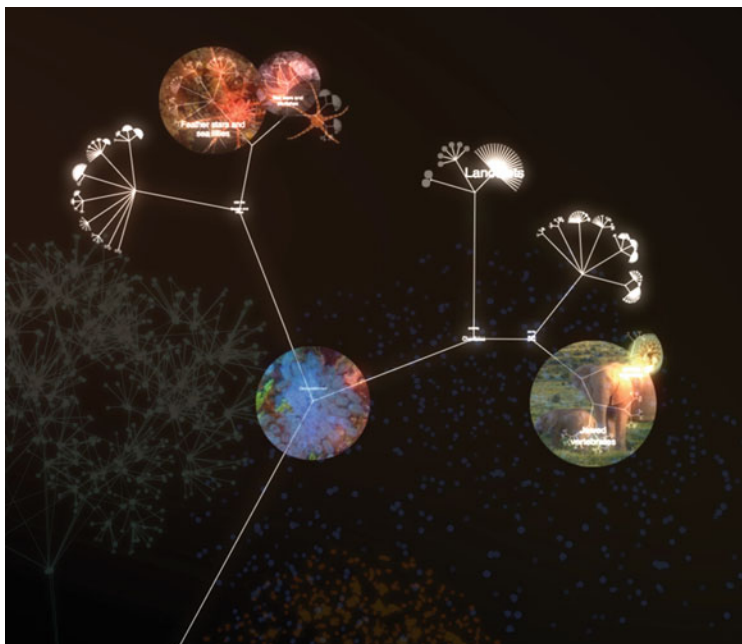


Fig. 3.10 A screenshot from the ‘One Tree, One Planet’ shown by artist Naziha Mestaoui, based on the polytomy view in OneZoom and projected onto the side of the Harn Museum in Florida

Tours Around the Tree of Life

We have noticed that when members of the public use OneZoom, some naturally get into it straight away and enjoy exploring for themselves, whereas others need to be explicitly shown around, at least to begin with. A highly useful outreach device would therefore be a collection of compelling tours around the tree – for example, a tour of medically important species, herbs used in cooking, or bioluminescent creatures. We envisage these tours as of a series of locations on the tree with extra imagery, text, or other media to show a scientifically interesting story associated with one particular route around the tree; they may also embellish the tree with easily digested information about the evolutionary links between groups. Tours may cross over each other by visiting the same locations on the tree, and the user might wish to switch between them or leave the tours altogether and start exploring the tree freely. The tours will have enriched the experience of exploring the tree and provided a more friendly way into something that can otherwise be mind-boggling. Earlier, we mentioned the rationale in biodiversity conservation for conserving evolutionarily distinctive species and overall phylogenetic diversity (see also Chaps. 2 and 4 in this volume). The installation of OneZoom in the ‘Wild Planet’ gallery at the Australian Museum in Sydney highlights imperilled species and branches as well as amounts of phylogenetic diversity that are imperilled. Visitors have enjoyed finding species

shown on the tree among over 400 specimens that are on display. In the future, a tour may help the visitor move through the gallery and better appreciate the links between the species' 'stories' and their positions on the tree of life.

A Place to Organise Evidence for Evolution

Having a complete tree of life will provide a framework to collect and organise explanations and evidence for evolution. This can then be connected with a much wider variety of species thus showing the true weight of evidence in support of evolution. Enabling people to explore a tree of all life on earth and find evidence for evolution wherever they go is much more compelling than to demonstrate a few pieces of evidence chosen by the science communicators. In an analogy, if we claimed there was a gold coin under every stone on a hillside and in support of this claim turned over a couple of stones of our choice to reveal coins, that is not very convincing; maybe they were planted there. However, if we invite a participant to wander freely around the hillside and turn over any stones of their choice to reveal coins underneath, this is much more compelling evidence for our claim.

More Fractal Projections of the Tree

The projections that have been discussed are only a fraction of what is possible; many more lie waiting to be discovered. One promising direction would be a more generic formula with parameters to indicate a desired position on the trade-off continua mentioned earlier. These parameters could be dynamically adjusted by the user (or more likely by a researcher) and tested to find which setting is optimal for different use cases. One example of this is the relationship between space allocated in the visualisation for a clade and its species richness which could take any functional form. Another approach is to build up a 'library' of graphical layouts for nodes of the tree so that large differences in child branch lengths, polytomies with many children, heavily unbalanced nodes, and any other cases are drawn from the library based on different rules and mixed together in a single visualisation. For example, a particularly long branch could twist around in a tight spiral before blossoming into the descending clade in the centre; this could use visually pleasing spirals while enabling longer branches to be clearly shown. The need for more projections of the tree will become greater with the incorporation of extinct species, as ideally they should be visually distinctive from extant species.

Connecting with Citizen Science Projects

We see great potential to integrate the tree of life with citizen science platforms, enabling people to explore the tree and easily record through a tree interface that they have seen a certain species, including any images they may have. We imagine that users could be presented with their own personalised tree of life featuring their own images and data showing when and where they have seen that species. This could also incorporate ways to connect with others that have an interest in a certain taxa, e.g. for research, viewing in the wild or horticulture. Such a platform would benefit from the tree structure so that people with interests in closely related taxa could easily be connected.

Sustainability of Tree of Life Community Resources

One concern that often arises in discussions around large-scale data curation projects or public web services is long-term project sustainability. The process by which science is supported through short grants with low success rates is not suited to maintain ongoing community resources. It would be wasteful for the scientific community to be left with a graveyard of projects that once consumed substantial resources but now are no longer maintained. Some are supported by institutions as part of their core activities: perhaps the best example of this in our field is the Encyclopedia of Life (see Section 2.1) which, after a substantial injection of initial funds, continues its core activities via staff secondments at major institutions. In the case of OneZoom, we set up a non-profit organisation and allow users to ‘sponsor’ leaves on the tree of life display, in exchange for donations to the upkeep of the project. This model has so far been sufficient to secure funding to continue operations but not to hire staff, and so volunteer time remains essential. We hope that further institutional-level support and changes to the way funding is allocated, as well as further innovations in funding models, will help support phylogeny-based community resource projects, such as the Open Tree of Life, ensuring them a more permanent future.

Hope for a Collaborative Future

Our work has only been made possible by its being a part of a much wider group of informally linked community projects such as the Open Tree of Life (Hinchliff et al. 2015) and the Encyclopedia of Life (eol.org) among others. The teams behind these

projects have put in a lot of work to collate wide ranging data from different sources and synthesise them into a huge and valuable data set. Nevertheless, we still worry that there are too many barriers in the way of a truly integrated and complete ‘ecosystem’ of biodiversity data projects, working together synergistically rather than in competition. These issues have been noted by others (Rees 2015), and there is little space to explore them here, but we would like to close our chapter with a call for greater collaboration. For example, we hope to see further steps towards an ‘official’ registry of all species. Who would have control of such a system? Our suggestion is no one: it should be in the form of a decentralised protocol that anyone can sign up to, whether it be a large institution with a rack full of servers or a school kid with a Raspberry Pi. Key properties of such a decentralised system would be preservation of historic states and easily verifiable proof of authorship. Another rather successful system with these properties has recently taken the world by storm: that is, Bitcoin and other Cryptocurrencies. If we can do this for economic currency, then surely we can manage the same for biodiversity, the most underappreciated global currency.

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Maze of Life Solution The solution to the problem in Fig. 3.1: human, does not fit anywhere (trick question); olympic torrent salamander, 3; tardigrade, 6; chocolate chip starfish, 5; Pacific dampwood termite, 1; house mouse louse, 2; blue-black spider wasp, 4; and cactus sea squirt, 7.

URLs for Resources Mentioned

Ancestor’s Trail	ancestorstrail.org.uk
DateLife	datelife.org
DeepTree	to.pbs.org/1J8YBfl
Encyclopedia of life	eol.org

(continued)

GlobalNames	globalnames.org
Lifemap	lifemap.univ-lyon1.fr/
OneZoom	onezoom.org
OneZoom code (GitHub)	github.com/OneZoom
Open Tree of Life	opentreeoflife.org
Tree of Life Web Project	tolweb.org
WikiProject Taxonomy	wikidata.org/wiki/Wikidata:WikiProject_Taxonomy
ZooBank	zoobank.org

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Chapter 4

Indicators for the Expected Loss of Phylogenetic Diversity



Daniel P. Faith, Simon Veron, Sandrine Pavoine, and Roseli Pellens

Abstract PD or “phylogenetic diversity” was proposed by Faith (Biol Conserv 61:1–10, 1992) as a measure of biodiversity “option value”, justifying its importance as a target of biodiversity conservation. The threats to phylogenetic diversity can be quantified by integrating PD with IUCN Red List categories and corresponding inferred extinction probabilities, to estimate amounts of threatened or imperilled PD. Practical conservation applications face decisions about which summaries of imperilled PD best provide priority setting among species and about how to use limited available data to estimate imperilled PD. Extensions of Weitzman’s phylogenetic “expected distinctiveness” provide a unifying foundation for many useful expected PD calculations. This reinforces the utility of expected PD calculations compared to methods in the EDGE programme. However, simplistic use of expected PD also has weaknesses. A resulting priority set of species may neglect other species that also could deserve conservation action. Improved priority setting might utilise conservative estimates of the reduction in extinction probability from conservation action. The priority ordering of a species has been equated with its order of selection in priority sets, but should instead reflect the lost opportunity in averting PD loss if there is no conservation action on that species. Species priorities can be estimated, under simple assumptions, even with the simple “evolutionary distinctiveness” (ED) information that is available for many species from the EDGE programme. This provides a simple approach to priority setting in which threatened

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species are simply ranked by their ED score, as an estimate of their averted PD loss if the species is conserved. The approximations also provide a tabulation of current total expected PD loss for a given taxonomic group. For biodiversity assessments by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, this approach indicated fractional imperilled PD varying from a low 8% for squamates to a high 65% for corals among the assessments for six major taxonomic groups.

Introduction

The phylogenetic diversity measure “PD” (Faith 1992) is now recognised as an informative measure of biodiversity (living variation), with applications extending across many taxonomic groups, including microbes (see Chap. 1). The PD of a subset of species (or other taxa) from a phylogenetic tree is simply the total length of the branches spanning that subset on the tree. This measure links to the “feature diversity” of the subset of species – effectively quantifying biodiversity at the level of features or characteristics of taxa. In the context of biodiversity conservation, such feature diversity has an intuitively reasonable link to an important core value of biodiversity. PD is recognised as capturing a value of biodiversity often called “option value” (for review and discussion, see Faith 2017). Option value captures the idea that living variation (biodiversity) is beneficial because it maintains options for future generations of people. We can think of those “options” as the ongoing provision of possible future uses and benefits to society.

This fundamental value of biodiversity provides the impetus to take conservation action to avert the loss of PD, over multiple taxonomic groups (see Chap. 1). At the species level, the IUCN Red List (e.g. see Brooks et al. 2016a, b) has been an important assessment of the degree of threat to species. The corresponding threats to phylogenetic diversity are indicated by integrating phylogeny with IUCN Red List categories to estimate threatened or imperilled PD (e.g. see Faith 2008). Such assessment of imperilled PD is well-established, for example, in the global EDGE of Existence programme (Isaac et al. 2007; see also <https://www.edgeofexistence.org/> and Chap. 2 in this volume). Challenges remain in deciding how best to quantify expected losses of PD and how to set priorities to avert such expected loss; these are discussed later in this chapter.

The idea of biodiversity as “maintaining” options for people is echoed in the work of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). IPBES is providing assessments of the state of knowledge about biodiversity and ecosystems and the related benefits, called “nature’s contributions to people” (“NCP”, Pascual et al. 2017). One of these NCPs, “maintenance of options”, refers to “benefits (including those of future generations)” associated with the continued existence of biodiversity as living variation. In this chapter, we describe how IPBES assessments of the status and trends in “maintenance of options” (Davies et al. 2018) have examined expected loss of PD, over multiple

taxonomic groups. We also discuss how this has presented challenges in making best-possible use of available information.

This chapter will address the challenges in global assessments and priority setting, such as those developed by EDGE and IPBES, by presenting some generally applicable approaches for assessing threatened PD. We will build on the key idea that PD offers an integrated way, through its various associated calculations, to produce quantifications or indices that are useful for assessment and priority setting. Another chapter of this book (see Chap. 1), in reviewing the history of PD work, noted the common misconception that PD is a single index, supposedly following the tradition of many individual ecological indices. In fact, Faith (1992) focused primarily on PD gains and losses, defining a “G” (for “gain”) index to capture the context-dependent distinctiveness of a species, reflecting how much PD it could add to an existing set of species (a “complementarity” value). In this chapter, we first provide some background history and explore how the IUCN Red List and probabilities of extinction provide a useful framework for working with “expected PD distinctiveness”, extending PD complementarity to the probabilistic case (e.g. following Witting and Loeschcke 1995).

This chapter then examines “expected PD distinctiveness”, in order to explore practical applications of expected PD. These practical considerations include how priority ordering of a species can best reflect the opportunity to avert PD loss, how such priorities might be estimated with the available simple “evolutionary distinctiveness” information, and how approximations also can provide a tabulation of current total expected PD loss for IPBES and other large-scale assessments.

Background

The IUCN (1980) inspired consideration of biodiversity option values and consideration of taxonomic distinctiveness as a factor relevant to conservation priorities (see Chap. 1). Equally importantly, IUCN (1980) also sets the stage for linking phylogeny, distinctiveness, and phylogenetic diversity to probabilities of extinction, in order to evaluate importance and urgency of priorities for action. A key table (Fig. A in IUCN 1980) showed how species having combinations of higher threat and higher amount of variation that would be lost (loss of a whole family counts more than loss of a genus) deserved higher conservation priority. Here, the principle was that “the greater the potential genetic loss the less imminent that loss need be to justify preventive action”.

Later, important foundational work built on these ideas in the context of PD. Witting and Loeschcke (1995) provided an early link between PD, probabilities of extinction, and expected diversity. Also, early work of Weitzman (1992) developed “expected diversity” approaches and later linked that to PD (Weitzman 1998). Weitzman (1998) adopted PD and Witting and Loeschcke’s (1995) expected PD, but without citation. Hartmann and Steel (2006) provided a review that makes clear the important links of PD to Weitzman’s framework for expected diversity:

Witting and Loeschke (1995) linked the phylogenetic diversity (PD) measure (Faith 1992) to extinction probabilities to obtain a method for minimizing the future loss of biodiversity. Weitzman (1998) proposed the ‘Noah’s Ark Problem’ (NAP), a framework based on PD.

Faith and Walker (1996a, b) further extended the Witting and Loeschke approach and provided software for selecting sets of species to maximise expected PD. Barker (2002) also explored other variants.

The EDGE programme (see Isaac et al. 2007) also provided an important implementation of the IUCN (1980) core ideas linking species’ distinctiveness and threat. Isaac et al. derived a measure of phylogenetic or evolutionary “distinctiveness” (their “ED” measure) from PD (they sometimes used “phylogenetic diversity”, and even “PD”, to refer to most any phylogenetic index, but their derivation of ED clearly used PD sensu Faith 1992):

In order to calculate ED scores for each species, we divide the total phylogenetic diversity of a clade amongst its members. This is achieved by applying a value to each branch equal to its length divided by the number of species subtending the branch.

Thus, the ED values of all species add up to the total PD. They integrated ED scores with IUCN Red List assessments of degree of threat (and implicitly extinction probabilities; for discussion, see also Redding and Mooers 2006; Mooers et al. 2008; Faith 2008). This programme continues to inspire people about the importance of our evolutionary heritage and the need for its conservation attention.

Unfortunately, the proposal of the EDGE method (Isaac et al. 2007), while listing Weitzman’s and Witting and Loeschke’s work in their background sections, appears not to have appreciated lessons from that work about how to link PD to extinction probabilities and about the use of complementarity among species. Almost immediately after publication of EDGE, on the EDGE website’s forum (www.edgeofexistence.org/forum), Faith (2007) documented how such links from PD to “probabilistic PD” were not yet appreciated, and he presented an example of “standard probabilistic PD” where selection for conservation action means the probability of extinction changes from p to 0. At the same time, this “HEDGE” approach, as a variant of expected PD, was fully described by Steel et al. (2007).

Thus, an added contribution of the EDGE programme is that it immediately highlighted the challenges in moving from the intuition that phylogenetically distinctive, threatened, species deserve priority to a programme with a clearer link to the goal of effectively averting the expected loss of phylogenetic diversity (for further discussions and review, see Steel et al. 2007; Faith 2008; Mooers et al. 2008; Faith 2015 and Chap. 2 in this volume).

A useful insight, tracing back to Faith (1992), is that we can avoid the Isaac et al. notion of distinctiveness as a static score, to consider a notion of distinctiveness that dynamically depends on what else is already protected. Distinctiveness in this dynamic sense is a form of “complementarity”; it reflects the PD that the species offers that is additional to what is already protected. When we think of protection as all or nothing, this is the “G” (“gain”) value. Here, the G value additional

contribution of a new species to a set reflects branches not already represented by other member species.

Later, Faith (2008), building on the important work of Witting and Loeschcke (1995) and Faith and Walker (1996a, b), showed that PD complementarity extends naturally to “expected” gains or losses when we are working with probabilities of extinction. In all this work, typically, independent probabilities among the species are assumed, so that the probability of loss of a given internal branch is the product of the probabilities of loss of its descendant species. For a newly conserved species, implying some decrease in its extinction probability, its complementarity value now reflects the resulting decrease in the extinction probability of all its ancestral branches (see below).

Expected PD and Expected Distinctiveness

It is useful to describe the relationship between expected PD and distinctiveness. Steel et al. (2007) documented a nice property of HEDGE and Mooers et al. (2008) noted that “HEDGE_i is also a product of two terms: the expected phylogenetic contribution of a species *i* given prob(extinction) of all other species, and prob(extinction)_{*i*}”. They add “HEDGE can also be formulated as the change in the total expected PD from the status quo to the expected PD if one sets the prob(extinction)_{*i*} = 0” (in other words, if species *i* is preserved).

Volkman et al. (2014) reviewed HEDGE and made the link to Weitzman’s work clear.

Weitzman [32] also arrived at this formulation ten years earlier, which he termed the ‘distinctiveness’ of a taxon, in the context of his ‘Noah’s Ark Problem’ of biodiversity preservation.

Weitzman (1998) characterised the important idea of “expected distinctiveness” of a species *i* within a set *Q*:

from its ‘nearest neighbor’ or ‘closest relative’ in *Q*. It seems natural to define the distinctiveness of *i*, denoted *D_i*, to be the Expected distance of *i* from its nearest neighbour or closest relative in *S*.

Here, we expand Weitzman’s expected distinctiveness (see also Steel et al. 2007) to support a general family of expected PD calculations.

The total PD is:

$$PD = \sum_k \{L_k\} \quad (4.1)$$

The *L* values are branch lengths indexed by *k*, where the branches summed up are those ancestral to the species in the set.

We now consider the formula for expected PD:

$$\text{Expected PD} = \sum_i \left\{ L_i \times \left[1 - \prod_{j=1}^{n_i} q_j \right] \right\} \quad (4.2)$$

Note that i counts over all branches and, for any one branch, j counts over all descendants of that branch, with q values indicating probabilities of extinction.

It follows that the simplest expression of the change in expected PD, when one or more probabilities change, is the difference in the two expected PD expressions (e.g. see Formula 4.7 in Faith 2013):

$$\text{Change in expected PD} = \left| \sum_i \left\{ L_i \times \left[1 - \prod_{j=1}^{n_i} q'_j \right] \right\} - \sum_i \left\{ L_i \times \left[1 - \prod_{j=1}^{n_i} q_j \right] \right\} \right| \quad (4.3)$$

where q' indicates the new probability of extinction values. The index i is over all branches on the tree, and for a given i , index j counts through all the descendants of that branch. We can combine the terms:

$$\text{Change in expected PD} = \left| \sum_i \left\{ L_i \times \left[\left(\prod_{j=1}^{n_i} q_j - \prod_{j=1}^{n_i} q'_j \right) \right] \right\} \right| \quad (4.4)$$

We also have:

Change in expected PD when a species extinction probability changes from p to p'

$$= \sum_{i=1} \left\{ L_i \times (p - p') \left[\prod_{j=2}^{n_i} q_j \right] \right\} \quad (4.5)$$

$$= (p - p') \sum_{i=1} \left\{ L_i \times \left[\prod_{j=2}^{n_i} q_j \right] \right\} \quad (4.6)$$

where the sum is now over branches ancestral to that nominated species. We can refer to the right-hand side of formula (4.6):

$$\text{PD expression of expected distinctiveness} = \sum_{i=1} \left\{ L_i \times \left[\prod_{j=2}^{n_i} q_j \right] \right\} \quad (4.7)$$

This makes it clear how the expression of Weitzman's (1998) "distinctiveness" (his formula 23) links to a general expected PD calculation. The multiplier, $(p - p')$, in formula (4.6), tells us the actual change in expected PD. Note that this difference in "before" versus "after" probabilities might be estimated simply as some overall gain, without knowing the actual p values.

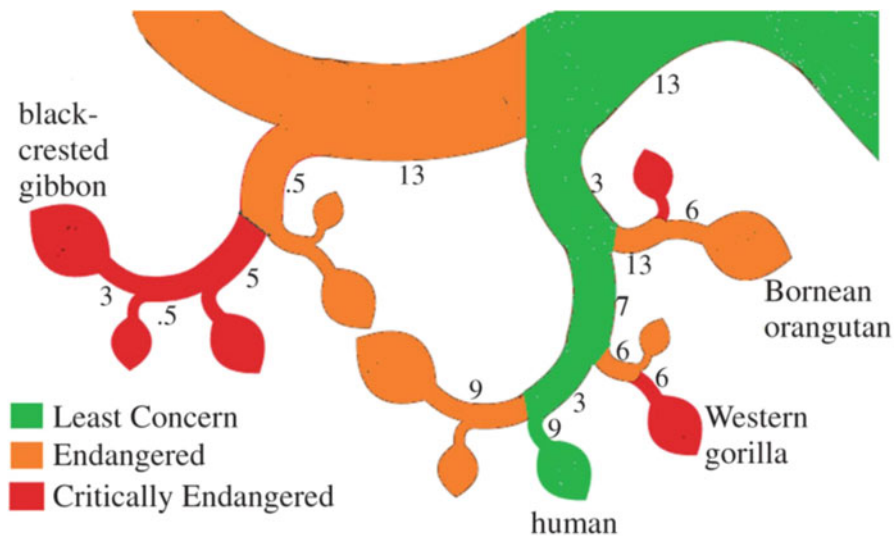


Fig. 4.1 An example from Forest et al. (2015), using their figure redrawn from OneZoom (Rosindell and Harmon 2012). Colours indicate Red List categories. Numbers along branches are approximate branch lengths in millions of years (“Ma”)

This is a nice general formula. Steel et al. (2007) provide a very useful description of this distinctiveness term but only discuss it in the context of the HEDGE method, where the multiplier is p . Mooers et al. (2008) review this special case, describing the term when used for the HEDGE score for a species as “the expected phylogenetic contribution of a species”, which is then multiplied by the probability of extinction of that species.

This can be written for the HEDGE special case:

$$\begin{aligned} &\text{Change in expected PD when } x \text{ is protected } (p' = 0) \\ &= p \sum_{i=1} \left\{ L_i \times \left[\prod_{j=2}^{n_i} q_j \right] \right\} \end{aligned} \quad (4.8)$$

Forest et al. (2015) provided a nice example of these various PD calculations, reproduced here (Fig. 4.1; Table 4.1).

Forest et al. used this example to illustrate how the different calculations would choose a second priority species, assuming that the human species is secure. ADEPD (Nunes et al. 2015) is the change in the total expected PD after a species is downlisted by one IUCN Red List category (so defining the p and p' values that would be used in formula 4.6). This contrasts with HEDGE strict assumption of p' probability of extinction equal to 0. These indices often highlight the importance of different species.

First, we will ignore the probabilities. Applying the ED criterion shows its failure to take into account the phylogenetic overlap of species. Here, it would select the

Table 4.1 An example of PD-related priorities, derived from Table 2 from Forest et al. (2015)

	Human	Western gorilla	Bornean orangutan	Black crested gibbon
ED	12.5	11.5	13.6	6.7
Unique PD	9.0	6.0	6.0	3.0
PD complementarity given the human species	0	12	19	22
EDGE	0.31	4.6	2.7	2.7
HEDGE	0	12	19	22
ADEPD	N/A	4.5	7.0	4.3

Index values in millions of years for named species in Fig. 4.1. Highest values for each metric are highlighted in bold

Bornean orangutan (for an ED total of 12.5 Ma + 13.6 Ma = 26.1 Ma). In contrast, a criterion to maximise PD uses PD complementarity (the “G” value of Faith 1992) given the human species. PD complementarity selects the black crested gibbon, as it adds 22 Ma, of PD.

Forest et al. also incorporated the probabilities of extinction inferred from the IUCN ratings (shown in Fig. 4.1). This provides a revealing comparison of EDGE, which does not incorporate complementarity, to expected PD variants, which do incorporate complementarity. EDGE indicates that the western gorilla has a higher priority than the black crested gibbon, because the secure status of the human species is ignored by the index. In contrast, HEDGE gives the black crested gibbon a higher priority, reflecting the 9 Ma + 13 Ma of PD at stake. ADEPD gives priority to the Bornean orangutan based on the change from the p value for “endangered” to the p' value for “vulnerable” (for further discussion and methods details see Forest et al. 2015).

From Single Species to Sets of Species

We will consider the challenge of selecting a set of priority species below. But first, it is useful to consider the expected PD of “given” sets of species, such as those found in a given area. We can consider the gain if that whole area is “selected” and conserved, say with all probabilities changing to 0. In this simple case, we can consider transformed branch lengths based on the probability of loss. As preparation, first note that the companion formula to (4.2) is that for the total expected PD loss:

$$\text{Expected PD loss} = \sum_i \left\{ L_i \times \left[\prod_{j=1}^{n_i} q_j \right] \right\} \tag{4.9}$$

Again, the L values are branch lengths over the whole tree indexed by i . The q values are probabilities of extinction for species indexed by j . We then can think of this product as rescaling the branches:

$$\text{Expected PD loss} = \sum_i \{\text{rescaled } L_i\} \quad (4.10)$$

where

$$\text{Rescaled } L_i = L_i \times \left[\prod_{j=1}^{n_i} q_j \right] \quad (4.11)$$

Thus, formula (4.10) for expected PD loss is just the same as calculating the total “PD”, but for a tree with rescaled branches. Such transformed branches are not a new idea; here the transformation uses probabilities of extinction, but interesting cases exist using transformed branches using probabilities of persistence, not extinction (e.g. see Thuiller et al. 2015). Using formula (4.10), the total “PD” is an amount that we would like to see made smaller – in effect we would like this tree to disappear.

Standard PD calculations look at changes, as when some q values change to q' in formula (4.4).

An informative calculation considers the scenario where selected threatened species (say, all those in a given area) are secured (e.g. assuming q' for those species is a probability of loss of 0, while all other species of course remain as q):

$$\text{Change in expected PD} = \sum_k \left\{ L_k \times \left[\left(\prod_{j=1}^{n_k} q_j \right) \right] \right\} \quad (4.12)$$

The L values are branch lengths indexed by k , so that we are summing up over only those branches ancestral to the set of species that have been secured (e.g. found in that area).

We again can think of this as rescaling the branches.

$$\text{Change in expected PD if secure that area} = \sum_k \{\text{rescaled } L_k\} \quad (4.13)$$

where

$$\text{Rescaled } L_k = L_k \times \left[\prod_{j=1}^{n_k} q_j \right] \quad (4.14)$$

Formula (4.13) is just the same as calculating the total “PD” of that area, but for the tree with the rescaled branches. This appears to be an unrealistic case, in assuming that probabilities for the set of threatened species go to 0. However, following Faith (2015), such a scoring may indicate a phylogenetic key biodiversity area (“PDKBA”) – an area that makes a big contribution, as long as it persists, to averting expected PD loss. This would serve Faith’s (2015) call for “a score for an area reflecting the total amount of threatened PD would allow prioritization among PDKBAs...” (see also Chap. 8). Use of the transformed tree also has a nice

property in this context: we can use commonly available software that calculates total PD of areas for an input tree, but we provide the software with a transformed tree.

This use of the transformed tree also serves to introduce issues, discussed in the next section, involved in selecting a set of priority species. Note that, if we start with the transformed tree and continue to assume that a selected species has new probability of 0.0, then we can apply to this tree the standard “greedy” methods (see Hartman and Steel 2006) for selecting maximal PD sets (because selecting a species changes the probability of loss of all its ancestral branches to 0.0), capturing that amount of “PD” from the transformed tree.

Challenges in Selecting Sets of Priority Species

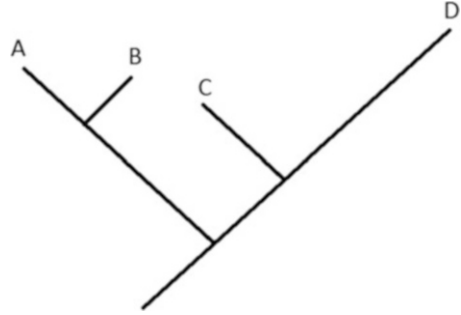
The primary contrast between expected PD and EDGE was based on the use of complementarity. In the example of Fig. 4.1 and Table 4.1, EDGE prioritised the western gorilla over the black crested gibbon, because complementarity is not used by EDGE; the secure status of the human species was ignored. The rationale for using complementarity seems to logically extend to the idea of selecting *sets* of priority species. After a given species is selected, its probability is changed to p' , and other species scores are recalculated so that the gain in selecting another species must be complementary. Faith (2008) described the approach:

A set of priority species for conservation action can be identified through the application of a simple expected-PD algorithm (derived from Faith and Walker 1996b) in which the species, h , with highest PD complementarity is selected and then the complementarity value for any remaining species, j , is updated on the basis of the recorded change from p_h to p'_h for species h . To update the complementarity of species j , Eq. 3 is reapplied, substituting p'_h for p_h . The process of identifying the species with highest complementarity and adding it to the priority set can continue until some budget constraint or other stopping rule is triggered.

In presenting ADEPD (see Table 4.1), Nunes et al. (2015) noted “In order to determine optimal sets of more than one species for down listing, we needed to take into account that down listing a species will change the ADEPD-cost scores for all the remaining species, an effect known as complementarity”. Their procedure for finding a set iteratively selected the species with the highest ADEPD score and then updated at each the scores for all species under the assumption that the selected species now had a new probability of extinction.

Jensen et al. (2016) proposed the same kind of approach, for the HEDGE special case. However, this study created some possible confusion. It set the new p value for a selected species to be some small non-zero value, but still called the approach “HEDGE”. Nevertheless, their method, in employing complementarity, is a special case of the basic strategy discussed in Faith and Walker (1996a, b) and Faith (2008); it provides a set of species, which has maximum gain in expected PD for the case where p goes to approximately 0.

Fig. 4.2 A hypothetical phylogeny for four species, with branches transformed by some hypothetical probabilities of extinction



The Jensen et al. (2016) study reveals two weaknesses in its potential applications of expected PD set selection. One weakness is the HEDGE adoption of p' as 0.0, followed by the application of complementarity. Consider, for example, the HEDGE selection of the black crested gibbon in the Table 4.1 example. After its selection, complementarity means that all the other threatened gibbons have low priority – the shared branches are now supposedly secure. The reality is that “selection” does not ensure “security” and a more risk averse approach may be useful. Expected PD in its more general form, with non-zero p' , should be preferred. In fact, p' needs not correspond strictly to that assigned (as in ADEPD) to the new category. Instead, future work may consider risk-based criteria to assign larger p' values to reflect these uncertainties. If priority selection of the black crested gibbon now is not assumed to absolutely secure those ancestral branches, then some priority may be given also to other gibbon species. Such an approach not only provides a more risk averse approach for averting expected PD loss, but it also may provide better communication to the public about priorities – avoiding impressions that choosing one species means that the others are not a priority.

The Jensen et al. (2016) study is noteworthy also because they advocate the order of selection of species as providing a conservation priority ordering. However, the priority assigned after the selection process should not be indicated by the score or order at the time of selection. A simple example shows why. In Fig. 4.2, suppose that the tree is a portion of a rescaled tree, as described earlier, and for simplicity, suppose we are assuming the case where $p' = 0.0$. Thus, a greedy PD algorithm captures the maximum improvement in expected PD with each selection.

By this process, species D is selected first, then A is selected in having the greatest complementarity contribution, then C is selected, and finally B is selected. Among the set of four species, A was selected before C, but which of these two species would yield the biggest regret if there was a failure to take actual conservation action on the species? The answer is C, even though it was selected later than A. The greater cost of non-action is indicated by the greater terminal length of the branch to C in our transformed tree (Fig. 4.2).

How is this priority calculated in practice? The formula for this priority for any species x in the set is based on our general formula (4.6), with priority:

$$= (p - p') \sum_{i=1} \left\{ L_i \times \left[\prod_{j=2}^{n_i} q_j \right] \right\} \quad (4.15)$$

where q_j remains unchanged for all species not in the selected set, and is equal to p' for all species in the set. However, for species x , the multiplier shows that the probability here is changed from the lower p' back to the higher p . Thus, for a selected set, the priority order of a given species is best indicated by the magnitude of lost opportunity for averted loss of PD if no action were taken on that species.

A recommendation is that optimum selected sets to maximise expected PD be used cautiously. Indeed, there are perhaps three major weaknesses of the Jensen et al. IHEDGE approach: (1) it is called HEDGE but assigns selected species a non-zero p' value; (2) it selects a set with such a low p' value that complementarity means that sister species may be viewed as neglected (never selected yet worthy of some attention); and (3) it provides an incorrect priority ordering for species if we are interested in best-possible averting of expected PD loss.

An attractive possibility is to move away for any idea of an optimal subset of priority species (unless we have a rare case of a fixed ensured budget). Instead, every threatened species is on the “list”, and a priority species is one where the failure to take action would imply the biggest missed opportunity to avert expected PD loss (using variants of formula 4.15). This includes interpretation of formula 4.8, not unrealistically as a gain in expected PD when $p'=0$ (HEDGE), but instead simply as the expected loss of PD that might be averted to some degree through action on that species. Here, there is flexibility in how the current extinction probabilities of the other species are defined. For example, those species where conservation action has been activated may be assigned some lower probability, with uncertainty of conservation success optionally also built in.

Approximations for Priorities

In the above discussion, the formulae for priority setting (6, 8, 14, 18) all depend on some form of expected distinctiveness (formula 4.7), although expected distinctiveness on its own is not adequate for priority setting in general. Formula (4.7) is, however, linked to a nice approximation. While EDGE was not designed to consider other related species probabilities, its dividing up of credit in the ED index mimics expected PD under specific conditions (see below).

First consider formula (4.6), which can be used to determine the priority species corresponding to the one proving the greatest gain in expected PD when the species probability of extinction changes from p to p' . The formula assumes that we know the estimated probabilities of extinction assigned to every species. If, as a rough approximation, we are willing to prioritise that species while simply assuming any other species is categorised as “vulnerable”, then there is a nice simple result: the

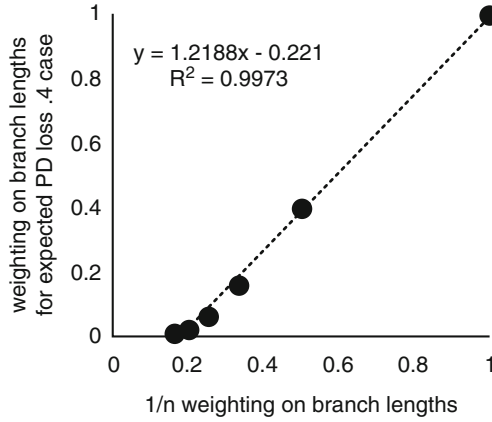


Fig. 4.3 This plot examines a single hypothetical branch of unit length and varies the assumed number of descendant species (so producing for the x -axis, ED-related values, where n is the number of descendants). The corresponding y -axis value is the expected PD gain if one descendant species is protected ($p' = 0.0$) and the other descendants are assumed all to be “vulnerable” with probability of extinction of about 0.4. The regression formula in the box shows a good linear fit

expected distinctiveness (formula 4.7) portion of formula (4.6) turns out to be roughly approximated by classic ED values (which are already calculated for many species) (Fig. 4.3).

Here, we assume “IUCN500” conversions of Red List categories to probabilities (see Mooers et al. 2008), where secure categories have approximate probability of extinction of 0, endangered categories have approximate probability of extinction of 1, and vulnerable categories have approximate probability of extinction of 0.4.

$$\text{Change in expected PD} \cong p \left(\sum_i \{L_i \times [1.22 * (1/n_i)]\} - 0.22 \sum_i \{L_i\} \right) \quad (4.16)$$

Formula (4.16) provides an approximation based on the linear relationship shown in Fig. 4.3. The second term is constant among all species for a timetree, so it can be ignored.

Thus, the change in expected PD when a species is protected ($p' = 0$) is approximately p times ED (or simply ED, when p is about 1.0).

$$\text{Change in expected PD} \cong p(ED) \quad (4.17)$$

So, using this approximation, we can rank all threatened species by their ED scores. Thus, if all other species are assumed to be vulnerable, then priorities can be roughly indicated by rankings of classic ED scores among threatened species. For this approximation, note that this does not escape the problem recognised for conventional ED scoring; an endangered species can get a high rank based on a long branch shared with one sister species, even if its sister is actually secure.

Total Expected PD Loss

The formula (4.18) for total expected PD loss is similar to that for expected distinctiveness of a given species – but it is a sum over *all* branch lengths (not just those ancestral to the given species).

$$\text{Total expected PD loss} = \sum_i \left\{ L_i \times \left[\prod_{j=1}^{n_i} q_j \right] \right\} \quad (4.18)$$

Again, i counts over all branches and for any one branch, j counts over all descendants of that branch. For any one species, the expected PD loss (or gain in expected PD if it is secured) is approximately its ED score, assuming all sister species are vulnerable. For a set of species, however, this approximation does not directly apply unless we assume that the set of species is not clumped on the tree.

Chaudhary et al. (2017) noted evidence, under random species loss in tree-pruning simulations, for a strong linear relationship between the summed ED of the set of lost species and the corresponding total PD loss. We note that, when the threatened species have a probability of extinction of about 1.0, the total *expected* PD loss naturally is about equal to the total PD loss when those species are pruned from the tree. It follows that total expected PD loss can be approximated by the sum of the ED values of the lost species. This suggests that the sum of the ED values of the threatened species (assuming p about 1.0) approximates the total expected PD loss.

This approximation is further supported by the observation that ED scores typically are dominated by terminal branch lengths. This terminal length is the minimum contribution to total expected PD loss by any threatened species. We conclude that a reasonable approximation for total expected PD loss is:

$$\text{Total expected PD loss} \cong \sum_t \{ \text{ED}_t \} \quad (4.19)$$

where the index t counts only among the threatened species, where the corresponding probability of extinction is assumed close to 1.0.

IPBES and Overall Expected PD Loss

The above approximation has been useful for IPBES assessments, where new analyses normally are not carried out but existing data can be tabulated to produce indicators. IPBES is an intergovernmental body providing assessments of the state of knowledge about biodiversity and ecosystems and the related nature's contributions to people ("NCP", Pascual et al. 2017). Among the NCP, IPBES has embraced the idea of the option values of biodiversity. For example, the IPBES Conceptual Framework (Diaz et al. 2015) referred to:

the option values of biodiversity as a reservoir of yet-to-be discovered uses from known and still unknown species and biological processes, and as a constant source, through evolutionary processes, of novel biological solutions to the challenges of a changing environment.

In accord with this, one of the NCPs is simply called “maintenance of options”. It makes reference to “benefits (including those of future generations) associated with the continued existence of a wide variety. . .”, thus properly linking to the core idea of “biodiversity” as living variation. The contribution of biodiversity in providing such “maintenance of options” therefore also accords well with the prehistory referred to (see Chap. 1) on the value of biotic diversity in maintaining options (e.g. following Haskins 1974; also reviewed in Faith 2017).

The IPBES phrasing drew on earlier discussion of option value and its natural link to phylogenetic diversity (Faith et al. 2010). It was natural, then, that the status and trends in biodiversity and “maintenance of option” (“NCP18”) would assess status and trends in PD for assessment of such global option values of biodiversity.

Given the constraint that generally there is no new research and analysis for IPBES assessments of our current state of knowledge, assessment of NCP18 based on PD therefore used the approximations described above, using available data over multiple taxonomic groups. The EDGE programme and the IUCN (Brooks et al. 2016a, b) and the related regional and global published studies (see below) provided the existing ED data and Red List assessments useful for this assessment of NCP18. These published studies, over different taxonomic groups, have integrated Red List assessments with estimates of evolutionary distinctiveness (ED) of species. The most useful summary of this available information, for a given taxonomic group, is simply the sum of the tabulated ED values of the threatened species (formula 4.19), as this approximates total threatened or “imperilled” PD. For IPBES, this was interpreted as one important way to link biodiversity status and change to status and change in NCP18.

Application of this approach (see Davies et al. 2018) indicated fractional imperilled PD varying from a low 8% for squamates to a high 65% for corals among the assessments for the six major taxonomic groups (Table 4.2).

Conclusions and Future Work

There is a strong intuitive appeal to integrating information about the tree of life and information about degree of threat to the different species at the tips of those trees. This intuition traces back at least to the early advocacy of priority setting by IUCN (1980). Since that time, there has been much constructive debate about methods for priority setting, including methods that can cope with inevitable uncertainties and lack of data. Now, robust methods based on phylogeny have focussed on PD and the idea of expected PD. Challenges remain, however, in how best to use the calculus to best address goals such as averting expected loss of PD.

Table 4.2 Status of NCP18 maintenance of options as indicated by amounts of imperilled phylogenetic diversity (PD), for six taxonomic groups

Taxonomic group	Cycads	Amphibians	Corals	Mammals	Birds	Squamates
Global non-imperilled PD	3081	105,803	4164	30,970	67,537	112,301
Imperilled PD	5776	44,145	1607	10,591	9613	10,177
Total imperilled PD/total PD	0.652	0.294	0.278	0.255	0.125	0.083

Data sources: Brooks et al. (2016a, b); EDGE of Existence (n.d.), <https://www.edgeofexistence.org/>, Safi et al. (2013); Tonini et al. (2016); Yessoufou et al. (2017); Daru et al. (2013)

The units of PD (Faith 1992) are millions of years. Taxonomic groups are ordered from left to right by the magnitude of total imperilled PD as a fraction of the total PD of the group. Global “non-imperilled” PD includes un-estimated portions of PD that cannot be identified as imperilled because the associated species are data deficient (DD)

Expected PD for priority setting does not offer one single best pathway; instead, its flexibility is an advantage. Not only can “downlisting” adopt conservative values for new probabilities, but also we can adjust estimated probabilities of extinction to reflect uncertainties about the success of conservation actions. An appealing overarching goal is the idea of “averted loss”. If “action” is distinguished from “no action” or “unsuccessful action” by corresponding assumed differences in consequent p values, then it is immediately clear which threatened species, if not acted upon, would imply a significant failure to avert expected future loss of PD. This provides a natural basis for priorities for conservation action.

Consistent with the framework are various approximations that are robust to common missing information. This proved helpful in integrating indicators of expected PD loss into IPBES assessments. The IPBES assessments show that methods are now robust enough that PD can provide policy-relevant “indicators” to address the challenges in linking biodiversity to one of the most difficult of “nature’s contributions to people”: the “maintenance of options”. However, for these assessments, reporting on status of “maintenance of options” requires more than the PD indicators. IPBES also has included anecdotal evidence about how the tree of life continues to deliver on the option values promise, producing often-unanticipated benefits for society. For example, the Asia-Pacific assessment (Davies et al. 2018) reported on the recent discovery (Chassagnon et al. 2017) of neuroprotection after stroke, derived from a spider venom peptide. These examples address the communication challenge in establishing a sense of value of the evolutionary heritage embodied in the tree of life.

Rosindell and Wong (Chap. 3 in this volume) describe the “OneZoom” system (Rosindell and Harmon 2012) for visualising aspects of the tree of life. In a gallery at the Australian Museum on biodiversity and the tree of life, the interactive OneZoom display of tetrapods, including many species on display, integrates PD and Red List information in a user-friendly way. The other end of the gallery presents examples of surprising benefits from the tree of life, highlighting why we should be concerned about loss of PD. However, there appears to be no place to readily find more examples, say, from a favourite region or taxonomic group. Future work therefore not only must produce an effective toolbox for expected PD but also must provide a clearing house of examples illustrating how phylogenetic diversity provides a storehouse of benefits for society.

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Chapter 5

Phylogenetic Resolution and Metrics of Biodiversity and Signal in Conservation



Nathan G. Swenson and Samantha J. Worthy

Abstract Over the past two decades, biodiversity assessments have moved beyond simply measuring species diversity and toward the utilization of phylogenetic information. Despite these major advances toward holistically estimating the similarity of species, challenges remain. Specifically, the large phylogenies utilized for such analyses often contain multiple polytomous nodes indicating a lack of information regarding the true relatedness of lineages. Unfortunately, relatively little attention has been paid to what impact these polytomies may have on downstream analyses of biodiversity. Here we begin with describing how polytomies should impact phylogenetic metrics of biodiversity, which is then followed by a simulation study examining 3360 phylogenies with seven different levels of resolution. The simulations show that an increase in polytomies increases the total branch length of the phylogenies, which generally results in an increase in phylogenetic metrics of biodiversity. Further, we show that measures of phylogenetic signal in species abundance increase when polytomies are introduced. The results are clear and consistent across phylogenies and are based on unavoidable realities regarding branch lengths in resolved versus unresolved phylogenies indicating that such results should be expected in all systems. We recommend that future research and biodiversity assessments clearly discuss these directional biases and attempt to quantify sensitivity in metrics via randomly resolving polytomies. These solutions will be needed as our exploration and documentation of biodiversity to promote better phylogenetic resolution across the tree of life continues.

Introduction

The biodiversity in assemblages of species has traditionally been characterized using some metric of species richness or diversity. While the number or diversity of species in an assemblage is likely the most immediately tangible axis of biodiversity

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93

for the field scientist and general public alike, they are relatively information poor metrics (Swenson 2011). Specifically, they convey little-to-no information regarding the similarity of species. The ecological similarity of species is a critical factor determining many ecological and evolutionary outcomes (e.g., MacArthur and Levins 1967; McGill et al. 2006). For example, experimental studies of grassland assemblages have demonstrated that these ecosystems are generally more productive and stable when the assemblage contains species with greater variation in their ecological or functional strategies (e.g., Tilman et al. 1997). In other words, equal numbers of species between two assemblages does not mean they are equal in their ecological diversity nor are they likely to be equal in their ecosystem functioning and ability to respond to future perturbations to the environment.

The implications of a lack of information about species similarity for conservation biology are, therefore, clear (Faith 1992, 1994). Because the degree of species similarity influences the current and future health of ecosystems, conservation decisions based solely on the number of species and their names in a region should be less preferred to those based on a metric or multiple metrics of the similarity of those species. The challenge for conservation biologists aiming to inform decision-makers in the near term, therefore, becomes how one can characterize the similarity of species using pragmatic approaches that do not require large amounts of new data collection that could take years to decades to amass (see Chap. 7 for discussion on this). For example, while the distribution of photosynthetic or hydraulic traits for an entire forest community would be extraordinarily valuable for forecasting the response and stability of that forest under future climate change, the measurement of these traits requires enormous effort even in species poor assemblages and likely could not be quantified in a timeframe necessary for making decisions (Swenson 2013). A further consideration is that we may not know those aspects of form and function that ultimately decide the fate of a species assemblage. Given this, it might be better to more holistically estimate the similarity of species rather than hope we have found the needle in the haystack of functions that will dictate the future health of an ecosystem.

Conservation biologists have recognized the above-outlined limitations since at least the early 1990s, and they have been leaders in proposing new metrics of biodiversity to deal with these limitations. Perhaps the most famous and lasting example of this is Faith's measure of phylogenetic diversity, which is now generally referred to simply as phylogenetic diversity (PD) (Faith 1992). The beauty of Faith's PD is in its simplicity and additive nature. Specifically, PD is the sum of the phylogenetic branch lengths connecting all species in a sample. Thus, as more species are added, more branch length is added and PD increases thereby capturing the importance of evolutionary history and species richness itself. Many other phylogenetic metrics of biodiversity do not have a monotonic relationship with species richness making them hard to interpret or weigh when making conservation decisions.

At face value, PD may appear to be an odd way of characterizing the ecological similarity of species. We are familiar with spectacular cases of evolutionary convergence between distantly related species and divergence between closely related

species. In such instances, the relatedness of species may have no relationship or the opposite relationship with their ecological similarity (Losos et al. 1998; Losos 2008). However, when considering deeper phylogenetic relationships, the relationship between ecological similarity and phylogenetic relatedness may become more reliable albeit approximate. In other words, palm trees may appear vastly different from one another and may not have reliable phylogenetic signal in their traits, but when they are compared in the context of pine trees, they are much more similar to one another than they are to a pine (Swenson 2011). Thus, phylogenetic relatedness as a proxy for ecological similarity is a matter of scale both in the breadth of the phylogenetic sample being considered (Cavender-Bares et al. 2009) but also the degree to which one is comfortable saying two species are more similar to one another than they are to a third species. Given the need for pragmatic and rapid approaches to quantifying biodiversity, it is obvious why phylogenetic metrics of biodiversity are often employed.

A Few Barriers to Quantifying Phylogenetic Metrics of Biodiversity

In the preceding, we have made a brief case for why phylogenetic metrics of biodiversity are of interest to conservation biologists, and we have noted the long history of these measures in conservation. So why, given these factors, have these measures been only recently widely utilized in the literature? Here, we highlight what we believe are two major reasons – the cataloging of species and constructing phylogenetic trees.

One approach to conservation may be to protect large tracts with the expectation that these large areas will contain large numbers of species and may sustain more stable populations of said species. However, in many cases hard decisions must be made regarding what areas should be a priority. Knowledge regarding the biodiversity each area holds greatly helps these decision-making processes. An approach to this issue is to estimate which area contains more habitat types and therefore likely more species. An alternative is to directly sample the biodiversity in both locations. Directly sampling biodiversity, we argue, is preferred, but the time and financial cost of such expeditions may be deemed by some to be too great. This results in a reduced or nonexistent catalog of the species in an area. Without such a list, a phylogenetic tree cannot be made, and phylogenetic measures of biodiversity cannot be generated.

A second and potentially more significant barrier to utilizing phylogenetic measures of biodiversity in conservation is the need for building phylogenetic trees for large numbers of species. It may seem, at this point in time, that there is so much genetic data in public repositories that a species-level phylogeny could be generated for just about any assemblage on the planet. However, this is, of course, far from true. For example, Kress et al. (2009) generated a DNA barcode dataset for the tree species in the 50 ha Barro Colorado Island (BCI) forest dynamics plot. The BCI

forest is the most heavily studied forest in the tropics and perhaps one of the most heavily studied in the world. When Kress et al. (2009) commenced their study, they found that fewer than 30% of the 300 tree species in the forest plot had any kind of sequence data in the NCBI database much less having the same genes sequenced thereby permitting sequence alignment and phylogenetic inference. Thus, even in the most heavily studied areas, we have exceedingly little molecular data at our fingertips, and it is likely that the areas of greatest conservation concern are presently greatly understudied. In other words, simply downloading DNA sequences for all the species in a region and inferring a resolved species-level phylogeny is a largely unreasonable expectation if one is studying anything lacking vertebrae.

At present, there are essentially two options for inferring phylogenetic relationships for large numbers of non-vertebrate species. The first is to go out and collect specimens, deposit them, and sequence their DNA. While this is challenging, large campaigns have been launched and successfully completed where 2- or 3-locus DNA barcodes have been sequenced for the species in biodiverse communities around the world. For example, Erickson et al. (2014) generated DNA barcode data for >1000 tree species worldwide to generate a large molecular phylogeny that was utilized to quantify phylogenetic metrics of biodiversity. Similarly, Muscarella et al. (2014) built off of previous work from Kress et al. (2010) to nearly complete a DNA barcode inventory of the trees of Puerto Rico thereby permitting phylogenetic analyses of biodiversity in different habitat types on the island. While such field and lab campaigns are becoming more common, they still represent significant barriers to many.

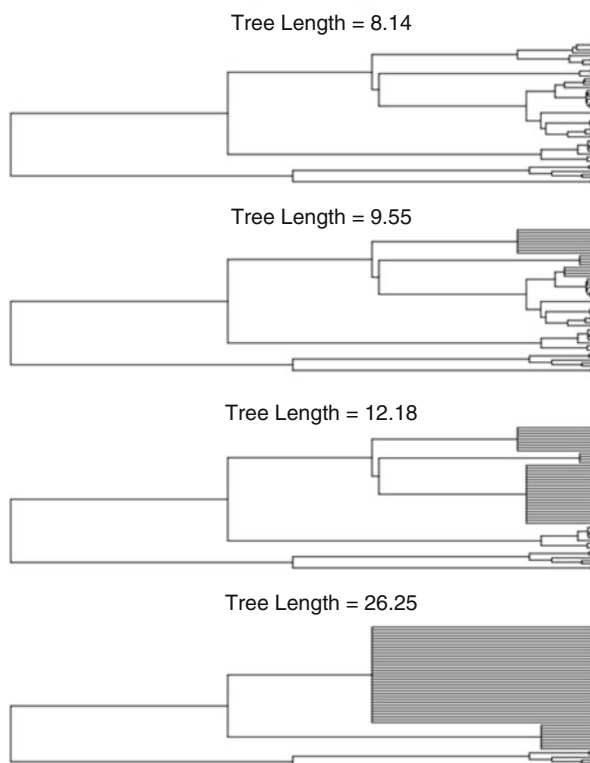
An alternative to the generation of new sequence information is to leverage existing information about phylogenetic relationships or existing sequence information to build a phylogenetic backbone upon which taxonomic relationships can be placed. The Phylomatic algorithm developed by Webb and Donoghue (2005) was the first to offer such an approach and is one of the key factors responsible for the large boom in phylogenetic analyses in ecology during the past decade. Specifically, Phylomatic utilizes a phylogenetic backbone (e.g., an Angiosperm Phylogeny Group phylogeny) and pastes on taxonomic relationships to terminal branches based upon the species list provided by the user. The result is a Frankenstein-esque tree of trees far from ideal, but it generally captures the phylogenetic relationships between major clades. A clear downside to this approach or any similar approach is that terminal relationships in the phylogeny are typically left unresolved (i.e., they contain many polytomies). Interestingly, the problem of widespread polytomies is not totally unique to Phylomatic trees. DNA barcode phylogenies often have many polytomies if more than two congeners are in the sample and/or certain loci (e.g., trnH-psbA; ITS) with faster rates of molecular evolution are not sequenced. Thus, polytomies are a reality of almost any modern-day analyses of phylogenetic biodiversity.

The importance of polytomies and the biases they may introduce in conservation or ecological studies has not been broadly examined. The main previous work that we are aware of has been conducted by Swenson (2009). Specifically, Swenson (2009) examined whether two measures of phylogenetic community diversity and dispersion were influenced by a lack of phylogenetic resolution. The motivation for

that study was that many community ecologists were utilizing poorly resolved Phylomatic phylogenies, and it was unclear how biased their results and inferences were or were not. Swenson (2009) showed that polytomies basally in the phylogeny had a large impact on pairwise and nearest neighbor metrics of phylogenetic biodiversity, but this effect was mitigated when polytomies were more terminally distributed. Additional work by Kress et al. (2009) showed that a lack of phylogenetic resolution likely results in elevated type I error rates in community phylogenetics where previous researchers using poorly resolved phylogenies (i.e., Kembel and Hubbell 2006) had reported weak to no phylogenetic structure in tropical tree communities when indeed there was strong structure when a fully resolved phylogeny was utilized (Kress et al. 2009). However, all of these previous studies have focused primarily on the results emanating from a null model analysis and have not focused on raw measures of phylogenetic biodiversity that are likely to be of more interest to conservation biologists. Beyond these community level analyses, additional work has focused on the impact of phylogenetic resolution on measures of phylogenetic signal. For example, Davies et al. (2012) demonstrated that a lack of resolution results in an overestimation of phylogenetic signal focusing on the K statistic from Blomberg et al. (2003). Munkemuller et al. (2012) came to a similar conclusion when analyzing multiple metrics of signal. These results are intuitive as polytomies artificially increase the amount of time for independent evolution by removing clade stems. Thus, the expected amount of evolution given a Brownian motion model in a phylogeny with polytomies is inflated making the signal in the observed trait data artificially inflated.

The impact phylogenetic resolution, or the lack thereof, has on measures of biodiversity may be clear to those that frequently analyze phylogenetic information. However, most ecologists and conservation biologists are not of this variety. Thus, we aim to first outline why phylogenetic resolution should alter measures of biodiversity, and we follow this with a simulation study to demonstrate these biases. In the following, when we say a lack of resolution, we are referring to soft polytomies. Soft polytomies are nodes in a phylogenetic tree that do not bifurcate due to a lack of knowledge regarding the relatedness of the branches emanating from that node. In other words, they represent ignorance or a lack of confidence. For example, if we have three species (species A, B, and C) and we do not know if species A and B are more closely related to each other than they are to C, or A and C than they are to B, or B and C than they are to A, we would represent this as a soft polytomy where all are equally related with the base of their branches all coming from the same node and ending at the same tip date on the phylogeny (present time). Thus, if the node is 10 million years old, then all species are 10 million years diverged from one another if there is no resolution. However, if species A and B are more related to each other having diverged 10 million years ago from species C and 2 million years ago from each other, then A and B are separated by 2 million years and A and C and B and C are each separated by 10 million years. In the first, unresolved, situation, we have three branches with lengths of 10 million years. Thus, the total branch length (i.e., tree length) is 30 million years. In the second, resolved, situation, we have a branch for species C that is 10 million years in length, a stem from the basal node to the split

Fig. 5.1 The relationship between phylogenetic resolution and tree length. In the top panel is a fully bifurcating phylogeny with 50 tips (i.e., species). The sum of all branch lengths (i.e., tree length) is 8.14. The same phylogeny is represented in the following three panels with less and less resolution to the point where two of the three most basal clades are completely collapsed into polytomies. At the final stage, the tree length is more than three times the original length



between species A and B that is 8 million years in length, and then two additional tip branches leading to species A and B that are each 2 million years. Thus, this second phylogeny has a tree length of 22 ($10 + 8 + 2 + 2$). In other words, the polytomy increases the total branch length equal to the length of the stem holding the two species that should be clustered together, which is 8 in this example.

Thus, all else being equal and assuming the phylogeny is ultrametric (i.e., all tips of the phylogeny end at the same place), a lack of resolution should increase the total branch length in a phylogeny and therefore likely bias phylogenetic metrics of biodiversity upward assuming some species in the assemblage under study are subtended by a polytomous node. In Fig. 5.1, we provide an example using a 50-species phylogeny where tree length is reported as major clades are collapsed into soft polytomies. Thus, from first principles we can see how and why phylogenetic resolution would influence phylogenetic metrics, but ideally this would be elucidated directly. In the following section, we utilize a series of simple simulations to quantitatively show how polytomies influence widely used measures of phylogenetic biodiversity and phylogenetic signal. We then end the chapter with a discussion of the results and the role of phylogenetic resolution, broadly, in conservation biology.

A (Simple) Simulation Study

To demonstrate the ways in which a lack of phylogenetic resolution can affect phylogenetic metrics of biodiversity, we carried out a simulation study. We acknowledge that any simulation study can be made more complex and more parameter space can be explored. In the present work, we explore parameter space related to the number of species in the phylogeny and the percent of terminal nodes left unresolved. Lastly, all simulated phylogenies were built using random coalescence. Thus, we did not set out to explore the role of phylogenetic imbalance or phylogenetic “tippy-ness” (i.e., the relative length of terminal or basal branches). Phylogenetic “tippy-ness” would be particularly interesting to explore in future work because phylogenies with shorter tips are likely to be most sensitive to issues of phylogenetic resolution.

We began by simulating phylogenies with 25, 50, 100, 250, 500, 1000, and 2000 tips (i.e., species). We replicated this 30 times per tip number leading to 210 phylogenies. Each phylogeny was generated using random coalescence in R (`rcoal()` function in the `ape` package in R (Paradis 2012)). Each of these trees was fully bifurcating and was deemed the “true” phylogeny for all subsequent analyses with 0% of the nodes unresolved. Next, we gradually unresolved phylogenies starting at 1% of the root to tip distance working from the tips toward the roots in increments of 2% until we reached 29%. For example, during the first step, we located all internal nodes in the phylogeny that were in the distal 1% of the phylogeny and collapsed them into polytomies if they were subtended by more than two species. If they were subtended by only two species, they couldn’t be “polytomized.” This phylogeny was saved. Next, we considered all nodes in the distal 3% of the phylogeny and polytomized them and saved the phylogeny and so on until we reached 29%. It is important to note that percent values here are based on root to tip distances and not the percent of nodes polytomized. This resulted in 15 additional phylogenies per tip number category. Thus, we had 480 phylogenies per tip number category (30 replicates by 16 levels of resolution) and 3360 phylogenies in the entire study (480 phylogenies for each of the 7 tip number categories). In Fig. 5.2, we report the total tree length (sum of all branch lengths) from each of these trees as the phylogeny becomes gradually unresolved. We see that tree length increases as expected. We can also see that in most cases, the increase in tree length is early and rapid. This is due to having many internal nodes clustered distally on the phylogeny leading to large increases in tree length and relative stasis in the number of nodes being polytomized tips.

Next, we generated random assemblages of species drawing from the original “true” phylogenies at random with a wide range of species richness values (10–50% of the total number of tips). We also conducted a small study sampling phylogenetically clustered species. The results were qualitatively similar and are therefore not shown. We also applied log-normal abundance distributions to the randomly assembled species. Again, those results were consistent with the presence-absence weighted results we show below indicating that the qualitative trends in phylogenetic biodiversity metrics with phylogenetic resolution are general phenomena, not

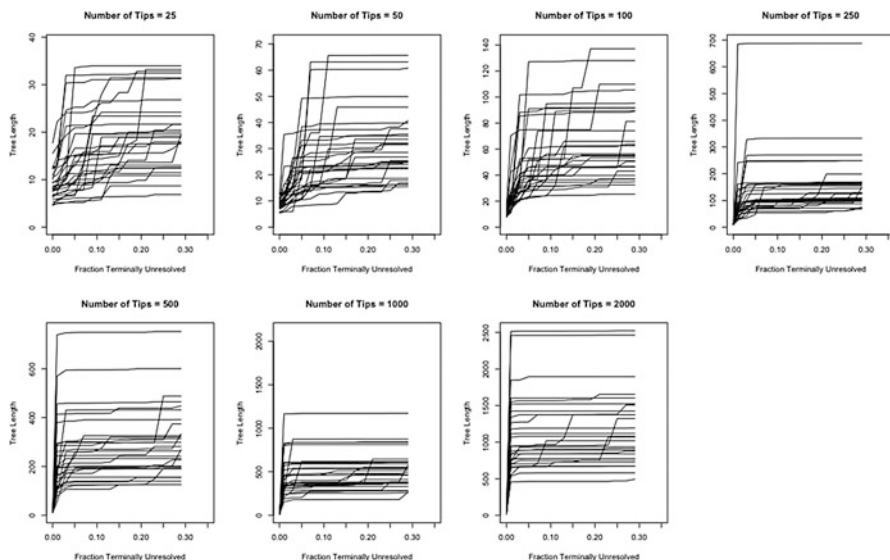


Fig. 5.2 The relationship between tree length and the fraction of terminal nodes left unresolved. The beginning of each line represents a “true” bifurcating phylogeny and each line is a replicate. While trees vary in their shape and distribution of nodes, we see an increase in tree length with decreasing resolution

peculiar to any system or situation. We then calculated several phylogenetic metrics of biodiversity that may be of interest to a conservation biologist. First, we calculated PD. The PD represents the sum of the branch lengths connecting species, and we have shown that tree length increases as resolution decreases; therefore, an increase in PD was expected. In Fig. 5.3, we show these results where PD grows substantially as the phylogeny becomes less and less resolved. Values do level off in most cases after 15% of the terminal part of the phylogeny is unresolved, but this only serves to indicate that resolving even the most terminal nodes in phylogenies could have major impacts on the level of PD estimated.

Next, we calculated the mean nearest phylogenetic neighbor distance (MNTD; Webb 2000) and the mean pairwise phylogenetic distance (MPD; Webb 2000) between the species in the assemblages (Swenson 2014). The MNTD results, which provide an average of the distances between each species and its nearest phylogenetic neighbor, are shown in Fig. 5.4 and show that the values tend to increase as resolution decreases. Again, this is intuitive because if there is a triplet of species and two are more closely related in the assemblage the MNTD will be low, but if that triplet is turned into a polytomy the length separating the two species in the assemblage increases by $1 \times$ the length of their stem. Similarly, MPD values, produced from a pairwise phylogenetic distance matrix between each species, also increase as the resolution decreases (Fig. 5.5) simply due to more branch length separating tips on average. However, contrary to other metrics, the magnitude of the

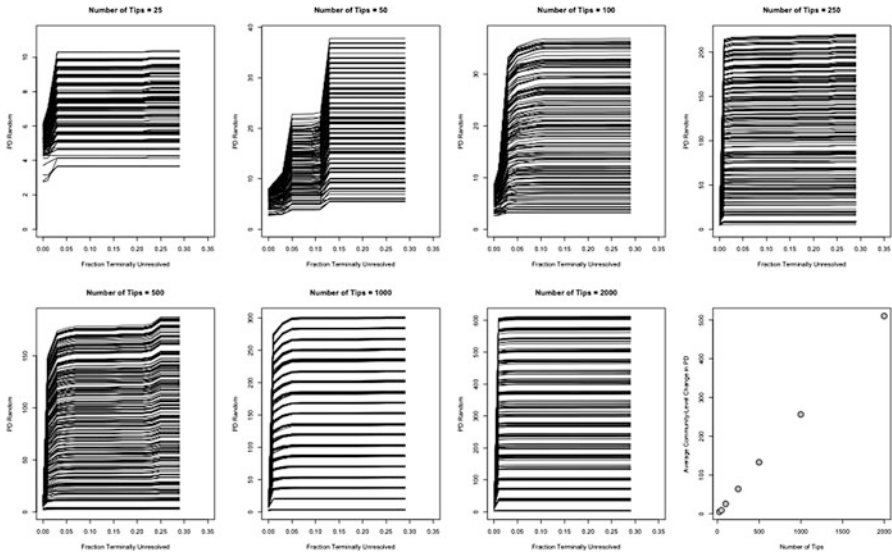


Fig. 5.3 The relationship between the phylogenetic diversity (PD) metric and the fraction of terminal nodes left unresolved. The beginning of each line represents a “true” community PD values using one exemplar phylogeny that is gradually unresolved. Assemblages of species were randomly assembled at different species richness values. In the final panel, we show the average change in PD from 0% to 29% unresolved across all assemblages averaged across all 30 phylogenies simulated for a given number of tips

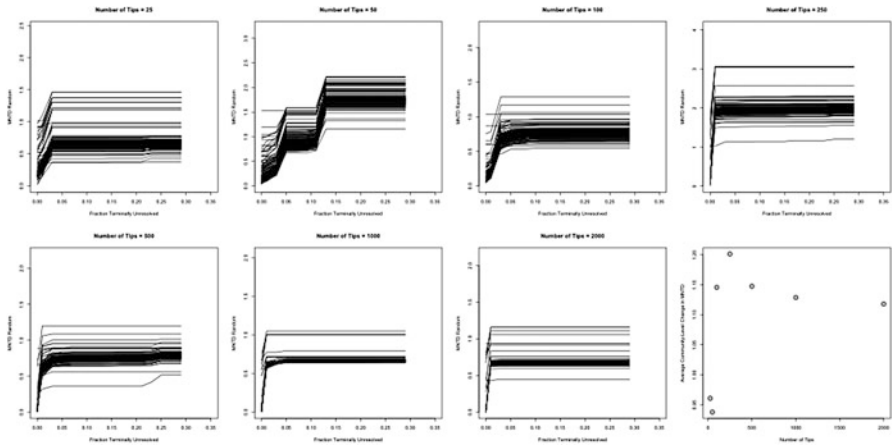


Fig. 5.4 The relationship between the mean nearest phylogenetic neighbor distance (MNTD) metric and the fraction of terminal nodes left unresolved. The beginning of each line represents a “true” community MNTD values using one exemplar phylogeny that is gradually unresolved. Assemblages of species were randomly assembled at different species richness values. In the final panel, we show the average change in MNTD from 0% to 29% unresolved across all assemblages averaged across all 30 phylogenies simulated for a given number of tips

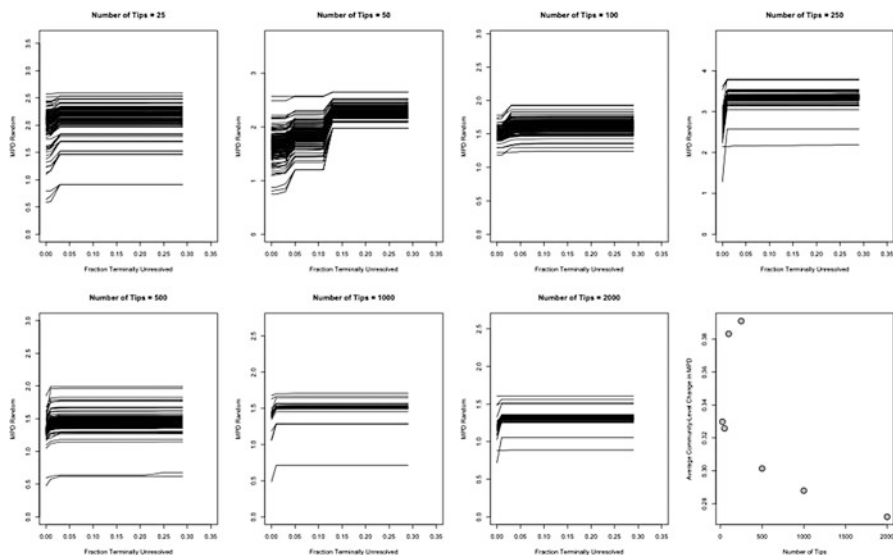


Fig. 5.5 The relationship between the mean pairwise phylogenetic distance (MPD) metric and the fraction of terminal nodes left unresolved. The beginning of each line represents a “true” community MPD values using one exemplar phylogeny that is gradually unresolved. Assemblages of species were randomly assembled at different species richness values. In the final panel, we show the average change in MPD from 0% to 29% unresolved across all assemblages averaged across all 30 phylogenies simulated for a given number of tips

change in MPD with loss of resolution is smaller in large phylogenies. In other words, loss of resolution has less of an impact when the phylogeny is large.

Next, for each of the “true” phylogenies, we calculated the evolutionary distinctiveness (ED) of species on the tree using Redding and Mooers (2006) equal-split approach. The ED of a species is, as the name would imply, a function of how distantly related it is to other tips on the phylogeny. Thus, as resolution decreases, the length of tips increases and ED should increase. In Fig. 5.6, we present changes in ED for an original phylogeny as it is gradually unresolved. Each line in this instance represents a species. In some instances, we can see that a species has the same level of ED throughout (i.e., a flat line). These are tips that arise from basal nodes in the phylogeny and are therefore not increased in their ED as the phylogeny is unresolved. However, other species do dramatically increase in their ED due to being derived from more terminal nodes in the original phylogeny.

The above considered diversity metrics, but a conservation biologist may also be interested in the degree of phylogenetic signal in species rarity. Phylogenetic signal can be defined broadly as the degree to which the similarity of species is dictated by their relatedness (i.e., shared branch lengths) (Blomberg et al. 2002, 2003). Here, we considered the abundances of species as a trait. Specifically, we arrayed a log-normal abundance distribution on the original phylogeny and measured two widely used metrics of phylogenetic signal – Blomberg’s K (Blomberg et al. 2003) and Pagel’s

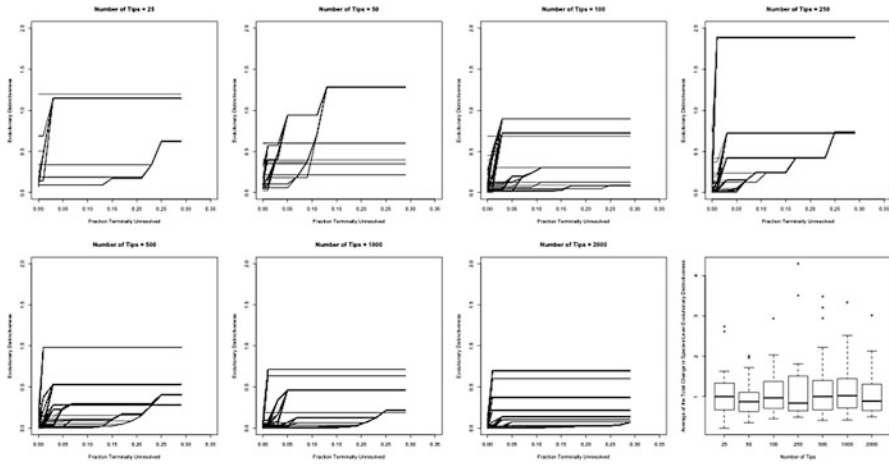


Fig. 5.6 The relationship between the evolutionary distinctiveness (ED) metric and the fraction of terminal nodes left unresolved. The beginning of each line represents the ED values for a species in the fully resolved phylogeny using one exemplar phylogeny that is gradually unresolved. In the final panel, we show boxplots of the average change in ED across species from 0% to 29% unresolved across all 30 phylogenies simulated for a given number of tips

lambda (Pagel 1999). Both metrics quantify the degree to which the variation in a trait between species, here $\log(\text{abundance})$, is explained by the shared branch lengths between those species. Using Brownian motion trait evolution as an expectation, it is expected that covariation in trait values between two species is related to their shared branch length (K or $\lambda = 1$). If the trait has less phylogenetic signal than expected from Brownian motion, both metrics should deviate lower toward zero. For both metrics, phylogenetic signal increases as phylogenetic resolution decreases (Figs. 5.7 and 5.8). In other words, a biologist may be prone to overestimate the phylogenetic signal in rarity when using a poorly resolved phylogeny. While the magnitude of changes in the metric is lower for λ than K , both metrics had the same general increase. We provide the following hypothetical example to elucidate why this occurs. Imagine two sister tips that split one million years ago and they have a moderate difference in their rarity. This moderate difference may be unexpected given the relatively short period of time since divergence (i.e., low phylogenetic signal in abundance). However, if we consider the sister tip to these two, which diverged 10 million years ago, and we now polytomize the triplet, then the two original tips are now separated by $10\times$ the original branch length. In this context, their differences in abundance may be as expected given their apparent 10 million years of independent evolution (i.e., high phylogenetic signal). Thus, through this general thought experiment and via computer simulations, we can see how and why phylogenetic signal would actually increase with a lack of phylogenetic resolution.

In summary, a decrease in terminal phylogenetic resolution leads to an increase in total branch lengths in a phylogeny (i.e., tree length) via a doubling of stem branch lengths. This increase in tree length leads to an increase in phylogenetic metrics of

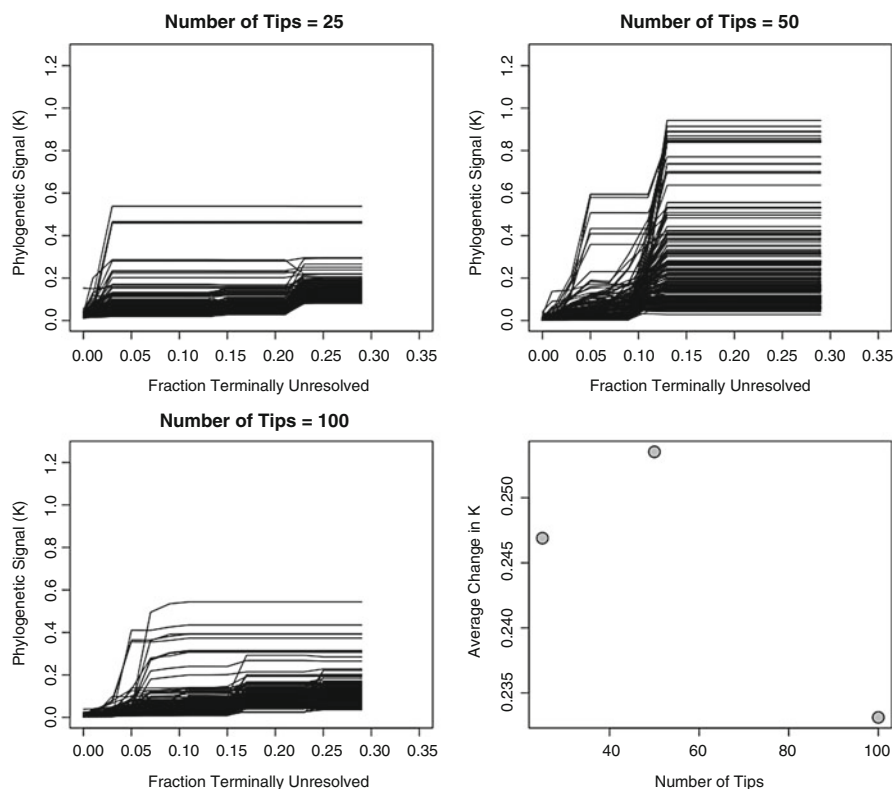


Fig. 5.7 The relationship between the phylogenetic signal in log(abundance) measured using Blomberg's K (Blomberg et al. 2003) and the fraction of terminal nodes left unresolved. The beginning of each line represents a “true” K value for assemblages using one exemplar phylogeny that is gradually unresolved. Assemblages of species were randomly assembled at different species richness values. In the final panel, we show the average change in K from 0% to 29% unresolved across all assemblages averaged across all 30 phylogenies simulated for a 25, 50, or 100 tips

biodiversity as well as the estimated phylogenetic signal in a quantitative trait such as abundance. The results were largely consistent across replicate simulations indicating that the trends reported are likely to be general in real-world datasets. In the following section, we will discuss the results in more detail and make recommendations for application of these metrics when phylogenies are incompletely resolved.

Discussion

During the past two decades, conservation biology and ecology have come to realize that holistic assessments of biodiversity require the consideration of multiple axes of biodiversity. Phylogenetic diversity was the first such axis to garner attention beyond

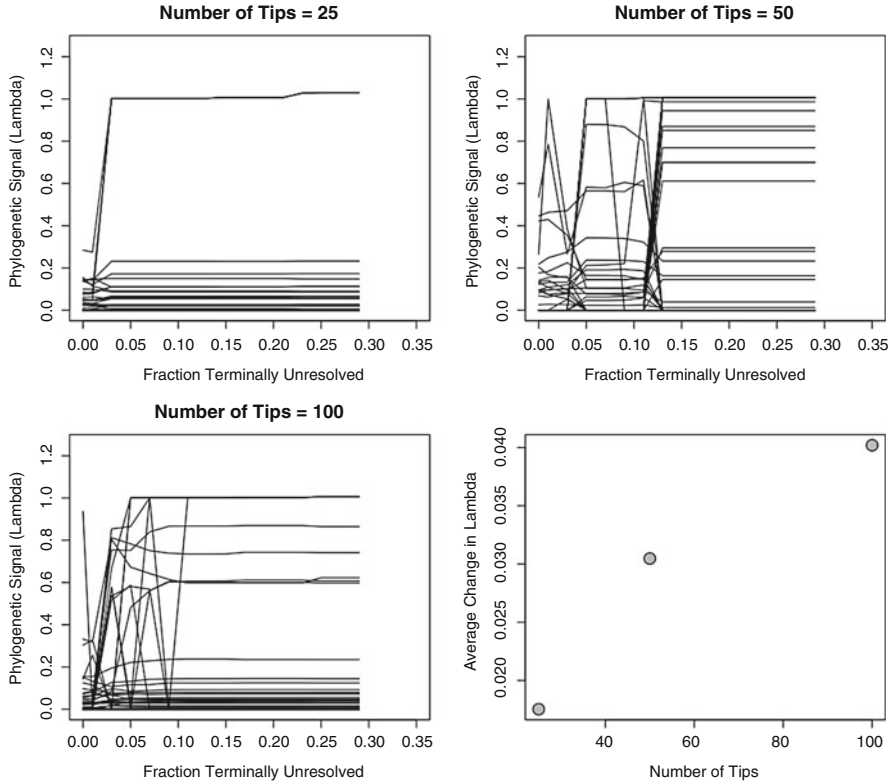


Fig. 5.8 The relationship between the phylogenetic signal in log(abundance) measured using Pagel’s lambda (Pagel 1999) and the fraction of terminal nodes left unresolved. The beginning of each line represents a “true” lambda value for assemblages using one exemplar phylogeny that is gradually unresolved. Assemblages of species were randomly generated at different species richness values. In the final panel, we show the average change in lambda from 0% to 29% unresolved across all assemblages averaged across all 30 phylogenies simulated for a 25, 50, or 100 tips

species diversity (Faith 1992, 1994) later followed by interest in functional diversity (e.g., Tilman et al. 1997). As the use of phylogenetic information continues to expand, being able to elucidate and quantifying where biases may arise using such information has become essential. In many instances, phylogenetic analyses in conservation biology require the use of large phylogenies. Such phylogenies often must be inferred using imperfect information that results in unresolved relationships (i.e., soft polytomies). While the use of phylogenies containing soft polytomies is widespread, the implications of this are greatly understudied. Here, we have utilized a series of simple simulations to show an increase in soft polytomies can influence phylogenetic measures of biodiversity. Specifically, we first showed that polytomies increase the total branch length (i.e., tree length) of phylogenies, and this generally results in a subsequent increase in PD values. We also showed that the MNTD and MPD (Webb 2000), two very popular metrics in community ecology, were also

affected by lack of resolution in the phylogeny where values tended to increase as resolution decreased. Lastly, when calculating the evolutionary distinctiveness (ED) of species, we found dramatic increases due to the species being derived from more terminal nodes in the original phylogeny. In addition to these diversity metrics, we also determined how lack of resolution would alter the degree of phylogenetic signal in a continuous trait. For the purposes of this study, we imagined this trait to be species rarity or abundance. We examined the sensitivity of two popular metrics of phylogenetic signal, Blomberg's K (Blomberg et al. 2003) and Pagel's lambda (Pagel 1999). We found that phylogenetic signal increased as phylogenetic resolution decreased. In other words, our simulation shows that biologists may overestimate phylogenetic signal in rarity if poorly resolved phylogenies are used for analysis. In the following, we discuss the results in more detail and provide suggestions for future work and applications.

Polytomies and Increases in Tree Length and Phylogenetic Measures of Biodiversity

As a general rule, the tree length of a phylogeny that is ultrametric will increase as polytomies are introduced (Fig. 5.2). Thus, when quantifying a phylogenetic measure of biodiversity that utilizes branch length information (i.e., essentially all of them), we should a priori expect a bias toward overestimating the true level of biodiversity. This general rule may be broken if the taxa in the area being studied are from a portion of the phylogeny that is fully resolved and the portions that are unresolved do not contain taxa in the area. In other words, in special cases where polytomies are nonrandomly distributed on the phylogeny and the taxa in the geographic area of interest are also nonrandomly absent from the region of the phylogeny that has polytomies, then the bias will be mitigated or nonexistent. However, this may be an unlikely scenario as we show here with simulations.

If there is a silver lining to this bias it is that understudied areas of the phylogeny or understudied geographic areas, that therefore likely contain taxa that are in unresolved areas of the phylogeny, will appear to have greater biodiversity when using a phylogenetic measure and may therefore attract more study. This is a far better direction of bias, in our opinion, than underestimating biodiversity in clades or geographic areas that have been traditionally understudied. However, we can imagine that others, perhaps those that study vertebrates with well-resolved phylogenies, would have a different perspective.

One may be tempted to say that a null model may magically make this bias disappear. Previous work has shown this not to be the case (Kress et al. 2009; Swenson 2009). Specifically, unresolved phylogenies tend to increase type I error in null model studies, but the direction of the type I error is not consistent. Specifically, polytomies cause researchers to falsely infer that their observed phylogenetic biodiversity measure is no different from that expected given the species richness when it

is in fact significantly higher or lower than expected. In other words, we know polytomies would cause someone to miss extreme or unusual levels of phylogenetic biodiversity, but they would not know if the true values were unusually high or low.

Polytomies and Measures of Phylogenetic Signal

Measures of phylogenetic signal have been popularized in ecology largely to justify the use of a phylogenetic proxy for ecological similarity (Swenson 2013). However, they are also used in comparative biology as a first step toward determining whether phylogenetically informed statistical methods are necessary, and they are utilized in macroecology to infer the importance of niche conservatism. In conservation biology, they have been utilized to quantify whether there is phylogenetic signal in conservation status (i.e., a binary trait) and other work has examined whether rare species tend to be distributed only in particular clades (Mi et al. 2012). In the later work, phylogenies with many polytomies were utilized as will be the case in many future studies. Thus, we ventured to quantify the magnitude of the problem using simulations.

A basic understanding of phylogenetic signal metrics and Brownian motion trait evolution can unveil likely biases. Specifically, polytomies will collapse clades and push the time since divergence deeper in time. Thus, the amount of time expected for traits to diverge via a random walk (i.e., Brownian motion) increases making the observed difference in traits between two species appear less and less abnormal. For example, if two species differ in 2 units of a trait and they diverged one million years ago versus 10 million years ago, we would clearly have different impressions in the pace of trait evolution and the amount of phylogenetic signal in the trait. Our simulations elucidate this, showing that a trait arrayed on a fully bifurcating phylogeny can have essentially no phylogenetic signal, but when 30% of that phylogeny is unresolved terminally, that same trait data can appear to have strong phylogenetic signal (Fig. 5.7).

If we place this finding into a conservation context, it would have the following importance. Assume, as we have done here, that the trait is abundance. When the phylogeny is fully resolved, we would say that rarity is not nonrandomly distributed on the phylogeny. However, when the phylogeny is less resolved, we would state that rare species are likely clustered in one to a few clades. In the first situation, one may choose to conserve a broader sampling of the phylogeny as all clades likely contain rare species. However, in the second situation, one may choose to overweight certain clades that (incorrectly) appear to have an unusual number of rare species. In other words, if we are in a position to consider whether certain clades have a relatively greater priority for conservation based upon a metric of phylogenetic signal, we would be wise to consider whether that clade or the overall phylogeny has many polytomous nodes. This problem is even more insidious than the bias for metrics of biodiversity. This is because phylogenetic signal is an overall metric from the traits and phylogeny data. Thus, it is not possible to simply

disentangle the bias with a thought experiment. For example, the polytomous nodes could occur in a portion of the phylogeny that contains common and rare species, and the longer branch lengths resulting from polytomies could result in more signal. Further, imagine a situation where the polytomies could occur in a region of the phylogeny that has many common species and the rare species are randomly distributed on the phylogeny. The polytomies in this case increase overall signal, but it would have nothing to do with rarity per se. Thus, the biases in phylogenetic signal metrics due to polytomies are obvious, but what to do about them is far less obvious.

Recommended Solutions and Where to go from Here?

While we have attended many recent seminars or symposia where a speaker happily exclaims that soon we will have a highly resolved tree of life, we are not quite so optimistic. Frequently, such speakers work on vertebrate groups and/or in the temperate zone. We argue that a more sober and pragmatic view recognizes that many species remain to be discovered and the waves of sequencing data being currently amassed will be similarly biased toward species poor vertebrate clades and the temperate zone. Simply put, conservation cannot sit around, wait, and hope that suddenly all taxa will be collected, identified, and sequenced in a useful and sensible manner. Given this, informatics solutions and sparse datasets will continue to be the norm in many phylogenetic analyses in conservation (see Chap. 7). The question is, therefore, how do we deal with the biases that will lurk in studies that cannot avoid utilizing phylogenies that contain polytomies?

The long-standing proposed solution to polytomies in comparative biology studies has been to randomly resolve the phylogeny many times and to rerun the analyses each time (Losos 1994). This would provide a distribution of possible outcomes that can be reported. While we fully support and advocate for such an approach, in reality it is not easy to implement or interpret. We suspect one major reason why this approach is not taken is that the main function in R utilized to turn polytomous nodes into a series of bifurcating nodes (i.e., the *multi2di()* function in the package *ape*) does not actually give new stem branch lengths. In other words, the branch lengths between species remain the same as in the original unresolved tree. If one were to utilize a program like Mesquite (Maddison and Maddison 2011) to randomly resolve polytomies, they would quickly find that a decision has to be made on where to place the new nodes in the phylogeny with the default being 50% of the distance between nodes. With enough time and computation, a distribution of distances between nodes could be explored. For example, the expected distribution of a metric could be calculated from 1000 randomly resolved phylogenies using 50%, and this could be repeated for 1000 randomly resolved phylogenies using 25% and 1000 for 75% and so on. When the analyses take seconds per phylogeny, this is a not-too-daunting exercise for a skilled analyst, but when the analyses take minutes to hours, it may mean waiting an enormous amount of time unless one has access to a

high-performance computing cluster and knows how to use it. So what to do? We would recommend, when possible, to at least randomly resolve a phylogeny 100 times placing new nodes 50% between nodes to provide a distribution of possible values thereby giving readers and decision-makers an idea of the range of possible biodiversity levels in a sample. To our knowledge, Mesquite is the only software that can quickly resolve phylogenies in this manner, but we hope that a similar function in R will be produced that avoids some of the issues that come with using the existing *multi2di()* function.

Along with quantitative solutions to dealing with a lack of phylogenetic resolution in conservation studies that utilize phylogenetic information, we would advocate a clear discussion of how a lack of resolution would, on average, affect the metrics utilized. Ideally, this discussion would be supplemented with information regarding the distribution of polytomies across taxa in the study. For example, if mammals are being compared to plants, it would be important for the authors of the study to highlight that the mammals, in most cases, were analyzed with a much more highly resolved phylogeny or whether certain clades within the plant phylogeny have substantially less resolution than others.

Finally, the ultimate solution will be provided by further exploration and documentation of biodiversity. Ideally, collections and further sequencing would result in perfectly resolved phylogenies, but we are realistic enough to know this will not be the case in the timeframe necessary. We, therefore, advocate pragmatic approaches where specimens are collected and low cost sequencing is performed (i.e., via DNA barcodes) so that the taxa under study can at least be roughly placed into a phylogenetic context. This will not result in resolved topologies, particularly for “younger” clades, but it would continue to propel the use of phylogenetic information in studies that measure biodiversity and help to reduce biases that may arise.

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Chapter 6

Phylogeny-Based Measurements at Global and Regional Scales



Shawn W. Laffan

Abstract The purpose of this chapter is to describe a number of methodological issues, considerations and opportunities when applying phylogeny-based measures at regional to global scales, a process recently labelled as “spatial phylogenetics”. These are divided into four sections: (1) the nature of the source data; (2) issues and effects of spatial aggregation; (3) some useful properties of phylogenetic diversity, endemism and rarity across landscapes; and (4) how to develop analysis tools that will complete in reasonable time for large data sets. The focus of the discussion is on measures of phylogenetic diversity, endemism and rarity, but the issues, considerations and opportunities raised have general application to other related indices across landscape and global scales.

Introduction

The purpose of this chapter is to describe a number of methodological issues, considerations and opportunities when applying phylogeny-based measures at regional to global scales, a process recently labelled as “spatial phylogenetics” (Scherson et al. 2017; Thornhill et al. 2017). Recent increases in the availability of spatial and phylogenetic data make such analyses possible, and likely routine in the near future, but only because the advances in data have been paralleled by developments in the suite of analysis tools (e.g. Kembel et al. 2010; Laffan et al. 2010; Webb et al. 2008).

Spatial phylogenetic analyses have been applied across a range of organismal groups, including angiosperms (González-Orozco et al. 2016; Mishler et al. 2014; Schmidt-Lebuhn et al. 2015; Thornhill et al. 2016), pteridophytes (Nagalingum et al. 2015), mammals (Rosauer and Jetz 2014) and combinations of multiple groups (Cassis et al. 2017; González-Orozco et al. 2015; Kooyman et al. 2013;

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Laity et al. 2015). The spatial extents of regions have ranged in size from subcontinental (Carta et al. 2017; González-Orozco et al. 2015; Scherson et al. 2017; Thornhill et al. 2017) to continental (Cassis et al. 2017; González-Orozco et al. 2016; Laity et al. 2015; Mishler et al. 2014; Schmidt-Lebuhn et al. 2015) to global (Rosauer and Jetz 2014). By using phylogenetic data, each of these analyses has provided insights into diversity that would not have been possible using taxon-level data alone.

As demonstrated in the other chapters in this book, phylogenetic diversity indices can be calculated for any sample. Spatially these can range from the small to the large, for example, microbial communities along an animal's gut (see Chap. 1), bird diversity across an archipelago, plant communities across a terrestrial biome (see Chap. 10), and a continental or oceanic fauna, to the global scale. The fundamental indices and how they are calculated are common across scales (e.g. see Chaps. 7, 8, 9, and 10), and many analyses developed for communities can be equally applied to regional data sets (Kooyman et al. 2012). However, the nature of the source data and how the biota is distributed across geographic space can impact on the interpretation of such indices, largely because of what the samples or assemblages represent at such scales.

This chapter is divided into four main sections. First the nature of the source data are described, along with several of the main issues that need to be considered when using them in analyses. Second, issues of aggregation are discussed, focusing on the choice of spatial unit and well-known spatial analysis effects. Third, some properties of phylogenetic diversity, endemism and rarity across landscapes are described. Finally, some approaches and examples of how to develop analyses that will complete in reasonable time for such data sets are given.

Sources and Issues of Distributional Data

With the exception of highly endemic and rare taxa, such as those found in one small locality, one will never have a complete census of a taxon at the species or genus rank. One might argue that higher levels such as families are likely to be well sampled for many groups, but the utility of such levels for conservation or in understanding evolutionary histories is often limited. Any biodiversity analysis, phylogenetic or otherwise, will thus use some form of sample or estimation of the geographic distribution of taxa used in an analysis.

Three general types of geographic data are used in spatial phylogenetics analyses (Fig. 6.1), observational data, range polygons and environmental niche model (ENM) derived. The nature of such data is an important consideration, and often a constraint, on how the analyses are applied and interpreted.

Each type of data has particular impacts on the mapped phylodiversity indices due to under- or overestimation of taxon ranges (Amboni and Laffan 2012), for example, through misidentification or incorrect georeferencing. These impacts are a function of the false negatives and false positives, i.e. how much of the tree is



Fig. 6.1 The main types of distributional data used for phylogenetic analyses are the observation records (left panel, black dots), range polygons (left panel, grey polygon) and environmental niche model (ENM) derived (right panel, grey region). The hachured region on the right panel shows a constrained subset of predicted areas that can be reached by dispersal. Data shown are for *Antechinus agilis*. (Source: Amboni and Laffan 2012)

missing from, or has been added to, a region's assemblage. These effects are constant across the data types, and the general principle is the same as for taxon-level analyses such as species richness except that for phylogenetic analyses, one is using the phylogenetic branches of the assemblage and not simply the tips of the tree.

If there is a false positive due to misidentification, but the misidentification is for a sister taxon, then the shared ancestors are already in the sample, and any effect is a function of the branch length differences of the two taxa. If a taxon is misidentified with one from a very distantly related lineage, then the effect on any analysis results would be substantial if that lineage is otherwise not in the assemblage, as different branches can be included and excluded from the sample. The effect is similar for false negatives (unsampled taxa). If a taxon is missing, but the assemblage includes a closely related taxon, then the missing branches might represent a small part of the overall phylodiversity. However, if a missing taxon is from a highly distinct clade, for which there are no samples in the assemblage, then clearly the effect will be greater.

A further consideration for georeferencing errors is that, if an incorrectly located record is retained in an analysis, then it will be located in a neighbouring region's assemblage as a false positive. The phylodiversity of the neighbouring region will therefore be inflated as a function of the incorrectly assigned branches.

Observational Data

Digitised observational data are increasingly available thanks to considerable and long-term efforts by organisations such as the Global Biodiversity Information Facility (GBIF), the Atlas of Living Australia (ALA) and the Australasian Virtual Herbarium (AVH). At the time of writing, there are approximately 850,000,000 records in GBIF, 73,000,000 in the ALA, and 8,000,000 in the AVH. A large

fraction of these records are georeferenced, making them usable for spatial phylogenetics analyses. One must, however, be mindful of duplicate records when combining data from multiple sources. Many of these repositories overlap, for example, the AVH records are also available through the ALA, which in turn is a node of GBIF.

Observational data can be categorised into two main types, occurrence records and plot data.

Plot data are generally small areas (e.g. 50×50 m quadrats) over which detailed surveys have been conducted for a specific type of organism such as beetles or woody plants. They are also typically georeferenced to a high degree of spatial accuracy, even if they long predate the use of global positioning systems (GPS). While well-surveyed, plot data can suffer from non-detection of ephemeral taxa, although this is amortised through repeat visits, and plots targeting such taxa will be designed with this in mind. Due to the intensive effort required, plot data are often spatially dispersed across landscapes, regulating the types of spatial analysis that can be applied using them (e.g. Rossetto et al. 2015).

Occurrence data are more commonly used in spatial phylogenetics analyses. They are typically in the form of georeferenced records from herbarium or museum collections, but an increasing amount of data is derived from citizen science observations using the now ubiquitous GPS-enabled devices. Augmenting observational data is a huge effort and requires collaboration by large teams of people (e.g. Guedes et al. 2018).

Quality issues with occurrence data are well known (Amboni and Laffan 2012; Chapman 1998; Haque et al. 2017; Newbold 2010), and services such as the ALA have implemented a variety of automated checks to identify potential georeferencing errors. However, no database will ever be completely error-free, and if working with data from a variety of sources, one must be aware of several issues related to taxon identification and geographic data quality.

Taxon identification can be affected by issues such as misidentification, ephemerality and sampling biases. Sampling biases are common in occurrence data, as such data are often derived from opportunistic collections. In many cases the purpose of sampling is to increase the taxonomic span of collections, for which repeated samples of the same taxon are unnecessary. One should therefore never expect such data to provide a complete census of an organism across its geographic range. Even detailed surveys of an area will usually focus on one group of organisms to the exclusion of all others, e.g. a pteridophyte survey is unlikely to include bryophytes. Misidentification is a relatively high risk for citizen science data but also applies to samples collected by experts when the taxa are cryptic. Similarly, ephemeral taxa can often be missed, even across multiple sampling campaigns. Experts for a particular group might simply lack the resources to sample a wildflower bloom in a remote region, even if they are aware of it.

The geographic accuracy of occurrence records can be affected by georeferencing errors and incorrectly specified coordinate systems. There can also be spatial variability of data quality. For example, a taxon might be well sampled in some regions, but poorly sampled elsewhere. Often this is a result of the cost of access, although

some regions are simply inaccessible, but could also be due to the levels of skill and experience of collectors.

Georeferencing errors for observational data can have greatly differing magnitude and are well documented (Chapman 1998). For example, a museum record might be reliable only to the nearest degree (~100 km) as the recorded coordinate did not include information about the minutes or seconds. Others have coordinates inferred from toponyms which are themselves ambiguous or duplicated. For example, there are three Fitzroy Rivers in Australia, in Western Australia, Queensland and Tasmania. These range in length from 58 to 733 km so, even if the name of the state is provided on a record, the geospatial accuracy of a sample described only as collected “near Fitzroy River” is limited.

Coordinate systems are used to describe a location on the surface of the Earth, but their effects are not often considered in analyses of biodiversity. The most commonly used coordinate system is geographic, in which degrees of latitude and longitude define a position on the surface of the Earth. Transposition of latitude and longitude in databases is common, as people are used to writing X and Y coordinates, not Y and X, although such issues are often relatively easy to identify. What is often missing is the datum used for the coordinates. The datum is the origin from which geographic coordinates are measured. The datum for the World Geodetic System, 1984 (WGS84), is the gravitational centre of the earth, around which the GPS satellites orbit, whereas that for Australian Geodetic Datum 1966 (AGD66; used for most maps in Australia prior the introduction of the Geocentric Datum of Australia 1994 – GDA94) is the centre of the Australian continent. Depending on where one is, the lateral offset between AGD66 and WGS84 is approximately 200 m for the same numerical coordinate value, whereas the offset for GDA94 and WGS84 is approximately 10 m (which is close to the accuracy of a hand-held GPS device). More recent datums are time dependent and take into account the movement of continental plates (Australia moved ~1.6 m northwards between 1994 and 2017), although such movements are irrelevant for most biodiversity analyses. Even the larger offsets matter little when data are analysed at resolutions such as 25×25 km grid cells. Nonetheless, researchers need to be cognisant of these issues as one would expect biodiversity analyses to be applied at increasingly fine resolutions as appropriate environmental data become available.

Range Polygons

Range polygons are often drawn by hand on the basis of expert advice or agreement. A good example of these is the data provided by the International Union for the Conservation of Nature (IUCN 2018). These are typically highly generalised and spatially imprecise, and the IUCN documentation is very clear that they are simplified representations. However, for continental extent analyses, such polygons are sometimes the best available data and will be an appropriate representation for highly mobile taxa such as bats (e.g. López-Aguirre et al. 2018).

An additional consideration is that, even though these polygons are derived from expert advice, it is worth remembering that experts can often disagree.

Some authors have encouraged the use of convex hulls derived from observation points (Guerin et al. 2015), but these have considerable limitations if not used with care. A key issue is that taxa with disjunct distributions will have convex hulls that span all the areas between the populations. In an extreme case, an Australian taxon found in both Tasmania and the Kimberley will result in a convex hull that includes all areas between the two populations, including the desert areas of central Australia. Such an issue is more likely to manifest when such polygons are applied to internal branches, where the point observations for the subtending terminal taxa are used to define the convex hull.

The issue of range polygons extending into areas where it is impossible for a taxon to occur can also affect contiguous populations. For example, polygons for terrestrial taxa can often span marine areas and vice versa. Similarly, a terrestrial taxon with a horse-shoe-shaped distribution around the coastline of a bay will result in a D-shaped polygon including marine areas within the bay. It is a simple GIS operation to clip such regions out, but automated analyses for data sets with many taxa will need a large amount of user intervention. More complex aggregation methods such as alpha hulls are an alternative approach (e.g. Gallagher 2016), as these allow for gaps and crenulations in distribution edges. Such approaches do not remove the need for quality control, though.

Environmental Niche Models

Environmental niche models are a common method to spatially extend point observation data (Franklin 2010), thereby obtaining an estimate of the geographic extent of a taxon. ENMs are more commonly referred to as species distribution models (SDM), but that is clearly an inappropriate term for rank-free phylogenetic data.

ENMs have considerable potential for spatial phylogenetic analyses (Amboni and Laffan 2012; González-Orozco et al. 2016; Rosauer et al. 2015; Schmidt-Lebuhn et al. 2015; Thornhill et al. 2017), but resource constraints have prevented their widespread application across large extents. A set of continental extent ENMs for 1000 taxa would require substantial computational resources. When one considers the need to develop scripts to run the models, their potential complexity and the sometimes bewildering choice of algorithms to use and parameters to specify, many researchers will avoid the task entirely. However, recent efforts are removing these barriers by providing access to simpler web-based user interfaces and high-performance computing back ends, allied with large repositories of environmental data (Hallgren et al. 2016; Williams et al. 2017).

A reality is that, as with any model, ENMs are only as good as their input data. If poor quality observations are used, or they do not adequately span the environmental range of a taxon, then one cannot expect the resultant model to be a good representation of the taxon's geographic distribution. Many of the spatial accuracy issues

noted above as having limited impact on analyses of observational data can manifest as subtle errors when used in an ENM. For example, a 200 m offset might shift a sample from a topographically exposed position to one that is highly sheltered. If the environmental data are sufficiently accurate and fine-grained to represent such a change, then the resultant model will be affected. Errors in the environmental data used also have potential effects on the modelled distributions (Van Niel and Austin 2007; Van Niel et al. 2004).

However, with good quality data and algorithms, and close attention to quality control, some important advances in our understanding of spatial phylogenetics are likely to be made in the near-term using such approaches.

The Choice of Spatial Unit: Issues of Aggregation and Division

The fundamental geographical representation for most organisms is a point. However, it is currently not practical to operate at the level of the specimen (or ENM cell) across global and regional scales. Even if one did have sufficient computational power, a set of biodiversity indices at a 5 m resolution for an entire continent is of limited practical use, regardless of which type of input data are used. At such a resolution, ENMs will contain errors such that one cannot be certain a taxon occurs at a location, polygons are clearly inappropriate and observational records are simply inadequate. Even then, practical conservation efforts at the scales considered here will not be applied at such a level of detail.

Biodiversity measures at the regional and larger scales considered here are generally applied to areal units (as distinct from points and lines which can be used to represent plots and transects, respectively, although lines can also be useful for riverine systems). The sample data are either aggregated to these areas or disaggregated in the case of polygonal range data. The geographic units can be irregular, such as for ecoregions, or regular and equal-sized, possibly contiguous, tessellations using squares, rectangles, hexagons or triangles (e.g. Fig. 6.2). Conceptually, they can all be treated as polygons, as each unit contains some number of samples.



Fig. 6.2 The square, hexagon and brick data structures are all regular tessellations that divide a landscape into equal-sized tiles

Hexagons have many advantages over square grid cells since the centroids of all neighbours are equidistant (e.g. Rosauer et al. 2018). Triangles are problematic since, unlike squares and hexagons, the orientation of adjacent triangles is not invariant (although one could argue it is essentially a matter of human perception as it is no less arbitrary than any other geometric shape). A method that has some potential for spatial analyses is the brick (Jasiewicz et al. 2016). The brick structure uses rectangular tiles, but those in alternate rows (or columns) are spatially offset by half a brick, as in a brick wall. This approach has some of the advantages of the hexagonal tessellation, such as near-equidistant centroids of neighbours, but uses a much simpler data structure and is thus easier to develop analysis algorithms for. Despite their potential advantages, the non-square tessellations are not commonly used, largely because there are few implementations that support them.

An unavoidable issue when aggregating spatial data is the modifiable areal unit problem (MAUP- Openshaw 1983). The simplest case is where a taxon's range spans the boundary between two areal units, increasing its geographic range after aggregation. If the areal units are large, then the taxon's range will be substantially overestimated. This is exacerbated if the range spans the corner of several tiles (leading to four cells for a rectangular tessellation and three for a hexagonal or brick tessellation). One could change the origin of the aggregation scheme to avoid such an overlap, but then it is likely to shift the problem to some other taxon's range elsewhere in the study region. In reality, all areal analysis units are arbitrary to some degree. Regular tessellations are affected by the choice of unit size or origin, while irregular polygons such as ecoregions might be defined on the basis of unrelated taxa or phenomena.

A broader implication of the MAUP is in range extents. Well-defined boundaries such as the marine/terrestrial edge will be appropriate for many taxa, but for most data sets one will expect taxon ranges to extend beyond the study area. For example, Crisp et al. (2001) identified approximately 30% of taxa from Cape York that were also located in New Guinea, and this was not included in the sample used to define the geographic ranges. If one extends this to phylogenetic analyses that include ranges, then deeper branches in the tree will always be missing some of their range unless the entire Earth is considered in the analysis. Ultimately one must be pragmatic and acknowledge that an analysis applies to a particular area (González-Orozco et al. 2015; Scherson et al. 2017; Thornhill et al. 2017).

An additional issue for many analyses is the choice of coordinate system. A regular tessellation using geographic coordinates will appear equal area when mapped in planar form, but in reality the true areas of the cells will decrease towards the poles.

Ultimately the choice of analytical unit is a pragmatic one. Aggregation is almost always needed when analysing observational data due to sampling biases and uneven sampling. Many of the spatial issues with observational data identified above can be mitigated by aggregation to a relatively large area, with 25 km and larger tiles typically used for continental analyses. Each tile can then be considered to represent an assemblage, although one must be cognisant that a 25 km cell can

span many different environments, and thus its contents are an assemblage and not a community.

One question is whether to spatially aggregate before an analysis or afterwards. In some cases it simply does not matter, and aggregates can themselves be aggregated using simple spatial relations (e.g. a polygon can contain a set of smaller cells). An alternative is to let the data define the aggregation. Agglomerative cluster analyses (González-Orozco et al. 2013, 2014a, b) and network clustering (Bloomfield et al. 2018; Vilhena and Antonelli 2015) can be used to derive regions based on some diversity metric, and these can then be used to analyse the data, preferably avoiding circularity. Such methods do, however, require initial aggregations to grid cells to generate assemblages used in the grouping analyses so are not immune from the considerations outlined above.

Some Properties of PD and Related Metrics

This section contains a brief description of some properties of PD and two related measures, phylogenetic endemism (PE) and phylogenetic rarity (PR). Compound measures, such as CANAPE (Mishler et al. 2014), are not dealt with in detail here, but many of the considerations can be equally applied to them.

Range and Abundance-Weighted PD: PE and PR

Phylogenetic diversity (PD; Faith 1992) for an area i is the sum of the branch lengths found in that area. The set of branches used comprises the minimum spanning path connecting the tips of the tree in an assemblage to either the root node or the most recent common ancestor of the tips, with the former being more commonly used. Each branch is counted only once:

$$PD_i = \sum_{\lambda \in \Lambda_i} L_\lambda \quad (6.1)$$

where L_λ is the length of branch λ in the set of all branches (Λ_i) found in area i .

The PD of a region is the sum of the branch lengths, but this does not indicate how important that region is for the PD score as a whole. To explore this, metrics have been developed that take into account the relative sizes of the branch ranges (phylogenetic endemism, PE) or abundances (phylogenetic rarity, PR), where PR is an adaptation of the EDt index described by Cadotte and Davies (2010). Both are weighted sums of the branch lengths in a sample. The weighting for each branch in both the PE and EDt calculations was originally described as the inverse of the branch's range or abundance (Cadotte and Davies 2010; Rosauer et al. 2009), but it is more flexibly described as the fraction of the branch's range or abundance that is

found in the sample (Laffan et al. 2016) (and has always been implemented in Biodiverse in this way).

Phylogenetic endemism (Rosauer et al. 2009) gives an estimate of the spatial restriction of PD for an area. It is calculated as a weighted variant of PD, where the branches are weighted by the fraction of their range found in the sample area:

$$PE_i = \sum_{\lambda \in \Lambda_i} L_\lambda \frac{r_\lambda}{R_\lambda} \quad (6.2)$$

where R_λ is the full geographic range of branch λ in the set of branches Λ_i located in area i , and r_λ is the local range of branch λ (that part of its range that occurs within area i). Note that the range of a branch is calculated as the union of the geographic range of the tips of the tree it subtends, so that a location containing more than one tip subtending an internal (ancestral) branch still counts only once for that branch. This ensures there is no double counting of areas when clade ranges are determined.

An important property of PE is that the ranges do not need to be calculated using units of grid cells. They can just as easily use the areas of irregular polygons or indeed any measure where the range of an internal branch is represented as the union of its descendants' ranges. The fact that polygonal approaches are not commonly used can be largely attributed to a lack of available implementations.

PR is an adaptation of the site level evolutionary distinctiveness (EDt) index described in Cadotte and Davies (2010). Its formulation closely parallels that of PE, with the difference being that branches are weighted by their fractional abundances instead of the fractional ranges. The abundance of an internal branch is calculated as the sum of its subtending terminal branch abundances, as an ancestral branch will be found in each of the organisms in which the terminal branch is found. The end result is an index of rarity, as distinct from endemism (which is geographic rarity):

$$PR_i = \sum_{\lambda \in \Lambda_i} L_\lambda \frac{a_\lambda}{A_\lambda} \quad (6.3)$$

where A_λ is the overall abundance of branch λ , and a_λ is the local abundance of branch λ in area i .

The weighting scheme used for PE and PR supports several properties. First, the individual scores for a collection of non-overlapping areas are additive, for example, the sum of PE scores for a set of areas will equal the PE score when calculated for that set of areas as a whole. It then follows that, for PE, if all terminal branches of the tree are in the geographic sample, and the geographic ranges are calculated only within the study region, then the PE scores will sum to the regional PD score. The same applies to the PR scores (as noted by Cadotte and Davies 2010 for the ED-related metrics, but for abundances instead of ranges). In such cases, the weights for all branches will equal 1, and thus the formulae simplify to that for PD.

The Contribution of Individual Branches, Clades and Lineages

The ED index was developed from the perspective of the terminal branches, primarily species. The calculation of that index, and those derived from it, is analogous to slicing a tree along its grain to determine what fraction of the tree is associated with each terminal. However, it is often better to operate from the perspective of branches, taking into account the evolutionary relationships represented by the phylogenetic tree.

The PD, PE and PR indices all take a collection of branches as a sample and return a result that can then be mapped, for example across cells, populations or even individuals. However, it is also possible to delve further into these results by mapping the results for subsets of branches.

One can map each individual branch, but branches should ideally not be considered in isolation. One of the main advantages of PD and related measures is that they use the topology of the tree of life. Any sample that contains a branch will, by definition, contain its descendant and ancestral branches. To understand the PE or PR of the clade or lineage as a whole, one need only use the subset of branches in a clade. A more refined approach also takes into account the contribution (and potential loss) of any ancestral branches of that clade that have no other descendants in the sample being considered. Such ancestral branches would be lost from an area (or sample) if a clade were to become locally extinct. One can then calculate the contribution of that clade to the phylodiversity of the whole sample to indicate the potential loss from a samples' phylodiversity if a clade is lost (Fig. 6.3).

Standardising and Transforming Results

A key issue with studies of PD, PE and PR across multiple trees is the comparison of the results. This is especially the case when the trees are not in comparable units, i.e. one tree is in units of base pairs, while another is in units of time.

The simplest means of standardising the results is to divide each score by the sum of all branches on the tree that was used (the total PD of the tree, as used by González-Orozco et al. (2015)). When done, the results are expressed as the fraction of the tree that occurs in the sample. This applies equally to the index results for full samples and for the clade-level subsets.

Standardising by the total PD is not free of issues, though, as it assumes that the root nodes are at relatively comparable positions, for example, those for time-calibrated trees are of similar ages. This is important when using indices that traverse the tree to the root branch. There are also issues related to inter-clade variability, for example, of divergence rates (Graham et al. 2018).

When working locally, it is also possible to transform PE and PR scores for a sample by dividing them by the PD of the same sample (Fig. 6.4). The transformed PE score is then the phylogenetic analogue of the corrected weighted endemism

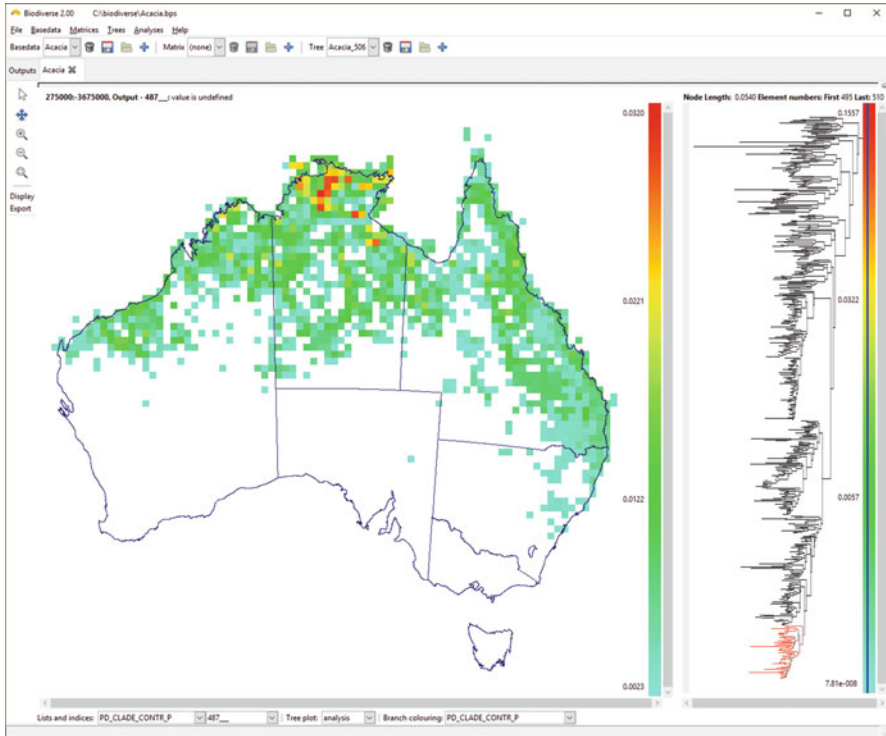


Fig. 6.3 The map shows the relative contribution of the clade highlighted in red on the tree at the right. Values are scaled as the fraction of the total PD of the tree, with red indicating a higher fraction. Data are the *Acacia* described in Mishler et al. (2014). Cells not containing the selected clade are not shown

index (Crisp et al. 2001; Laffan and Crisp 2003; Laffan et al. 2013) and can be interpreted as the average fraction of the ranges represented by the sample's PD score. That for PR is the fraction of the abundances. Values closer to 1 represent greater endemism or rarity of the sample. The transformed PE index has been used by Millar et al. (2017) in an analysis of the New Zealand vascular flora. The same transformation can also be applied when working with subsets of branches but dividing by either the PD of the full sample or the subsample. Each transformation aids the interpretation of the contribution of that branch to the tree or to the respective score at that location.

Plotting Results on the Tree

The phylogenetic metrics above have all been described for a geographic distribution. However, if one is to work with individual branches or clades, possibly with an

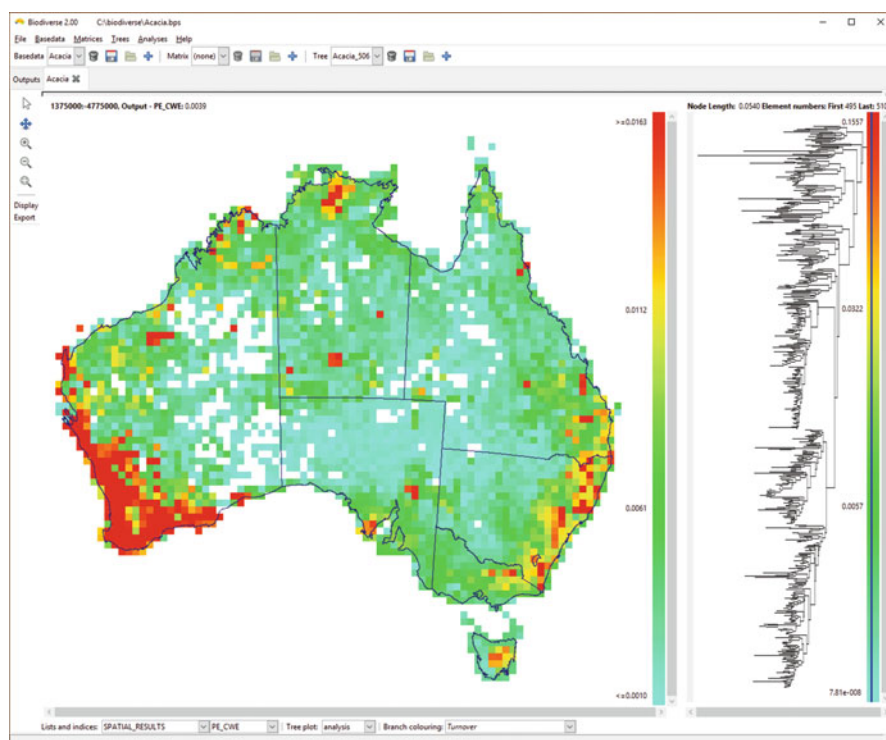


Fig. 6.4 Each cell is coloured by the PE score divided by the PD score for the sample in the cell. Areas in red have relatively high fractions, although the absolute scores are $<2\%$. Larger window sizes will result in higher fractions (Laffan and Crisp 2003)

ancestral component, then the interpretation of the results is almost impossible due to the sheer volume of generated maps. A bifurcating tree with 500 terminal branches will have 999 branches in total, and thus 999 branch level maps, 999 clade maps and 999 ancestral component maps, and these numbers are multiplied by the number of indices calculated (e.g. perhaps PD and PE are calculated). Understanding the change in branch contributions to the diversity metrics will be extremely difficult, even if an animation is developed.

An alternative is to display the branch properties for a sample on the tree itself, e.g. by colouring branches by their phylodiversity score (or that of their clade; Fig. 6.5). This can generate as many tree plots as there are sample locations, but this can be alleviated through interactive visualization systems, where users can select which samples to explore from a map of any relevant index.

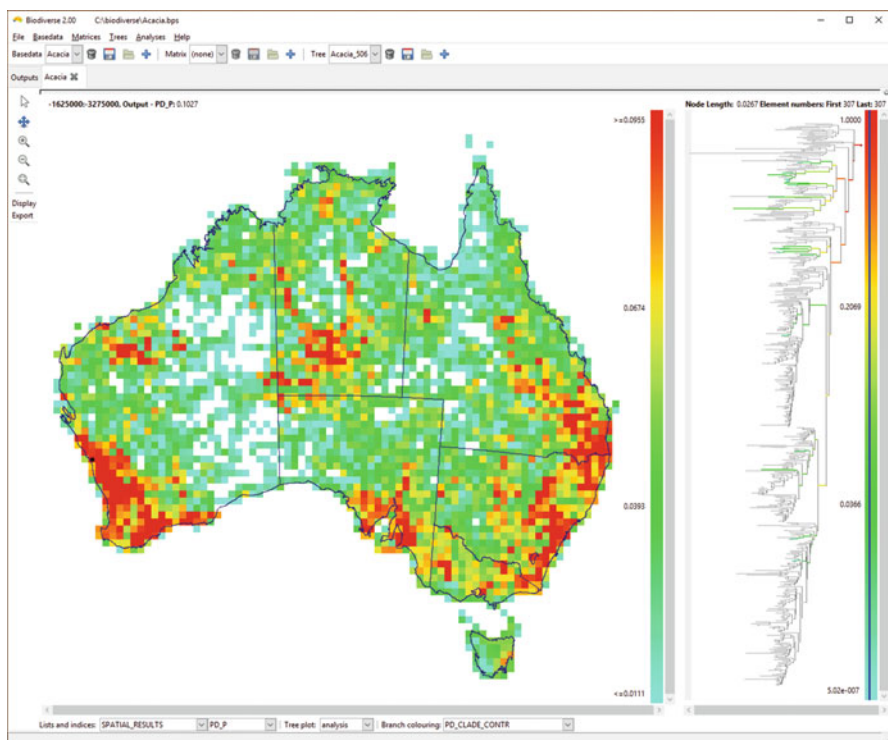


Fig. 6.5 The map shows the PD score for each cell, rescaled to be the fraction of the tree. Branches in the tree plot are coloured by their contribution to the PD score (log scaled) in the cell in south-west Western Australia highlighted with the black dot. Grey coloured branches are not found in the selected cell. The implementation in Biodiverse is interactive, and the tree colours are updated as cells are hovered over with the mouse

Improving Processing Speeds

The speed of a calculation is an important consideration when scaling analyses to continental or global extents, especially when using SDM outputs for large numbers of taxa. This is due to the sheer volume of calculations required, as one will realistically have tens to hundreds of thousands of locations for which to calculate phylogenetic indices, potentially for thousands to tens of thousands of branches. This issue is further exacerbated when randomisation analyses are used (e.g. there might also be calculations for 999 randomisation iterations). Even with faster computers and massive parallelisation, user expectations of analysis complexity and data set sizes will expand to consume available resources.

Optimisation of code has been a major focus of development for the Biodiverse project over recent years, and this section draws on the experiences gained while doing so.

Processes that are relatively fast on test data sets used in development will often show quadratic or exponential run-times when applied to data sets of increasing size. For example, an analysis that compares each item with each other item (e.g. pairwise comparisons of branches) requires $n \times n$ comparisons in a brute-force search. Other approaches might visit each branch and calculate some function based on its descendants, for example, the mean length to the terminals, or simply return a list of the descendant branches.

Many readers will immediately identify issues with the above examples and perhaps have ideas for how to accelerate them. However, a commonly used adage in computer science is that “premature optimisation is the root of all evil (or at least most of it) in programming” (Knuth 1974). It is far more effective to use a code profiler in tandem with benchmarking. It is a sobering experience to realise that several days coding and debugging have led to no meaningful improvement in analysis times because of incorrect assumptions about the causes of speed bottlenecks. One also needs to have tests of the code in place to ensure that the same result is obtained for a given input before and after a code change and that other analyses are not impacted as side-effects. One must also use realistically sized data sets to reveal the relevant bottlenecks.

Once speed bottlenecks are identified and tests are in place, there are several general approaches that can be taken. One is to avoid repeatedly calling expensive functions that always return the same result for the same inputs, instead calling them once and caching their results. This is a commonly used memory-for-speed trade-off.

The level at which caching is applied is the next consideration. If one caches within an analysis, then that analysis run will be faster, but not subsequent analyses using the same data. For example, a spatial analysis using a phylogenetic tree might cache some of the internal results on the analysis object, but then this cache would be discarded if the analysis object is not persistent, or if it is persistent, then its cache might be inaccessible by subsequent analyses. For phylogenetic analyses, caching on a tree object itself will apply across subsequent analyses using that tree, as the cache is reused each time. Care must be taken, though, to ensure caches are cleared when the tree object is changed in some way, for example, through pruning (which itself might involve quadratic speed issues).

Other techniques can also be applied when working with tree data structures. For example, the last common ancestor (LCA) of a pair of terminal branches is needed for local PD analyses, as branches ancestral to the LCA should not be considered. It is also used for indices such as the mean pairwise distance in PhyloCom (Webb et al. 2008) as the pair-to-pair path must pass through the LCA.

The search for the LCA can be improved using several steps. First, the path of branches from each terminal to the root is obtained (caching them so they do not need to be searched for again). A brute-force search would apply a pairwise comparison of the branches from the end of each list (the root branch). The last common ancestor is the last pair of list items that are the same branch. However, this can involve many redundant comparisons when the last common ancestor is near the terminals, something that is exacerbated when the tree has a deep topology and when many thousands of such comparisons need to be made. One can reduce the search

space by only considering branches with comparable depths (the number of ancestral branches until the root branch). For example, if branch1 has a depth of 100 and branch2 has a depth of 70, then the first 30 items in the path for branch1 to the root can be ignored since it is not possible for them to be ancestral to branch2 (this approach is implemented in PhyloCom; Webb et al. 2008). A binary search can then be applied to the subsets. The search starts at the mid-point of the two lists (depth 35 in the example). If the branches at that point in the two lists match, then the last common ancestor must be towards the tip of the tree; otherwise it must be towards the root. The next iteration moves the comparison point halfway towards the tip or root as appropriate (i.e. depth 52 or 17) and repeats the comparison. The process is continued until there are no more branches to check. If the last check is false, then the LCA is the parent branch of the final comparison pair. The advantage of binary searches is that they complete in $O(\log n)$ time for the worst case, as opposed to linear searches that would take $O(n)$ time. For small data sets, such a difference will not be noticeable, but it can make a substantial difference for real-world data sets such as from the Tree of Life project (compare 70 vs $\log(70) = 4.2$). In some cases profiling might show that the binary search itself takes meaningful amounts of time, in which case caching could be considered if the same searches are repeatedly applied.

Conclusions

This chapter has presented a brief overview of some of the considerations, opportunities and inner workings of phylogenetic diversity analyses when applied at regional to global scales. While the available indices are common across scales, the nature of the data and how they are handled influence how analysis results are treated and interpreted.

The broader field is enormous, so several aspects of such analyses have not been covered. Examples include compound measures such as CANAPE (Mishler et al. 2014), spatial and temporal turnover (Laffan et al. 2016) and complementarity (Laity et al. 2015) and factors such as the choice of analysis window (moving beyond single grid cells; Di Virgilio et al. 2017; Laity et al. 2015). These can be explored using existing literature and the references therein.

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Chapter 7

Phylogeny-Based Measures of Biodiversity When Data Is Scarce: Examples with the Vascular Flora of Chile and California



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Abstract The urgency for conserving biodiversity has elicited much research on how to determine its “value” for conservation. The use of phylogenetic diversity (PD) has provided a quantitative framework to assess biodiversity at different taxonomic levels and spatial scales. PD assessments at any spatial resolution need a reasonably well-resolved phylogeny and distribution data for the target taxa. Because of the explosive growth of phylogenetic work, it is becoming easier to get phylogenetic information for a given group. However, distribution data is much harder to obtain. Using examples from world flora, this study explores the correlation between PD-related studies and availability of distribution data. We found that most PD studies in plants have been done in a handful of countries, which correlate with the amount of available distribution information. In order to address the question of whether PD studies should be done in places where information is scarce, we took two recent studies in which PD was calculated – the flora of Chile and California – as examples of poor and good sampling efforts of herbaria data, respectively. We randomly pruned the California database to see if and how spatial patterns of PD change with data depletion. We show that if redundancy (a measure

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of sampling) is kept at reasonable levels, meaning using larger grid sizes, PD patterns could still be inferred even with 25% of the original data. We argue that these studies are worth doing even with poor data sets, since even coarse PD patterns can point at places where future studies, and conservation efforts, should be focused.

Introduction

Human population is rapidly increasing, causing several environmental impacts such as increase in invasive species, changes in land use, pollution, and climate change (Millennium Ecosystem Assessment 2005). To ensure the maintenance of biodiversity, and its value, we need to maintain a range of taxa that would maximize feature diversity in a changing environment (Davies et al. 2008). Recently, a set of nine human-induced phenomena were assessed to define a “safe operating space” for humanity, providing thresholds beyond which human welfare and the resilience of ecosystems would be affected (Steffen et al. 2015). Three of these parameters or planetary boundaries have been exceeded beyond the safe operating space, being biodiversity loss the most seriously affected (Rockström et al. 2009a).

Even though the importance of biodiversity and of its urgent need for conservation is clear, there are still many questions regarding the portion of it that requires the most urgent actions. This is especially true when resources for conservation are limited (Wilson et al. 2006), for example, in developing countries. This has been known for a long time as “the agony of choice,” as proposed by Vane-Wright et al. (1991) or the “resource allocation problem” (Wilson et al. 2006). Biodiversity hotspots and other prioritization initiatives have somehow responded to this need, defining priority areas by their number of species, endemism, and human intervention (Myers et al. 2000) or by characteristics such as their capacity to provide ecosystem services and resilience to global change phenomena among others (Brooks et al. 2015).

One of the main challenges facing biodiversity conservation today is the uncertainty as to the conditions, especially climatic, that taxa would have to face in the future. Global change phenomena have intensified with respect to historic registries, and several potential scenarios have been put forward to describe the new environments that taxa and ecosystems will have to persist (Midgley et al. 2002). This poses a new challenge to conservation strategies, in which evaluation processes must consider attributes, which are directly related to the diversity that those taxa and ecosystems have accumulated over evolutionary time (Vane-Wright et al. 1991; Faith 1992a; Sechrest et al. 2002; Fisher and Owens 2004; Faith and Baker 2006; Forest et al. 2007; Pio et al. 2011).

In this context, considering that the patterns of distribution of taxa, their processes, and interactions cannot be decoupled from their evolutionary history (Faith 1992a; Forest et al. 2007; Pio et al. 2011), indices that consider the evolutionary processes are a powerful homogenizing criterion, since it is accepted that ancestry relationships make species that are more evolutionary related, share certain

ecological and life history traits. This suggests that measures based on evolutionary history can be used as surrogates for feature diversity.

Phylogenetic diversity (PD) is a phylogeny-based index that considers the accumulated evolution of a group of taxa and as such provides an idea of their evolutionary potential (Faith 1992b). PD is a measure that has proven robust to taxonomic change and, importantly, allows direct comparisons among different areas. Recently, PD has been suggested as the most adequate measure for the biodiversity loss planetary boundary (Faith et al. 2010; Mace et al. 2014). In an extinction scenario, sequential loss of taxa will eventually lead to the loss of long phylogenetic branches of the “tree of life” representing the biota of a certain area, allowing the identification of tipping points (Faith 2015a). A system that considers evolutionary limits and tipping points (both measured using PD) can be referred as to an “evosystem,” that not only includes ecosystem services but also future option values (Faith et al. 2010; Mace et al. 2014; Faith 1992b).

When considering PD, one can identify areas that harbor a larger than expected evolutionary potential. As proposed by Brooks et al. (2015), this concept could be included in the definition of key biodiversity areas (KBAs), defined by the IUCN as “sites that contribute significantly to the global persistence of biodiversity.” KBAs could easily consider biodiversity in a comprehensive manner, including PD or related measures of evolutionary potential. This applies to any portion of the tree of life, and as proposed by Sgrò et al. (2011), they would allow identifying evolutionary refugia that could maintain biodiversity features in the face of global change.

Studies of this type have helped to further develop the concept of biodiversity “hotspot” (Myers et al. 2000) to include criteria other than species richness (Marchese 2015). Currently, hotspots can also be considered areas of concentration of evolutionary history or concentration of geographically restricted evolutionary history (phylogenetic endemism) (Rosauer et al. 2009; Mishler et al. 2014; Thornhill et al. 2016). It is then appropriate to ask whether these new concepts will change the priorities for allocating resources, which is an important question especially for areas in which funds for conservation are scarce and politically difficult to obtain.

The Data Problem

One of the main problems for any study of biodiversity is that we do not yet know where all species are distributed, a problem that has been known as the “Wallacean shortfall” (Lomolino 2004; Cardoso et al. 2011). This, together with the paucity of georeferenced data, is especially prevalent in developing countries (Dunn et al. 1997; Mennecke and West 2001; Hortal et al. 2007). In many countries, for example, herbaria lack geographic data for their collections, and most do not have online databases freely available. Sgrò et al. (2011), in their proposal for the identification of evolutionary important areas in the face of global change, recognize that there are many areas in which scarcity of data is an important issue. When data is poor,

occurrence data it is often also spatially non-homogeneous. This is particularly problematic for large-scale studies (Hortal et al. 2007). Most of data heterogeneity stems from collection bias, a problem that prevails even for densely sampled areas (Phillips et al. 2009; Daru et al. 2018). In this sense, the use of species distribution models (or environmental niche models for rank-free analyses; see Chap. 6) could be a way of overcoming poor collection effort. Species distribution models have been used for assessing PD-related measures, with estimated occurrences (Pio et al. 2011, 2014; Thuiller et al. 2011; Buerki et al. 2015). However, there could be species whose distributions are controlled by factors other than the macro-environment, for example, soil characteristics or the presence of symbiotic taxa, which could limit the use of these models when working with large numbers of taxa (Patsiou et al. 2014). For additional discussion, see Chap. 6 in this volume.

There are two other problems related to the amount of data available to the PD calculation or related measures of evolutionary potential. One has to do with taxonomic issues and taxon delimitation, and the other is the lack of available DNA sequences. Ideally, lineage sampling should be performed at the finest scale possible. However, the lack of DNA data and locality information for many species, combined with poor understanding of fine-scale taxonomy in many taxa, makes it easier and likely more taxonomically accurate to work at higher taxonomic levels such as genera. Working at these higher levels has proven very useful in outlining general spatial patterns of biodiversity (Forest et al. 2007; Thuiller et al. 2011; Thornhill et al. 2016). However, finer-scale operational units should be the general goal in order to increase accuracy in PD estimations. This problem is directly related to the amount of phylogenetic work that has been carried out in the different groups. This in turn is related to where the taxa are located, how much material is available for DNA extraction in museums and herbaria, and the accessibility to necessary resources to carry out molecular studies. While DNA sequence data is becoming more available with time, the rate at which this happens, however, depends on the geographical location of the region under study.

Phylogeny-Based Studies and Data Availability: A Case Study with World Flora

Allocating resources for conservation is challenging for all countries. However, the availability of Internet-based data has made it easier for scientists and the informed community to study patterns of biodiversity distribution at several levels, something that has been recently called “citizen science” (Wood et al. 2011; Sullivan et al. 2014). For phylogeny-based measures, the information freely available has been crucial, both for obtaining DNA sequences and/or phylogenetic trees and for information on localities for target taxa. Many valuable studies have been performed using only information obtained online (Mishler et al. 2014; Rosauer and Jetz 2015; Zhang et al. 2015; Guerin et al. 2016).

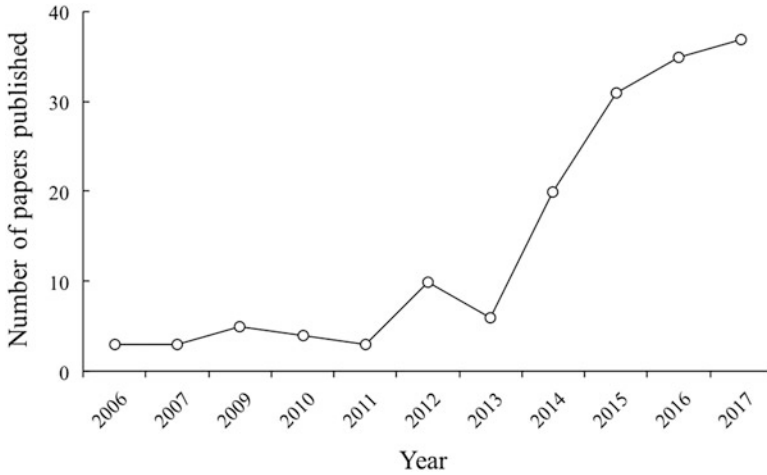


Fig. 7.1 Number of scientific publications reporting studies of spatial patterns of PD in plants, listed in the ISI Web of Science database in the last 12 years

The issue regarding the availability of free information for these types of studies is not trivial. To demonstrate how the availability of information influences the number of studies in different countries, we conducted an analysis of published studies in the last 12 years (2006–2017) that used phylogeny-based measures of biodiversity in plants. We searched the Thomson Reuters Web of Science database using the keywords “phylogenetic diversity,” from 2006 to 2017 (both years included), and selected all studies that involved both plant PD measures and databases of georeferenced locality data. We found three patterns of interest. The first one worth mentioning is the drastic increase in publications on the topic from 2013 to 2016, shown in Fig. 7.1.

The second pattern has to do with the geographical location of the floras studied. Most papers published are studies done in only four countries, China, the United States, Australia, and Brazil, which together represent 58% of all the papers published using PD measures of plants during this period of time (Fig. 7.2).

For the correlation between availability of information and amount of studies performed, we analyzed the entire Global Biodiversity Information Facility (GBIF) database (<https://www.gbif.org>), filtered by Plantae and sorted by country. We then correlated the amount of studies on PD for plants shown in Figs. 7.1 and 7.2, with the information available for each of those countries in GBIF. As predicted, there is a high correlation between the information available and the studies done per country (Fig. 7.3). Interesting exceptions to this trend are China (CN), the country with most records of publications in the subject, clearly an outlier with respect to the amount of GBIF records. Brazil (BR) also has more publications than predicted by the amount of GBIF records. Countries such as Mexico (MX) and Colombia show the opposite trend, with fewer publications than predicted by their number of plant records published on GBIF.

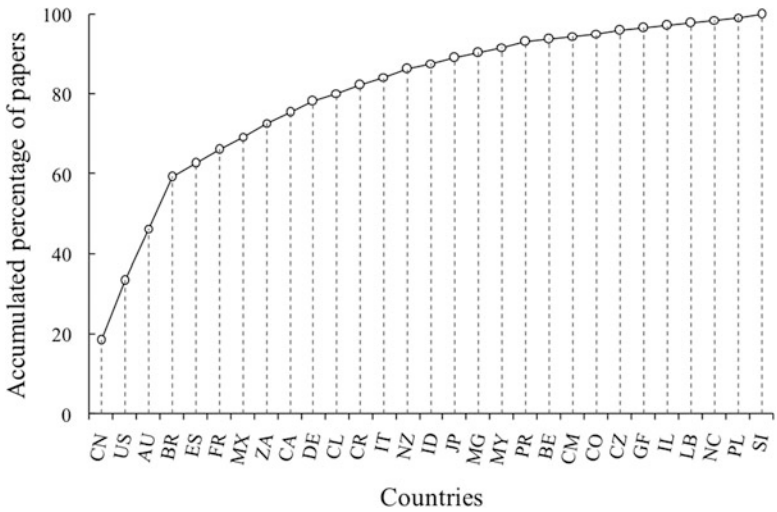


Fig. 7.2 Total scientific publications reporting studies of spatial patterns of PD in plants, listed in the ISI Web of Science database in the last 12 years, sorted by countries which were the topic of the study. Country codes follow the ones used in the GBIF database, as per ISO-3166-1. CN, China; US, United States of America; AU, Australia; BR, Brazil; MX, Mexico; ES, Spain; ZA, South Africa; FR, France; CA, Canada; DE, Germany; CR, Costa Rica; NZ, New Zealand; CL, Chile; IT, Italy; JP, Japan; MG, Madagascar; ID, Indonesia; MY, Malaysia; PR, Puerto Rico; CO, Colombia; NC, New Caledonia; CM, Cameroon; GF, French Guiana; PL, Poland; IL, Israel; CZ, Czechia; BE, Belgium; SI, Slovenia; LB, Lebanon

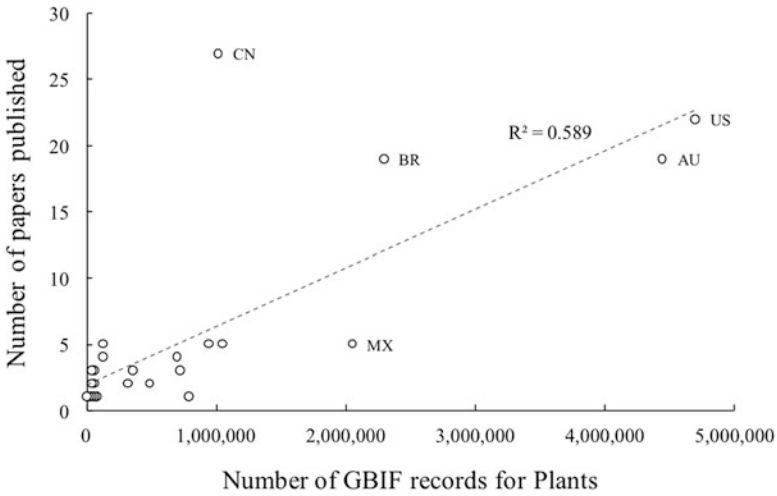


Fig. 7.3 Correlation between number of data records for Plantae published in the GBIF database per country and the number of scientific publications reporting studies of spatial patterns of PD in plants, listed in the ISI Web of Science in the last 12 years. Residuals are shown with circles. Countries that show distinctive deviations from the trend are marked: China (CN), Brazil (BR), Mexico (MX), Australia (AU), United States (US)

Phylogeny-Based Studies When Data Is Scarce: Chile Versus California

Chile is a very interesting territory in terms of its biodiversity distribution patterns. It is considered a “biogeographic island,” given its geographic isolation caused by geologic events such as the breakup of Gondwana, the separation of Antarctica generating the Humboldt Current, the onset of Arid Diagonal, and the rise of the Andes mountains. All these events created a space surrounded by natural barriers (Villagrán and Armesto 2005; Moreira-Muñoz 2011). Regarding its flora, Chile is the country of South America with the highest number of strict endemic plant genera, plus four endemic families – Gomortegaceae, Francoaceae, Thyrsopteridaceae, and Lactoridaceae (Moreira-Muñoz 2011). The most recent revision counted 86 endemic vascular plant genera, 10% of the total number of genera, with 69 genera located on the continent and 17 on the islands (Urbina-Casanova et al. 2015). As a comparison, Argentina has only one family and four endemic genera (Zuloaga et al. 1999); Perú, harboring a flora of more than 17,000 species, has 51 endemic genera and no endemic family (Brako and Zarucchi 1993).

The highest floral richness is concentrated in the central area of the country (Bannister et al. 2011), within the Chilean Winter Rainfall-Valdivian Forests biodiversity hotspot (Myers et al. 2000; Arroyo et al. 2004, 2008). This is worrying, given that 80% of the population in the country is concentrated in this area, as well as the main industries and agricultural land (INE 2002; CAPP 2008). Despite the evident threats, only a small proportion of this area is protected by the National System of Protected Areas (CONAMA 2008; Plischoff and Fuentes-Castillo 2011; Luebert and Plischoff 2017).

Despite the fact that studies that have applied phylogeny-based methods to the study of biodiversity patterns are very few in Chile, the areas with higher than expected evolutionary history are found consistently toward the south of the country (Scherson et al. 2014, 2017). In addition, important areas of phylogenetic neo-endemism have been found in the north of the country, while the south harbors areas with significant phylogenetic paleo-endemism (Scherson et al. 2017). These studies can have an implication in conservation strategies, especially when looking at future environmental change.

As explained in the first part of this chapter, a major problem for carrying out these types of studies is the lack of data sets that are dense enough to allow a detailed analysis of biogeographic patterns. In our latest study (Scherson et al. 2017), the largest database that we were able to assemble consisted of ca. 150,000 records, and the smallest grid cells that we were able to use were 75 km grid squares, to keep a moderate redundancy of the data. As a comparison, a recent study of the phylogenetic patterns of the native vascular flora of California, a system that is very comparable with Chile given that they both share an extensive Mediterranean area, was analyzed using almost 4 million records of individual species (Thornhill et al. 2017). Obviously, the level of detail in the analyses that a study with a denser database is capable of obtaining is much higher. The California analysis was done using 15 Km equal grid squares.

This leads to the central questions of this chapter; how do we perform phylogeny-based analyses on areas that are data deficient? Should we wait the number of years that it would take to produce the necessary data for these kinds of analyses to be performed? Can we trust broad biodiversity patterns even in the absence of dense data sets?

The above-mentioned study of spatial patterns of PD and related indices for the California floristic province used one of the more complete data sets published to date for this type of study (Thornhill et al. 2017). Using random depletions of their published data, we conducted an analysis aimed at finding out the influence of data density on the spatial patterns of PD.

The complete data set of the study by Thornhill et al. (2017) was downloaded from the DASH repository at the University of California, Berkeley (<https://dash.berkeley.edu/stash>). From the complete data set of more than 4 million records, they built a phylogenetic tree for 1,395,079 operational taxonomic units (OTUs), defined by the authors as clades at the lowest possible resolution available in previous studies.

The complete data set was randomly depleted, meaning that OTUs were removed at random, to generate four sets containing: (a) 75% of the records, (b) 50% of the records, (c) 25% of the records, and (d) 10% of the records. For each of these new data sets, PD was calculated using the software Biodiverse 2.0 (Laffan et al. 2010). Two types of data sets were generated:

- (i) Adjusted grid size: The grid size was adjusted in each set to keep the average redundancy of the original set. Redundancy is calculated as $1 - [\text{richness}/(\text{number of records})]$. A value close to one represents a good sampling of taxa, while a value of zero means that there is only one sample per taxon, therefore, poor sampling. The original data set used a grid size of 15 km equal squares, with a redundancy of 0.7. For each of the sequentially depleted data sets, we increased the size of the grid in 5 km increments, until the average redundancy was as equal to the original as possible.
- (ii) Equal grid size: For the second set, we kept the grid size constant, at 15 km, regardless of the redundancy, for each data set.

Results and Discussion

- (i) Adjusted grid size: Fig. 7.4a–e presents the spatial patterns of PD for the flora of California using a grid size that keeps the redundancy value constant among sequentially depleted data sets. In order to keep a redundancy of 0.7, the grid squares needed to be enlarged in the following way:
 - (a) 75% of the data, using a grid of 20 km equal squares.
 - (b) 50% of the data, using a grid of 30 km equal squares.
 - (c) 25% of the data, using a grid of 50 km equal squares.
 - (d) 10% of the data, using a grid of 100 km equal squares.

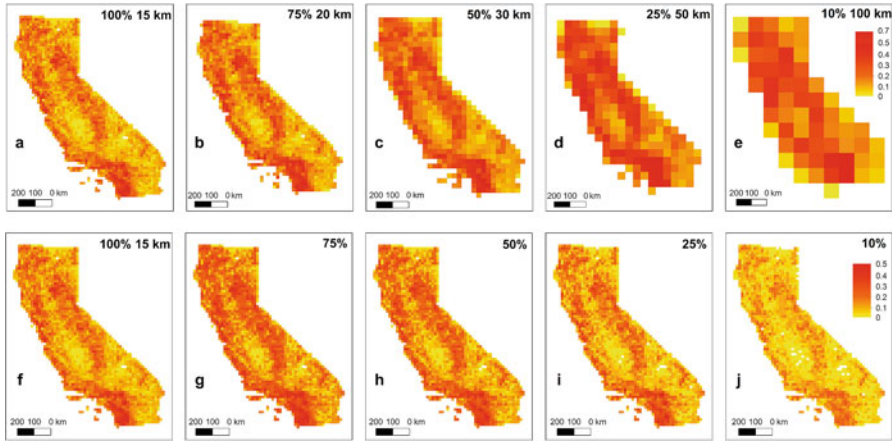


Fig. 7.4 Spatial patterns of PD for the vascular flora of the California floristic province, according to the data provided by Thornhill et al. 2017. A sequential random depletion of the original data set in 25% increments was performed until only 10% of the data was left. Maps (a–e) show increments of the grid cell sizes such that the redundancy of the data was kept equal. Maps (f–j) show equal size grids, regardless of the redundancy values

For PD, the general distribution patterns are kept fairly constant even with 25% of the data and a 50 km grid. Basically, as stated by Thornhill et al. (2017), the mountainous areas of the California floristic province show a concentration of PD, compared to the valley areas. The data set with 10% of the original data and 100 km squares is clearly coarser, and patterns are more difficult to infer. The information provided by a grid this size will of course depend on the size of the area itself. Maybe for California the resolution might be too broad; however, it might still be useful to point at patterns that could be studied in the future when more data becomes available. This is especially true for larger areas in the world.

- (ii) **Equal grid size:** In this analysis, each grid was kept at a constant size, regardless of the redundancy. This means that the amount of information per grid varies depending on sampling effort. As shown in Fig. 7.4f–j, even with 50% of the information, the patterns keep fairly constant. However, when the amount of data drops to 25%, the spatial patterns lose their accuracy, showing a stronger effect than seen in Fig. 7.4d.

In the equal grid size analysis, the lack of data can be seen as an increasing number of blank cells and also a reduction in the estimation of PD in some areas.

In terms of data accuracy, this analysis shows that redundancy should be a more important criterion than grid size, at least for total PD calculation, and that even when data sets are not as complete, important spatial patterns can still be observed.

Going back to the case of Chile, we showed recently that there are areas in the north and south of the country that are centers of paleo- and neo-endemism (Scherson et al. 2017). Even though our grid size was large, we can still foresee areas that will require further attention. In a very simple spatial analysis, we overlapped these paleo- and neo-endemism areas with a map of the protected areas in the country. As shown in Fig. 7.5, it is clear that in the south of the country, many paleo-endemic areas fall within the state-protected areas. However, for the neo-endemic areas in the north there is very little protection. This points clearly to a need for further studies of the protected areas in the north of Chile and their relationship with the neo-endemic flora.

In the face of global change, it is worth asking whether this protection will suffice for a future distribution of taxa. If we consider the centers of neo- and paleo-endemism as important areas of biodiversity concentration, then we should focus our efforts in increasing the current protection of these areas and studying how they will react to global change phenomena. For that, models of species distribution could give us an estimation of if and how our protected area system can harbor this important biodiversity in the future.

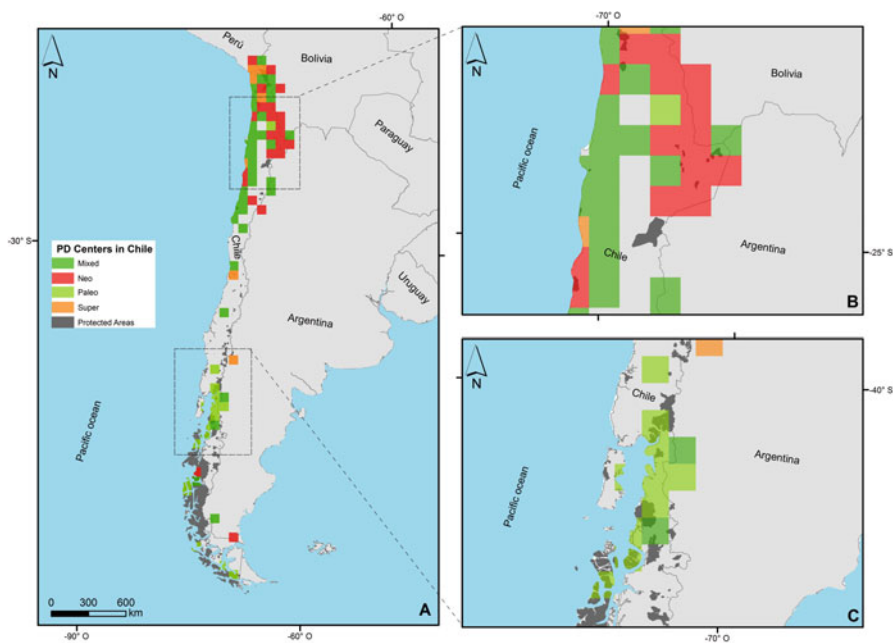


Fig. 7.5 The left panel shows a map of Chile highlighting significant areas of neo (red), paleo (light green), mixed (green), and super (orange) centers of endemism. (From Scherson et al. 2017). Two areas are enlarged in the right panels, in which the state-protected areas are shown in grey

Conclusions and Future Directions

In this chapter, we showed that there are important differences in the amount of data availability for different countries and that, at least for plants, there is a direct correlation between freely available data and the amount of spatial phylogenetic studies published, with some exceptions. Even though the amount of studies incorporating spatial patterns of PD in flora has increased exponentially in the last 10 years, only a handful of countries account for most of the studies carried out. This bias leaves many interesting places on earth understudied, many of which probably harbor very important concentrations of biodiversity and the potential to be refuges in the face of global change.

However, we also have shown that even with poor data sets, it is possible to still distinguish interesting spatial patterns of PD. Even though these patterns might not be conclusive now, they can still be used to point out future directions in biodiversity studies. The urgent need for conservation demands immediate actions. Our data suggests that even incomplete data sets might give us important insights on where these efforts could be focused.

Information deficiency has been considered the main issue in conservation. Mace et al. (2003), for example, considered the lack of information as the main obstacle to proper biodiversity conservation. However, as stated by Forest et al. (2015), “a less complete overview of the situation with some potential biases is probably better than waiting to get the full picture and obtaining it too late to take effective action.” We believe that, if managed properly, even scarce databases have the potential to contribute to biodiversity assessment and should be used at least for a first approach especially in highly threatened areas.

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Chapter 8

Priority Areas for Phylogenetic Diversity: Maximising Gains in the Mediterranean Basin



Simon Veron, Daniel P. Faith, Roseli Pellens, and Sandrine Pavoine

Abstract Phylogenetic diversity (PD) is a valuable component of biodiversity, reflecting variation produced by past evolutionary processes and providing options for future generations. A challenge is to design and implement effective conservation actions for the preservation of PD for multiple taxonomic groups. We here proposed a region-level approach to this aim because some previous research at this scale may have been misleading. We focused on a biodiversity hotspot, the Mediterranean Basin, as a case study. We assessed how much PD does not yet receive any protection for three vertebrate groups: amphibians, squamates and terrestrial mammals. We found that a very low proportion of the PD of the region was protected, justifying the urgent need to set up conservation actions to safeguard PD. We identified sites whose protection would maximise the persistence of the PD of the Mediterranean region. Our methodology focused on two criteria for selecting sites that were prioritised for protection: the gain in PD representation and the gain in securing threatened PD. Our results show that potential protected areas were widely distributed across the basin and were generally different across the three taxonomic groups. However, several common priority sites were concentrated in islands the Maghreb and Spain and in the Eastern part of the basin. Our study attempts to further develop an effective framework for priority settings based on PD. We also discuss a risk approach where conservation decisions depart from the idealised case of a minimum set of areas that would protect the whole PD of the region.

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Introduction

Human activities cause great threats to biodiversity and could be at the origin of a sixth extinction crisis (Millennium Assessment 2005; Barnosky et al. 2011; Ceballos et al. 2015). Facing the limited resources for conservation and the intensity and diversity of threats, it is unlikely that all biodiversity can be protected (Vane-Wright et al. 1991). Preserving evolutionary history, quantified by phylogenetic diversity (PD; Faith 1992), has been proposed as one important focus of biodiversity conservation (Faith 1992; Barker 2002; Mace et al. 2003; MacLaurin and Sterelny 2008; Rolland et al. 2012; Lean and MacLaurin 2016). The PD of a subset of taxa is measured from a phylogeny that depicts evolutionary relationships between species. Phylogenetic diversity is the sum of the branch lengths on the minimum path that joins the taxa on a phylogenetic tree (Faith 1992). Preserving PD may have several advantages compared to traditional measures used for conservation such as species richness (Margules and Pressey 2000; Arponen and Zupan 2016). In addition to capturing more of the diversity of life that came about through evolution (Pellens and Grandcolas 2016), it may represent unexpected future benefits for human societies, i.e. option values (Faith 1992; Forest et al. 2007; Cadotte and Davies 2010; Faith et al. 2010; Larsen et al. 2012; Jetz et al. 2014; Laity et al. 2015; Mouillot et al. 2016; Veron et al. 2016a; Pollock et al. 2017). The basis of PD as a measure of biodiversity option values is the assumption that shared ancestry accounts for shared features among species; thus, higher PD will be expected to correspond to higher feature diversity (for discussion, see Chap. 1 in this book). In contrast, measures of species richness consider all species as making equal contributions to biodiversity option values (Faith et al. 2010).

In spite of those benefits for conservation, PD is still highly threatened and poorly protected (Zupan et al. 2014; Thuiller et al. 2015; Mouillot et al. 2016; Veron et al. 2016a). Sites representing high amounts of PD are rarely covered by protected areas (Sitas et al. 2009; Santamaría and Mendez 2012; Veron et al. 2016a). This may be because conservation has mainly focused on species richness (e.g. Myers et al. 2000 [richness of endemic species]), but areas of threatened species richness and phylogenetic diversity may differ (Tucker and Cadotte 2013; Pollock et al. 2017). A limitation of the above studies is that they examine the richness or PD of individual sites. Biodiversity conservation planning, however, is interested not in diversity of sites but in how they complement an overall set of protected areas. Thus, the species richness or the PD of the overall set is the criterion, not richness scores for individual sites. For regional planning, the contrast between PD and species richness criteria is even more dramatic (Forest et al. 2007). The conservation of PD thus needs specific attention.

The core framework for PD conservation was set out by Faith (1992) for regional to global conservation planning. An important aspect of such planning is the use of complementarity – an area to be added to a network of protected areas is evaluated based on the marginal gain in overall PD that the area offers. Surprisingly, this basic PD planning framework, which parallels that for more general systematic conservation planning (Margules and Pressey 2000), has been ignored in many regional PD

studies that consider conservation priorities. Total PD rather than PD complementarity has been the typical focus. One reason has been the distraction of community ecology, where the total PD within a locality is of interest. In fact, the total PD of a site has been wrongly interpreted in many recent papers as the conservation criterion proposed by Faith (1992) to prioritise among sites. A second challenge in incorporating PD into regional planning arises in the need to consider that different species have different estimated probabilities of extinction.

The work of Witting and Loeschke (1995) and Faith (2008) proposed in effect that marginal gains or complementarity should be extended to reflect gains in “expected PD”. A third related challenge lies in how best to integrate PD with these probabilities. Faith (2008) demonstrated that the common calculation of “distinctiveness” scores, followed by multiplication by extinction probabilities, is not an effective basis for priority setting among species. This problem is likely to extend to priority setting among areas. Thus, given these challenges, work remains to be done in setting out a clear framework for integrating PD, probabilities of extinction into forms of priority setting for areas. A survey of recent work (Faith et al. 2010; Mace et al. 2014; Brooks et al. 2015; Faith 2015) highlights the fact that there are two slightly different but related problems in PD conservation priority setting for areas. One is the “representativeness” problem – do we have a system of protected areas that represents the whole PD of a group? The second is the threatened PD problem: can we identify areas, or sets of areas, whose protection would safeguard much of currently threatened PD? These two problems sometimes merge together because the representativeness problem, for example, may require viable representation of species and branches. The challenge of achieving PD representativeness and the challenge of overcoming threatened PD also adopt similar perspectives on how conservation planning can look at both individual sites and at sets of sites. Our planning goal may be an effective set of protected sites, but few predefined priority sets are ever implemented as a set; consequently, it remains essential to note individual places that are likely to be important to include over many possible sets. In the case of representativeness, these may be sites with lots of restricted-range PD. In the case of threatened PD, the precedents are unclear, but here we suggest it may be key biodiversity areas (KBAs) based on PD, i.e. sites which significantly contribute to the conservation of PD (Eken et al. 2004; Brooks et al. 2015; Faith 2015; Veron et al. 2016b).

Our aim in this study is to propose a region-level prioritisation methodology using PD. First, we identify sets of priority areas that maximise the gain in PD and optimise the securing of threatened PD (Pollock et al. 2017). Phylogenetic diversity gain is defined as the amount of branch length a species adds to the PD already represented by the given set of species (also known as species’ PD complementarity value; see Faith 1992). Especially, in a conservation strategy, PD gain is the additional branch length which is secured after the protection of species and/or sites. Second, we propose that the amount of threatened PD that is secured when a site is protected, independently of others, may represent its KBA value based on PD. The KBAs do not define a set of sites that collectively maximise PD; they are only a collection of sites likely to be important in selecting a set (IUCN 2016). A key

perspective in this paper is that such KBAs might be highlighted as important for future planning and also might enter into the first iteration of selecting an optimal set of sites that, as a set, maximises gain in PD.

As a case study, we propose the identification of priority sites based on PD gain and on how much threatened PD they secure in the Mediterranean Basin, a region identified as the second largest biodiversity hotspot in the world. The Mediterranean Basin stretches west to east from Morocco to Turkey and north to south from northern Italy to the Canary Islands. The wildlife of the region is strongly influenced by a climate characterised by hot, dry summers and cool, humid winters (Cuttelod et al. 2008). In addition to that, the intense human activity has resulted in landscapes characterised by complex patchworks of habitats, generating a high diversity of species (Cuttelod et al. 2008). The Mediterranean Basin is the third richest hotspot in the world in terms of its plant diversity (Blondel and Aronson 1999; Mittermeier 2004), and 26% of mammals, 48% of squamates and 64% of amphibians occurring in this region are endemic (Cuttelod et al. 2008). Yet, 19% of the species found in the basin are threatened with extinction mainly because of habitat degradation, overexploitation of natural resources, invasive species and pollution (Cuttelod et al. 2008; Veron et al. 2016b). The ongoing economic development of some countries, an increasing human population, the strong touristic attraction (32% of the world's international tourists visit this region) and threats from climate change are expected to cause an even greater pressure on biodiversity in the near future (Coll et al. 2010; Apostolopoulos et al. 2014). The network of protected areas in the region is however insufficient to protect its biodiversity (Underwood et al. 2009).

To guide the design of new protected areas in the Mediterranean region, KBAs have already been identified in dragonflies, butterflies, freshwater fish, amphibians, *reptiles*, birds and mammals. These KBAs have been identified based on species richness or the presence of particular species, but do not include information on the evolutionary history they represent (e.g. Darwall et al. 2014). This is not surprising; overall, few studies to date have considered evolutionary history and its conservation in the region. Moreover, existing studies have focused on the conservation of evolutionary history at local scale or in marine environment (Mouillot et al. 2011; Guilhaumon et al. 2015; Veron et al. 2016b). We thus aim to conduct the first study that could guide the establishment of conservation actions to preserve PD at a regional scale. To do so, we first assessed the amount of PD that was not protected in the region. We then identified the sites that could help to define potential protected areas and the PD expected to be gained if they were protected. We focused on amphibians, mammals and squamates due to their high rates of endemic and threatened species.

Method

Data

We used phylogenetic data from Veron et al. (2016b) who updated phylogenies from Rolland et al. (2014), Pyron and Burbrink (2014) and Pyron and Wiens (2011) for

mammals, squamates and amphibians, respectively. With the update, missing species were included as polytomies. These had very low impact on the identification of priority areas for conservation (Veron et al. 2016b). Species whose threat status had not been assessed were removed from the analysis, whereas the status of data-deficient species was imputed thanks to a methodology based on relative range size, as suggested by Veron et al. (2016c). Our final data set comprised 270 terrestrial mammals, 107 amphibians and 237 squamates. All known Mediterranean terrestrial mammals and amphibians assessed by the IUCN were included, whereas some information about the distribution of squamates was missing (342 squamates were assessed in the region [IUCN 2015]).

The Mediterranean Basin was delimited as the biome “Mediterranean forests, woodlands and scrub” of the Palearctic realm (Olson et al. 2001). Distribution ranges of species were downloaded from the IUCN extent of occurrence maps (IUCN 2015). All maps were projected in a World Behrman projection. Local sites were defined thanks to a $1^\circ \times 1^\circ$ resolution grid to the Mediterranean Basin map for squamate and amphibian analyses (corresponding to approximately 100×100 km) and a $0.5^\circ \times 0.5^\circ$ grid for mammal analyses (50×50 km). This choice allows us to have a sufficient number of species per grid cell, reliable data, a precise resolution and to decrease omission errors, i.e. when a species is mistakenly thought to be absent (Barbosa et al. 2012; Veron et al. 2016b). A site is thus a $1^\circ \times 1^\circ$ cell for squamates and amphibians and a $0.5^\circ \times 0.5^\circ$ cell for mammals.

Coverage of Regional PD by Protected Areas

The aim of this analysis is to assess the proportion of PD intersecting protected areas for Mediterranean terrestrial mammals, squamates and amphibians. We only considered protected areas of categories I, II and IV as other categories are not specifically designed to the conservation of species (Ornat and Reynés 2004). In order to generate a set of sites that maximise PD of the whole taxonomic group (the representativeness problem) and to minimise the amount of threatened PD outside protected areas (the threatened PD problem) we proceeded as follows. First, for each species, we assessed the proportion of its distribution range that did not intersect protected areas. We here assume that the proportion of the range size that is unprotected is linearly related to probability of extinction (Crooks et al. 2017; but see Keil et al. 2018). The probability of survival of a branch was 1 minus the product of the proportion of non-protected range for all species supported by that branch. A phylogenetic branch had thus a probability to survive of 1 if at least one of its descending species had a proportion of non-protected range equal to 0, i.e. if at least one species was fully protected. We then estimated the regional PD protected as the sum of branch lengths weighted by probabilities of survival for each branch (Table 8.1). Two scenarios were used to estimate the regional PD protected. First, we considered that a species was fully protected (proportion of non-protected range equal to 0) when its coverage by protected areas was higher than 10% of its total

range as proposed by Mouillot et al. (2016). A 10% threshold was set to consider that protected areas may not always efficiently conserve species (Mouillot et al. 2016). From this, we estimated the PD viably represented in protected areas as defined in Table 8.1 (measure $protectedPD_{10\%}$).

In the second scenario, as threshold values for species conservation may be controversial (Alagador et al. 2011), we removed the 10% threshold and attributed to each species the original proportion of their range non-protected, i.e. even for species intersecting with protected areas on more than 10% of their range. We then measured $protectedPD$ as the sum of branch lengths weighted by the probabilities that at least one species survives (equal to 1 minus the product of the proportions of the range non-protected for each species supported by that branch, Table 8.1). The assumption here is that the probability of survival of a species is equal to its fractional protected range (Crooks et al. 2017).

The above analyses consider a viable representation of PD. Next, we assessed whether the threatened PD of the whole region was covered by protected areas. To calculate threatened PD (ThPD, Table 8.1), we weighted branch lengths by the probability that all the species it supports go extinct, based on their IUCN threat status (product of their probabilities of extinction; $ExpPD_{loss}$ in Faith 2008). We estimated probabilities of extinction within 50 years based on the transformation of the IUCN categories into extinction probabilities as described by Mooers et al. (2008). We then repeated the same methodology as above for this transformed tree and measured $protectedThPD_{10\%}$ and $protectedThPD$ (Table 8.1). This analysis thus estimates how much of the threatened PD is expected to be secured thanks to existing protected areas.

Identification of Priority Areas Based on Evolutionary History

We then aimed to identify Mediterranean sites whose protection would correspond to a significant gain in PD at a regional scale. We did not consider, for this study, the presence of protected areas because we first defined priority sites to insure the persistence of Mediterranean PD independently of protected areas (IUCN 2016). Our analysis used an iterative approach. First, we considered that all species had a probability of survival of 0. The regional PD safe at this step is 0. Then, we selected sites one by one and simulated their protection independently of the protection of others, i.e. the new probability to survive of a species given that a site i is protected became equal to the proportion of the range of the species in site i . When a species became protected on more than 10% of its range, we considered that the species was safe, i.e. its probability of survival became equal to 1. We did not define a scenario without a fractional threshold as it would mean that the full range of each species had to be protected, i.e. the whole region. Using thresholds described above, i.e. that a species becomes fully protected when 10% of its range is covered by protected areas, we aimed to propose a minimum set of priority areas that achieved these targets. For each site i whose protection was simulated, we weighted branch lengths of the

Table 8.1 Measures of the proportion of PD protected

Description	Without protection	Scenario 1 of protection	Scenario 2 of protection
Phylogenetic diversity (PD)	$PD = \sum L_{-j}$	$\text{protectedPD}_{10\%} = \sum L_{-j} \left(1 - \prod_{d_j} \varepsilon_{d_j} \right)$	$\text{protectedPD} = \sum L_{-j} \left(1 - \prod_{d_j} r_{d_j} \right)$
Proportion of PD protected		$\frac{\text{protectedPD}_{10\%}}{PD}$	$\frac{\text{protectedPD}}{PD}$
Threatened PD	$\text{ThPD} = \sum L_{-j} \prod_{d_j} p_{d_j}$	$\text{protectedThPD}_{10\%} = \sum L_{-j} \prod_{d_j} \left(1 - \varepsilon_{d_j} \right) p_{d_j}$	$\text{protectedThPD} = \sum L_{-j} \prod_{d_j} \left(1 - r_{d_j} \right) p_{d_j}$
Proportion of threatened PD protected		$\frac{\text{protectedThPD}_{10\%}}{\text{ThPD}}$	$\frac{\text{protectedThPD}}{\text{ThPD}}$

L_j is the length of branch j on the phylogenetic tree; p_{d_j} is the probability of extinction of the d th descendant of branch j within a defined period of time, given the IUCN Red List; ε_{d_j} is the proportion of the distribution range of species d_j that is NOT protected if the proportion of the distribution range of species d_j that is protected is less than or equal to 10% and 0 if it is more; and r_{d_j} is the proportion of the distribution range of species d_j that is NOT protected. We considered two scenarios. In the first, we assumed that a species was fully protected when more than 10% of its total range was in protected areas. In the second, we did not integrate any threshold to estimate whether a species was protected or not

phylogenetic tree by the new probability that at least one of the descending species survives and measured protectedPD10% (Table 8.1, Fig. 8.1). This method allows to account for the shared responsibility of species to protect the internal branches of the phylogenetic tree (Faith 2008). We then estimated the new safe regional PD given that site i is protected by summing the weighted branch lengths over the regional phylogenetic tree.

The gain in PD (Faith 1992) due to the protection of site i was then calculated as the difference between the regional PD which is safe if site i was protected and the regional PD that was secured before site i was protected (Table 8.2).

$$\text{PDgain}_i = \text{protectedPD}_{10\%}^{\{S\} \cup i} - \text{protectedPD}_{10\%}^{\{S\}}$$

where $\{S\}$ is the initial set of protected sites and $\{S\} \cup i$ is the union of S and i .

We calculated the gain in PD due to the protection of each site independently from each other (we thus had a number of PDgain values equal to the number of sites). The priority site was the one with the highest gain in PD (Fig. 8.1). In case of ties of highest PDgain, we selected the site where was found the species with the smallest range size, assuming that range restricted species are more vulnerable and thus need to be protected in priority. Once a single priority site was identified, we considered it as secured and calculated the new regional PD which is safe given the updated set of protected sites (Fig. 8.1).

We repeated the procedure from the new regional safe PD (i.e. we simulated the protection of sites one by one, defined a site with the highest PDgain as priority, and calculated the new regional safe PD considering the priority site as protected) until the set of priority sites identified resulted in the protection of the total PD of the region. All priority sites were considered as potential future protected areas. We finally defined an order among identified priority sites from the highest to the lowest PDgain, representing the scenario of protection in which the gains would be the highest by setting up a minimum number of protected areas.

We repeated this procedure to assess sites that would maximise the protection of threatened PD (ThPDgain), in which the phylogenetic branches were initially weighted by species probabilities of extinctions. The set of priority sites in ThPDgain are the set of sites which, if protected as a set, would most efficiently preserve threatened PD. An interesting prospect is that the ThPDgain of each site calculated at the first iteration of our calculations roughly corresponds to the definition of KBAs of phylogenetic diversity as described by Faith (2015). Although those scores do not reflect the spatial complementarity of sites for conservation, they may correspond to a risk approach where one tries to anticipate all the sites that may be critical. This risk approach may be especially useful when, in practice, conservation strategies depart from adoption of the ideal minimum set, which maximises PD.

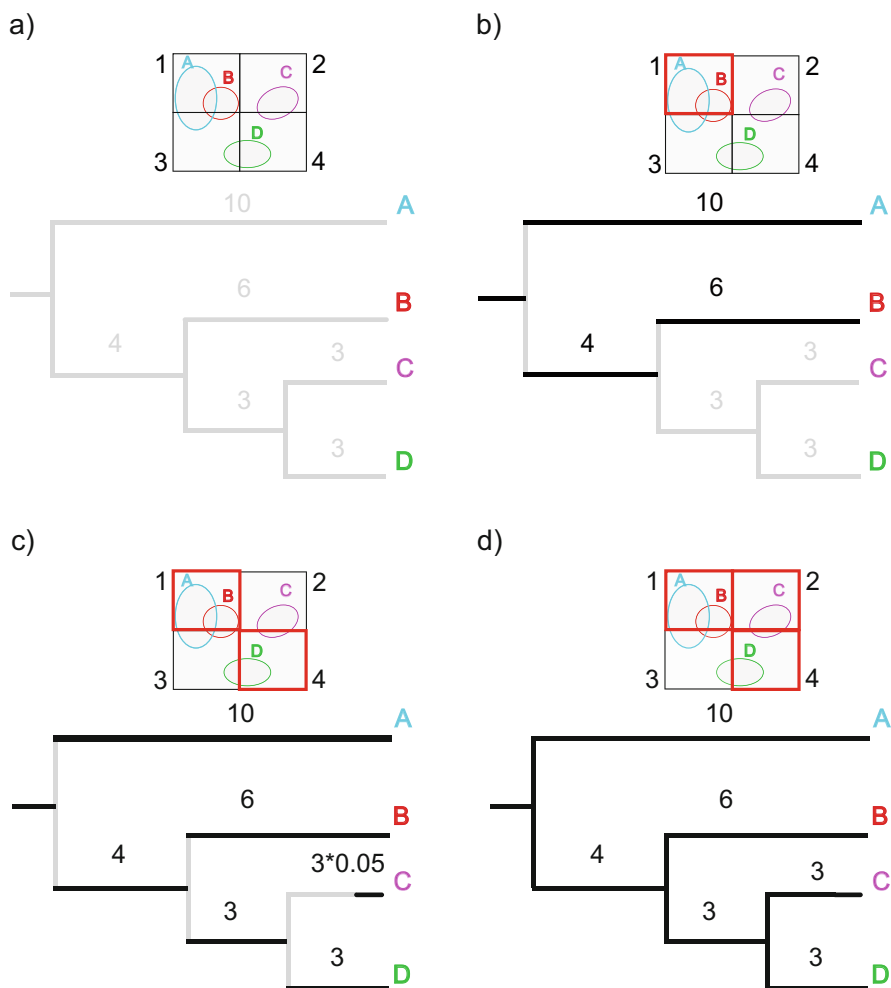


Fig. 8.1 Illustration of how we defined the order of priority to maximise cumulative PDgain. Circles represent species distribution areas of species A, B, C and D. Squares 1, 2, 3 and 4 represent sites. A red square means that the site is protected. The phylogenetic tree represents evolutionary relationships between species A, B, C and D. Bold branches represent branches which are safe following the protection of a given site. **(a)** No site is protected and the PDgain is equal to 0. The order of priority of these sites would be the following: **(b)** protection of site 1 enables the highest gain of PD, species A and B are protected on more than 10% and their terminal branches are thus fully protected. Moreover, as B is fully protected, the branches supporting species A, B and C and species B, C and D are safe (PDgain = 20 Myears). **(c)** Protection of site 4. This stage enables to fully protect the terminal branch supporting species D and thus deep branches supporting species C, D and C, D, B (new PD = 26.15 Myears; PDgain = 26.15 - 20 = 6.15 Myears). **(d)** Protection of site 2 enables to protect the terminal branch supporting species C, which was the last branch not fully protected (new PD = 29 Myears; PDgain = 29 - 26.15 = 2.85 Myears)

Table 8.2 Measures of PD gain when protecting a new site i given a set of preliminarily protected sites $\{S\}$

Description	Without protection	Scenario1 of protection
Phylogenetic diversity (PD)	Secured PD = 0	$\text{protectedPD}_{10\%}^{\{S\}} = \sum_j L_j \left(1 - \prod_{dj} \epsilon_{dj}^{\{S\}}\right)$
Gain in protected PD (PDgain)	PDgain = 0	$\text{protectedPD}_{10\%}^{\{S\} \cup i} - \text{protectedPD}_{10\%}^{\{S\}}$
Threatened PD	Secured ThPD = 0	$\text{protectedThPD}_{10\%}^{\{S\}} = \sum_j L_j \prod_{dj} \left(1 - \epsilon_{dj}^{\{S\}}\right) p_{dj}$
Gain in threatened PD protected (ThPDgain)	ThPDgain = 0	$\text{protectedThPD}_{10\%}^{\{S\} \cup i} - \text{protectedThPD}_{10\%}^{\{S\}}$

L_j is the length of branch j on the phylogenetic tree; p_{dj} is the probability of extinction of the dth descendant of branch j within a defined period of time, given the IUCN Red List; $\epsilon_{dj}^{\{S\}}$ equals to the proportion of the distribution range of species dj that does not fall in cells of set $\{S\}$ if the proportion protected is less than or equal to 10% and 0 if it is more; and $\{S\} \cup i$ contains all cells of set $\{S\}$ plus cell i . We did not define a scenario without a fractional threshold as it would mean that the full range of each species had to be protected, i.e. the whole region

Results

Proportion of the Mediterranean PD Protected

For all groups, only a small proportion (between 4% and 30%) of their PD was protected, even if this proportion was logically higher in the analysis considering a 10% threshold. Mammals PD was the least protected, whereas amphibian PD was the most secured (Table 8.3). When branches covered on more than 10% of their range by protected areas were considered safe, the proportion of threatened PD protected was of 14.4%, 12.1% and 12.8% for mammals, squamates and amphibians, respectively (Table 8.3). Those values decreased to 4.9%, 7.6% and 5.9% when we removed the 10% threshold value.

Identification of Potential KBAs and the Minimum Set of Areas Maximising the Gain in PD

The priority sites that maximise mammal PDgain were widely distributed across the Mediterranean basin (Figs. 8.2, 8.3 and 8.4). One hundred and fourteen sites were needed to cover the whole PD of Mediterranean mammals (Figs. 8.2 and 8.5). Considering the priority order, the sites that would enable to gain the greatest part of total PD were found in Israel, Egypt, the Maghreb, the French Alps, Turkey and Albania (Fig. 8.2a). To squamates, 75 priority sites were needed to protect the whole

Table 8.3 Proportion of regional phylogenetic diversity (PD) and threatened PD which currently overlaps with protected areas of categories I, II and IV

		Terrestrial mammals	Squamates	Amphibians
Proportion of PD protected	10% threshold	15.2	22.9	30.1
	No threshold	4.1	14.3	17.9
Proportion of threatened PD protected	10% threshold	14.4	12.1	12.8
	No threshold	4.9	7.6	5.9

We considered two scenarios. In the first, we assumed that a species was fully protected when more than 10% of its total range was in protected areas. In the second, we did not integrate any threshold to estimate whether a species was protected or not

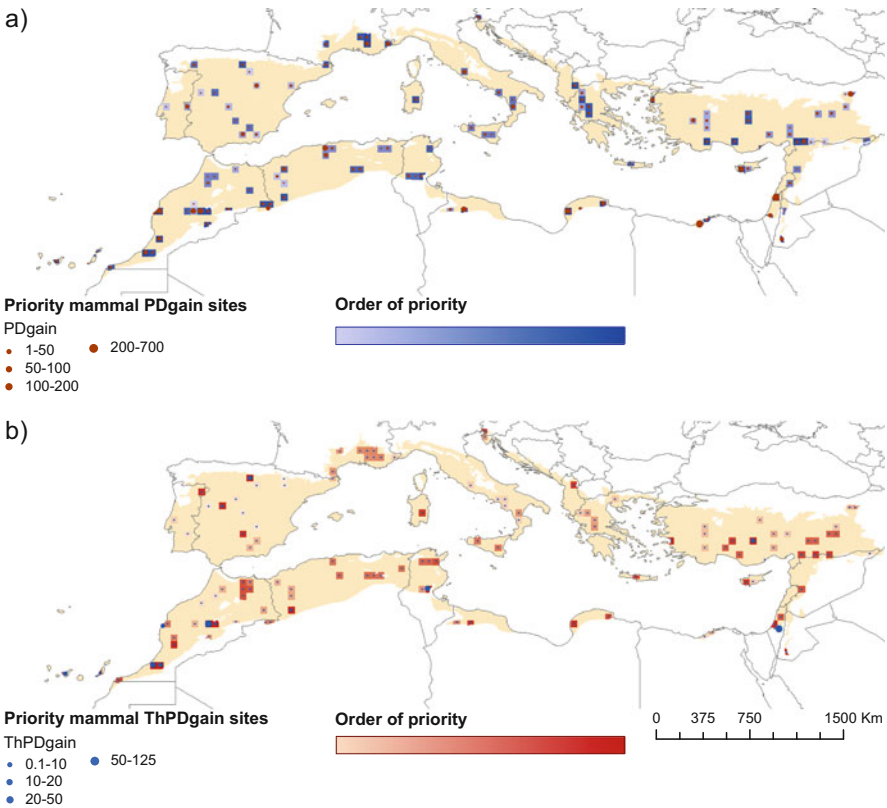


Fig. 8.2 Spatial distribution of mammal priority sites based on (a) PDgain and (b) ThPDgain and their order of priority

PD of the region (Fig. 8.3). Moreover, 95% of the PDgain could be achieved with the protection of only 42 sites (Fig. 8.5). The sites with highest priority were located in southern Greece, southern Turkey, Morocco and Israel (Fig. 8.3a). To amphibians, 46 sites would be necessary to protect the whole PD, but the first 31 sites in the

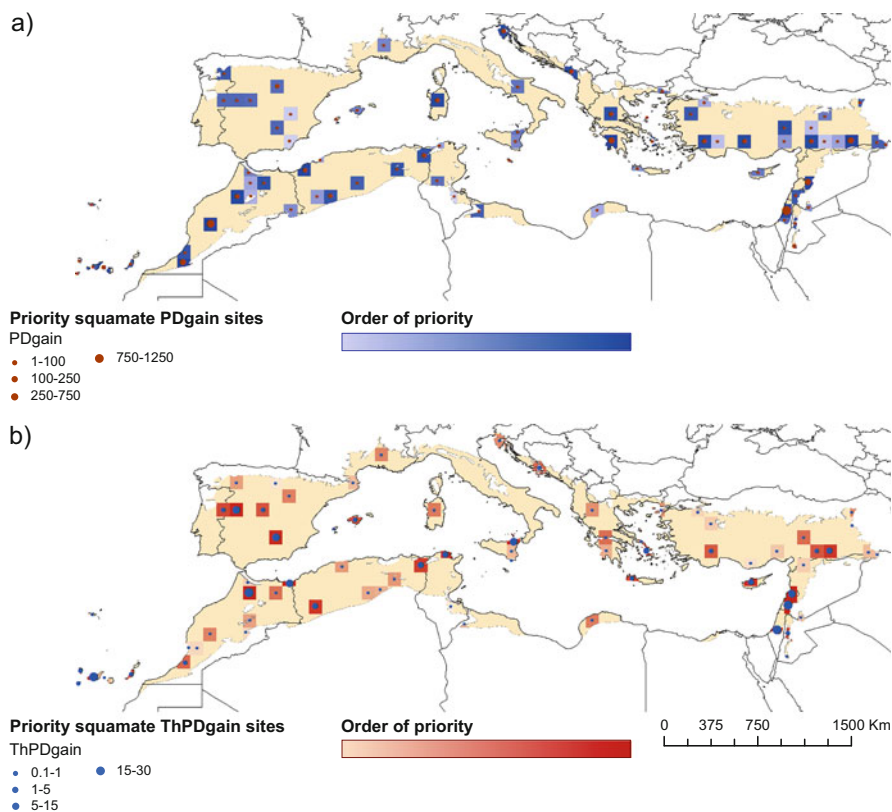


Fig. 8.3 Spatial distribution of squamate priority sites based on (a) *PDgain* and (b) *ThPDgain* and their order of priority

ranking could safeguard 95% of PD (Figs. 8.4 and 8.5a). The sites whose protection would result in highest PDgain were in Italy, Corsica, Croatia, Sardinia and Southern Turkey (Fig. 8.4a). Although priority sites tended to be found in the same regions among taxa (Figs. 8.2, 8.3 and 8.4), squamate priority sites intersected with a higher proportion of priority sites of other taxa than amphibian and mammal priority sites did (Table 8.4).

The sites that contribute more to the protection of threatened PD (*ThPDgain*) matched well with the priority sites of *PDgain*. However, the ranking order differed. For mammals, sites with the highest priority were in northern Spain, Tunisia, Canary Islands, Morocco and Israel (Fig. 8.2b). For squamates, priority sites were in Israel, Spain, Canary Islands, Morocco and Syria (Fig. 8.3b). As for amphibians, the highest gains in threatened PD were found in southern Turkey, Sardinia, Israel and Spanish Pyrenees (Fig. 8.4b). Not surprisingly, those sites matched with the independent *ThPDgain*, calculated at the first iteration of selecting an optimal set of sites, that would roughly correspond to the definition of PD KBAs as defined by Faith (2015) (Fig. 8.6).

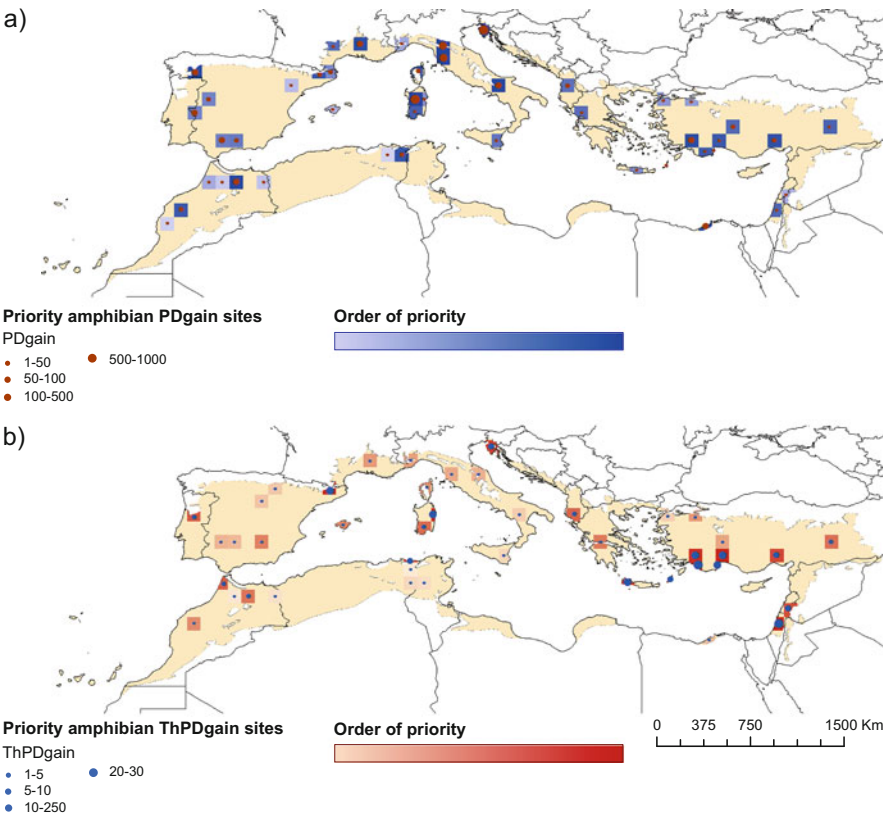


Fig. 8.4 Spatial distribution of amphibian priority sites based on (a) PDgain and (b) ThPDgain and their order of priority

Table 8.4 Proportion of intersecting priority sites between taxon. The table provides the proportion of priority sites of a given taxon which intersect with the priority sites of another taxon

	Terrestrial mammals	Squamates	Amphibians
(a) <i>PDgain</i>			
Mammals	NA	59.9%	14.1%
Squamates	58.6%	NA	21.3%
Amphibians	26.6%	35.5%	NA
(b) <i>ThPDgain</i>			
Terrestrial mammals	NA	53.27%	27.8%
Squamates	55.7%	NA	12.3%
Amphibians	50%	27.7%	NA

Priority sites are identified based on (a) PDgain and (b) ThPDgain. For example, the first line specifies that 59.9% and 14.1% of mammal PDgain priority sites intersect with squamate and amphibian PDgain priority sites, respectively

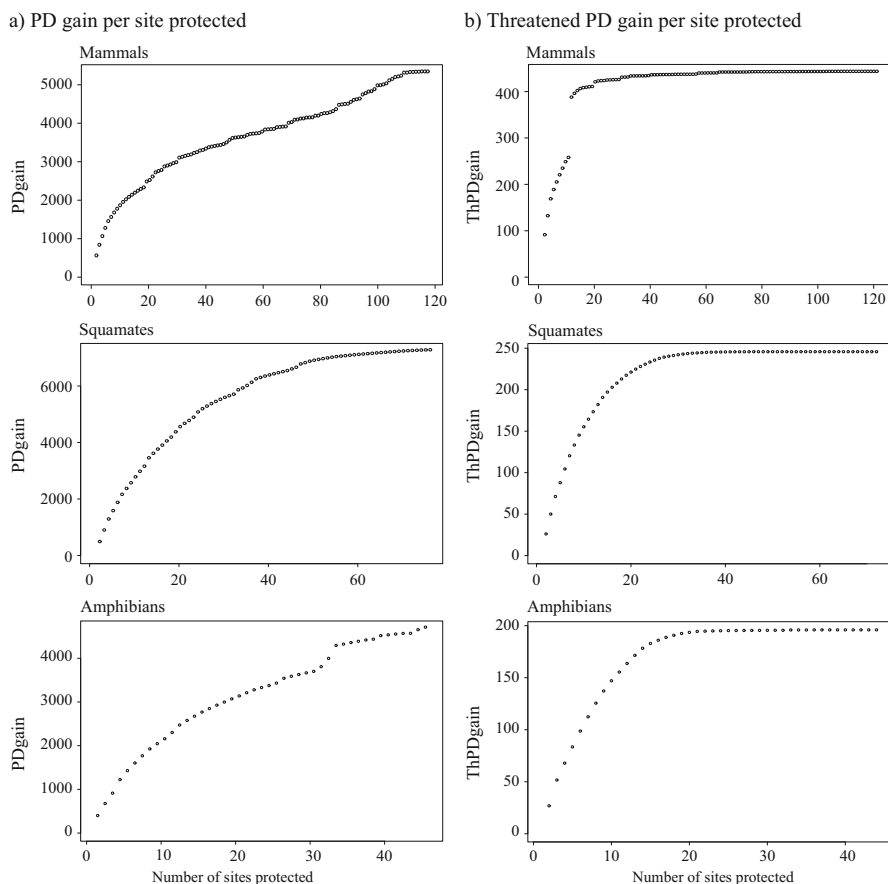


Fig. 8.5 Accumulated PDgain and ThPDgain with the number of identified priority sites protected. Key biodiversity areas are consecutively protected depending on their order of priority

The gain in ThPDgain per site protected indicates important differences among these three groups of vertebrates. For squamates and amphibians, the accumulation curve is nearly asymptotic, indicating that 19 and 14 sites, respectively, would be sufficient to protect 95% of their threatened PD, but several additional sites would be necessary to reach 100% (Fig. 8.5b). To mammals, 95% of threatened PD can be protected with 19 sites. But, the increase in this first part of the curve is not linear, and the tenth site situated in Morocco is crucial to this gain, probably reflecting the protection of evolutionary distinct and threatened Mediterranean mammals such as the Barbary macaque (*Macaca sylvanus*), the leopard (*Panthera pardus*) or the caracal (*Caracal caracal*).

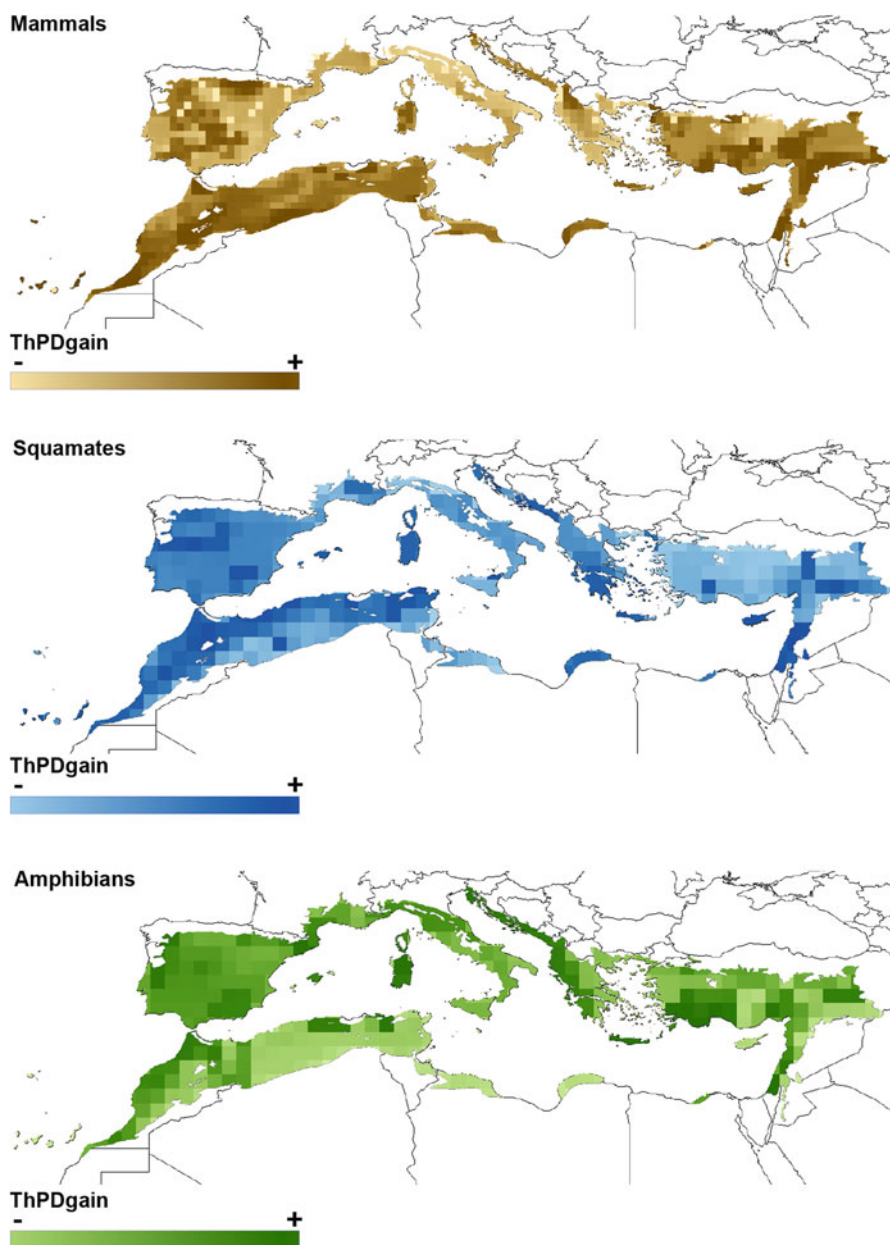


Fig. 8.6 Maps of ThPDgain when sites are selected independently of the protection of other sites. The priority sites defined using ThPDgain roughly correspond to the definition of PD KBAs as defined by Faith (2015)

Discussion

In this study, we assessed the extent to which the protected areas in the Mediterranean region contribute to protect the phylogenetic diversity of mammals, squamates and amphibians. We identified a minimum set of priority areas, which, if protected, would maximise the gain in PD. We also defined the priority order among those sites and showed that a small set of them could provide a maximum conservation of PD. Moreover, we calculated a KBA score for each site, representing how much threatened PD would be secured if such sites were protected independently.

We showed that only small proportions of the regional PD were protected in terrestrial mammals, amphibians and squamates, and threatened PD was even less well protected. Amphibians and squamates were the best protected groups for PD and threatened PD, respectively, whereas mammals constituted the least protected group for both measures. This is probably because a lot of mammal PD is concentrated in the southern part of the basin where protected areas are less numerous (Veron et al. 2016b). Conserving the whole PD of the region would require to set up protection measures in both the northern and the southern part of the Mediterranean basin. Islands (e.g. Canary Islands, Sardinia), Israel, Turkey and the Maghreb are regions where there are concentrated priority areas for all taxa. The proportion of intersecting priority sites was moderate with squamate priority sites capturing the most of priority sites of other taxa while amphibians' captured the least. Using a taxa as a proxy of others to conserve the PD of several taxa may thus not be an appropriate strategy. Sites with the highest priority may enable to protect the most evolutionary distinct species, closely related species sharing long branches and/or endemic species. For example, in amphibians, the protection of sites located in Croatia and with identified high priority for conservation would secure the most evolutionary distinct Mediterranean amphibian, the olm (*Proteus anguinus*), which is endemic to the region. In mammals, the protection of priority sites situated in Algeria, Morocco and Gibraltar may enable to secure the only primate of the region, the Barbary macaque (*Macaca sylvanus*), also an endemic species of the Mediterranean region.

Conservation strategies in the Mediterranean Basin, especially KBAs, have been defined using criteria based on threatened and geographically restricted species but do not yet consider evolutionary history (Brooks et al. 2015; IUCN 2016). Nonetheless, a strategy relying on PD in the region may be more valuable for conservation than considering species richness alone (Faith 1992; Polasky et al. 2001; Sechrest et al. 2002; Mace et al. 2003; Gascon et al. 2015; Veron et al. 2016a). Indeed, maintaining PD may allow the preservation of the evolutionary heritage, trait diversity and unexpected future services for human well-being in the Mediterranean Basin (Faith 1992; Mace et al. 2003; Mooers et al. 2005; Faith et al. 2010; Pellens and Grandcolas 2016; Faith 2017). Evolutionary processes have provided services in the past and are likely to result in current and future benefits for biodiversity and societies (Faith et al. 2010). Thus, it has been argued that preserving phylogenetic diversity would preserve known but also unknown future services or benefits

(biodiversity option values; Faith 1992; Forest et al. 2007; Faith et al. 2010; Faith 2017; Lean and MacLaurin 2016). Option values are unanticipated future benefits with a high cost of replacement. They are crucial for conservation because we still and will probably always understand little about the effect of biodiversity change on human societies, and decisions have to be taken in spite of those uncertainties (Oliver 2016). Faith et al. (2010) argued that PD may be an effective surrogate for option values because PD can capture a diversity of features, many of them unknown, which may provide future benefits to society (see also Faith 1992). Lean and MacLaurin (2016) hypothesised that a PD-like measure would be the best strategy to maximise option values because it would exploit the different numerous evolutionary processes that caused functional and morphological divergence within lineages.

The Mediterranean Basin is in urge for new conservation actions to preserve a highly threatened and endemic biodiversity (Underwood et al. 2009; Cox and Underwood 2011). In a recent study, Veron et al. (2016b) identified the threatened and endemic PD of the Mediterranean Basin at a local scale. They showed that sites where local PD was the most threatened in terrestrial mammals, squamates and amphibians were situated in the eastern part of the Mediterranean Basin, in the Maghreb and in islands. Those sites may represent the expected local loss of PD and of associated future unexpected localised services for human well-being (Veron et al. 2016b). Moreover, over the region, the proportion of those sites that were protected was low. Yet, increased protection of those sites is not necessarily a good way to protect overall regional PD (see Pollock et al. 2017 for a related contrast between local and regional PD priorities). The reason is that those hotspots of local threatened PD may capture redundant information for a regional conservation strategy, i.e. a branch may be present in several sites. Further, a branch that is threatened for one locality may be secure regionally because the regional (but not local) phylogeny includes a secure descendant of that branch.

Here we proposed an alternative strategy, which minimises the number of sites to insure the preservation of PD at a regional, instead of local, scale. This regional approach may thus be more valuable to identify future protected areas that conserve biodiversity option value (Faith 1992), in contrast to localised services. The network approach we used enabled us to better allocate resources for conservation when the aim is to preserve regional biodiversity (e.g. Kati et al. 2004). In our approach, we considered complementarity at the level of sites but also at the level of phylogenetic branches by estimating PDgain: the proportion of a deep branch that is safe is the proportion of the safe range of all the species it supports. Phylogenetic complementarity is a key principle for the conservation of PD: it is the additional branch lengths represented by a species, which is not spanned by an existing set of species (Faith 1992). If phylogenetic complementarity is not considered, the regional or global conservation of PD may be incorrectly assessed. Measuring PDgain insures that the highest proportion of PD is protected at each new identification and protection of priority sites while minimising the number of sites protected. Including probabilities of extinctions in this framework may then allow to maximise the gains in securing threatened PD, which may change the order of priority among sites.

One drawback of this approach is that those priority sites are defined as a set. Yet, in practice, the minimum set rarely has any guaranteed policy relevance. In this context, hotspots defined by Veron et al. (2016b) may be useful because their value for conservation did not depend on the protection of other sites. Following both site and phylogenetic complementarity, some priority sites we identified were congruent with the local hotspots identified in Veron et al. study (2016b). Especially, similar priority sites were found in islands, Turkey, the Maghreb and the Balkans. Those sites may locally concentrate high amounts of PD which may contribute to a high proportion of the regional PD. They are thus likely to be of crucial importance for conservation. Another possibility to account for the risk that the minimum set of areas to protect PD is not selected in practice is to estimate the independent ThPDgain of sites, which would roughly correspond to Faith (2015)'s definition of KBAs. The logic can be described this way: if a site has many threatened elements of biodiversity, then, as long as that site remains secure, we can say that the site is making a large contribution to global persistence of biodiversity. Those values of ThPDgain may help to anticipate all the sites that may be critical in future sets of protected areas, even if some of them are similar: we cannot predict which sites may become unavailable or of prohibitive cost, so we are interested in a suite of sites that individually are likely key contributors to PD gain. This will extend previous suggestion on PD KBAs by Brooks et al. (2015); Faith (2015) and Veron et al. (2016b). In our study, we found that many sites with the highest independent ThPDgain values were also selected in the minimum set of areas that would protect the whole PD of the region.

In spite of those similarities, some advances are still required to refine the identification of PD-based protected areas in the future. An increased knowledge on the biodiversity of the region and the range distribution of species will improve the definition of protected areas. Especially, initiatives such as the Global Assessment of Reptile Distribution (GARD 2016) will help to have more data on the distribution of squamates which were missing in our study. Including those missing species in a regional conservation strategy may conduct to the identification of new priority areas even if it is likely that the sites we identified will remain important for conservation. Another knowledge gap in our study was the absence of conservation status (degree of threat) for a number of species. In assessing PD gains, we included data-deficient species thanks to a methodology developed by Veron et al. (2016c). Although this may be a better strategy than excluding species to refine assessments of PD gain, data on the distribution of those species may be uncertain. As for species not assessed by the IUCN, their data may be even poorer or missing (IUCN 2015). We had to exclude them from the analysis. Naturally, a missing secure species could imply that an apparent threatened ancestral branch was in fact secure. A missing threatened species at very least is missing information about its terminal branch. An alternative analysis might attempt some sort of bracketing approach where both optimistic (secure) and pessimistic scenarios for non-assessed species are investigated.

We considered that a species was secure, i.e. had a probability of survival of 1, when 10% of its range was protected. However, in more realistic approaches,

estimating the probability of a species to survive using the proportion of its range protected may be more or less appropriate depending on the species and on its rarity. A rare species would need a higher proportion of its range inside protected areas than a common species. Although species range was shown to be a strong predictor of extinction risks, some other factors such as body size may influence species vulnerability (Cardillo et al. 2008; Böhm et al. 2016; Veron et al. 2016c) and might be considered to estimate how the protection of a site impacts species probability of extinctions. Further, protected areas are not always the best solution to preserve species and conservation should also focus on the protection of species outside protected areas, for example, by maintaining habitats and biological connectivity (Hockings and Phillips 1999; Cox and Underwood 2011). Additional factors to consider the establishment of future protected areas from priority sites would be conservation costs and probabilities of success. Indeed, for some species, it may be necessary to allocate huge amounts of resources with only little chance of success. This knowledge could then guide conservationists in their decisions to protect phylogenetic diversity. Our framework for calculating PD gains allows for the integration of algorithms that consider such costs and constraints when they become available.

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Chapter 9

Evolutionary Diversity Patterns in the Cape Flora of South Africa



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Abstract The Greater Cape Floristic Region of South Africa is a relatively small area found in the south-western corner of the African continent. It is characterised by a Mediterranean-type climate and harbours an incredible floral diversity with almost 11,500 species of plants, of which more than three-quarters are found nowhere else in the world. The floristic composition of the region is very distinctive with families not dominant in other floras, such as Iridaceae, Aizoaceae, Ericaceae, Proteaceae and Restionaceae. In addition, small-leaved, sclerophyllous low shrubs and geophytes are by far the predominant growth forms with a low proportion of tree and annual species present. Understandably, this flora has attracted a lot of interest, and many have pondered what environmental and ecological factors could be responsible for its richness and uniqueness and the resulting biodiversity patterns observed today. In this chapter, the current state of knowledge of large-scale evolutionary patterns, investigated using phylogenetic diversity metrics, within the Cape region are reviewed. As well as published works, we report on ongoing studies examining biodiversity patterns using phylogenetic diversity, phylogenetic endemism and phylobetadiversity, both in plants and animals. We also review how these phylogeny-based approaches have been used to address applied questions (e.g. biodiversity surrogates, phytogeographical delimitations).

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Introduction

Nestled in the south-western corner of the African continent lies one of the most diverse floral hotspots on the planet, the Cape region of South Africa. Bordered to the west and south by the Atlantic and Indian oceans, respectively, and to the north by the arid Succulent Karoo biome, it is crisscrossed by multiple mountain ranges covered in the prominent vegetation type of the region, Fynbos. The uniqueness of the Cape region has been recognised for decades. It was considered by many as one of only six floral kingdoms of the world, on par with much larger areas such as the Holarctic and Neotropical kingdoms (e.g. Good 1974; Takhtajan 1986; Marloth 1908; but see Cox 2001), and has been identified as one of the world's biodiversity hotspots (Myers et al. 2000). Although the delimitation of its boundaries has varied according to different authors (e.g. the inclusion of the Kamiesberg mountains; Marloth 1908), traditionally the region has been broadly defined as the Cape Floristic Region (CFR; Bond and Goldblatt 1984; Goldblatt and Manning 2000) or Core Cape Flora (Manning and Goldblatt 2012).

While the CFR is largely associated with Fynbos, the main vegetation type (and biome) found in the region as currently defined, it also includes four other very distinct biomes, the Renosterveld, Succulent Karoo, Afromontane Forests and Albany Thicket (Bergh et al. 2014). It has been argued for some time that the delimitation of the Cape region should be expanded to include the neighbouring Succulent Karoo biome, found to the north of the Cape Floristic Region (e.g. Jürgens 1991, 1997; Bayer 1984). The intimate links between the CFR (with Fynbos, its main component) and the Succulent Karoo were demonstrated more formally by Born et al. (2007) based on an extensive analysis of the shared floristic composition of the regions at the genus and species level, the identification of the dominant families and maximising endemism. Their analyses supported the concept of the Greater Cape Floristic Region (GCFR), which has since become more widely accepted and further developed (e.g. Verboom et al. 2014; Colville et al. 2014) (Fig. 9.1).

Wherever lie its boundaries or how these are interpreted, it is undeniable that the Cape is one of the most biodiversity-rich regions of the world (Allsopp et al. 2014). We will refer here predominantly to the Cape Floristic Region (Goldblatt and Manning 2000; or Core Cape Flora, Manning and Goldblatt 2012) but will include reference to a wider Cape region incorporating the Greater Cape Floristic Region (Born et al. 2007). Both the CFR and GCFR have been used in many different studies investigating the diversity patterns found in the Cape region and the potential factors and processes that are responsible for these patterns.

The Cape region is home to an incredible 11,423 species of vascular plants, of which more than three-quarters are endemic to the region (77.9%); 9,383 species are found within the CFR, of which just over 68% are endemic. This high level of endemism is also found at the genus level, with 22.2% of the 1,119 genera found only in the region, as well as at the family level, with four endemic families (Penaeaceae, Grubbiaceae, Roridulaceae, Geissolomataceae; Snijman 2013). The

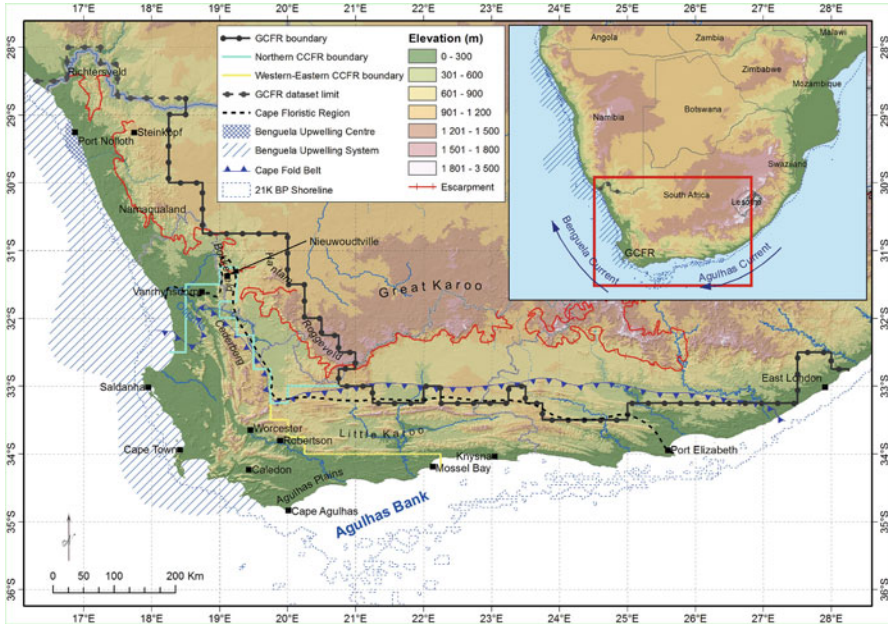


Fig. 9.1 The boundaries of the Greater Cape Floristic Region (GCFR) and the Core Cape Floristic Region (Core CFR). Topography and the position of the Palaeo-Agulhas Plain are indicted. (Reproduced with permission from Cowling et al. (2017) (see Acknowledgements))

floristic composition is also very distinctive. While one would expect Asteraceae and Fabaceae, two of the world’s largest families, to be well-represented in the Cape region (representing together ca 20% of the flora), many of the largest families present in the region are not dominant in floras from elsewhere in the world (e.g. Iridaceae, Aizoaceae, Ericaceae, Proteaceae and Restionaceae). Finally, small-leaved, sclerophyllous low shrubs and geophytes are by far the predominant growth forms in the Cape region, with a low proportion of tree and annual species present, the former mostly restricted to the Afromontane Forests and Albany Thicket biomes.

Several evolutionary, ecological and environmental processes have been suggested as the key drivers responsible for the remarkable diversity found in the Cape region; these include (but are not limited to) shifts in pollinators, low gene flow, climatic stability, topography, rainfall regime, edaphic factors, fire regimes and geomorphic dynamics (e.g. Cowling 1987; Cowling et al. 2009; Goldblatt 1978; Goldblatt and Manning 2002; Johnson 1996, 2010; Linder 1985, 1991, 2003; Verboom et al. 2015). Although some of these factors have been favoured by various authors, none alone can explain the diversification patterns seen in many Cape-centred groups. The increased analytical possibilities offered by molecular phylogenetic approaches have shown that the diversity found in the Cape has evolved from several independent radiations interspersed over a long period of time (since

the early Miocene or even earlier), from various geographical sources and involving a mixture of processes (e.g. Cowling et al. 2009; Linder 2005; Verboom et al. 2009, 2014). The biodiversity patterns resulting from these evolutionary processes are discernible in the Cape region of today, and phylogenetic diversity and associated metrics offer new ways to explore these.

In this chapter, we review the current state of knowledge of large-scale evolutionary patterns based on phylogenetic information within the Cape region and discuss briefly some of the ongoing work aimed at refining our current understanding of the potential processes that led to them. We strive to be as exhaustive as possible to cover the various studies that have used phylogenetic information to investigate evolutionary patterns in the Cape region, including discussion on faunal studies. We also review how these phylogeny-based approaches have been used to address various more applied questions, such as the development of surrogates for other components of biodiversity in the region, help delimiting phytogeographical zones and how these methods can provide information useful for the protection of the biodiversity of the Cape region.

Plant Regional Phylogenetic Diversity Patterns

One of the first investigations of evolutionary patterns in the Cape region using phylogenetic diversity as a metric was a study of the differences between vegetation characteristics of four biomes, namely, Fynbos, Grassland, Albany Thicket and Nama Karoo (Procheş et al. 2006); Forest and Renosterveld biomes were not represented in this study. For this comparison, Procheş and colleagues used 16 plot surveys from each of the 4 biomes and a previously published, family-level, angiosperm phylogenetic tree (Davies et al. 2004). They calculated PD for each plot and found that phylogenetic diversity in Thicket was not significantly different than expected from a random sample of species. On the other hand, species were phylogenetically clustered in the other three biomes, and in the Nama Karoo biome, the clustering increased in plots with greater numbers of species (Procheş et al. 2006). The clustering observed was attributed to the presence of several species from particular families (or even genera), such as Poaceae and Asteraceae in grassland and Karoo and Restionaceae and Ericaceae in Fynbos (Procheş et al. 2006). The lack of clustering (or overdispersion) in Thicket was assumed to be the result of the presence of an assemblage of phylogenetically isolated and dispersed species, as well as a greater frequency of exchange with neighbouring biomes than in other vegetation types (Procheş et al. 2006).

More recently, Verboom et al. (2014) used lists of species identified as characteristic of biome-specific vegetation types (Mucina and Rutherford 2006), in combination with a genus-level phylogenetic tree of the flora of the Cape region (Forest et al. 2007), to re-examine the phylogenetic diversity present in the Cape region's biomes and how it is distributed within each of them. Verboom et al. (2014) compiled phylogenetic diversity and PD per genus for each of the five Cape region's

biomes and assessed if the taxa present in each biome were phylogenetically clustered, overdispersed or present in expected levels given the generic diversity present. They found that PD was relatively similar in each biome, except in the Forest biome, where it was significantly lower. However, the amount of PD per genus was about twice as high in forests than in other biomes (Verboom et al. 2014). This discrepancy can be explained by the fact that the forests of the Cape region comprise far fewer species than the other four surrounding biomes, but that these are more phylogenetically diverse, an assertion supported by the overdispersed nature of phylogenetic diversity in forests compared to three of the other four biomes (Verboom et al. 2014). The exception is Thicket. Despite showing levels of PD and PD per genus similar to those found in Fynbos, Renosterveld and Succulent Karoo, the PD in Thicket is not clustered, which indicates a greater proportion of elements that are phylogenetically isolated, supporting the findings of Procheş et al. (2006). In general, these results indicate that genera found in Fynbos, Renosterveld and Succulent Karoo are more closely related than those found in Forest and Thicket and hint at either a more recent origin or a more stable and isolated evolutionary history (Verboom et al. 2014).

Large-scale phylogenetic diversity (PD) patterns in the CFR were first investigated using a genus-level, molecular-based phylogenetic tree comprising 735 of the ca 950 genera of angiosperms found in the region, together with distribution data (presence/absence) at the quarter-degree square (QDS) scale (Forest et al. 2007). Although the scale was relatively coarse, some clear patterns emerged from this work. The comparison of genus diversity and phylogenetic diversity suggested a close correlation between the two metrics, both distributed in a similar manner with a concentration of diversity in the south-western part of the region and in some areas along the southern coast, the Cederberg mountains and the Bokkeveld plateau. Further analyses, however, found that phylogenetic diversity is more phylogenetically clustered in the western part of the Cape, which experiences a winter-rainfall regime, while the eastern part (non-seasonal rainfall) mostly showed QDS with expected phylogenetic diversity levels given the number of genera they contain. Likewise, a regression of PD on number of genera identifies a concentration of QDS with negative residuals in the western part, while the eastern part of the region is dominated by QDS with positive residuals. Thus, QDS have generally lower amounts of PD per genus in the western part of the region (Fig. 9.2; Forest et al. 2007; Verboom et al. 2014). These patterns can be explained by the presence in the western part of the region of a greater number of closely related genera on relatively shorter terminal branches (i.e. less unique PD per genus), while the flora of the eastern region is composed of a greater number of more phylogenetically isolated genera on longer terminal branches (i.e. more unique PD per genus). The fact that such a configuration of phylogenetic diversity is evident across the Cape region at the genus level could lead to the assumption that the same analysis performed at the species level would yield similar patterns, although much accentuated given the distribution of species diversity in the region with high numbers of species in the south-western winter-rainfall areas. Moreover, the concentration of diversity in a relatively small number of families and genera would further emphasise this pattern

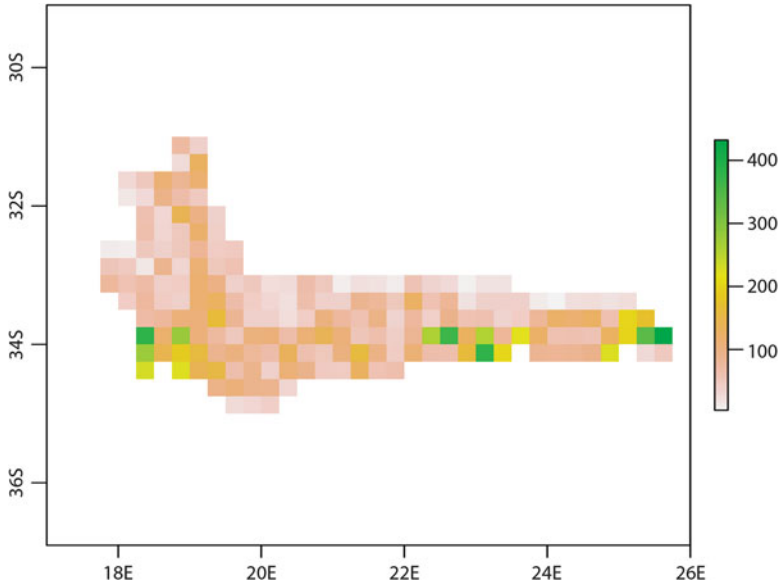


Fig. 9.2 Phylogenetic endemism patterns in the Cape region calculated with the function *phylogenetic.endemism.R* of Guerin and Lowe (2015) using the published phylogenetic tree of the Cape flora of Forest et al. (2007) and presence/absence distribution data per QDS

(e.g. the 20 largest families in the Cape, out of 178, account for over 75% of the species diversity; Manning and Goldblatt 2012).

With the same genus-level phylogenetic tree of the Cape flora (Forest et al. 2007), but this time using species distribution modelling (Mecenero et al. 2015), which produced probability of occurrence of species ($n = 8,347$ taxa) in 2 min square grid cells, subsequently compiled to produce genus-level distributions, we were able to produce a much-refined map of the phylogenetic diversity patterns for the region. While this analysis still identifies the south-western part of the Cape region and most areas along the southern coast as harbouring high amounts of PD, it differs in two main aspects from the analysis of Forest et al. (2007). First, the mountains along the north-south axis in the west of the region, including the Cederberg and the Groot Winterhoek, are not highlighted as hotspots of PD, contrary to what is observed in the previous analysis (Forest et al. 2007; Fig. 9.2). Second, and most interestingly, a large gap is observed in the Agulhas Plain, in what would otherwise have been an uninterrupted band of high PD along the western and southern coasts of the Cape region, from Vredenburg to Port Elizabeth (Fig. 9.2a); this gap is not present in the analysis of Forest et al. (2007) (Fig. 9.2b). It is not yet entirely clear what environmental or ecological factors could explain the lower amounts of PD in the Agulhas Plain. The region is renowned for the presence of high numbers of putatively neoendemic species resulting from a complex mosaic of different edaphic conditions that has led to high level of edaphic specialisation (Cowling and Holmes 1992).

Large numbers of these young species would have resulted in relatively low amounts of PD and PD per genus in the area, explaining the observed gap in PD. Since this gap is found more or less at the boundary between the winter and summer rainfall regimes, it can also be hypothesised that climatic fluctuations in the Pleistocene (Dynesius and Jansson 2000) may have affected the geographic boundary between the two rainfall regimes, creating a relatively unstable environment. Here, speciation would have been diminished or extinction would possibly have been more prominent than in other parts of the Cape region.

Additional work is underway to further dissect the phylogenetic diversity patterns in the Cape region of South Africa and to shed light on the potential processes that are responsible for their existence. As mentioned above, examining PD patterns at the species level rather than at the genus level for the entire flora would most likely produce a much more accentuated cline in PD values between the western and eastern parts of the regions. Preliminary studies using species-level phylogenetic data with species distribution modelling have found generally similar patterns to the genus-level patterns, but with some noticeable discrepancies (Colville et al. [In prep](#)). To assess the phylogenetic diversity patterns at the species level within the region, given that producing a species-level tree of the whole flora would be a mammoth task, Colville et al. ([In prep](#)) restricted their analysis to a set of large clades that are assumed to have first diversified in the region and for which at least half of species are found in the Cape (21 clades, representing 3,585 species, 38% of the CFR flora). These groups have been dubbed the “Cape clades”, a concept initially proposed by Linder (2003) for the CFR and later modified and expanded by Verboom et al. (2014) for the broader Cape region, who defined Cape clades as having a Cape origin and comprising ten or more species endemic to the GCFR. This definition does not, however, disregard groups with a reasonable number of endemic species that also comprise a large number of species that diversified outside the Cape region following dispersal from the Cape (e.g. Gnaphalieae, Asteraceae; Verboom et al. 2014). Results from these analyses corroborate the idea that more accentuated patterns are found at the species level than those first observed at the genus level (Colville et al. [In prep](#)). Furthermore, after controlling for species richness, Colville et al. ([In prep](#)) showed that the core winter-rainfall CFR has the highest phylogenetic diversity values.

There is a panoply of phylogeny-based metrics that have been developed based on the concept of phylogenetic diversity (Tucker et al. 2017). Phylogenetic endemism (PE) integrates weighted endemism (species weighted by the inverse of its distribution range) and phylogenetic diversity and is the sum of the branch length divided by the clade range for all the branches connecting a set of taxa to the root on a phylogenetic tree (Rosauer et al. 2009). We calculated phylogenetic endemism, as implemented in the R function *phylogenetic.endemism.R* (Guerin and Lowe 2015), for the Cape region using again the available genus-level phylogenetic tree (Forest et al. 2007) and QDS presence/absence distribution data. Three hotspots of phylogenetic endemism were identified: the Cape Peninsula and the Hottentots Holland mountains; the boundary between the Outeniqua and Tsitsikamma mountain ranges; and near Port Elizabeth (Fig. 9.3). The presence of QDS with high PE values around

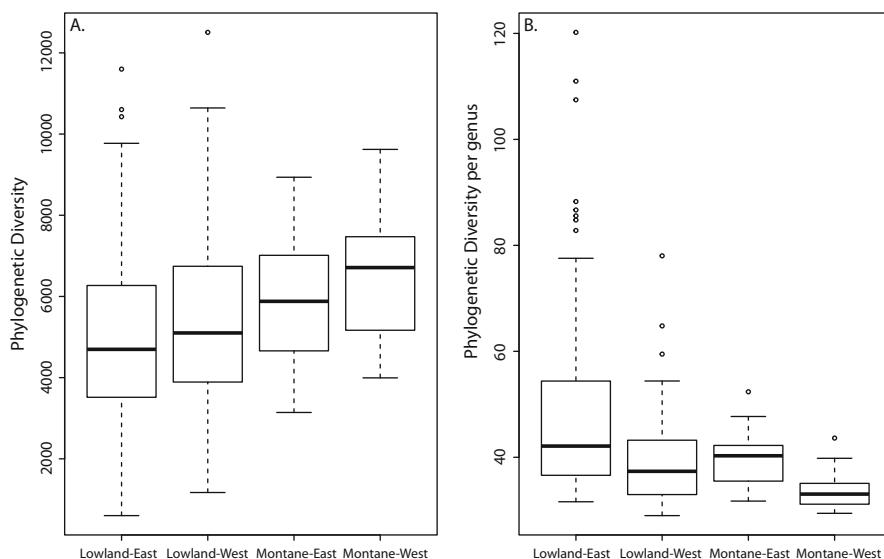


Fig. 9.3 Distribution of (a) PD and (b) PD per genus in QDS dominated by either eastern lowland ($N = 72$), western lowland ($N = 71$), eastern montane ($N = 30$) or western montane ($N = 28$) habitats. Habitat types were assigned to QDS as in Cowling and Lombard (2002), and PD is as calculated in Forest et al. (2007). LE, lowlands (east); LW, lowlands (west); ME, montane (east); MW, montane (west)

the Cape Peninsula and the Hottentots Holland mountains was to be expected given the high numbers of narrow endemics found there (e.g. Cowling et al. 1992; Helme and Trinder-Smith 2006) and the high numbers of species from few families and genera (Manning and Goldblatt 2012). While the genera found in the two other PE hotspots of the eastern part of the Cape are generally restricted to these areas within the Cape region, they also occur outside the Cape region. These two high PE centres could therefore be likely artefacts resulting from the delimitation of the study area in the case of the CFR, although some argue that the eastern boundary of the CFR may lay further east (Colville et al. 2014). A comprehensive analysis at the country scale would likely provide a more definite answer regarding the validity of these two PE hotspots found in the eastern part of the Cape region.

The use of species distribution modelling to investigate phylogenetic diversity patterns is a relatively novel approach that has not been extensively adopted. Unsurprisingly perhaps, some studies have shown that using distribution ranges assessed through species distribution modelling can result in a very different outcome in terms of phylogenetic diversity patterns. For example, more than half of the legume (Fabaceae) collections of Madagascar recorded in a database comprising almost 12,000 accessions have been made within 10 km of major roads. Consequently, if this distribution data is taken at face value, the highest concentration of PD for this group of plants in Madagascar is found along the major roads and cities

of the island (Buerki et al. 2015). When phylogenetic diversity patterns in Malagasy legumes are examined using modelled distribution data, high PD is found on the central plateau and in the wet tropical areas of the north of the island, quite a different picture than the patterns uncovered using traditional presence/absence data (Buerki et al. 2015). The use of modelled species distribution also allows the prediction of future species distributions using climate change projections, which in turns allows the study of the effects of climate change on phylogenetic diversity. A study focussing on South African animal and plant groups has found that losses in PD are not significantly higher or lower under a climate change scenario than if extinction occurred randomly (Pio et al. 2014), which has obvious consequence for the prioritisation of potentially climate-threatened species.

One particular pattern within the Cape region that has received attention is the longitudinal (west to east) gradient of declining plant species diversity, particularly in Cape clades. About twice as many species per unit area are found in the western part of the region compared to those occurring in the eastern part (Cowling and Lombard 2002). Likewise, higher concentrations of species assigned to the largest genera or clades found in the Cape region are found in the western part of the region (e.g. Cowling and Lombard 2002; Levyns 1964; Ojeda 1998; Oliver et al. 1983). The obvious environmental factor that is correlated with this longitudinal diversity gradient is the different rainfall regimes; the western region is dominated by a winter-rainfall regime, while the eastern region has a non-seasonal regime (Cowling and Lombard 2002; Procheş et al. 2005). A winter-rainfall, Mediterranean-type climate likely persisted in the western region throughout the Pleistocene (Braun et al. Submitted) with considerable northwards expansion during the long glacial periods that prevailed (Engelbrecht et al. Submitted). This stability could have facilitated the persistence and even expansion of the flora during the Pleistocene (Dynesius and Jansson 2000; Midgley et al. 2001; Midgley et al. 2005). However, Pleistocene climatic stability was much lower in the eastern region where rainfall amount and seasonality showed dramatic shifts, leading to rapid biome switches (Bar-Matthews et al. 2010; Braun et al. Submitted; Engelbrecht et al. Submitted). Changes unfavourable for Cape clades would have produced potentially higher rates of extinction than in the more stable western region where higher diversification rates produced a richer flora, thus explaining the west-east geographical gradient (Cowling and Lombard 2002; Cowling et al. 2017).

This gradient has been recognised for decades (e.g. Oliver et al. 1983; Weimarck 1941), but one botanist, Margaret R. Levyns, has been instrumental in assessing its importance and the potential factors that have influenced its development (Levyns 1952, 1964). The importance of her work in identifying this pattern has been recognised with the formulation of the Levyns' Law: "all clades endemic (or nearly so) to the GCFR show a westwards concentration in regional scale species diversity and a decline eastward, concomitant with a decline in the proportion of winter rainfall" (Cowling et al. 2017). Cowling et al. (2017) explore three main hypotheses to try to explain the pattern depicted by Levyns' Law. The first one is the metabolic theory, which states that assemblages in the eastern part of the Cape region would have higher diversity due to higher temperatures and metabolic rates, and

shorter generation times, but the clearly higher number of species assigned to Cape clades in the west compared to the east refute this hypothesis (Cowling et al. 2017). The second hypothesis, ecological opportunity, postulates that high speciation rates (coupled to low extinction rates) due to higher amounts of available resources and greater niche axes will promote adaptive radiation and thus higher species diversity. Under these premises, one would expect that the eastern part of the Cape region is more diverse as a result of lower seasonality and higher productivity (Cowling et al. 2017). Again, this is not upheld by Levyns' Law because there is no significant difference between the west and the east among any of the potential ecological factors examined to date that could affect speciation in the Cape (e.g. topography, pollination). Finally, the age and area hypothesis assume that the longer a biome has been in existence and the larger its area occupancy is, the more diverse it will be, due to its geographical and temporal stability allowing more opportunities for speciation and less extinction (Cowling et al. 2017). Of the three hypotheses, age and area is the only one with some evidence that could explain Levyns' Law, but current data supporting this remain limited (Cowling et al. 2017). Nevertheless, a recent study based on phylogenetic and modelled distribution data for several Cape clades (> 9,000 taxa) showed that biome stability, a component of the age and area hypothesis, had a strong effect on most diversity metrics examined but that some aspects of the ecological opportunity hypothesis also had strong effects on several metrics of diversity. This suggests that the explanation for Levyns' Law might require a contribution from both hypotheses (Colville et al. [In prep](#)).

While the west-east longitudinal gradient in the Cape region has attracted a great deal of attention, the elevation gradient in diversity has been less studied. However, the complex topographic heterogeneity of the Cape region has been identified by several studies as a key factor promoting speciation and responsible for the high species richness of the region (e.g. Thuiller et al. 2006; Linder 1985; Verboom et al. 2015; Britton et al. 2014). Verboom et al. (2015) further showed that species found in high-elevation habitats generally have more narrow distribution ranges than species found in lowland areas and that the ranges of sister species in these mountainous areas are more often non-overlapping (i.e. allopatric; see also Britton et al. 2014). There is also evidence that the montane ecological niche in the Cape is phylogenetically conserved (Verboom et al. 2015).

A topographical pattern, i.e. greater diversity in montane than lowland habitats or vice-versa, was recorded for rare species in the western part of the Cape where there is greater diversity in mountain areas, while it was not uncovered for rare species found in the eastern part (Cowling and Lombard 2002). We revisited this pattern by assigning each QDS as either lowland or montane based on the dominant habitat present in each, following Cowling and Lombard (2002), and examine if phylogenetic diversity (obtained from Forest et al. 2007) varied significantly in this admittedly coarse dichotomy of high- and low-altitude environments. We find that PD is higher and more narrowly spread in the montane environments, with ca 90% of the QDS in montane environments falling within a relatively narrow range of total phylogenetic diversity (Fig. 9.4), while ca 60% of the QDS identified as lowlands are also found in this range. This indicates higher variability in PD among QDS

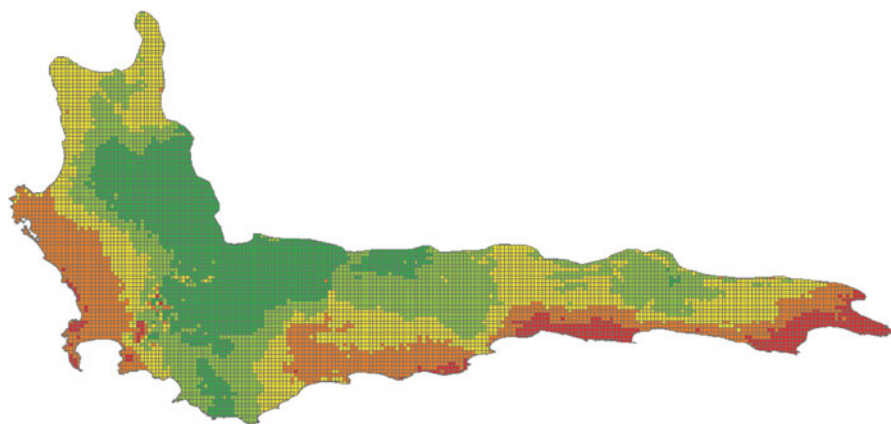


Fig. 9.4 Phylogenetic diversity patterns in the Cape region using the published phylogenetic tree of the Cape flora of Forest et al. (2007) with species distribution models for all Cape clade species found in the area at a resolution of 2 min (see text for details)

found in lowland areas, with some QDS having little PD and others with much higher amounts (Fig. 9.4). While PD per genus in montane habitats across the Cape is lower than in lowland habitats (mean of 35.3 Ma vs 40.4 Ma), the difference between montane and lowland habitats in the west is slightly larger than in the east (average PD per genus: east lowland, 42.1 Ma; east montane, 40.3 Ma; west lowland, 37.4 Ma; west montane, 33.1 Ma). As for PD, the average amount of PD per genus in lowland habitats is more widely spread than in montane habitats. Montane environments in the western part of the Cape region have much less PD per genus than any of the other three categories (Fig. 9.4). This supports the findings of Cowling and Lombard (2002) who observed a topographical pattern only in the west, where the mountains harbour more rare species. Likewise, the niche conservatism inferred in mountain species (Verboom et al. 2015) could be the factor responsible for the lower PD per genus found in montane areas of the western part of the Cape because mountain assemblages would comprise more closely related genera and species than those found in lowland areas. However, given that the difference in PD per genus between montane and lowland habitats is not as significant in the east, topography alone might not be able to explain these discrepancies. Additional studies investigating phylogenetic diversity patterns at the species level with finer-scale distribution data (or species distribution models; see above) would be needed.

It is also unclear if patterns recorded for plants would be matched in faunal groups. For example, the large number of paleorelictual invertebrates representing often deep phylogenetic nodes and confined to Cape montane regions (e.g. Switala et al. 2014; Picker and Samways 1996; Sharratt et al. 2000) suggests that these would be important areas of phylogenetic diversity and endemism. Western coastal areas of the broader Cape region are also considered to house several paleorelictual

invertebrate groups with high niche conservatism (Endrödy-Younga 1978), and therefore faunal patterns of phylogenetic diversity along altitudinal gradients may show similar and contrasting patterns to that of plants.

Phylogenetic Diversity Patterns in Animals

While much work has taken place (and is ongoing) on the phylogenetic diversity patterns of the Cape region flora, fewer studies have investigated similar patterns in faunal groups, and none in fungi that we are aware of. Phylogenetic diversity patterns were explored in the chameleon genus *Bradypodion* (Tolley et al. 2008). This work identified areas with higher *Bradypodion* phylogenetic diversity in the Cape region, with a few half-degree grid squares scattered in the western part of the region and an area covering a large part of the eastern part of the Cape. The other main centre of evolutionary diversity for this group is found in the Maputaland-Pondoland-Albany hotspot. While species richness and phylogenetic diversity were found to be correlated in South Africa for this group, several grid squares showed greater PD values compared to their associated species richness (Tolley et al. 2008).

Peron and Altwegg (2015) explored what they called “taxonomic dispersion” in South African passerine birds using a phylogenetic tree derived from multiple phylogenetic studies and taxonomic classifications; branch lengths were defined based on taxonomic levels. They calculated this taxonomic dispersion using the Near Relatedness Index (NRI; Webb et al. 2002). Negative NRIs indicate taxonomically clumped assemblages, while positive NRIs identify assemblages that are more taxonomically dispersed; assemblages with NRIs close to zero are considered “taxonomically even” (Peron and Altwegg 2015). Generally, they found that the NRI values in the western part of the Cape region were more often negative, while those found in the eastern part of the region were more often positive. With passerine assemblages in the western part of the Cape region overclustered compared to the overdispersed ones found in the eastern part, these findings support those obtained from phylogenetic diversity studies of plants. Peron and Altwegg (2015) also investigated the taxonomic dispersion within biomes and found that Grassland and Karoo biomes had the highest levels of taxonomic clumping under a null model assuming that the source pool comprises all the species on the subcontinent. Under the other null model considered, i.e. the species pool spans the species found within a 100 km radius of the focal grid, most biomes are more taxonomically dispersed, in particular the Karoo biome. The Fynbos is taxonomically clumped under both models, while Thicket and Forest biomes are taxonomically overdispersed (Peron and Altwegg 2015), which again support findings obtained for the flora (Procheş et al. 2006; Verboom et al. 2014). The studies described above summarise the extent of the published works examining phylogenetic diversity patterns in the fauna of the Cape region.

South Africa is also a hotspot for reptiles, with more than 400 species found in the country. Using a DNA data set comprising 91% of terrestrial reptiles found in

South Africa and modelled distribution data based on more than 140,000 distribution records, Smid and Tolley (2017) produced the first phylogenetic diversity map for a major group of vertebrates in South Africa. They investigated specifically skinks and lacertids and identified a centre of ancient diversity (“museum”) in north-eastern South Africa for both groups, while recent diversity centres (“cradles”) were found in the arid west and north-west of the country for skinks and in the Great Karoo for the lacertids (Smid and Tolley 2017). They expect that the north-east part of the country will act as a museum for most groups of reptiles but that cradles will be found across the country depending on the groups. The Cape does not appear to be a centre of phylogenetic diversity for reptiles based on the current available information, but data for other faunal groups, such as frogs (Holt et al. 2013), birds (Colville et al. 2014) and invertebrates (see above), suggest that the Cape could harbour important phylogenetic diversity. Other important large-scale studies on patterns of phylogenetic diversity for various groups of insects such as bees, beetles and butterflies are ongoing (J. F. Colville, pers. comm.).

Surrogates for Faunal Diversity

Conservation practitioners, faced with the problem of scarce resources and a sense of urgency (i.e. “the agony of choice”; Vane-Wright et al. 1991), have been looking for approaches and indices that can facilitate the selection of areas and species that should be prioritised for conservation. The identification of surrogates that can represent total biodiversity, or one of its component, has received a fair amount of attention in the literature (e.g. Faith and Walker 1996; Grantham et al. 2010; Hanson et al. 2017; Nic Lughadha et al. 2005), and this is particularly appealing to many as a large part of the Earth’s diversity remain undiscovered, especially in certain groups such as fungi and insects. Furthermore, the search for surrogates is fuelled by the anticipation that these will save time and resources compared to more exhaustive approaches such as in-depth and repeated biological surveys. Phylogenetic diversity metrics have also been touted as potential surrogates for biodiversity and several of its aspects, including species diversity and functional diversity.

Given the extensive knowledge accumulated to date on the flora of the Cape region coupled to our relatively limited knowledge of its insect diversity and the generally recognised close association between plants and insects, one would anticipate that plant diversity may be a good proxy of insect diversity in the Cape region. Although Fynbos has often been thought to have a particularly poor insect fauna compared to its floral diversity (Giliomee 2003; Johnson 1992; Braschler et al. 2012), insect diversity in this biome is equivalent to the diversity found in other biomes and maybe even higher (Procheş and Cowling 2006). If a component of plant diversity in a given area could adequately represent the insect diversity in the same area, it would have important implications for biological surveys, especially in the context of conservation planning. Procheş et al. (2009) investigated the efficiency of various plant biodiversity metrics (i.e. species richness, genus richness, phylogenetic

diversity) as potential predictors of insect diversity in the Cape region. While the relationships between plant biodiversity metrics and insect richness varies depending on taxonomic ranks and across spatial scales, plant phylogenetic diversity was found to be the best surrogate for insect diversity among the evaluated metrics, although genus richness was also an overall good predictor (Procheş et al. 2009). This was somewhat contradicted by results from a study focussing on family Restionaceae, which show that species diversity was a better surrogate of herbivorous insect diversity across a series of 30 plots positioned in the western Cape region than plant phylogenetic diversity (Kemp and Ellis 2017). Despite the limited amount of studies that have investigated the links between insect and plant diversities in the Cape region, both with and without a phylogenetic framework, it appears that this relationship is not entirely due to a similar response to environmental pressures (e.g. Kemp and Ellis 2017). Clearly, a lot more work is needed to dissect the association between plants and insects, including a broader phylogenetic framework than at the genus level or limited to a single family, as well as exploring different guild of insects such as pollinators, for example.

Phytogeographical Zone Definition

Delimitation of phytogeographical centres within southern Africa has a long history (Werger 1978), although these early biogeographical studies generally did not relate to distribution patterns in terms of ecology and history and rather confined their approaches to historical biogeography. A further shortcoming of these early studies was the largely intuitive approach in examining species distribution maps with the lack of empirical techniques, essentially making these studies unrepeatable (Bradshaw et al. 2015). Nonetheless, these early studies were invaluable in subdividing Africa into a number of clearly defined phytogeographical zones (e.g. White 1976). The development of modern analytical techniques has made it possible to analyse large taxonomic and distribution data sets more objectively to identify biogeographical patterns (e.g. Linder et al. 2012). Advanced clustering techniques for delimiting centres of endemism have further been undertaken for several plant groups within southern Africa (e.g. Bradshaw et al. 2015).

More recently, studies have turned to phylogenetic tools to incorporate phylogenetic information directly in the delimitation of biogeographical regions. Using a large phylogenetic tree comprising more than 1400 species of trees from southern Africa (Maurin et al. 2014) and associated distribution data at a coarse scale of 50 km 2 grid cells, Daru et al. (2016) used a phylogenetic beta diversity approach to identify distinct biogeographical units within the woody flora of southern Africa. They identified 15 “phyloregions”, or biogeographically distinct units, that largely correspond to recognised biomes and vegetation types within southern Africa but also that emphasise the potential existence of additional previously unrecognised phytogeographical regions. Their approach also identifies an area that broadly corresponds to the GCFR (although with more limited coverage of the eastern part of the GCFR and

despite the bias of their data towards woody species) and identifies it as the most evolutionary distinct biome in this study (Daru et al. 2016). These authors evaluate in a subsequent paper how these phyloregions evolved through time and space to “determine the phylogenetic depth at which the signal for their present-day structure emerges” (Daru et al. 2018). These analyses support a Miocene origin for their Fynbos phyloregion (which corresponds broadly to the GCFR), although many elements typical of this vegetation type are known to be much older (e.g. Linder 2005).

Although not specifically aimed at identifying phytogeographical regions, Daru et al. (2015) compared the distribution of diversity across southern Africa using a suite different metrics, including phylogenetic diversity and phylogenetic endemism, based on the same data set of tree species (Maurin et al. 2014), and found a large level of incongruence of the areas recovered as hotspots between these various metrics. Almost three-quarters of the hotspots were found to be unique to a particular diversity metric, and only one, the mean species endemism, which measures the distribution of restricted endemic species, identifies the western part of the Cape as a biodiversity hotspot (Daru et al. 2015). Phylogenetic diversity and phylogenetic endemism do not identify the Cape region as a biodiversity hotspot in the context of southern Africa. However, such a result might be expected given that this analysis is based on tree species and that the Cape region, especially the western part, comprises very few tree species, as the region is dominated by Fynbos, which is mostly composed of sclerophyllous shrubs.

Concluding Remarks

The Cape region is an ideal case study for the investigation of biodiversity patterns, particularly in plants, and offers unique opportunities to test hypotheses related to the evolutionary processes that are responsible for these patterns. The relatively small size of the region makes it comparatively easy to survey in its entirety. Consequently, decades of persistent efforts by generations of botanists have resulted in very exhaustive checklists and group revisions of the flora of the region (e.g. Manning and Goldblatt 2012; Snijman 2013). Some believe that less than one percent of the flora remain to be discovered, mainly comprising narrow endemic species of high conservation value (Treurnicht et al. 2017). Many components of its fauna (e.g. reptiles, birds, butterflies) are also well-known. These comprehensive lists are partly possible due to the wealth of comprehensive species-level distribution data for the region (a large part of which is georeferenced although with varying levels of accuracy), which is particularly remarkable compared to what has been accumulated for many regions of the world. This abundance of distribution data in turns allows for more reliable species distribution modelling, resulting in finer-scale evolutionary patterns, which are, in principle, better suited for their integration in conservation planning.

We refer above to a number of ongoing studies investigating the evolutionary patterns in plants and animals from the Cape region. These new surveys expand on the previous works in terms of coverage (some extending to the whole country), finer-scale distribution data and the use of novel approaches integrating species distribution modelling and occurrence probabilities into phylogenetic diversity metrics. Nevertheless, several aspects of the regional evolutionary patterns in the Cape will require more in-depth investigations. Despite some recent or ongoing projects on reptiles and certain insect groups, faunal phylogenetic diversity patterns remain scarce compared to those produced for the flora, and it is still unclear if the Cape region would represent an area of important PD for faunal groups. While phylogenetic diversity patterns within the CFR have been examined, as reported here, patterns in the Succulent Karoo are yet to be investigated; this would lead to a comprehensive survey of evolutionary patterns for the GCFR, which is currently not available. Other phylogenetic diversity-based metrics such as phylobetadiversity (Graham and Fine 2008), phylogenetic endemism (Rosauer et al. 2009) and expected phylogenetic diversity (Faith 2008) have not been investigated (or only scarcely) within the Cape or in relation to surrounding biomes.

The advent of relatively new high-throughput sequencing technologies will bring a new suite of tools, such as metagenomics (the investigation of genetic material recovered from environmental samples such as soil and water), allowing a more in-depth investigation of biodiversity patterns within the region. These, coupled to remote-sensing and spatial analysis approaches, as well as large-scale digitisation initiatives of natural history collections, will offer a plethora of new opportunities to explore biodiversity patterns in the Cape and elsewhere and consequently help protect biological diversity in the face of the environmental changes it is facing.

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Chapter 10

Effects of Phylogenetic Diversity and Phylogenetic Identity in a Restoration Ecology Experiment



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Abstract Our understanding of the effects of plant biodiversity on ecosystem function rests in large part on experiments that have disentangled environmental variables from local diversity. Yet phylogenetic diversity (PD) effects can be confounded by phylogenetic identity effects in such experiments if assemblages with low or high PD tend to be dominated by a single clade. We illustrate this problem in a 127-species experiment designed to test the effects of angiosperm PD and trait diversity on tallgrass prairie restoration outcomes. In this experiment, the taxon pool exhibits a phylogenetic bias: if species were randomly assigned to experimental assemblages, low PD plots would frequently be dominated by a single

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clade (the sunflower or daisy family, Asteraceae). We present a visualization tool for examining phylogenetic experiments for this bias and propose a taxonomically constrained experimental design to reduce the most egregious causes of bias. We then present the experimental design we developed using the constrained approach and summarize initial findings from this large-scale restoration experiment. Entanglement of phylogenetic diversity and phylogenetic identity is an underappreciated and likely widespread challenge for PD experiments, particularly those that draw upon a large number of candidate species. By recognizing, quantifying, and counteracting this bias, researchers can better differentiate the effects of PD *per se* from phylogenetic identity effects.

Phylogenetic diversity (PD; Faith 1992) as a conservation criterion has been hypothesized to help maintain biodiversity (Rosauer and Mooers 2013), preserve evolutionary potential and future economic benefits in biodiversity hotspots (Forest et al. 2007), protect communities with greater functional diversity (Mouquet et al. 2012), and deliver greater net ecosystem function (Srivastava et al. 2012). In contrast, phylogeny has yet to be embraced as a tool in ecological restoration (Hipp et al. 2015). A review of 190 restoration ecology studies (Brudvig 2011 and pers.comm.) found that 88% assessed biodiversity in some way. Of those, the vast majority focused on species-level diversity, 11% investigated functional diversity, and only one assessed genetic diversity; none assessed phylogenetic diversity. Moreover, few researchers have even discussed the potential for phylogeny to influence restoration outcomes (though see Cavender-Bares and Cavender 2011; Montoya et al. 2012; Verdú et al. 2012). This is a critical gap, given strong relationships between phylogenetic diversity and ecological function at the community level (e.g., Ch. 9 in Cadotte and Davies 2016).

Moreover, ecological restoration is expensive. Any savings in maintenance efficiency and other costs that can be made through improvements in design have the potential to compound over many years and many ecological restorations. For example, in our study system, tallgrass prairie of the upper Midwest, the cost of establishing and maintaining tallgrass prairie in an existing 10-acre turfgrass field is approximately \$160,000 USD for a 10-year period (url: <http://pizzonorsergy.com/wp-content/uploads/Turf-Prairie-Cost-Comparison-2.pdf>, accessed 2017-08-28). Incorporating phylogenetic considerations into restoration planning and design has the potential to increase the resilience and function of ecological restorations, potentially saving money and reducing the risk of failure.

With this in mind, we are investigating the effects of phylogenetic diversity on ecological restoration outcomes—maintenance of species diversity over time, resistance to invasion, productivity, and effects on soil quality and soil microbial diversity—in a tallgrass prairie restoration experiment in northeastern Illinois (USA). Ecological experiments allow us to isolate local species diversity from fine-scale environmental variation, and such experiments have been instrumental in our understanding of the effects of plant biodiversity on ecosystem function

(Tilman et al. 2012; Wright et al. 2017; Weisser et al. 2017). Tallgrass prairie of the upper Midwest was a natural choice for this experiment for several reasons. It is one of the most endangered ecosystems in North America: less than 0.01% of the original tallgrass prairie remains intact in the Illinois (White 1978; Iverson 1988), and < 1% of the original tallgrass prairie and savanna remains in the upper Midwest overall (Leach and Givnish 1996; Kline 1997). Tallgrass prairie is native to the area where we work and supports high diversity at both landscape and local scales, where average plant species richness may exceed 17 species in a single 0.25 m² plot (Bowles and Jones 2004). Due to the high species density at the plot scale (<1 m), tallgrass prairie offers the opportunity to investigate the outcomes of competitive interactions at fine spatial scales, the scales at which restoration ecologists and landowners in our area generally operate. Grasslands are globally important in terms of biodiversity, food production, and ecosystem services such as water cycling and carbon sequestration (Risser 1988; Jones and Donnelly 2004; Li et al. 2017). Finally, prairie restoration research has been an important contributor to our understanding of North American restoration since the 1930s, especially in densely populated suburban and urban environments (Court 2012). Yet restored prairies in general are lower in phylogenetic and native species diversity than nearby remnants (Barak et al. 2017), suggesting that we have more to learn about how to restore them. Lessons learned in the tallgrass prairie system may have broad applicability to global grassland restoration efforts.

In this chapter, we describe our experiment from concept to implementation. We lay out first the conceptual design and our concern with separating phylogenetic diversity from phylogenetic identity, an issue that is amplified as more species are incorporated into experiments. We then describe the experimental design, including species selection, site preparation, soil attributes, and blocking, and assignment of species mixes to treatment plots. Finally, we report on first outcomes of the experiment: survivorship and species abundance change in planted plots and seedling emergence in seeded plots.

Quantifying Phylogenetic and Functional Diversity and Varying These Experimentally

Assembled from the flora of the eastern deciduous forests, Rocky Mountains, and Great Plains, the tallgrass prairie draws broadly from both the angiosperm tree of life and phytogeographic regions of the USA (Gleason 1922; Larkin et al. 2015). Its plant diversity disproportionately arises from the large plant families Asteraceae (the sunflower or daisy family), Poaceae (the grass family), Fabaceae (the bean family), Rosaceae (the rose family), and Lamiaceae (the mint family) (Larkin et al. 2015).

In remnant prairies of Illinois, there is variance among families in their effects on phylogenetic diversity. For example, occurrences of species from the orders Asterales, Apiales, and Lamiales tend to have negative effects on mean pairwise phylogenetic distance (MPD) of remnant prairie plots, while species of Cyperales and Rosales tend to increase MPD (Larkin et al. 2015). Thus an experiment manipulating phylogenetic diversity by drawing from a representative pool of tallgrass prairie species could have treatments in which low or high PD plots are disproportionately associated with certain clades, confounding PD (how broadly is the tree of life is sampled?) with phylogenetic composition or identity (which portions of the tree of life are sampled?).

We initially designed our experiment based on a suite of 129 species readily available to tallgrass prairie restorationists, with particularly strong representation of the Asteraceae (41 species), Fabaceae (13 spp.), Poaceae (11 spp.), and Lamiaceae (10 spp.; Fig. 10.1; see Supplemental Methods for phylogenetic methods). It quickly became apparent that drawing random plots would produce the desired wide ranges of PD, but that resultant low PD plots would be dominated by the Asteraceae. Furthermore, we suspected this was a general issue related to phylogenetic distinctiveness, with more evolutionarily distinct species being more strongly associated with high PD. To test this, we generated 2×10^6 mixes, each comprising 15 species to reasonably emulate the diversity of a young restored tallgrass prairie, by drawing at random from our species pool. We calculated phylogenetic diversity (PD, sensu Faith 1992) of each mix. For each species, we then calculated the average PD of all plots in which the species occurs and regressed this against w , a phylogenetic distinctiveness metric we derived from the weights utilized in phylogenetic generalized least squares regression methods (Fig. 10.1; Box 10.1). As expected, w is a highly significant predictor of average PD in this experiment (Fig. 10.2; $r^2 = 0.807$, $p < 0.0001$); that is, phylogenetically distinctive species (from clades that were underrepresented in our experimental species pool, e.g., *Lilium michiganense* [Liliaceae] or *Thalictrum dasycarpum* [Ranunculaceae]) occurred disproportionately in plots with higher than average PD, while the Asteraceae dominated plots with lower than average PD. As a consequence, it would be difficult to tease apart the effects on a plot of being dominated by Asteraceae from the effects of having a low PD species assemblage. Even using standard methods of partitioning the contribution of selection and complementarity to overall diversity effects (Hector 1998; Loreau 1998; Loreau and Hector 2001), one would be left with the question of whether any inferred complementarity effects were actually artifacts of the influence of Asteraceae relative to other families.

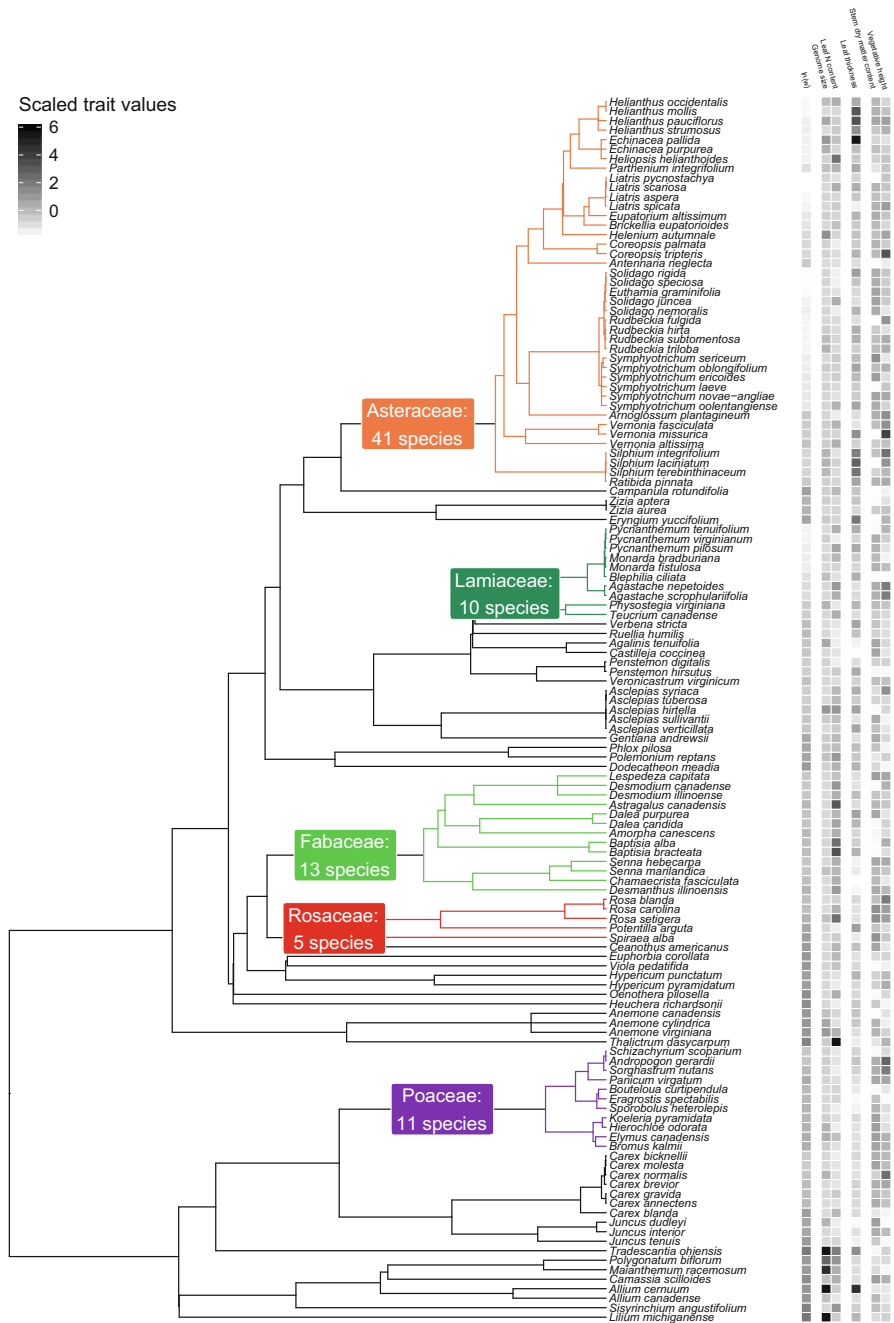


Fig. 10.1 Phylogenetic tree of species included in experiment, with traits and phylogenetic distinctiveness (w). This phylogenetic tree was generated using the ca. 32,000 species tree of Zanne et al. (2014) as a backbone, then splicing taxa, and pruning as needed (see Barak et al. 2017 for full methods) in R (R Core Team 2016), using the ape and phytools packages (Paradis et al. 2004; Revell 2012) and custom functions weldTaxa and make_matAndTree (<http://github.com/andrew-hipp/morton>). Traits are all rescaled to a mean of zero and unit variance for purposes of visualization

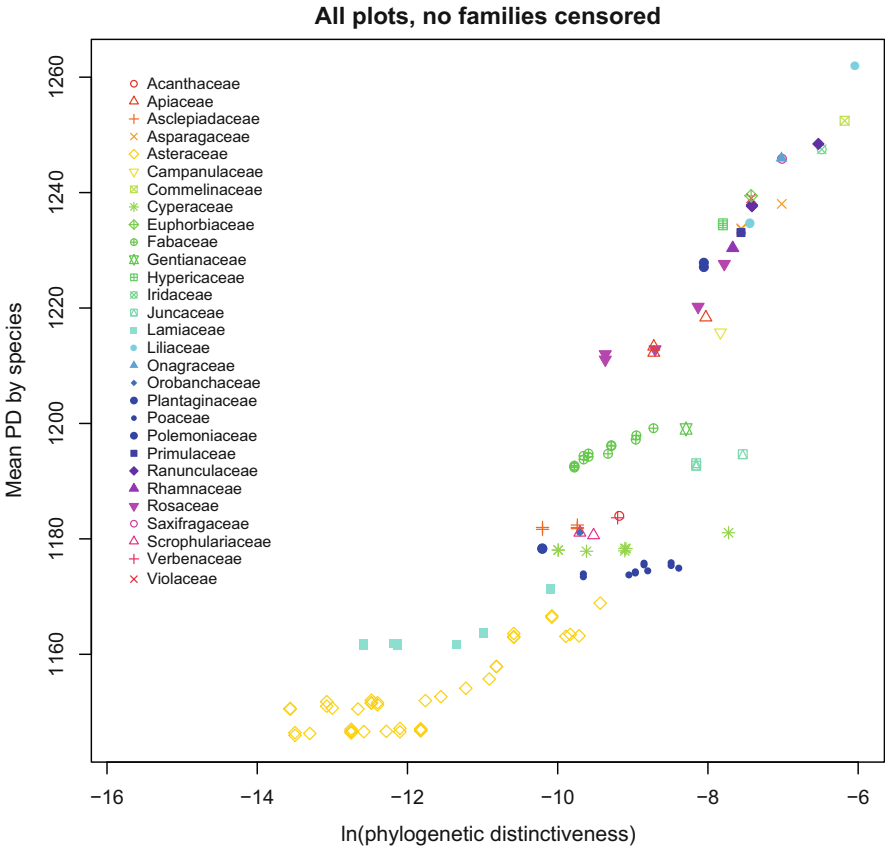


Fig. 10.2 Plot of mean phylogenetic diversity (PD) of all random 15 species assemblages including each of the species in the experiment as a function of phylogenetic distinctiveness, $\ln(w)$. Phylogenetic distinctiveness is here calculated as the natural log of w , a weight matrix derived from the phylogenetic covariance matrix (see Box 10.1): species with higher phylogenetic distinctiveness have, on average, fewer close relatives in our sample of the angiosperm tree of life. Phylogenetic diversity (PD) for each assemblage is calculated as the sum of branch lengths for the tree pruned just to the 15 species in that assemblage. Mean PD for each species is then calculated as the mean PD of all species assemblages in which that species occurs among the 2×10^6 random assemblages of 15 species each

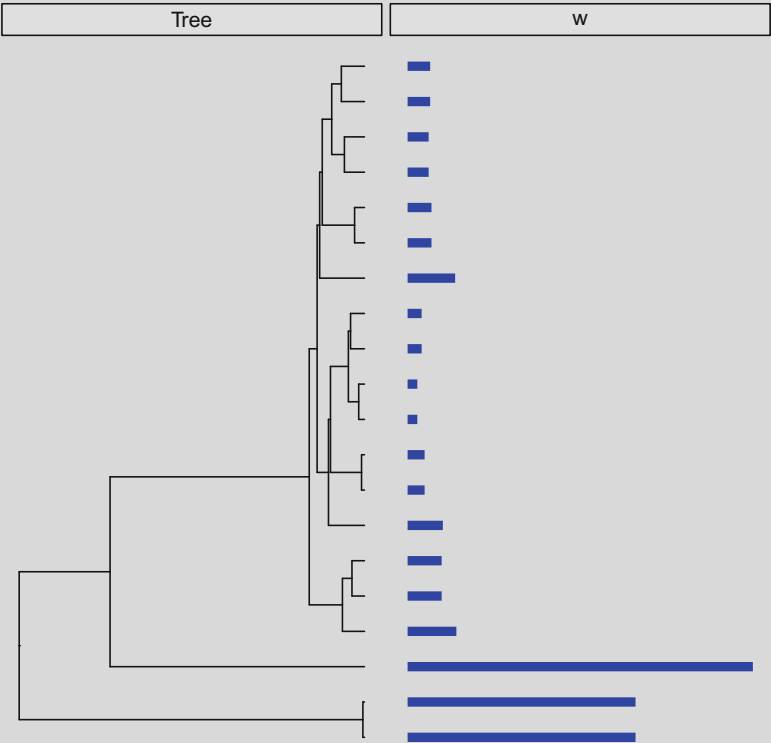
Box 10.1: w , an Estimator of Phylogenetic Distinctiveness Based on Generalized Least Squares

There are multiple metrics of phylogenetic distinctiveness (citations in ch. 9 of Cadotte and Davies 2016), all of which rely on some combination of branch length and/or number of nodes between the root of the tree and the tip being considered. Here we utilize a metric of phylogenetic distinctiveness that we derive from the generalized least squares (GLS) estimator of the root state used in trait evolution, for it is the evolution of traits themselves on the tree of life

(continued)

Box 10.1 (continued)

that makes phylogenetic history a target of interest in understanding functional diversity. The maximum likelihood estimator of the root state of a character evolving according to a Brownian motion process is equivalent to the weighted mean of the tips, using the column sums of the inverted phylogenetic covariance matrix (**C**) as the weights. **C** is the pairwise (square) phylogenetic covariance matrix, in which the value of each off-diagonal cell [i, j] equals the amount of time taxa i and j evolved together before divergence at their most recent common ancestor, and the value of each cell on the diagonal equals the distance from the root of the tree to the tip of the tree corresponding to the taxon intersecting that cell (O’Meara et al. 2006, pg. 923). The higher covariance for taxa that have a longer history of shared ancestry reflects the fact that we expect closer relatives to be more similar to each other than distant relatives. Taking the column sums of the inverted matrix (**C**⁻¹) therefore yields the lowest values for taxa with high average covariance (many taxa with whom they share a long period of evolutionary history prior to divergence). The vector $w = \mathbf{1}' \mathbf{C}^{-1}$, in other words, reflects the proportional contribution we expect of a modern-day species to our understanding of the ancestral condition for the clade we are considering, and thus it strikes us as an apt estimator for phylogenetic distinctiveness (Figure: Box 10.1).



To reduce this correlation, we first tried sampling taxa with a probability that is either an exponential or a linear function of phylogenetic distinctiveness (w). These sampling strategies had the expected effect of increasing mean PD of plots, but the unexpected (and undesirable) effect of maintaining a strong positive correlation between w and mean PD within the monocots plus Ranunculaceae and within the core eudicots, while disproportionately decreasing the mean PD of the monocots plus Ranunculaceae (Supplement, Fig. 10.S1). We then tried a simpler approach of constraining our plots for just the families most strongly associated with low plot-level PD. We conducted trials, in which we constrained Poaceae, Lamiaceae, Asteraceae, and mixes of these taxa to a maximum of three to five taxa. We found through these trials that the simplest constraint that maximally reduced the correlation between w and mean PD, while maintaining the maximum range of PD, entailed limiting our plots to a maximum of three Asteraceae (Table 10.1; Fig. 10.S2).

Because such a large percentage of our species were in the Asteraceae, constraining all families to a maximum of three species per plot had little effect on the relationship between w and mean PD, but doing so slightly narrowed the range of PD in the simulated plots. We therefore generated plots with a constraint only on the number of Asteraceae species (maximum 3). Given the desire to maximize the species number and distribution in this experiment, using species drawn from the angiosperm phylogeny in proportion to species actually present in tallgrass prairie communities, it was impossible to entirely disentangle phylogenetic diversity from phylogenetic identity. Our approach, however, sufficed to ensure that there were multiple ways of obtaining low phylogenetic diversity in the experiment, though this balancing of family diversity may be offset if some families (e.g., Poaceae, Asteraceae) tend to reach higher abundances within plots.

Examining the effects of phylogenetic diversity alone, however, ignores the fact that both convergent and non-convergent traits evolve on the tree of life (see also chap. 1). Thus phylogeny cannot be assumed to track ecological divergence at all phylogenetic scales. It may succeed in integrating over the history of trait evolution when the traits that structure communities are phylogenetically conserved (Kraft and Ackerly 2010) and even outperform trait-based predictors of ecological interactions (Cadotte et al. 2009; Pearse and Hipp 2009). However, phylogeny is not guaranteed to capture meaningful variation in the traits that structure communities. Assessing whether it does requires studying the traits themselves (Gerhold et al. 2015).

We designed our experiment to explicitly examine both PD, as discussed in the previous paragraphs, and functional diversity. Functional diversity was assessed by measuring 6 categorical traits, 12 continuous leaf traits, 8 binary root traits, seed mass, a categorical habitat moisture trait, and genome size (Supplement, Table 10.S1). Traits were selected to emphasize competition for resources that would influence growth, not reproduction, because we wanted to assess diversity effects within the first generation. Thus such traits as flower size and color were excluded; genome size, flowering phenology, and seed mass, however—the latter two of which certainly are reproductive traits—were included as potential proxies for unmeasured life history traits. These traits may of course have excluded functional variation key to understanding competitive interactions in our experiment, and our choice not to

Table 10.1 Effects of different plot constraints on the range of among-plot PD retained and the correlation between w and mean PD of taxa

Constraint	Proportion plots retained	Proportion species unaffected	PD (95% CI)	Width of PD CI	Cor (PD, w)
Asparagaceae constrained	1	0.98	1184.3 (897.4,1462.4)	565	0.893
No families constrained	1	1	1184.3 (897.4,1462.4)	565	0.893
Ranunculaceae constrained	1	0.97	1184.3 (897.4,1462.4)	565	0.893
Rosaceae constrained	0.999	0.96	1184.3 (897.4,1462.4)	565	0.8929
Asclepiadaceae constrained	0.999	0.96	1184.4 (897.6,1462.4)	564.8	0.893
Cyperaceae constrained	0.996	0.95	1184.8 (898.1,1462.6)	564.5	0.8945
Lamiaceae constrained	0.983	0.92	1186.4 (900.8,1463.3)	562.6	0.8886
Poaceae constrained	0.976	0.91	1186.8 (900.5,1463.8)	563.3	0.9042
Fabaceae constrained	0.955	0.9	1184.5 (896.5,1463.5)	566.9	0.8913
Asteraceae constrained	0.232	0.68	1292 (1035.2,1533.6)	498.3	0.6343
Ast + Lam constrained	0.225	0.6	1297 (1046.5,1535.1)	488.6	0.6188
Ast + Poa constrained	0.221	0.6	1298.3 (1047.3,1535.9)	488.6	0.6588
Ast + Lam + Fab constrained	0.205	0.5	1300.7 (1049,1538.5)	489.5	0.6111
Ast + Lam + Fab + Poa constrained	0.195	0.42	1307.7 (1063.8,1541)	477.3	0.6395
All families constrained	0.193	0.23	1309.4 (1068.5,1541.5)	473	0.6454

explicitly weight these characters may mean that traits are not represented in our final estimate of functional diversity in proportion to their importance for community interactions. In our final trait data matrix, 20.9% of data cells were empty. We filled in missing trait data using multiple imputation based on chained equations (MICE; reviewed in van Buuren and Groothuis-Oudshoorn 2011) using predictive mean matching in the mice R package (van Buuren and Groothuis-Oudshoorn 2011). Trait imputation had minimal effect on species clustering in trait space (Supplement, Fig. 10.S3; cf. especially panels C, D). We estimated functional diversity of each plot using functional dispersion, the mean Gower distance of all species from the centroid of the species-level ordination, as implemented in the R package FD (Laliberté and Legendre 2010). Families largely clustered together in trait space

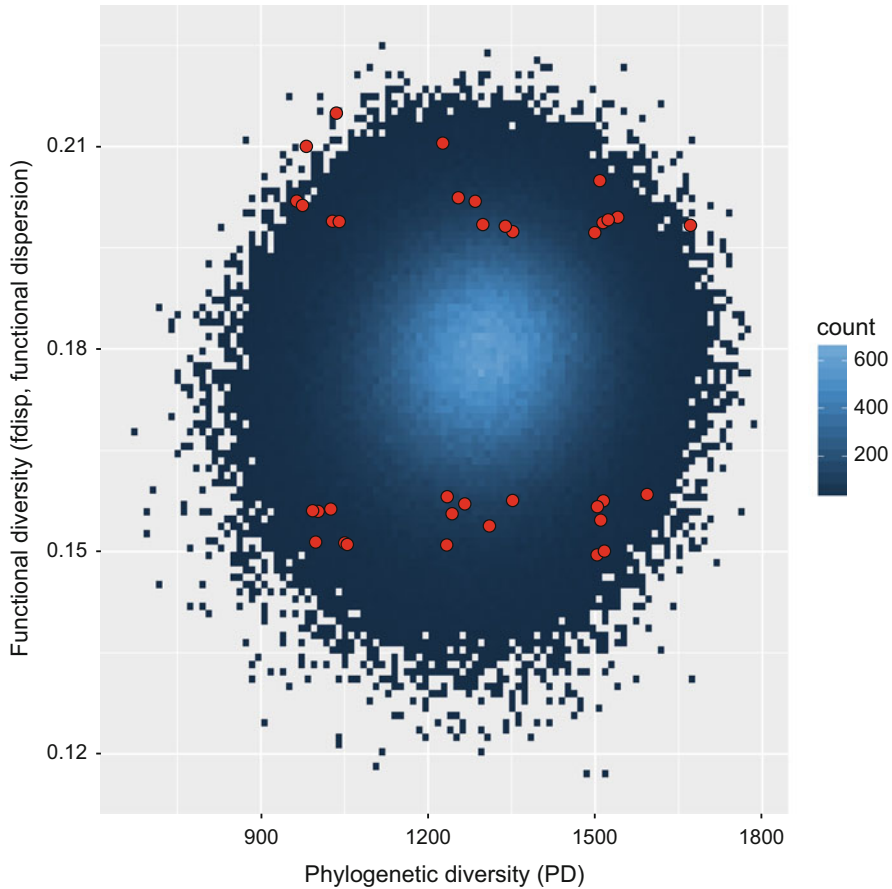


Fig. 10.3 All simulated plots with a maximum of three Asteraceae, plotted in PD/functional diversity space. Red points indicate the 36 species mixes selected for the experiment, at three PD levels and two functional diversity levels. Due to space and design constraints, no plots were sampled from intermediate levels of functional diversity

(Supplement, Fig. 10.S3), and there were two strongly phylogenetically structured categorical traits—life form, which includes the graminoid condition that is unique to sedges, grasses, and rushes (the Cyperaceae, Poaceae, and Juncaceae, respectively); and nitrogen fixation, the predisposition to which appears to have evolved just once in angiosperms (Soltis et al. 1995; Werner et al. 2014), represented in our dataset by Fabaceae and Rhamnaceae. Nonetheless, plotting all simulated 15 species plots in functional diversity and phylogenetic diversity space showed that these two dimensions of biodiversity were significantly but not strongly correlated ($r = 0.235$, $p < 0.0001$ when all 2×10^6 simulated plots are included; $r = 0.058$, $p < 0.0001$ when only the 464,044 plots with a maximum of three Asteraceae are included; Fig. 10.3).

Experimental Methods

The experiment is a six-block, fully factorial design with three levels of phylogenetic diversity and two levels of trait diversity, for a total of six treatments, each represented in the experiment by 6 species mixes, for a total of 36 species mixes. The 36 species mixes included in our experiment were drawn from quadrants of the simulated plots distribution (Fig. 10.3) representing low, medium, and high phylogenetic diversity crossed with low and high trait diversity. In the species mixes utilized, only 127 of the original 129 species included in the design were actually included in any of the experimental plots. Fifteen species were used in each mixture to maintain species richness while varying phylogenetic and functional diversity. Each species mixture was replicated four times, twice in plots planted from seed and twice in plots planted as plugs, allowing us to assess how the same species mix performs under different establishment conditions. Prior to planting the plug plots, all 15 species were assigned random positions within each quarter of the 4 m² plot, so that each quarter plot (1 m²) contains a full set of 15 species. Additionally, each of the 127 species in the experiment was planted in two separate monoculture plots as plugs only so that we could directly estimate species-specific effects in all treatment plots.

Plots are partitioned into six blocks in total, three blocks in each of the two superblocks. Blocks and superblocks were delineated based on preplanting measurements of soil A-horizon depth (Fig. 10.4). In each of the three blocks of each superblock, two species mixes of each of the six treatments were drawn at random and planted as both seed and plug plots, with positions randomized within the block; each superblock thus is a three-block replicate of the entire suite of treatment plots (Fig. 10.4). Within each superblock, all 127 monocultures were planted in random positions; these monoculture plots being evenly apportioned among the three blocks on each side but randomly placed within those blocks. Details of site preparation and installation of the experiment are included in the supplemental methods, along with a photo of the site at approximately the time of vegetation sampling (Supplement, Fig. 10.S4).

Outcomes at End of Year 1

We surveyed the experiment for species diversity twice in its second season. The first survey, conducted ca. 10 months after planting, was an exhaustive survey of planted plugs for survivorship, in which plants were coded as alive vs. dead (a more nuanced coding for plant health was made at the same time but is not analyzed here, as it was found to be observer-sensitive). Seed plots were not surveyed at this time. The second survey, conducted ca. 12 months after planting, was a sample of vegetation cover in 1 m² quarters of the plug plots and seed plots, estimating cover by percentage for all planted species.

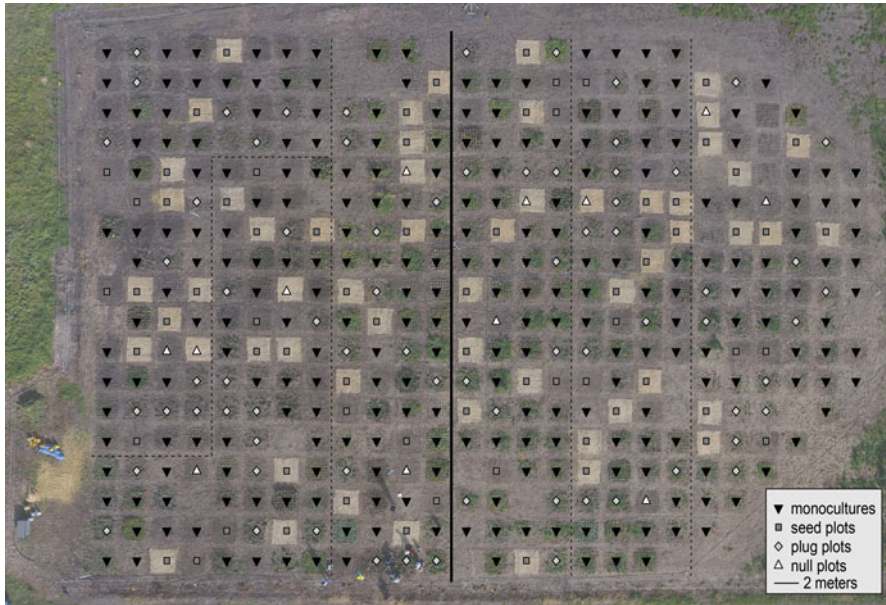


Fig. 10.4 Experiment as designed. Blocks are demarcated by dashed lines and superblocks by the solid line. The background image was taken by drone in November 2016, ca. 3 months after initiation of the experiment. The design is oriented with north at the top of the page. (Photo by Lane Scher, 2016–11-09)

Survivorship in midsummer was 83% in both the monocultures and the mixed plots, and mean survivorship was not affected by block or by whether observations were made in mixture plots or monocultures (two-way ANOVA results for block ($F_{5,300} = 1.623$) and plot species mixture status ($F_{1,300} = 0.002$) were both nonsignificant). Our initial observations suggested that smaller species might be performing more poorly in mixed plots than in the monocultures, and we thought this would manifest itself in a significant selection effect in assessing biodiversity. However, the selection effect (following the approach of Loreau and Hector (2001) as implemented in Tobner et al. (2016)) was weakly negative ($t = -1.907$, $df = 47$, $p = 0.063$), exactly the opposite of what we expected given our observations, while the complementarity effect was weakly positive ($t = 2.057$, $df = 47$, $p = 0.045$). The net diversity effect, however, was not different than 0, and phylogenetic and functional diversity among treatments (analyses not shown) similarly had no effect on survivorship through the winter.

However, abundance/cover and presence-absence data analyzed later in the summer showed an effect of phylogenetic diversity. Percent cover of each of the 15 species planted in each mixture plot or of the single species planted in monoculture plots was estimated by pairs of trained observers using cover classes ten percentage points in width between 10% and 90% and 1–5% wide at below 10% and above 90%. Cover classes were converted to their midpoint percentages for

analysis (e.g., a plant recorded as being in the 10–20% cover class was analyzed as 15%), and data were formatted as binary matrices (presence-absence) and as continuous (abundance/cover) matrices.

Most species planted as plugs were found to be alive during the sampling period. However, we excluded 19 species from analyses presented here, six because they are spring ephemerals that were poorly represented in the August sampling and 13 because they established poorly and had to be planted a second time soon after our August sampling. This is perhaps overly conservative for the seed plots, as many of these species performed well in the seed plots even if they did poorly in the plug plots; however, for comparison between the plug and seed plots, we excluded the same 19 species from all analyses. We also performed sensitivity analyses excluding three genera that were difficult to identify to species in the seed plots (*Carex*, *Senna*, and *Penstemon*), but these analyses did not affect conclusions and are not reported here. For each plot, the phylogenetic diversity and trait diversity as the experiment was designed were treated as predictors known without error, and the following diversity estimates were measured: phylogenetic diversity sensu Faith (Faith 1992; PD), abundance-weighted mean pairwise distance (MPD), abundance-weighted mean nearest taxon distance (MNTD), and the standardized effect size (SES) of each of these; species richness; and both the absolute difference and ratio between observed vs. planted diversity for each measure, conditioning on the 108 species that were not excluded from analysis. Data were analyzed by linear regression and by ANOVA, with block, trait diversity class as designed (low or high), and phylogenetic diversity class as designed (low, medium, or high) as factors. In all analyses, block and trait diversity were nonsignificant factors and are not discussed further here.

The rank order of phylogenetic diversity standardized effect size as planted was maintained in both the plug plots and the seed plots, as estimated using either PD ($r^2 = 0.49$ – 0.86 , $P < 0.0001$) or MNTD ($r^2 = 0.30$ – 0.65 , $P < 0.0001$) (Fig. 10.5). Observed standardized effect size of PD correlated with planted PD significantly. Observed MPD, on the other hand, showed no correlation with planted PD ($r < 0.05$, $p > 0.3$), and planted MPD showed weaker correlation with PD than MNTD did ($r = 0.388$ and 0.773 , respectively, $P < 0.001$; results not shown in Fig. 10.5).

Observed species richness, however, correlated negatively with phylogenetic diversity as planted in both the plug plots ($r^2 = 0.171$, $F_{2,69} = 7.12$, $P = 0.0015$) and the seed plots ($r^2 = 0.153$, $F_{2,69} = 6.23$, $P = 0.0033$). In other words, plots planted with high phylogenetic diversity at the start of the experiment lost a greater percentage of their species richness in their first year. Observed SR correlated positively with observed PD in both plug plots and seed plots ($r = 0.381$ and 0.588 , respectively, $P \leq 0.001$), as expected, but weakly negatively with the standardized effect size of observed PD in the seed plots ($r = -0.238$, $P = 0.0440$). This decrease in species richness held even when starting diversity—which varies among plots due to the fact that we conditioned our analyses on the 108 species that established well and were evident at the time of August surveys—was corrected for by calculating the proportion of SR maintained after excluding 19 species (Fig. 10.6). In seed plots, the lowest PD plots maintained the

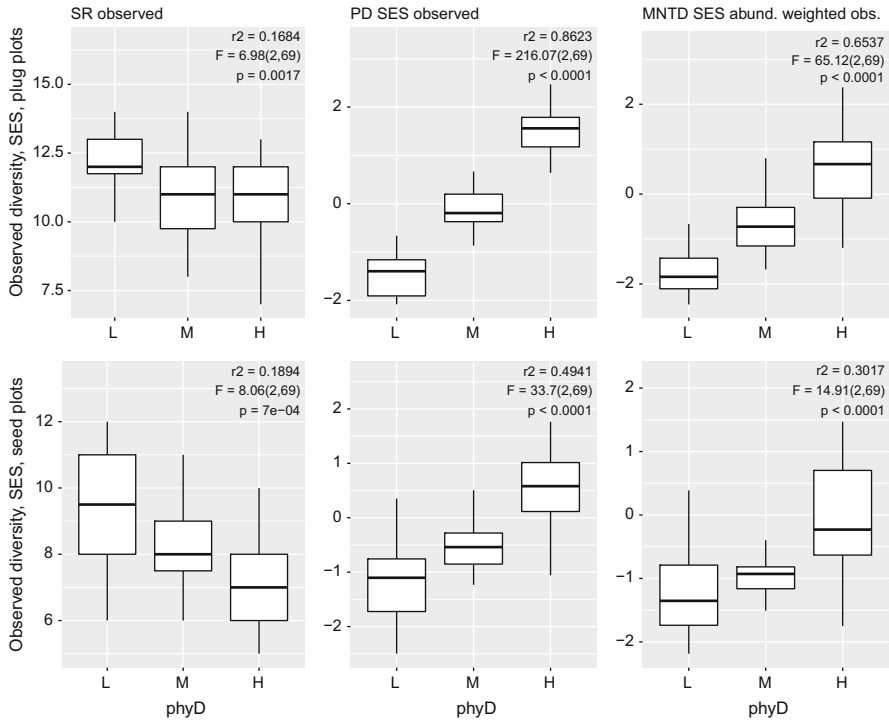


Fig. 10.5 Species richness and standard effect size of Faith's PD and mean nearest taxon distance plotted as a function of designed phylogenetic diversity. Abbreviations: L, low; M, medium; H, high

highest proportion of species richness ($F_{2,69} = 4.91$, $P = 0.0101$), PD ($F_{2,69} = 13.51$, $P < 0.0001$), and MNTD ($F_{2,69} = 4.28$, $P = 0.0176$). A post hoc Tukey test shows that the lowest PD seed plots differ significantly from the high-diversity plots in proportion of SR and of PD maintained ($P < 0.01$) and from the medium PD seed plots in proportion of PD and MNTD ($P < 0.05$).

The selection model used to partition diversity effects into selection vs. complementarity effects was formulated for biomass and is not well suited to cover class data, as cover for multiple species can easily exceed 100% of the plot area, whereas cover for a single species cannot. Thus cover class data should demonstrate a significant bias toward diversity effects. This fact notwithstanding, we employed the diversity effect equation of Loreau and Hector (2001) to detect whether among-plot variation in selection effects, complementarity effects, or net diversity effects were predicted by phylogenetic diversity, trait diversity, or block. None of these were significant predictors of among-plot variation in diversity effects. The mean selection effect was 0.1125 ± 0.1248 (sd), and the mean complementarity effect was 0.7616 ± 0.2505 (sd); both are significantly greater than zero ($t_{71} > 7$, $P < 0.001$),

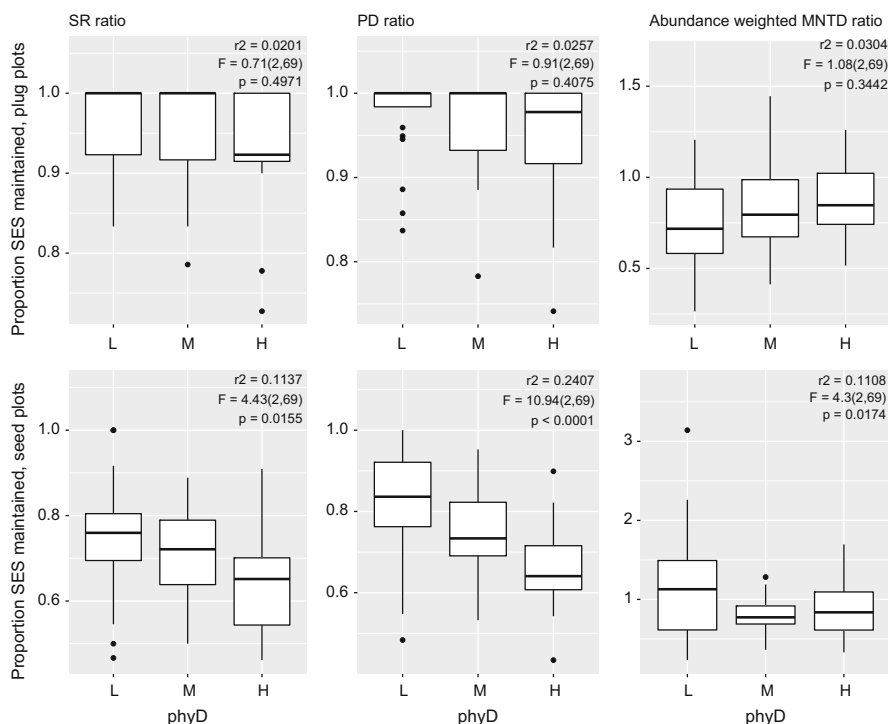


Fig. 10.6 Ratio of observed to planted species richness, Faith's PD, and mean nearest taxon distance (MNTD) plotted as a function of designed phylogenetic diversity. Abbreviations: L, low; M, medium; H, high

though the implication of this for the net diversity effect is uncertain due to the bias we note in applying this approach to cover data.

The positive selection effect suggests that strong competitors are disproportionately affecting compositional outcomes in our experiment, which may be seen in the correlation between overall cover by species in monoculture plots and the relative cover in mixture plots (normalized by monoculture cover). Among-species variance in mixture plot relative abundance normalized by monoculture abundance is significantly correlated with coverage in the monocultures ($r = 0.3695$, $P = 0.0001$; Fig. 10.7). This is a result of the fact that numerous strong competitors (red-labeled species in Fig. 10.7) perform disproportionately well in the mixture plots, even after taking into account their high cover in the monocultures.

There is a strong negative correlation between $\ln(w)$ and proportional abundance in mixture plots relative to expectation in the monocultures ($r = 0.410$, $P < 0.001$), exemplified by the fact that most of the strongest competitors are in the Asteraceae and have low phylogenetic distinctiveness (Fig. 10.7, dot size). Oddly, despite the importance of Asteraceae in our experiment, there is no evidence of phylogenetic signal (using the K statistic of Blomberg et al. 2003) in abundance in the mixture

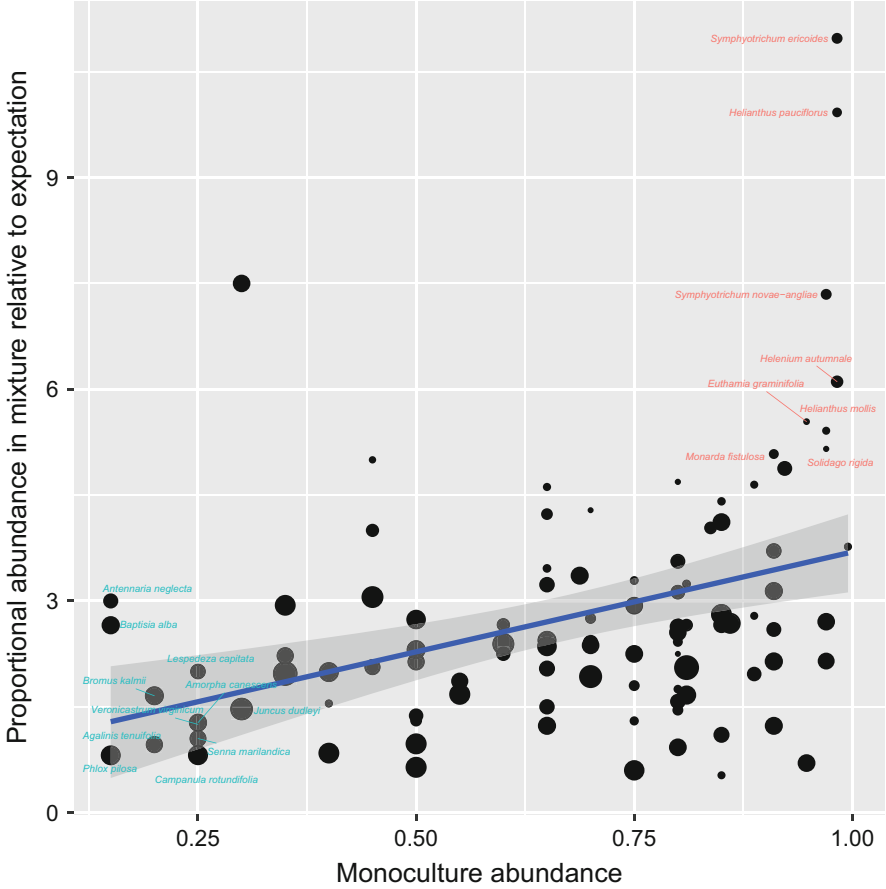


Fig. 10.7 Proportional abundance in mixture plots, normalized by abundance in monoculture plots, plotted against monoculture abundance. The y-axis is the ratio of proportional abundance in the mixture plots to expected abundance based on the monoculture; the positive correlation suggests that strong competitors (some of the most prominent are annotated in red) are disproportionately abundant in the mixture plots and may be strongly affecting experimental outcomes. This is analogous to the species selection effect estimated in the Loreau and Hector (2001) equation. Dot size is proportional to $\ln(w)$, log-transformed phylogenetic distinctiveness

plots relative to the expectations in the monocultures ($K = 0.0061$, $P = 0.2170$) and very weak phylogenetic signal in abundance in the monocultures ($K = 0.0127$, $P = 0.0010$).

However, both the measured traits after phylogeny is accounted for and functional variation captured by the phylogeny after the measured traits are accounted for are shaping year 1 maintenance of biodiversity in the experiment. When we take into account both phylogeny and traits by regressing either abundance measure on the axes of a two-dimensional nonmetric multidimensional (NMDS) trait ordination or on the axes of a two-dimensional NMDS ordination of pairwise phylogenetic

Table 10.2 Phylogenetic diversity and family-level diversity of plots in the high, medium, and low phylogenetic diversity bins included in design

	Number of families	Mean species per family	Max species per family	Phylogenetic diversity (PD)
Low PD plots	8.1 ± 1	1.9 ± 0.26	3.8 ± 0.58	1011.8 ± 30.8
Medium PD plots	9.5 ± 1.09	1.6 ± 0.21	3.3 ± 0.49	1282.9 ± 47.1
High PD plots	12.2 ± 1.11	1.2 ± 0.12	2.6 ± 0.51	1534.1 ± 50.4

distances, the first axes of both the phylogenetic and trait ordinations are important predictors of abundance ($P < 0.01$ for the first axis of each ordination, $R^2 = 0.105$ for multiple regression). Importantly, we only see this effect in multiple regressions. While the simple regression is nonsignificant for the first axis of both the trait ordination ($r^2 = 0.013$, $P = 0.248$) and the phylogenetic ordination ($r^2 = 0.032$, $P = 0.070$), the partial regression of each in a two-predictor model is strongly significant ($r^2 = 0.072$, $P = 0.006$ and $r^2 = 0.090$, $P = 0.002$, respectively, in both cases holding the other predictor fixed using partial regression). Species presence/absence in plugs and in seed plots shows no correlation with phylogeny or traits (multiple $R^2 < 0.06$, $P > 0.15$).

As suggested by the results of simple vs. multiple regression, the effect of traits on abundance in the treatment plots relative to abundance in the monocultures is not predicted by the phylogeny in a consistent manner (Table 10.2). The three most strongly conserved traits phylogenetically, for example, leaf nitrogen content, genome size, and nitrogen fixation ability, are among the least strongly correlated with abundance ($|r| < 0.5$, $P > 0.65$). Conversely, two of the best predictors of abundance—stem dry matter content and vegetative height; $r > 0.20$, $P < 0.04$)—exhibit no phylogenetic signal (Pagel’s (1997) $\lambda = 0$). In a linear model, phylogenetic signal (λ) does not predict the correlation between abundance and trait value ($r^2 = 0.0262$, $P = 0.47$). Given that some of these data are imputed, these results are tentative, pending additional trait data gathering.

Discussion

Our experiment demonstrates the potential importance of distinguishing between (1) diversity effects and clade effects and (2) functional diversity and phylogenetic diversity in both designing and analyzing biodiversity experiments. In principle, we agree with Tilman, who wrote, “To separate [the effects of changes in composition. . . from those of diversity] requires replication and randomization that assures the species compositions are unbiased by species diversity, and vice versa” (Tilman 1997 p. 185). In practice, however, it may be impossible to disentangle composition effects from diversity effects, especially when large numbers of species are involved. As our goal was to address diversity effects in a relatively realistic restoration context, inclusion of a large number of species was essential.

The universe of all possible treatments in such a study entangles phylogenetic position and phylogenetic diversity, such that taxa at all levels—species, genera, families... indeed, any clade—are apportioned non-randomly among diversity treatments. It is thus desirable to subsample selectively from the possible treatments so as to reduce the interaction between clade identity and phylogenetic diversity.

In designing our experiment, we found it impossible to altogether disentangle these effects, though we reduced them to the point that our low-diversity plots present multiple ways of being low in phylogenetic diversity. In an unconstrained experiment, the lowest diversity plots would have been dominated by Asteraceae and would have tended to cluster in phylogenetic space. We developed a visualization approach that helps to identify the degree to which the phylogenetic uniqueness (w) of taxa predicts the phylogenetic diversity (PD) of the plots in which they occur, either in the actual planting or in the universe of possible plots. We recommend this or similar kinds of visualizations to researchers undertaking such an experimental design.

The first year of our experiment has yielded at least one unexpected outcome: higher-diversity plots have suffered higher than expected diversity attrition. This negative relationship between planted diversity and maintenance of diversity is evident in plots planted by plug as well as in those planted by seed, but is it truly a diversity effect? The established way of quantifying diversity effects is to take the difference between an observed outcome and the expectation from a weighted average of the monocultures of species planted in the experiment. A positive difference indicates a positive diversity effect: species growing together outperform the expectation based on their performance alone. Maintenance of biodiversity is not readily studied in this framework, however, as loss of species can't be measured in monocultures. We are obliged instead to look at a gradient of diversity treatments to measure diversity effects. When we do so, we find that the higher-diversity plots lose more species and more phylogenetic diversity (Fig. 10.6). Thus there is a diversity effect, but species are not impacted equally: each species' coverage in mixture plots normalized by its coverage in monocultures (i.e., how it performs in a high-diversity plot relative to expectation based on monoculture performance) is a function of both traits and phylogenetic history ($R^2 = 0.105$, $P < 0.05$). Success of species in the mixture plots is determined both by the phylogenetic diversity of the plots in which species occur and the phylogenetic and trait identity of those species.

Our expectation going into this project was that increasing either trait diversity or phylogenetic diversity would positively influence survivorship and maintenance of biodiversity, among other restoration outcomes not measured in our first-year data. It seems, however, that the diversity we sampled in our species pool for this experiment varies in response traits (Lavorel and Garnier 2002) that are not uniformly phylogenetically conserved. The traits influencing species abundance in the treatment plots relative to the monocultures mostly seem to be reasonable proxies for competitiveness: presence of rhizomes ($r = 0.194$, $P = 0.047$), flowering start date ($r = 0.202$, $P = 0.039$), leaf carbon content ($r = -0.207$, $P = 0.034$), vegetative height ($r = 0.210$, $P = 0.032$), stem dry matter content ($r = 0.239$, $P = 0.014$), presence of adventitious roots ($r = 0.259$, $P = 0.008$), and leaf thickness ($r = 0.280$, $P = 0.004$). Three of these—presence of rhizomes ($\lambda = 0.238$), vegetative height,

and stem dry matter content ($\lambda = 0$)—exhibit little phylogenetic signal. These traits have perhaps evolved in parallel, and we would expect them to be under strong selection, which generally has the effect of eroding phylogenetic signal (Hansen 1997; Blomberg et al. 2003) except when species closely track an optimum predicted by a selective regime that itself exhibits phylogenetic signal (Hansen et al. 2008). Very possibly too, they would exhibit higher phylogenetic signal in the context of a more densely sampled phylogeny, one in which we had not handpicked species from the broader phylogeny based on their suitability to mesic prairies of northern Illinois.

The combined influence of both convergent (homoplastic) and conserved traits thus appears to produce the effect we observe: neither phylogeny nor traits alone influence species performance in the treatment plots relative to the monocultures, but in combination, both do. Moreover, our low-diversity plots sample disproportionately from clades that exhibit highest species richness in the tallgrass prairie: these species may simply be the best prairie specialists and the most effective at surviving in a young prairie restoration. The experiment consequently demonstrates that phylogenetic diversity alone is unlikely to drive outcomes in a complex community diversity experiment. Rather, species and clade identity drive many of the outcomes we observe, and these are necessarily entangled with phylogenetic diversity in our experiment as in real communities. From a plant's perspective, it is both who you are and who you are growing with that determine how well you perform.

Conclusions

Our experiment has gone through only one season and is not yet fully established. Our initial finding that higher phylogenetic diversity plots suffer higher rates of species attrition has yet to be probed to determine whether this is a transitory (early in the experiment) phenomenon; an artifact of our resampling effort in August, which excluded the spring ephemerals and species that had not fully established in the first year; or a robust observation. Moreover, this finding needs to be compared with productivity and invasion resistance data that will be available a year hence to determine whether diversity effects on cover and biomass will be dominated by the most competitive species in the long run.

Our results may point to a reason that restorations have, on average, lower phylogenetic diversity than remnants (Barak et al. 2017): the species that perform best in young restoration plantings (and at 1 year old, our experiment is certainly that) draw disproportionately from a limited number of angiosperm clades. In the long run, and in prairies larger than our 2 m $\times$ 2 m plots, more phylogenetically distinctive species may be able to find room to establish and persist. In the meantime, our results suggest that the long view of evolutionary history (Barak et al. 2015) offers restoration practitioners a perspective on species that goes beyond the traits and life history attributes we have measured to date. Phylogeny and traits of species go hand-in-hand with phylogenetic and trait diversity of plots to predict the outcomes of at least young prairie restorations.

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Index

A

ADEPD, 79, 80, 82
 Agulhas Plain, 172
 Alzheimer's disease (AD), 14
 APG classification of plants, 42
 Amphibians, 157, 160
 Angiosperm phylogeny, 96, 170
 Autism, 14

B

Barro Colorado Island (BCI), 95
 Binary search, 126
 Biodiversity, 134–138
 data problem, 133, 134
 georeferenced data, 135
 global change, 132, 133, 140, 141
 KBAs, 133
 paleo- and neo-endemism, 139, 140
 PD, 134
 phylogeny and data availability
 Chile vs. Californi, 137, 138
 DNA sequences, 134
 GBIF, 135, 136
 internet-based, 134
 Biogeographic island, 137
 Blomberg's *K*, 102, 104
Bradypodion, 178
 Brownian motion model, 97, 195

C

California, 137–139
 Cape clades, 173, 175

Cape Floristic Region (CFR), South Africa,
 170–176
 animals PD, 178
 boundaries, 169
 defined phytogeographic zones, 180, 181
 ecological and environmental processes,
 169
 endemism, 168
 Fynbos, 168
 GCFR, 168
 plant PD
 Agulhas Plain, 172
 biomes, 170
 Cape clades, 175
 climatic fluctuations, 173
 core winter-rainfall, 173
 eastern region, 171
 Madagascar, 174
 overdispersed nature, 171
 PE, 173
 QDS, 171, 176
 species distribution modelling, 172–174
 species richness, 176
 Thicket, 171
 topographical pattern, 176
 surrogates, 179, 180
 Cape montane regions, 177
 Chile, 8
 vs. Californi, 137, 138
 endemism, 140
 Cladograms, 5
 Climate change, 175
 Clustering techniques, 180
 Code profiling, 126

Code benchmarking, 125
 Complementarity, 5, 35, 76, 77, 146
 Conservation biology, 28, 94, 98, 104, 107
 characterized, 28
 EDGE (*see* Evolutionary Distinct Globally
 Endangered (EDGE))
 Conservation planning, 146, 147, 179, 181
 Conservation status uncertainty, 32
 Cost-benefit analysis, 28

D

Data availability
 dates, 49
 identifiers, 47
 names, 46
 outreach, 47, 48
 phylogenies, 48
 robust mapping, 48
 synthetic tree, 49
 taxonomy, 48
 TNRS, 47
 DeepTree, 56
 Defined phylogeographic zones, 180, 181
 Diet and microbial PD, 16
 Distributional data
 effects, 113
 georeferencing errors, 113
 impacts, 112
 misidentification, 113
 sources, 112
 types, 113
 Diversity attrition, 206
 DNA
 barcode data, 96
 extraction, 134
 sequences, 134

E

Ecological and environmental processes, 169
 Ecological integrity, 11
 Ecological opportunity, 176
 Ecological restoration, 190
 Ecophylogenetics, 18
 Elevation gradient, 176
 Environmental niche model (ENM), 112
 advantages, 118
 biodiversity measures, 117
 geographic distribution, 116
 issues of aggregation, 118, 119
 MAUP, 118
 potential complexity, 116

 quality control, 117
 as SDM, 116
 Evolutionary Distinct Globally Endangered
 (EDGE), 32–35
 applications to birds, 30
 complementarity, 35
 conservation prioritization strategies, 37
 EDGE-like metrics, 31
 Existence program, 30, 31
 scaling, 36
 score calculation, 29
 uncertainty
 complementarity, 35
 extinction risk, 33, 34
 phylogenetic, 32, 33
 practical choice, 34
 randomizations, 32
 Evolutionary distinctiveness (ED), 102,
 103, 106
 Evolutionary distinctiveness index
 (EDt), 120
 Evolutionary history, 42, 46, 49, 94, 132, 137,
 146, 148, 160, 195
 Evolutionary patterns, 170, 181
 Evosystem, 133
 “Expected diversity” approaches, 75
 Experimental methods
 ANOVA, 201
 Asteraceae, 203, 206
 blocks and superblocks, 199, 200
 diversity and clade effects, 205, 206
 diversity attrition, 206
 family-level diversity, 205
 functional variation, 204
 MNTD, 201–203
 monoculture plots, 199, 200
 monocultures, 204, 206
 MPD, 201
 NMDS, 204
 PD, 201, 205
 phylogenetic distinctiveness, 194
 phylogenetic signal, 207
 seed plots, 201
 selection model, 202, 203
 SES, 201
 simple vs. multiple regression, 205
 simulated plots, 198, 199
 site preparation and installation, 199
 SR, 201
 survey, 199
 survivorship, 200, 206
 trait diversity, 201
 Extinction probabilities, 147

F

Family-level diversity, 205

Feature diversity, 5, 6

Fractals, 55

 DeepTree, 55

 dimension, 55

 Lifemap, 55

 OneZoom, 55

 panning and zooming, 53, 54

 testing self-similarity, 56

Frankenstein-esque tree of trees, 96

Functional diversity (FD), 2, 195

 life form, 198

 nitrogen fixation, 198

 R package, 197

 reproductive traits, 196

Fynbos, 179

G

Generalized least squares, 194, 195

Geographic isolation, 137

Global Assessment of Reptile Distribution
(GARD), 162

Global Biodiversity Information Facility
(GBIF), 135, 136

Global change, 132, 133, 140, 141

Greater Cape Floristic Region (GCFR), 168,
169, 180

Gut microbiomes, 15

H

Homoplasy, 6

Human inflammatory bowel disease
(IBD), 12, 13

I

Indices, 17, 19

Insurance value

 biotic diversity, 9, 10

 ecological factors, 10

 IUCN, 9

 precautionary principle, 10

 soil biodiversity, 10

 supply vs. demand, 9

Interactive Fractal Inspired Graphs, 54

Intergovernmental Science-Policy Platform on
Biodiversity and Ecosystem Services
(IPBES), 8, 74

International Union for Conservation of Nature
and Natural Resources (IUCN)

EDGE programme, 76

 option values, 75

 taxonomic distinctiveness, 75

Iterative-HEDGE (I-HEDGE), 35

K

Key biodiversity areas (KBAs), 147

 definition, 133

 PD, 147

L

Last common ancestor (LCA), 125, 126

Levyns' Law, 175

 age and area hypothesis, 176

 ecological opportunity, 176

 metabolic theory, 175

Longitudinal gradient, 176

M

Macrobial community

 convex hull measure, 20

 feature diversity, 18

 functional trait recovery, 19, 20

 phylogenetic measures, 18, 19

Madagascar, 174

Malagasy legumes, 175

Map, 62, 63

 Ancestor's Trail, 44

 GE scores, 36

 panning and zooming approach, 53

 PD endemism, 8

 projection, 56

Maputaland-Pondoland-Albany hotspot, 178

Mammals

 EDGE, 29, 31

 Mediterranean, 154

 PD calculations, 4

 spatial distribution, 155

 terrestrial, 155

Maze of life' puzzle, 51

Mean nearest phylogenetic neighbor distance
(MNTD), 100, 101

Mean nearest taxon distance (MNTD), 201, 203

Mean pairwise phylogenetic distance (MPD),
102, 192, 201

Mediterranean Basin, 150–160, 162

 amphibians and squamates, 160

 climatic condition, 148

 conservation strategies, 160, 161

 economic development, 148

Mediterranean Basin (*cont.*)

- evolutionary processes, 160
- KBAs, 148, 160
- mammals, 160
- option values, 161
- phylogenetic complementarity, 161
- phylogenetic data, 148
- priority sites
 - amphibian, 157
 - iterative approach, 150
 - KBA score, 160
 - mammal, 155
 - maximise mammal PDgain, 154
 - PD gain, 152–154, 156, 158
 - as set, 162
 - squamate, 156, 160
 - taxon, 157
 - ThPDgain, 152, 156, 158, 159, 162
- probability of extinctions, 163
- probability of survival, 149
- protected areas, 149
 - categories I, II and IV, 155
- protectedPD
 - probability of survival, 150
 - proportion measurement, 151
 - ThPD, 150

Mesquite, 108

Metabolic theory, 175

Metrics, evolutionary isolation, 28

Microbial communities

- AD, 14
- autism, 14
- diet, 16
- gut microbiome, 15
- IBD, 12, 13
- PD calculus, 17

Modifiable areal unit problem (MAUP), 118

Multiple imputation based on chained equations (MICE), 197

N

Nature's contributions to people (NCP), 74, 86, 87

Near Relatedness Index (NRI), 178

Noah's Ark Problem (NAP), 76

Nonmetric multidimensional (NMDS), 204

Null model analysis, 97

O

Occurrence data, 114

OneZoom, 56

- aesthetic appeal, 64
- botanical tree, 64
- fractals, 55
- identifiers, 48
- 'natural' projection model, 56, 58
- node dates, 49
- 'polytomy' projection, 56
- 'Wild Planet' gallery, 65

Operational taxonomic units (OTUs), 138

- Option values, 74
 - biotic diversity, 7
 - complementarity, 9
 - conservation groups, 8
 - and feature diversity, 6, 8
 - global biodiversity, 7, 8
 - IUCN, 7
 - Mediterranean Basin, 161
 - PD calculus, 8
 - species richness, 146

P

Pagel's lambda, 105

Paleo- and neo-endemism, 139, 140

Permutation probability test (PTP), 5, 6

Phylobetadiversity, 182

Phylodiversity indices, 112

- processing speeds, 124–126

Phylogenetic analyses/approach

- advantages, 43
- conservation biology, 105
- ecology, 96
- predictions of traits, 42

Phylogenetic beta diversity approach, 180

Phylogenetic context, 109

Phylogenetic distinctiveness, 192, 194, 195

- Asteraceae, 192, 196
- generalized least squares
- Brownian motion process, 195
- trait evolution, 194
- PD plot calculation, 192
- phylogenetic covariance matrix, 194
- plot constraints, 197
- Ranunculaceae, 196
- sampling strategies, 196
- tree of life, 193, 194

Phylogenetic diversity (PD), 122, 124

- ADEPD, 79
- approximations for priorities, 84, 85
- branches, 119
- branches contribution, 121
- clade contribution, 121
- complementarity, 76, 77

- definition, 2
 - ecological similarity, 95
 - EDGE, 76, 80
 - “ED” measure, 76
 - expected distinctiveness, 77
 - expected PD, 78
 - expected PD loss, 80, 81
 - floristic province, California, 138, 139
 - gain, 147
 - grid size, 138, 139
 - HEDGE, 77, 79
 - hypothetical probabilities of extinction, 83
 - IUCN Red List, 74, 75
 - microbes, 74
 - NAP, 76
 - NCP18, 87, 88
 - option value, 74
 - phylogenetic relatedness, 95
 - probabilities of extinction, 80
 - properties, 119
 - ranges, 119, 120
 - set of priority species, 82–84
 - species richness values, 101
 - standardising, 121
 - taxonomic change, 133
 - total expected PD loss, 80, 81, 86
 - transforming results, 121, 122
 - tree plots, 122, 123
 - vascular flora, California, 137, 139
 - Phylogenetic diversity at risk, 152, 162
 - Phylogenetic endemism (PE), 119, 172, 173, 181
 - PD, 4
 - property, 120
 - Phylogenetic identity, 206
 - Phylogenetic key biodiversity area (“PDKBA”), 81
 - Phylogenetic rarity (PR), 119
 - Phylogenetic resolution and metrics, 95, 97
 - barriers
 - BCI forest, 95
 - cataloging of species, 95
 - constructing phylogenetic tree, 95
 - K statistic, 97
 - type I error rate, 97
 - Blomberg’s *K*, 106
 - measures of signal, 107, 108
 - NCBI database, 96
 - null model, 106
 - Pagel’s lambda, 106
 - polytomies, 96, 106–107
 - pragmatic approach, 109
 - pragmatic view recognized, 108
 - sequencing data, 108
 - simulation study, 99–104
 - soft polytomies, 97, 98, 105
 - Phylogenetic signal
 - Blomberg’s *K*, 104
 - Pagel’s lambda, 105
 - Phylogeny, 28
 - Phylomatic algorithm, 96
 - Phytogeographical zones, 180–181
 - Pragmatic view recognized, 108
 - Process-based model, 4
 - Projection
 - children of nodes, 58
 - DeepTree, 56
 - geographic maps, 56
 - OneZoom, 56
 - species richness, 59
 - zoomable canvas, 56, 60
- Q**
- Quarter-degree square (QDS)
 - scale, 171–173, 176
- R**
- Range polygons, 115, 116
 - Reptiles, 178
 - “Representativeness” problem, 147
 - Restoration ecology
 - environmental variation, 190
 - resilience and function, 190
 - tallgrass prairie, 190 (*see also* Tallgrass prairie restoration)
 - “Robust proxy”, 8
- S**
- Safe operating space, 132
 - Scales of application, 2
 - Scaling, 36
 - Science communication, 63, 64, 66
 - Selection vs. complementarity effects, 202
 - Selection model, 202, 207
 - Simple vs. multiple regression, 205
 - Spatial phylogenetics, 114, 115
 - data, 112
 - distributional data, 112
 - diversity, 112
 - ENM, 116
 - observational data
 - coordinate systems, 115
 - geographic accuracy, 114

Spatial phylogenetics (*cont.*)

- georeferencing errors, 115
- occurrence data, 114
- occurrence data issues, 114
- plot data, 114
- taxon identification, 114
- types, 114
- organismal groups, 111
- range polygons, 115, 116
- Species-based conservation, 28
- Species distribution models (SDM), 116, 134, 173, 174
- Species prioritization, 3, 7, 28
- Species richness
 - conservation, 146
 - option values, 146
 - regional planning, 146
- Squamates, 148, 149, 154–158, 160–162
- Standardized effect size (SES), 201
- Surrogates, 179, 180
- Survivorship, 200, 206

T

- Tallgrass prairie restoration, 199
 - Asteraceae, 192
 - experiment (*see* Experimental methods)
 - MPD, 192
 - phylogenetic distinctiveness, 192
 - restoration planning and design, 190
 - spatial scales, 191
 - species richness, 191
 - tree of life and phylogeographic, 191
- Taxa and ecosystems, 132
- Taxon delimitation, 134
- Taxonomic dispersion, 178
- Taxonomic distinctiveness
 - biodiversity's value, 2
 - and complementarity, 3, 4
 - IUCN, 3, 4
 - PD calculations, 4
- Taxonomic Name Resolution Services (TNRS), 47
- Tessellations, 117, 118
- Topographical pattern, 176
- Total expected PD loss, 81, 86
- Tree of life, 46–50
 - citizen science platforms, 67
 - complete tree usage, 44, 45
 - conservation, 45, 46
 - conveying knowledge, 43
 - data availability and curation dates, 49

- identifiers, 47
- names, 46
- outreach, 47, 48
- phylogenies, 48
- robust mapping, 48
- synthetic tree, 49
- taxonomy, 48
- TNRS, 47

- evidence for evolution, 66
- fractal projections, 57, 66
- 'official' registry, 68
- OneZoom, 65
- phylogeny, 42, 43
- sustainability, 67
- visualisation, 43, 45 (*see* Visualizing phylogenetic tree)
- Tree of Life Web Project (ToLweb), 48, 50
- Trichoplax adhaerens*, 59
- Type 1 error rate, 6

U

- Uncertainty
 - extinction risk, 33, 34
 - phylogenetic, 32, 33
 - practical choice, 34
 - randomizations, 32
- "Unique fraction", 12

V

- Vascular flora, 8, 122, 137, 139
- Vertebrates, 179
- Visualization approach, 206
- Visualizing phylogenetic tree, 50–53, 56, 58–60, 62
 - aesthetic appeal, 63, 64
 - DensiTree, 50
 - existing methods
 - 2D representations, 52
 - 3D representations, 51
 - principles, 52
 - two-dimensional representations, 50
- fractals, 53, 55, 56
- issues, 50
- polytomies, 57
- projection
 - children of nodes, 58
 - DeepTree, 56
 - geographic maps, 56
 - OneZoom, 56
 - species richness, 59
 - zoomable canvas, 56, 60

- ToLweb, 50
- topology
 - vs. species richness, 59, 60
 - vs. time, 62
- very large trees
 - focus and context approach, 53
 - large media approach, 53

- panning and zooming approach, 53
- simplification approach, 53

W

- Wallacean shortfall, 133
- World flora, 134, 135